



INDIANA DEPARTMENT OF ENVIRONMENTAL MANAGEMENT

We Protect Hoosiers and Our Environment.

100 N. Senate Avenue • Indianapolis, IN 46204
(800) 451-6027 • (317) 232-8603 • www.idem.IN.gov

Eric J. Holcomb
Governor

Bruno L. Pigott
Commissioner

NOTICE OF 30-DAY PERIOD FOR PUBLIC COMMENT

Preliminary Findings Regarding a Significant Modification to a Part 70 Operating Permit
for Louis Dreyfus Company Agricultural Industries LLC in Kosciusko County

Significant Source Modification No.: 085-39532-00102

Significant Permit Modification No.: 085-39729-00102

The Indiana Department of Environmental Management (IDEM) has received an application from Louis Dreyfus Company Agricultural Industries LLC, located at 7344 State Road 15 South, Claypool, Indiana, for a significant modification of its Part 70 Operating Permit issued on January 5, 2017. If approved by IDEM's Office of Air Quality (OAQ), this proposed modification would allow Louis Dreyfus Company Agricultural Industries LLC to make certain changes at its existing source. Louis Dreyfus Company Agricultural Industries LLC has applied to modify the permit as follows:

- (a) to establish a 326 IAC 8-1-6 VOC BACT limit for the Biodiesel Manufacturing Operation which uses soybean oil to produce a biodiesel fuel.
- (b) Clarify the applicability of NESHAP Subpart YYYY for the existing gas turbine, identified as GT-1. This unit was originally constructed in 1999 in New Jersey and was then relocated to its current location in 2016. Due to this construction year, it is now classified as an existing stationary combustion turbine under Subpart YYYY.

This draft Significant Permit Modification does not contain any new equipment that would emit air pollutants; however, some conditions from previously issued permits/approvals have been corrected, changed, or removed. These corrections, changes, and removals may include Title I changes (e.g., changes that add or modify synthetic minor emission limits). This notice fulfills the public notice procedures to which those conditions are subject. IDEM has reviewed this application and has developed preliminary findings, consisting of a draft permit and several supporting documents, which would allow for these changes.

A copy of the permit application and IDEM's preliminary findings are available at:

Warsaw Community Public Library
310 East Main Street
Warsaw, IN 46580

and

IDEM Northern Regional Office
300 North Dr. Martin Luther King Jr. Boulevard, Suite 450
South Bend, IN 46601-1295

A copy of the preliminary findings is available on the Internet at: <http://www.in.gov/ai/appfiles/idem-caats/>.

A copy of the preliminary findings is also available via IDEM's Virtual File Cabinet (VFC.) Please go to: <http://www.in.gov/idem/> and enter VFC in the search box. You will then have the option to search for permit documents using a variety of criteria.

How can you participate in this process?

The date that this notice is published in a newspaper marks the beginning of a 30-day public comment period. If the 30th day of the comment period falls on a day when IDEM offices are closed for business, all comments must be postmarked or delivered in person on the next business day that IDEM is open.

You may request that IDEM hold a public hearing about this draft permit. If adverse comments concerning the **air pollution impact** of this draft permit are received, with a request for a public hearing, IDEM will decide whether or not to hold a public hearing. IDEM could also decide to hold a public meeting instead of, or in addition to, a public hearing. If a public hearing or meeting is held, IDEM will make a separate announcement of the date, time, and location of that hearing or meeting. At a hearing, you would have an opportunity to submit written comments and make verbal comments. At a meeting, you would have an opportunity to submit written comments, ask questions, and discuss any air pollution concerns with IDEM staff.

Comments and supporting documentation, or a request for a public hearing should be sent in writing to IDEM at the address below. If you comment via e-mail, please include your full U.S. mailing address so that you can be added to IDEM's mailing list to receive notice of future action related to this permit. If you do not want to comment at this time, but would like to receive notice of future action related to this permit application, please contact IDEM at the address below. Please refer to permit number SSM 085-39532-00102 and SPM 085-39729-00102 in all correspondence.

Comments should be sent to:

Daniel W. Pell
IDEM, Office of Air Quality
100 North Senate Avenue
MC 61-53 IGCN 1003
Indianapolis, Indiana 46204-2251
(800) 451-6027, ask for Daniel W. Pell or (317) 234-8532
Or dial directly: (317) 234-8532
Fax: (317) 232-6749 attn: Daniel W. Pell
E-mail: dpell@idem.IN.gov

All comments will be considered by IDEM when we make a decision to issue or deny the permit. Comments that are most likely to affect final permit decisions are those based on the rules and laws governing this permitting process (326 IAC 2), air quality issues, and technical issues. IDEM does not have legal authority to regulate zoning, odor, or noise. For such issues, please contact your local officials.

For additional information about air permits and how the public and interested parties can participate, refer to the IDEM Air Permits page on the Internet at: <http://www.in.gov/idem/airquality/2356.htm>; and the Citizens' Guide to IDEM on the Internet at: <http://www.in.gov/idem/6900.htm>.

Air Permit Legal Notices

On November 14, 2018, the State of Indiana Environmental Rules Board adopted rule amendments to 326 IAC 2-1.1-6, 326 IAC 2-7-13, 326 IAC 2-7-17, 326 IAC 2-8-13, 326 IAC 2-8-18, and 326 IAC 2-12-1 (LSA #17-395), concerning legal notice provisions for air permits issued under the NSR and Title V permit programs and other air permits for which newspaper notices are published by IDEM OAQ. The adopted rule amendments require that IDEM OAQ provide electronic public notices on IDEM's website as the primary and consistent method for communicating air permit notices to the public. IDEM anticipates that the final (effective) rule amendments will be promulgated on or about March 14, 2019. The status of these rule amendments (LSA #17-395) and the final effective date will be posted on the following website: <https://www.in.gov/idem/legal/2351.htm>.

Until the rule amendments to 326 IAC 2-1.1-6, 326 IAC 2-7-13, 326 IAC 2-7-17, 326 IAC 2-8-13, 326 IAC 2-8-18, and 326 IAC 2-12-1 are promulgated final (effective), IDEM OAQ will publish both newspaper public notices and electronic public notices on IDEM's website. Once the rule amendments are

promulgated final (effective), IDEM OAQ will no longer publish newspaper public notices and will only publish electronic public notices on IDEM's website.

Electronic public notices, including permitting, rulemaking, meeting, and hearing notices, are posted on IDEM's website at: <https://www.in.gov/idem/5474.htm>. Public notices posted on IDEM's webpage will be accessible for the duration of the public comment period.

IDEM OAQ provides alternative methods for receiving public notices, such as the interested parties mailing list. The IDEM OAQ interested parties mailing list consists of people who have asked to be notified by email list or direct mail delivery of air permit actions related to a specific source or multiple sources, or for all air permit actions in a certain county or multiple counties. If you would like to be added to the IDEM OAQ interested parties mailing list, call Patty Pear at (317) 233-6875 or call (800) 451-6027, select option 4, and ask for the "Permits Administration Section".

Citizens and interested parties can also subscribe to IDEM's regional public notice pages and receive an e-mail or text message to your phone every time IDEM adds information to a subscribed region at the following website: https://public.govdelivery.com/accounts/INDEM/subscriber/new?gsp=INDEM_3

What will happen after IDEM makes a decision?

Following the end of the public comment period, IDEM will issue a Notice of Decision stating whether the permit has been issued or denied. If the permit is issued, it may be different than the draft permit because of comments that were received during the public comment period. If comments are received during the public notice period, the final decision will include a document that summarizes the comments and IDEM's response to those comments. If you have submitted comments or have asked to be added to the mailing list, you will receive a Notice of the Decision. The notice will provide details on how you may appeal IDEM's decision, if you disagree with that decision. The final decision will also be available on the Internet at the address indicated above, at the local library indicated above, at the IDEM Regional Office indicated above, and the IDEM public file room on the 12th floor of the Indiana Government Center North, 100 N. Senate Avenue, Indianapolis, Indiana 46204-2251.

If you have any questions, please contact Daniel W. Pell of my staff at the above address.



Heath Hartley, Section Chief
Permits Branch
Office of Air Quality



INDIANA DEPARTMENT OF ENVIRONMENTAL MANAGEMENT

We Protect Hoosiers and Our Environment.

100 N. Senate Avenue • Indianapolis, IN 46204

(800) 451-6027 • (317) 232-8603 • www.idem.IN.gov

Eric J. Holcomb
Governor

Bruno L. Pigott
Commissioner

DRAFT

Mr. David Selig
Louis Dreyfus Company Agricultural Industries, LLC
7344 State Road 15 South,
Claypool, Indiana 46510

Re: 085-39729-00102
Significant Permit Modification

Dear Mr. Selig:

Louis Dreyfus Company Agricultural Industries, LLC was issued Part 70 Operating Permit Renewal No. T085-36889-00102 on January 5, 2017 for a stationary refined bleached soybean oil (RB Oil), soybean salad oil, soybean meal, and biodiesel manufacturing plant located at 7344 State Road 15 South, Claypool, Indiana. An application requesting changes to this permit was received on January 23, 2018. Pursuant to the provisions of 326 IAC 2-7-12, a Significant Permit Modification to this permit is hereby approved as described in the attached Technical Support Document.

Please find attached the entire Part 70 Operating Permit as modified. The permit references the below listed attachments. Since these attachments have been provided in previously issued approvals for this source, IDEM OAQ has not included a copy of these attachments with this modification:

- Attachment A: 40 CFR Part 60.40b, Subpart Db, NSPS for Industrial-Commercial-Institutional Steam Generating Units
- Attachment B: 40 CFR Part 60.300, Subpart DD, NSPS for Grain Elevators
- Attachment C: 40 CFR Part 60.480, Subpart VV - NSPS for Equipment Leaks of VOC in the Synthetic Organic Chemicals Manufacturing Industry for which Construction, Reconstruction, or Modification Commenced After January 5, 1981, and on or Before November 7, 2006.
- Attachment D: 40 CFR Part 60.660, Subpart NNN - NSPS for Volatile Organic Compound (VOC) Emissions from Synthetic Organic Chemical Manufacturing Industry (SOCMI) Distillation Operations.
- Attachment E: 40 CFR Part 60.700, Subpart RRR - NSPS for Volatile Organic Compounds Emissions from Synthetic Organic Chemical Manufacturing Industry (SOCMI) Reactor Processes.
- Attachment F: 40 CFR Part 60.4200, Subpart IIII - NSPS for Stationary Compression Ignition Internal Combustion Engines
- Attachment G: 40 CFR Part 63.2430, Subpart FFFF - NESHAP: Miscellaneous Organic Chemical Manufacturing
- Attachment H: 40 CFR Part 63.2830, Subpart GGGG - NESHAP: Solvent Extraction for Vegetable Oil Production
- Attachment I: 40 CFR 63.6580, Subpart ZZZZ - NESHAP for Stationary Reciprocating Internal

DRAFT

Combustion Engines

- Attachment J: 40 CFR 63.7480, Subpart DDDDD - NESHAP for Industrial, Commercial, and Institutional Boilers and Process Heaters
- Attachment K: 40 CFR Part 60.480a, Subpart VVa, NSPS for Equipment Leaks for VOC in the Synthetic Organic Chemicals Manufacturing Industry for which Construction, Reconstruction or Modification Commenced after November 7, 2006
- Attachment L: 40 CFR Part 60.330, Subpart GG, NSPS for Stationary Gas Turbines
- Attachment M: 40 CFR Part 63.6080, Subpart YYYY, NESHAP for Stationary Combustion Turbines

Previously issued approvals for this source containing these attachments are available on the Internet at: <http://www.in.gov/ai/appfiles/idem-caats/>.

Previously issued approvals for this source are also available via IDEM's Virtual File Cabinet (VFC.) Please go to: <http://www.in.gov/idem/> and enter VFC in the search box. You will then have the option to search for permit documents using a variety of criteria.

Federal rules under Title 40 of United States Code of Federal Regulations may also be found on the U.S. Government Printing Office's Electronic Code of Federal Regulations (eCFR) website, located on the Internet at: http://www.ecfr.gov/cgi-bin/text-idx?tpl=/ecfrbrowse/Title40/40tab_02.tpl.

A copy of the permit is available on the Internet at: <http://www.in.gov/ai/appfiles/idem-caats/>. A copy of the permit is also available via IDEM's Virtual File Cabinet (VFC.) Please go to: <http://www.in.gov/idem/> and enter VFC in the search box. You will then have the option to search for permit documents using a variety of criteria. For additional information about air permits and how the public and interested parties can participate, refer to the IDEM Air Permits page on the Internet at: <http://www.in.gov/idem/airquality/2356.htm>; and the Citizens' Guide to IDEM on the Internet at: <http://www.in.gov/idem/6900.htm>.

This decision is subject to the Indiana Administrative Orders and Procedures Act - IC 4-21.5-3-5.

If you have any questions regarding this matter, please contact Daniel W. Pell, Indiana Department Environmental Management, Office of Air Quality, Permits Branch, 100 North Senate Avenue, MC 61-53 IGCN 1003, Indianapolis, Indiana 46204-2251, or by telephone at (317) 234-8532 or (800) 451-6027, and ask for Daniel W. Pell or (317) 234-8532.

Sincerely,

Heath Hartley, Section Chief
Permits Branch
Office of Air Quality

Attachments: Modified Permit and Technical Support Document

cc: File - Kosciusko County
Kosciusko County Health Department
U.S. EPA, Region 5
Compliance and Enforcement Branch
IDEM Northern Regional Office



INDIANA DEPARTMENT OF ENVIRONMENTAL MANAGEMENT

We Protect Hoosiers and Our Environment.

100 N. Senate Avenue • Indianapolis, IN 46204

(800) 451-6027 • (317) 232-8603 • www.idem.IN.gov

Eric J. Holcomb
Governor

Bruno L. Pigott
Commissioner

DRAFT

Part 70 Operating Permit (Renewal) OFFICE OF AIR QUALITY

**Louis Dreyfus Company Agricultural Industries, LLC
7344 State Road 15 South
Claypool, Indiana 46510**

(herein known as the Permittee) is hereby authorized to operate subject to the conditions contained herein, the source described in Section A (Source Summary) of this permit.

The Permittee must comply with all conditions of this permit. Noncompliance with any provisions of this permit is grounds for enforcement action; permit termination, revocation and reissuance, or modification; or denial of a permit renewal application. Noncompliance with any provision of this permit, except any provision specifically designated as not federally enforceable, constitutes a violation of the Clean Air Act. It shall not be a defense for the Permittee in an enforcement action that it would have been necessary to halt or reduce the permitted activity in order to maintain compliance with the conditions of this permit. An emergency does constitute an affirmative defense in an enforcement action provided the Permittee complies with the applicable requirements set forth in Section B, Emergency Provisions.

This permit is issued in accordance with 326 IAC 2 and 40 CFR Part 70 Appendix A and contains the conditions and provisions specified in 326 IAC 2-7 as required by 42 U.S.C. 7401, et. seq. (Clean Air Act as amended by the 1990 Clean Air Act Amendments), 40 CFR Part 70.6, IC 13-15 and IC 13-17.

Operation Permit No.: T085-36889-00102	
Master Agency Interest ID.: 14308	
Issued by: Original Signed Jason R. Krawczyk, Section Chief Permits Branch, Office of Air Quality	Issuance Date: January 5, 2017 Expiration Date: January 5, 2022

First Significant Source Modification No. 085-37444-00102, issued on December 15, 2016.

First Administrative Amendment No. 085-38449-00102, issued on June 16, 2017.

First Significant Permit Modification No.: 085-39729-00102	
Issued by: Heath Hartley, Section Chief Permits Branch Office of Air Quality	Issuance Date: Expiration Date: January 5, 2022

DRAFT

TABLE OF CONTENTS

SECTION A	SOURCE SUMMARY	9
A.1	General Information [326 IAC 2-7-4(c)][326 IAC 2-7-5(14)][326 IAC 2-7-1(22)]	
A.2	Emission Units and Pollution Control Equipment Summary [326 IAC 2-7-4(c)(3)][326 IAC 2-7-5(14)]	
A.3	Specifically Regulated Insignificant Activities [326 IAC 2-7-1(21)][326 IAC 2-7-4(c)][326 IAC 2-7-5(14)]	
A.4	Part 70 Permit Applicability [326 IAC 2-7-2]	
SECTION B	GENERAL CONDITIONS	23
B.1	Definitions [326 IAC 2-7-1]	
B.2	Permit Term [326 IAC 2-7-5(2)][326 IAC 2-1.1-9.5][326 IAC 2-7-4(a)(1)(D)][IC 13-15-3-6(a)]	
B.3	Term of Conditions [326 IAC 2-1.1-9.5]	
B.4	Enforceability [326 IAC 2-7-7][IC 13-17-12]	
B.5	Severability [326 IAC 2-7-5(5)]	
B.6	Property Rights or Exclusive Privilege [326 IAC 2-7-5(6)(D)]	
B.7	Duty to Provide Information [326 IAC 2-7-5(6)(E)]	
B.8	Certification [326 IAC 2-7-4(f)][326 IAC 2-7-6(1)][326 IAC 2-7-5(3)(C)]	
B.9	Annual Compliance Certification [326 IAC 2-7-6(5)]	
B.10	Preventive Maintenance Plan [326 IAC 2-7-5(12)][326 IAC 1-6-3]	
B.11	Emergency Provisions [326 IAC 2-7-16]	
B.12	Permit Shield [326 IAC 2-7-15][326 IAC 2-7-20][326 IAC 2-7-12]	
B.13	Prior Permits Superseded [326 IAC 2-1.1-9.5][326 IAC 2-7-10.5]	
B.14	Termination of Right to Operate [326 IAC 2-7-10][326 IAC 2-7-4(a)]	
B.15	Permit Modification, Reopening, Revocation and Reissuance, or Termination [326 IAC 2-7-5(6)(C)][326 IAC 2-7-8(a)][326 IAC 2-7-9]	
B.16	Permit Renewal [326 IAC 2-7-3][326 IAC 2-7-4][326 IAC 2-7-8(e)]	
B.17	Permit Amendment or Modification [326 IAC 2-7-11][326 IAC 2-7-12]	
B.18	Permit Revision Under Economic Incentives and Other Programs [326 IAC 2-7-5(8)][326 IAC 2-7-12(b)(2)]	
B.19	Operational Flexibility [326 IAC 2-7-20][326 IAC 2-7-10.5]	
B.20	Source Modification Requirement [326 IAC 2-7-10.5]	
B.21	Inspection and Entry [326 IAC 2-7-6][IC 13-14-2-2][IC 13-30-3-1][IC 13-17-3-2]	
B.22	Transfer of Ownership or Operational Control [326 IAC 2-7-11]	
B.23	Annual Fee Payment [326 IAC 2-7-19][326 IAC 2-7-5(7)][326 IAC 2-1.1-7]	
B.24	Credible Evidence [326 IAC 2-7-5(3)][326 IAC 2-7-6][62 FR 8314][326 IAC 1-1-6]	
SECTION C	SOURCE OPERATION CONDITIONS.....	34
	Emission Limitations and Standards [326 IAC 2-7-5(1)]	34
C.1	Particulate Emission Limitations For Processes with Process Weight Rates Less Than One Hundred (100) Pounds per Hour [326 IAC 6-3-2]	
C.2	Opacity [326 IAC 5-1]	
C.3	Open Burning [326 IAC 4-1][IC 13-17-9]	
C.4	Incineration [326 IAC 4-2][326 IAC 9-1-2]	
C.5	Fugitive Dust Emissions [326 IAC 6-4]	
C.6	Stack Height [326 IAC 1-7]	
C.7	Asbestos Abatement Projects [326 IAC 14-10][326 IAC 18][40 CFR 61, Subpart M]	
	Testing Requirements [326 IAC 2-7-6(1)].....	36
C.8	Performance Testing [326 IAC 3-6]	
	Compliance Requirements [326 IAC 2-1.1-11]	36
C.9	Compliance Requirements [326 IAC 2-1.1-11]	

DRAFT

Compliance Monitoring Requirements [326 IAC 2-7-5(1)][326 IAC 2-7-6(1)]	36
C.10 Compliance Monitoring [326 IAC 2-7-5(3)][326 IAC 2-7-6(1)][40 CFR 64][326 IAC 3-8]	
C.11 Instrument Specifications [326 IAC 2-1.1-11][326 IAC 2-7-5(3)][326 IAC 2-7-6(1)]	
Corrective Actions and Response Steps [326 IAC 2-7-5][326 IAC 2-7-6]	37
C.12 Emergency Reduction Plans [326 IAC 1-5-2][326 IAC 1-5-3]	
C.13 Risk Management Plan [326 IAC 2-7-5(11)][40 CFR 68]	
C.14 Response to Excursions or Exceedances [40 CFR 64][326 IAC 3-8][326 IAC 2-7-5][326 IAC 2-7-6]	
C.15 Actions Related to Noncompliance Demonstrated by a Stack Test [326 IAC 2-7-5][326 IAC 2-7-6]	
Record Keeping and Reporting Requirements [326 IAC 2-7-5(3)][326 IAC 2-7-19]	40
C.16 Emission Statement [326 IAC 2-7-5(3)(C)(iii)][326 IAC 2-7-5(7)][326 IAC 2-7-19(c)][326 IAC 2-6]	
C.17 General Record Keeping Requirements [326 IAC 2-7-5(3)][326 IAC 2-7-6]	
C.18 General Reporting Requirements [326 IAC 2-7-5(3)(C)][326 IAC 2-1.1-11] [40 CFR 64][326 IAC 3-8]	
Stratospheric Ozone Protection	42
C.19 Compliance with 40 CFR 82 and 326 IAC 22-1	
SECTION D.1 EMISSIONS UNIT OPERATION CONDITIONS	43
Emission Limitations and Standards [326 IAC 2-7-5(1)]	48
D.1.1 PSD Minor Limit for Particulate [326 IAC 2-2]	
D.1.2 Particulate Emissions Limitations [326 IAC 6-3-2]	
D.1.3 Significant Source Modification Avoidance Limit [326 IAC 2-7-10.5(f)]	
D.1.4 Preventive Maintenance Plan [326 IAC 2-7-5(12)]	
Compliance Determination Requirements [326 IAC 2-7-5(1)]	51
D.1.5 Particulate Control	
D.1.6 Testing Requirements [326 IAC 2-1.1-11]	
Compliance Monitoring Requirements [326 IAC 2-7-5(1)][326 IAC 2-7-6(1)]	52
D.1.7 Visible Emissions Notations [40 CFR 64]	
D.1.8 Parametric Monitoring [40 CFR 64]	
D.1.9 Broken or Failed Bag Detection [40 CFR 64]	
Record Keeping and Reporting Requirements [326 IAC 2-7-5(3)][326 IAC 2-7-19]	53
D.1.10 Record Keeping Requirements	
D.1.11 Reporting Requirements	
SECTION D.2 EMISSIONS UNIT OPERATION CONDITIONS	55
Emission Limitations and Standards [326 IAC 2-7-5(1)]	58
D.2.1 PSD Minor Limit for Particulate [326 IAC 2-2]	
D.2.2 Particulate Emissions Limitations [326 IAC 6-3-2]	
D.2.3 Preventive Maintenance Plan [326 IAC 2-7-5(12)]	
Compliance Determination Requirements [326 IAC 2-7-5(1)]	60
D.2.4 Particulate Control	
D.2.5 Testing Requirements [326 IAC 2-1.1-11]	
Compliance Monitoring Requirements [326 IAC 2-7-5(1)][326 IAC 2-7-6(1)]	61
D.2.6 Visible Emissions Notations [40 CFR 64]	
D.2.7 Parametric Monitoring [40 CFR 64]	
D.2.8 Broken or Failed Bag Detection [40 CFR 64]	
D.2.9 Cyclone Failure Detection [326 IAC 2-7-6(1)][326 IAC 2-7-5(1)]	
Record Keeping and Reporting Requirements [326 IAC 2-7-5(3)][326 IAC 2-7-19]	63
D.2.10 Record Keeping Requirements	

DRAFT

SECTION D.3 EMISSIONS UNIT OPERATION CONDITIONS	64
Emission Limitations and Standards [326 IAC 2-7-5(1)]	65
D.3.1 PSD Minor Limits for Particulate [326 IAC 2-2]	
D.3.2 PSD Minor Limit for VOC [326 IAC 2-2]	
D.3.3 Particulate Emissions Limitations [326 IAC 6-3-2]	
D.3.4 Volatile Organic Compounds (VOC) [326 IAC 8-1-6]	
D.3.5 Preventive Maintenance Plan [326 IAC 2-7-5(12)]	
Compliance Determination Requirements [326 IAC 2-7-5(1)].....	68
D.3.6 Particulate Control	
D.3.7 Volatile Organic Compounds (VOC)	
D.3.8 Testing Requirements [326 IAC 2-1.1-11]	
D.3.9 Leak Detection and Repair (LDAR) Program [326 IAC 8-1-6]	
Compliance Monitoring Requirements [326 IAC 2-7-5(1)][326 IAC 2-7-6(1)].....	69
D.3.10 Visible Emissions Notations [40 CFR 64]	
D.3.11 Monitoring for Mineral Oil Absorber and Mineral Oil Stripping Column [40 CFR 64]	
D.3.12 Cyclone Failure Detection [326 IAC 2-7-6(1)][326 IAC 2-7-5(1)][40 CFR 64]	
Record Keeping and Reporting Requirements [326 IAC 2-7-5(3)][326 IAC 2-7-19]	71
D.3.13 Record Keeping Requirements	
D.3.14 Reporting Requirements	
SECTION D.4 EMISSIONS UNIT OPERATION CONDITIONS	73
Emission Limitations and Standards [326 IAC 2-7-5(1)]	75
D.4.1 PSD Minor Limits for VOC [326 IAC 2-2]	
D.4.2 VOC Best Available Control Technology (BACT) Limits [326 IAC 8-1-6]	
D.4.3 Preventive Maintenance Plan [326 IAC 2-7-5(12)]	
Compliance Determination Requirements [326 IAC 2-7-5(1)].....	76
D.4.4 Volatile Organic Compounds (VOC)	
D.4.5 Testing Requirements [326 IAC 2-1.1-11]	
Compliance Monitoring Requirements [326 IAC 2-7-5(1)][326 IAC 2-7-6(1)].....	76
D.4.6 Monitoring for mineral oil absorber and water absorber [40 CFR 64]	
Record Keeping and Reporting Requirements [326 IAC 2-7-5(3)][326 IAC 2-7-19]	78
D.4.7 Record Keeping Requirements	
D.4.8 Reporting Requirements	
SECTION D.5 EMISSIONS UNIT OPERATION CONDITIONS	79
Emission Limitations and Standards [326 IAC 2-7-5(1)]	79
D.5.1 Particulate [326 IAC 6-2-4]	
D.5.2 Preventive Maintenance Plan [326 IAC 2-7-5(12)]	
Compliance Monitoring Requirements [326 IAC 2-7-5(1)][326 IAC 2-7-6(1)].....	79
D.5.3 Continuous Emission Monitoring [326 IAC 3-5]	
SECTION D.6 EMISSIONS UNIT OPERATION CONDITIONS	81
Emission Limitations and Standards [326 IAC 2-7-5(1)]	81
D.6.1 PSD Minor Limit for Particulate [326 IAC 2-2]	
D.6.2 Particulate Emissions Limitations [326 IAC 6-3-2]	
D.6.3 Preventive Maintenance Plan [326 IAC 2-7-5(12)]	
Compliance Determination Requirements [326 IAC 2-7-5(1)].....	82
D.6.4 Particulate Control	
Compliance Monitoring Requirements [326 IAC 2-7-5(1)][326 IAC 2-7-6(1)].....	82
D.6.5 Visible Emissions Notations	
D.6.6 Broken or Failed Bag Detection	

DRAFT

Record Keeping and Reporting Requirements [326 IAC 2-7-5(3)][326 IAC 2-7-19]	83
D.6.7 Record Keeping Requirements	
SECTION D.7 EMISSIONS UNIT OPERATION CONDITIONS	84
Emission Limitations and Standards [326 IAC 2-7-5(1)]	84
D.7.1 PSD Minor Limits for Particulate [326 IAC 2-2]	
D.7.2 Particulate Emissions Limitations [326 IAC 6-3-2]	
D.7.3 Particulate [326 IAC 6-2-4]	
D.7.4 Preventive Maintenance Plan [326 IAC 2-7-5(12)]	
Compliance Monitoring Requirements [326 IAC 2-7-5(1)][326 IAC 2-7-6(1)]	86
D.7.5 Visible Emissions Notations	
D.7.6 Broken or Failed Bag Detection	
Record Keeping and Reporting Requirements [326 IAC 2-7-5(3)][326 IAC 2-7-19]	86
D.7.7 Record Keeping Requirements	
SECTION D.8 Reserved	87
SECTION E.1 NSPS	88
New Source Performance Standards (NSPS) Requirements [326 IAC 2-7-5(1)]	88
E.1.1 General Provisions Relating to New Source Performance Standards [326 IAC 12-1][40 CFR Part 60, Subpart A]	
E.1.2 Industrial-Commercial-Institutional Steam Generating Units NSPS [326 IAC 12][40 CFR Part 60, Subpart Db]	
SECTION E.2 NSPS	89
New Source Performance Standards (NSPS) Requirements [326 IAC 2-7-5(1)]	
E.2.1 General Provisions Relating to New Source Performance Standards [326 IAC 12-1][40 CFR Part 60, Subpart A]	
E.2.2 Grain Elevators NSPS [326 IAC 12][40 CFR Part 60, Subpart DD]	
SECTION E.3 NSPS	93
New Source Performance Standards (NSPS) Requirements [326 IAC 2-7-5(1)]	
E.3.1 General Provisions Relating to New Source Performance Standards [326 IAC 12-1][40 CFR Part 60, Subpart A]	
E.3.2 Equipment Leaks of VOC in the Synthetic Organic Chemicals Manufacturing Industry for which Construction, Reconstruction, or Modification Commenced After January 5, 1981, and on or Before November 7, 2006 NSPS [326 IAC 12][40 CFR Part 60, Subpart VV]	
SECTION E.4 NSPS	96
New Source Performance Standards (NSPS) Requirements [326 IAC 2-7-5(1)]	
E.4.1 General Provisions Relating to New Source Performance Standards [326 IAC 12-1][40 CFR Part 60, Subpart A]	
E.4.2 Volatile Organic Compound (VOC) Emissions From Synthetic Organic Chemical Manufacturing Industry (SOCMI) Distillation Operations NSPS [326 IAC 12][40 CFR Part 60, Subpart NNN]	
SECTION E.5 NSPS	98
New Source Performance Standards (NSPS) Requirements [326 IAC 2-7-5(1)]	
E.5.1 General Provisions Relating to New Source Performance Standards [326 IAC 12-1][40 CFR Part 60, Subpart A]	
E.5.2 Volatile Organic Compound Emissions From Synthetic Organic Chemical Manufacturing Industry (SOCMI) Reactor Processes NSPS [326 IAC 12][40 CFR Part 60, Subpart RRR]	

DRAFT

SECTION E.6 NSPS	100
New Source Performance Standards (NSPS) Requirements [326 IAC 2-7-5(1)]	
E.6.1 General Provisions Relating to New Source Performance Standards [326 IAC 12-1][40 CFR Part 60, Subpart A]	
E.6.2 Stationary Compression Ignition Internal Combustion Engines NSPS [326 IAC 12][40 CFR Part 60, Subpart IIII]	
SECTION E.7 NESHAP	102
National Emission Standards for Hazardous Air Pollutants (NESHAP) Requirements [326 IAC 2-7-5(1)]	
E.7.1 General Provisions Relating to National Emission Standards for Hazardous Air Pollutants under 40 CFR 63 [326 IAC 20-1][40 CFR 63, Subpart A]	
E.7.2 Miscellaneous Organic Chemical Manufacturing NESHAP [40 CFR Part 63, Subpart FFFF][326 IAC 20-84]	
SECTION E.8 NESHAP	106
National Emission Standards for Hazardous Air Pollutants (NESHAP) Requirements [326 IAC 2-7-5(1)]	
E.8.1 General Provisions Relating to National Emission Standards for Hazardous Air Pollutants under 40 CFR 63 [326 IAC 20-1][40 CFR 63, Subpart A]	
E.8.2 Solvent Extraction for Vegetable Oil Production NESHAP [40 CFR Part 63, Subpart GGGG][326 IAC 20-60]	
SECTION E.9 NESHAP	114
National Emission Standards for Hazardous Air Pollutants (NESHAP) Requirements [326 IAC 2-7-5(1)]	
E.9.1 General Provisions Relating to National Emission Standards for Hazardous Air Pollutants under 40 CFR 63 [326 IAC 20-1][40 CFR 63, Subpart A]	
E.9.2 Stationary Reciprocating Internal Combustion Engines NESHAP [40 CFR Part 63, Subpart ZZZZ][326 IAC 20-82]	
SECTION E.10 NESHAP	115
National Emission Standards for Hazardous Air Pollutants (NESHAP) Requirements [326 IAC 2-7-5(1)]	
E.10.1 General Provisions Relating to National Emission Standards for Hazardous Air Pollutants under 40 CFR 63 [326 IAC 20-1][40 CFR 63, Subpart A]	
E.10.2 Industrial, Commercial, and Institutional Boilers and Process Heaters NESHAP [40 CFR Part 63, Subpart DDDDD][326 IAC 20-95]	
SECTION E.11 NSPS	118
New Source Performance Standards (NSPS) Requirements [326 IAC 2-7-5(1)]	
E.11.1 General Provisions Relating to New Source Performance Standards [326 IAC 12-1][40 CFR Part 60, Subpart A]	
E.11.2 Equipment Leaks for VOC in the Synthetic Organic Chemicals Manufacturing Industry for which Construction, Reconstruction or Modification Commenced after November 7, 2006 NSPS [326 IAC 12][40 CFR Part 60, Subpart VVa]	
SECTION E.12 NSPS	120
New Source Performance Standards (NSPS) Requirements [326 IAC 2-7-5(1)].....	
E.12.1 General Provisions Relating to New Source Performance Standards [326 IAC 12-1][40 CFR Part 60, Subpart A]	120
E.12.2 Stationary Gas Turbines NSPS [326 IAC 12][40 CFR Part 60, Subpart GG]	

DRAFT

SECTION E.13 NESHAP	121
National Emission Standards for Hazardous Air Pollutants (NESHAP) Requirements	
[326 IAC 2-7-5(1)].....	121
E.13.1 General Provisions Relating to National Emission Standards for Hazardous Air Pollutants under 40 CFR Part 63 [326 IAC 20-1][40 CFR Part 63, Subpart A]	
E.13.2 Stationary Combustion Turbines NESHAP [40 CFR Part 63, Subpart YYYYY][326 IAC 20-90]	
D.13.3 Preventive Maintenance Plan [326 IAC 2-7-5(12)]	
CERTIFICATION	122
EMERGENCY OCCURRENCE REPORT	123
Part 70 Quarterly Report.....	125
Part 70 Quarterly Report.....	126
Part 70 Quarterly Report.....	127
Part 70 Quarterly Report.....	128
Part 70 Quarterly Report.....	129
Part 70 Quarterly Report.....	130
Part 70 Quarterly Report.....	131
Part 70 Quarterly Report.....	132
QUARTERLY DEVIATION AND COMPLIANCE MONITORING REPORT	133
 Attachment A: 40 CFR Part 60.40b, Subpart Db, NSPS for Industrial-Commercial-Institutional Steam Generating Units	
 Attachment B: 40 CFR Part 60.300, Subpart DD, NSPS for Grain Elevators	
 Attachment C: 40 CFR Part 60.480, Subpart VV - NSPS for Equipment Leaks of VOC in the Synthetic Organic Chemicals Manufacturing Industry for which Construction, Reconstruction, or Modification Commenced After January 5, 1981, and on or Before November 7, 2006.	
 Attachment D: 40 CFR Part 60.660, Subpart NNN - NSPS for Volatile Organic Compound (VOC) Emissions from Synthetic Organic Chemical Manufacturing Industry (SOCMI) Distillation Operations.	
 Attachment E: 40 CFR Part 60.700, Subpart RRR - NSPS for Volatile Organic Compounds Emissions from Synthetic Organic Chemical Manufacturing Industry (SOCMI) Reactor Processes.	
 Attachment F: 40 CFR Part 60.4200, Subpart IIII - NSPS for Stationary Compression Ignition Internal Combustion Engines	
 Attachment G: 40 CFR Part 63.2430, Subpart FFFF - NESHAP: Miscellaneous Organic Chemical Manufacturing	
 Attachment H: 40 CFR Part 63.2830, Subpart GGGG - NESHAP: Solvent Extraction for Vegetable Oil Production	
 Attachment I: 40 CFR 63.6580, Subpart ZZZZ - NESHAP for Stationary Reciprocating Internal Combustion Engines	
 Attachment J: 40 CFR 63.7480, Subpart DDDDD - NESHAP for Industrial, Commercial, and Institutional	

DRAFT

Boilers and Process Heaters

Attachment K: 40 CFR Part 60.480a, Subpart VVa, NSPS for Equipment Leaks for VOC in the Synthetic Organic Chemicals Manufacturing Industry for which Construction, Reconstruction or Modification Commenced after November 7, 2006

Attachment L: 40 CFR Part 60.330, Subpart GG, NSPS for Stationary Gas Turbines

Attachment M: 40 CFR Part 63.6080, Subpart YYYY, NESHAP for Stationary Combustion Turbines

DRAFT

SECTION A

SOURCE SUMMARY

This permit is based on information requested by the Indiana Department of Environmental Management (IDEM), Office of Air Quality (OAQ). The information describing the source contained in conditions A.1 through A.3 is descriptive information and does not constitute enforceable conditions. However, the Permittee should be aware that a physical change or a change in the method of operation that may render this descriptive information obsolete or inaccurate may trigger requirements for the Permittee to obtain additional permits or seek modification of this permit pursuant to 326 IAC 2, or change other applicable requirements presented in the permit application.

A.1 General Information [326 IAC 2-7-4(c)][326 IAC 2-7-5(14)][326 IAC 2-7-1(22)]

The Permittee owns and operates a stationary refined bleached soybean oil (RB Oil), soybean salad oil, soybean meal, and biodiesel manufacturing plant.

Source Address:	7344 State Road 15 South, Claypool, Indiana 46510
General Source Phone Number:	(574) 566-2100
SIC Code:	2075 (Soybean Oil Mills), 2079 (Shortening, Table Oils, Margarine, and Other Edible Fats and Oils, Not Elsewhere Classified), 2869 (Industrial Organic Chemicals, Not Elsewhere Classified)
County Location:	Kosciusko
Source Location Status:	Attainment for all criteria pollutants
Source Status:	Part 70 Operating Permit Program Minor Source, under PSD and Emission Offset Rules Major Source, Section 112 of the Clean Air Act Nested Source with Chemical Process Plant (biodiesel) as 1 of 28 Source Categories, within a non-listed source

A.2 Emission Units and Pollution Control Equipment Summary [326 IAC 2-7-4(c)(3)][326 IAC 2-7-5(14)]

This stationary source consists of the following emission units and pollution control devices:

(a)

Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
A030000	Truck Dump No. 1 *2006	600	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A020000	Truck Dump No. 2 *2006	600	Grain Receiving /Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
Truck Dump No. 3 (with doors)	Truck Dump No. 3 *2013	360	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A030100	Discharge Conveyor No. 1 *2006	600	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A020100	Discharge Conveyor No. 2 *2006	600	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD

DRAFT

Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
A040000	Bean Receiving Leg No. 1 *2006	600	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A050000	Bean Receiving Leg No. 2 *2006	600	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A010000	Rail Dump and Rail Collection Conveyor *2006	600	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A010100	Rail Scale Discharge Conveyor *2006	360	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A150100	Cross Bin No 1 thru 3 *2006	600	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A120100	Cross Bin No 4 thru 6 *2006	600	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A151000	Discharge Bin No 1 thru 3 *2006	360	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A121000	Discharge Bin No 4 thru 6 *2006	360	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A153000	Day Bin Leg *2006	360	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
AF-2 A200000	Grain Receiving/Meal Loadout Baghouse *2006, **2010	38,000 acfm @ 0.005 grain/acf outlet gr loading		Stack AF-2	Yes under NSPS DD
A152000	West Bin Cross Conveyor 1-3 *2006, **2010 and 2011	360	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A122000	East Bin Cross Conveyor 4-6 *2006, **2010	360	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A130100	West Bin Feed Conveyor *2006, **2010	600	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A100100	East Bin Feed Conveyor *2006, **2010	600	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD

DRAFT

Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
G020500	Meal Storage Feed Conveyor *2006, **2010, **2011, ** 2015	200	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
G070300	Truck Meal Loadout Feed Conveyor *2006, **2010, ** 2015	300	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
G290000	Truck Collection Conveyor *2010, ** 2015	300	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
G280000	Truck Loader No.1 *2010, ** 2015	330	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
G270000	Truck Loader No.2 *2010, ** 2015	330	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
G080000	Truck Pelleted Hull Loadout Bin *2006, **2010, ** 2015	148	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
G070000	Truck Meal Loadout Bin *2006, **2010, 2012, ** 2015	300	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
G180000	Rail Pelleted Hull Loadout Bin *2010, ** 2015	148	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
G130000	Rail Meal Loadout Bin *2006, **2010, 2012, **2015	300	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
G160000	Pellet Hulls Conveyor to Loadout *2006, **2010, 2012, ** 2015	17.0	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
G170000	Rail Car Collection Conveyor *2006, **2010, ** 2015	300	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
G220000	Rail Car Loadout *2010, ** 2015	330	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
G010100	Meal Reclaim Conveyor *2006, **2010, ** 2015	200	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
G010200	Meal Reclaim Leg *2010, ** 2015	200	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
AF-1	Meal Loadout Baghouse *2015	22,125 cfm @ 0.005 grain/acf outlet gr loading		Stack AF-1	
Rail Receiving Leg	Rail Receiving Leg *2013	360	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
Truck Dump No. 3 Receiving Conveyor	Truck Dump No. 3 Receiving Conveyor *2013	360	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD

DRAFT

Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
Truck Dump No. 3 Receiving Leg	Truck Dump No. 3 Receiving Leg *2013	360	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
Scalperator Leg	Scalperator Leg *2013	210	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
Scalperator Feed Conveyor	Scalperator Feed Conveyor *2013	210	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
Scalperator Jack Leg	Scalperator Jack Leg *2013	210	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
Scalperator	Scalperator *2013	210	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A060000	Screener *2006, **2011 and 2012	264	Prep Exhaust Baghouse	Stack AF-3	Yes under NESHAP GGGG
A160300	VSC Leg Feed Conveyor *2006, **2010 and 2012	264	Prep Exhaust Baghouse	Stack AF-3	Yes under NESHAP GGGG
A170000	Screenings Tank *2006	5	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
A170300	Screenings Recycle Leg *2006	5	Prep exhaust baghouse	Stack AF-3	Yes under NSPS DD and NESHAP GGGG
B011300	Bean Weigh Scale *2006, **2012	264	Prep exhaust baghouse	Stack AF-3	Yes under NSPS DD and NESHAP GGGG
B310000	Screenings Weight Belt *2006	5	Prep exhaust baghouse	Stack AF-3	Yes under NSPS DD and NESHAP GGGG
AF-7	Pod rinder/Screener Baghouse *2011	5,000 acfm		Stack AF-7	Yes under NESHAP GGGG
B310200	Pod rinder/Destoner *2006, **2010	5	Pod Grinder/ Screener Baghouse	Stack AF-7	Yes under NESHAP GGGG
B011200	VSC Feed Leg *2006, **2012	264	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
A060400	Screener Feed Conveyor *2010, **2012	264	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
B010100	Whole Bean Aspirator No 1 *2006, **2012	264	Prep exhaust baghouse	Stack AF-3	Yes under NSPS DD and NESHAP GGGG
B020100	Whole Bean Aspirator No 2 *2006, **2012	264	Prep exhaust baghouse	Stack AF-3	Yes under NSPS DD and NESHAP GGGG
B010900	Whole Bean Aspirator Cyclone *2006, **2012	264	Prep exhaust baghouse	Stack AF-3	Yes under NSPS DD and NESHAP GGGG
B030800	Conditioned Bean Feed Conveyor *2006, **2012	264	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG

DRAFT

Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
B030900	Hull Collection Conveyor *2006, **2012	17.0	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
E130000	Hull Screener No.1 *2006	9.6	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
E150000	Hull Screener No.2 *2006	9.6	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
B440000	Secondary Hull Collection L-Path *2010, **2012	17.0	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
B430000	Secondary Hull Collection Conveyor *2010, **2012	17.0	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
E070300	4 Hour Hull Tank *2006, **2012	17.0	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
E080000	Pellet Cooler *2006, **2012	17.0	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
E090000	Pellet Cooler Cyclone *2006, **2012	17.0	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
E050200	Hull Hammer Mill Feeder *2006, **2012	17.0	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
E050000	Hull Hammer Mill *2006, **2012	17.0	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
E050100	Hull Hammer Mill Plenum *2006, **2012	17.0	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
G050100	Pelleted Hulls Leg *2006, **2012	17.0	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
G050300	Pelleted Hulls Storage Conveyor *2006, **2010, 2012, **2015	17.0	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
E050400	Hulls Addition Screw *2011, **2012	17.0	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
B010300	Conditioner Bean Loop Path *2006, **2012	264	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
AF-3 G100000	Prep exhaust baghouse *2006	28,900 acfm @ 0.005 grain/acf outlet grain loading		Stack AF-3	Yes under NESHAP GGGG
	Bean Storage Bins #2, #3, #6, and #7 *2006	600 tons/hr and each Bin has a maximum storage capacity of 500,000 bushels	None	None	Yes under NSPS DD

DRAFT

Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
	Bean Storage Bins #4 and #8 *2013	600 tons/hr and each Bin has a maximum storage capacity of 500,000 bushels	None	None	Yes under NSPS DD
	Bean Storage Silos #1 and #5 *2008	600 tons/hr and each Bin has a maximum storage capacity of 500,000 bushels	None	None	Yes under NSPS DD

Note *Approved in the year indicated above for construction.

Note **Approved in the year indicated above for modification.

(b)

Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
Piles #1 and #2 *2008	Two (2) covered seasonal grain storage piles	each with a maximum storage capacity of 1,000,000 bushels of soybeans	None	None	Yes under NSPS DD

Note *Approved in the year indicated above for construction.

(c)

Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
C200100	Flaker Feed Loop Conveyor *2010, **2012	247	Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
C010600	Flake Collection Conveyor (12 flakers) *2006, **2010 and 2012	247	Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
C010000	Flaking Roll No. 1 *2013	22.9	***Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
C020000	Flaking Roll No. 2 *2006	22.9	Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
C030000	Flaking Roll No. 3 *2006	22.9	Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
C040000	Flaking Roll No. 4 *2012	22.9	***Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
C050000	Flaking Roll No. 5 *2006	22.9	Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
C060000	Flaking Roll No. 6 *2006	22.9	Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
C070000	Flaking Roll No. 7 *2012	22.9	***Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
C080000	Flaking Roll No. 8 *2006	22.9	Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG

DRAFT

Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
C090000	Flaking Roll No. 9 *2006	22.9	Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
C100000	Flaking Roll No. 10 *2013	22.9	***Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
C0110000	Flaking Roll No. 11 *2009	22.9	***Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
C0120000	Flaking Roll No. 12 *2009	22.9	***Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
AF-4 C110000	Flaker aspiration baghouse *2006	24,000 acfm @ 0.005 grain/acf outlet grain loading		Stack AF-4	Yes under NESHAP GGGG
B040000	Hulloosinator No. 1 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B080100	Hulloosinator No. 2 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B130000	Hulloosinator No. 3 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B170000	Hulloosinator No. 4 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B050000	Cascade Dryer No. 1 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B090000	Cascade Dryer No. 2 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B140000	Cascade Dryer No. 3 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B180000	Cascade Dryer No. 4 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B210000	CCD Cyclone *2006, **2010	42,000 cfm	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B060000	Cracking Roll No.1 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B100000	Cracking Roll No.2 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B150000	Cracking Roll No.3 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B190000	Cracking Roll No.4 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B070000	Cascade Conditioner No. 1 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B110000	Cascade Conditioner No. 2 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B160000	Cascade Conditioner No. 3 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B200000	Cascade Conditioner No. 4 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B230000	CCC Cyclone *2006, **2010	42,000 cfm	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
E130100	Secondary Aspirator No 1 *2006, **2010	9.6	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG

DRAFT

Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
E150100	Secondary Aspirator No 2 *2006, **2010	9.6	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
E160000	Secondary Aspirator Cyclone *2006, **2010	9.6	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
A160100	Feed Day Tank Conveyor *2006	600	Hot dehulling baghouse	Stack AF-5	Yes under NSPS DD
A160000 (Day Tank) A160500 (Aspirator) B420000 (Cyclone)	Day Tank (with Aspirator and cyclone) *2006, **2010, **2012, and **2014	264	Hot dehulling baghouse	Stack AF-5	Yes under NSPS DD
AF-5 B260000	Hot dehulling baghouse *2006	43,000 acfm @ 0.005 grain/acf outlet grain loading		Stack AF-5	Yes under NESHAP GGGG
E020300	Grinding Discharge Conveyor *2011, **2012	198	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E020400	Hammer Mill Mixing Conveyor *2006, **2011 and 2012	198	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E010100	Meal L-Path Conveyor *2006, **2012	198	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E010300	Meal Hammer Mill Feed Conveyor *2006, **2012	198	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E020200	Meal Hammer Mill Feeder No. 1 *2006	74.0	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E030200	Meal Hammer Mill Feeder No. 2 *2006	74.0	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E040200	Meal Hammer Mill Feeder No. 3 (switch) *2006	74.0	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E020000	Meal Hammer Mill No. 1 *2006	74.0	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E030000	Meal Hammer Mill No. 2 *2006	74.0	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E040000	Meal Hammer Mill No. 3 (switch) *2006	74.0	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E020100	Meal Hammer Mill Bin No. 1 *2006	74.0	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E030100	Meal Hammer Mill Bin No. 2 *2006	74.0	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E040100	Meal Hammer Mill Bin No. 3 (switch) *2006	74.0	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG

DRAFT

Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
E230200	Meal Hammer Mill Feeder No. 5 *2012	74.0	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E230000	Meal Hammer Mill No. 5 *2012	74.0	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E230100	Meal Hammer Mill Bin No. 5 *2012	74.0	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
G010300	Meal Leg *2006, **2010 and 2012	198	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
G150000	Meal Conveyor to Loadout *2006, **2012, ** 2015	198	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
AF-6 E110000	Mill Grinding Baghouse *2006	18,000 acfm @ 0.005 grain/acf outlet grain loading		Stack AF-6	Yes under NESHAP GGGG
B010000	VSC No. 1 *2006, **2012	132	VSC Cyclone No. 1	Stack S-1	Yes under NESHAP GGGG
B020000	VSC No. 2 *2006, **2012	132	VSC Cyclone No. 2	Stack S-6	Yes under NESHAP GGGG
B010500	VSC Air Heater No. 1 *2006, **2012	264	VSC Cyclone No. 1	Stack S-1	Yes under NESHAP GGGG
B020500	VSC Air Heater No. 2 *2015	264	VSC Cyclone No. 2	Stack S-6	Yes under NESHAP GGGG
B010700	VSC Cyclone No. 1 *2006, **2010	42,000 cfm		Stack S-1	Yes under NESHAP GGGG
B020700	VSC Cyclone No. 2 *2015	42,000 cfm		Stack S-6	Yes under NESHAP GGGG
B120000	Jet Dryer No. 1 *2006, **2010 and 2012	132	Jet Dryer Baghouse AF-8	Stack S-1	Yes under NESHAP GGGG
B030000	Jet Dryer No. 2 *2006, **2012	132	Jet Dryer Baghouse AF-9	Stack S-1	Yes under NESHAP GGGG
B120100A	Jet Dryer Baghouse AF-8	74,000 acfm		Stack S-1	Yes under NESHAP GGGG
B120100B	Jet Dryer Baghouse AF-9	74,000 acfm		Stack S-1	Yes under NESHAP GGGG

Note *Approved in the year indicated above for construction.

Note **Approved in the year indicated above for modification.

Note ***The Flaker aspiration baghouse has been determined to be integral to the process for this unit.

(d)

Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
D010000	Soybean oil extractor *2006, **2010 and 2012	264	Mineral oil absorber	Stack S-4	Yes under NESHAP GGGG
	One (1) set of evaporators *2006		Mineral oil absorber	Stack S-4	Yes under NESHAP GGGG
D020000	One (1) Desolventizer/toaster *2006, **2010		Mineral oil absorber	Stack S-4	Yes under NESHAP GGGG

DRAFT

Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
	One (1) set of water separators *2006		Mineral oil absorber	Stack S-4	Yes under NESHAP GGGG
D060000	Main Vent Condenser *2006		Mineral oil absorber	Stack S-4	Yes under NESHAP GGGG
	Five (5) hexane storage tank *2006 for original tank, and 2010 for other tanks **2010 for original tank	20,690 gallons each	Mineral oil absorber	Stack S-4	Yes under NESHAP GGGG
I220000	One (1) soybean oil pre-treat Tank *2010	35,170 gallons			Yes under NESHAP GGGG
	Three (3) soybean oil storage tank (Degummed Oil Tanks #1 and #2 and Crude Oil Tank #3) *2006 for original tank and 2010 for other tanks, **2010 for original tank	725,000 gallons each			Yes under NESHAP GGGG
D070000	Mineral oil absorber *2006			Stack S-4	Yes under NESHAP GGGG
D310000-1	DC Deck No. 1 *2006, **2010 and 2012	208	DC Deck Cyclone No. 1	Stack S-2	Yes under NESHAP GGGG
D310000-2	DC Deck No. 2 *2006, **2010, 2011, and 2012	208	DC Deck Cyclone No. 2	Stack S-2	Yes under NESHAP GGGG
D310000-3	DC Deck No. 3 *2006, **2010 and 2012	208	DC Deck Cyclone No. 3	Stack S-2	Yes under NESHAP GGGG
D310000-4	DC Deck No. 4 *2006, **2010 and 2012	208	DC Deck Cyclone No. 4	Stack S-2	Yes under NESHAP GGGG
D310700	DC Deck Cyclone No. 1 *2006, **2010 and 2011	18,000 scfm		Stack S-2	Yes under NESHAP GGGG
D310800	DC Deck Cyclone No. 2 *2006, **2010 and 2011	18,000 scfm		Stack S-2	Yes under NESHAP GGGG
D310900	DC Deck Cyclone No. 3 *2010, ** 2011	18,000 scfm		Stack S-2	Yes under NESHAP GGGG
D311000	DC Deck Cyclone No. 4 *2010, **2011	18,000 scfm		Stack S-2	Yes under NESHAP GGGG

Note *Approved in the year indicated above for construction.

Note **Approved in the year indicated above for modification.

(e)

Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
CL-5045	1st Primary Transester Column *2006, **2007 and 2010	12,960 gals/hr	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS NNN, NSPS RRR, NSPS VV, and NESHAP FFFF
CL-5046	1st Secondary Transester Column *2006, **2007 and 2010	12,960 gals/hr	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS NNN, NSPS RRR, NSPS VV, and NESHAP FFFF

DRAFT

Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
CL-5063	2nd Primary Transester Column *2006, **2007 and 2010	12,960 gals/hr	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS NNN, NSPS RRR, NSPS VV, and NESHAP FFFF
CL-5064	2nd Secondary Transester Column *2010	12,960 gals/hr	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS NNN, NSPS RRR, NSPS VV, and NESHAP FFFF
R-8171	Esterification Reactor *2010	12,960 gals/hr	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS NNN, NSPS RRR, NSPS VV, and NESHAP FFFF
VU010000	Vacuum group package *2006, **2007 and 2010	12,960 gals/hr	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS NNN, NSPS VV, and NESHAP FFFF
	Biodiesel Mineral Oil Absorber *2010			Stack S-5	Yes under NSPS VV, and NESHAP FFFF
	Biodiesel Water Absorber *2006, **2007	0.448 gpm		Stack S-5	Yes under NSPS VV, and NESHAP FFFF
Biodiesel Distillation	Biodiesel Distillation *2013	15	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS VVa, NSPS NNN, & NESHAP FFFF
I040000	Biodiesel Storage Tank #4 *2006, **2007	725,000 gals			Yes under NSPS VV, and NESHAP FFFF
I050000	Biodiesel Storage Tank #5 *2006, **2007	725,000 gals			Yes under NSPS VV, and NESHAP FFFF
I060000	Biodiesel Storage Tank #6 *2009	325,000			Yes under NSPS VV, and NESHAP FFFF
I070000	Biodiesel Storage Tank #7 *2006, **2007	325,000 gals			Yes under NSPS VV, and NESHAP FFFF
I080000	Biodiesel Storage Tank #8 *2006, **2007	325,000 gals			Yes under NSPS VV, and NESHAP FFFF
I090000	Biodiesel Storage Tank #9 *2006, **2007	325,000 gals			Yes under NSPS VV, and NESHAP FFFF
I100000	Biodiesel Storage Tank #10 *2006, **2007	325,000 gals			Yes under NSPS VV, and NESHAP FFFF
I110000	Biodiesel Storage Tank #11 *2007	325,000 gals			Yes under NSPS VV, and NESHAP FFFF
I140000	Biodiesel Storage Tank #0 *2009	735,000			Yes under NSPS VV, and NESHAP FFFF
I120000	Glycerin Tank #12 *2006, **2010	360,000 gals			Yes under NSPS VV, and NESHAP FFFF

DRAFT

Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
I130000	Glycerin Tank #13 *2006, **2010	360,000 gals			Yes under NSPS VV, and NESHAP FFFF
I250000	Methanol Storage Tank #1 *2006, **2007 and **2010	38,850 gallons	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS VV, and NESHAP FFFF
I260000	Methanol Storage Tank #2 *2006, **2007 and 2010	38,850 gallons	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS VV, and NESHAP FFFF
I270000	Methanol Storage Tank #3 *2006, **2007 and 2010	38,850 gallons	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS VV, and NESHAP FFFF
I280000	Methanol Storage Tank #4 *2006, **2007 and 2010	38,850 gallons	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS VV, and NESHAP FFFF
I290000	Methanol Storage Tank #5 *2006, **2007 and 2010	38,850 gallons	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS VV, and NESHAP FFFF
I300000	Methanol Storage Tank #6 *2007	38,850 gallons	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS VV, and NESHAP FFFF
I230000	Sodium Methylate (catalyst) Storage Tank #1 *2006, **2007 and 2010	38,850 gallons	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS VV, and NESHAP FFFF
I240000	Sodium Methylate (catalyst) Storage Tank #2 *2007, **2010	38,850 gallons	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS VV, and NESHAP FFFF
Rail Rack	Loading Rack (Rail) *2006, **2007, **2010 and **2011	500 gallons per minute			Yes under NSPS VV, and NESHAP FFFF
Truck Rack #1	Loading Rack (Truck) *2006, **2007, **2010 and **2011	430 gallons per minute			Yes under NSPS VV, and NESHAP FFFF
Truck Rack #2	Loading Rack (Truck) *2011	430 gallons per minute			Yes under NSPS VV, and NESHAP FFFF

Note *Approved in the year indicated above for construction.

Note **Approved in the year indicated above for modification.

(f)

Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
B-1	Main Boiler, natural gas fired *2006	220 MMBtu/hr	Low NOx burner and Flue gas recirculation	Stack S-3	Yes under NSPS Db and NESHAP DDDDD
B-S	Temporary Mobile Boiler, Firing Natural Gas, *2013	40 MMBtu/hr	None	Stack S-MB	-

Note *Approved in the year indicated above for construction.

(g)

Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
G010000	Meal Bin No. 1*** *2006, **2010, 2011, and 2012	198	Meal Bin Filter No.1	Stack MBF-1	Yes under NESHAP GGGG

DRAFT

G020000	Meal Bin No. 2*** *2006, **2010, 2011, and 2012	198	Meal Bin Filter No.2	Stack MBF-2	Yes under NESHAP GGGG
G030000	Meal Bin No. 3*** *2006, **2010, 2011, and 2012	198	Meal Bin Filter No.3	Stack MBF-3	Yes under NESHAP GGGG
G040000	Meal Bin No. 4*** *2006, **2010, 2011, and 2012	198	Meal Bin Filter No.4	Stack MBF-4	Yes under NESHAP GGGG
G050000	Meal Bin No. 5*** *2010, **2011, 2012, and 2012	198	Meal Bin Filter No.5	Stack MBF-5	Yes under NESHAP GGGG

Note *Approved in the year indicated above for construction.

Note **Approved in the year indicated above for modification.

Note ***There are five meal bins. However, the plant is only physically capable of loading one meal bin at a time. Thus, the PTE for these units is calculated at a rate of 198 tons/hr for all five meal bins combined.

(h)

Unit ID	Description	Output Power Rating	Control	Discharging to Stack	Affected Facility?
GT-1	Natural Gas-fired power generator *2016 (Originally constructed in 1999)	4.6 MW	none	outside	Yes, under NSPS GG and NESHAP YYYY

Note *Approved in 2016 for construction. Original construction of the Natural Gas-fired power generator, GT-1, occurred in New Jersey in 1999.

A.3 Specifically Regulated Insignificant Activities [326 IAC 2-7-1(21)][326 IAC 2-7-4(c)][326 IAC 2-7-5(14)]

This stationary source also includes the following insignificant activities which are specifically regulated, as defined in 326 IAC 2-7-1(21):

Description	Capacity	Control	Affected Facility?
Kaolin Receiving Tank [326 IAC 2-2][326 IAC 6-3-2] *2006, **2010	10,800 and 40 tons per hour	Bin Filter	
Hull Overflow Tank [326 IAC 2-2][326 IAC 6-3-2] *2006, **2010 and 2012	13,900 cu. ft and 17 tons per hour	None	
diesel/#2 fuel oil storage tank *2006, **2011	44,839 gallons	None	
Cooling tower with a maximum drift rate of 0.005% *2006	11,000 gpm	None	
Glycerin cooling tower with a maximum drift rate of 0.005% *2016	875 gpm	None	
Three (3) Emergency Diesel Fire Pumps *2006	575 BHP each	None	Yes under NSPS IIII and NESHAP ZZZZ
Three (3) diesel-fired emergency generators *2016	One (1) 250 kW and Two (2) 150 kW	None	Yes under NSPS IIII and NESHAP ZZZZ
Two (2) natural gas-fired space heaters *2013	0.25 MMBtu per hour, each	None	

DRAFT

Diatomaceous Earth (DE) Storage Bin [326 IAC 6-3-2] *2009, **2011	767 tons per year	Filter	
Hot Oil Heater *2013	6 MMBtu/hr	None	Yes, Under NESHAP DDDDD
Natural Gas-Fired Glycerin Steam Boiler *2014	1.0 MMBtu/hr	None	Yes, Under NESHAP DDDDD
Glycerin Refinery *2013, **2014	7.0 tons per hour	None	Yes, Under NSPS VVa, NSPS NNN, and NESHAP FFFF
Glycerin Truck/ Rail Loadout	7.0 tons per hour	None	Yes, Under NSPS VVa and NESHAP FFFF
Glycerin Yellow Tank	8,000 gallons	None	Yes, Under NSPS VVa and NESHAP FFFF
Glycerin Yellow Tank 1	50,000 gallons	None	Yes, Under NSPS VVa and NESHAP FFFF
Glycerin Yellow Tank 2	50,000 gallons	None	Yes, Under NSPS VVa and NESHAP FFFF
Glycerin Yellow Tank 3	50,000 gallons	None	Yes, Under NSPS VVa and NESHAP FFFF
Glycerin Yellow Tank 4	50,000 gallons	None	Yes, Under NSPS VVa and NESHAP FFFF
Enzyme Degumming Reactor 1 (EDG Reactor 1)	29,000 gallons	None	No
Enzyme Degumming Reactor 2 (EDG Reactor 2)	29,000 gallons	None	No
Enzyme Degumming Reactor 3 (EDG Reactor 3)	29,000 gallons	None	No

Note *Approved in the year indicated above for construction.

Note **Approved in the year indicated above for modification.

A.4 Part 70 Permit Applicability [326 IAC 2-7-2]

This stationary source is required to have a Part 70 permit by 326 IAC 2-7-2 (Applicability) because:

- (a) It is a major source, as defined in 326 IAC 2-7-1(22);
- (b) It is a source in a source category designated by the United States Environmental Protection Agency (U.S. EPA) under 40 CFR 70.3 (Part 70 - Applicability).

DRAFT

SECTION B GENERAL CONDITIONS

B.1 Definitions [326 IAC 2-7-1]

Terms in this permit shall have the definition assigned to such terms in the referenced regulation. In the absence of definitions in the referenced regulation, the applicable definitions found in the statutes or regulations (IC 13-11, 326 IAC 1-2 and 326 IAC 2-7) shall prevail.

B.2 Permit Term [326 IAC 2-7-5(2)][326 IAC 2-1.1-9.5][326 IAC 2-7-4(a)(1)(D)][IC 13-15-3-6(a)]

- (a) This permit, T085-36889-00102, is issued for a fixed term of five (5) years from the issuance date of this permit, as determined in accordance with IC 4-21.5-3-5(f) and IC 13-15-5-3. Subsequent revisions, modifications, or amendments of this permit do not affect the expiration date of this permit.
- (b) If IDEM, OAQ, upon receiving a timely and complete renewal permit application, fails to issue or deny the permit renewal prior to the expiration date of this permit, this existing permit shall not expire and all terms and conditions shall continue in effect, including any permit shield provided in 326 IAC 2-7-15, until the renewal permit has been issued or denied.

B.3 Term of Conditions [326 IAC 2-1.1-9.5]

Notwithstanding the permit term of a permit to construct, a permit to operate, or a permit modification, any condition established in a permit issued pursuant to a permitting program approved in the state implementation plan shall remain in effect until:

- (a) the condition is modified in a subsequent permit action pursuant to Title I of the Clean Air Act; or
- (b) the emission unit to which the condition pertains permanently ceases operation.

B.4 Enforceability [326 IAC 2-7-7][IC 13-17-12]

Unless otherwise stated, all terms and conditions in this permit, including any provisions designed to limit the source's potential to emit, are enforceable by IDEM, the United States Environmental Protection Agency (U.S. EPA) and by citizens in accordance with the Clean Air Act.

B.5 Severability [326 IAC 2-7-5(5)]

The provisions of this permit are severable; a determination that any portion of this permit is invalid shall not affect the validity of the remainder of the permit.

B.6 Property Rights or Exclusive Privilege [326 IAC 2-7-5(6)(D)]

This permit does not convey any property rights of any sort or any exclusive privilege.

B.7 Duty to Provide Information [326 IAC 2-7-5(6)(E)]

- (a) The Permittee shall furnish to IDEM, OAQ, within a reasonable time, any information that IDEM, OAQ may request in writing to determine whether cause exists for modifying, revoking and reissuing, or terminating this permit, or to determine compliance with this permit. Upon request, the Permittee shall also furnish to IDEM, OAQ copies of records required to be kept by this permit.
- (b) For information furnished by the Permittee to IDEM, OAQ, the Permittee may include a claim of confidentiality in accordance with 326 IAC 17.1. When furnishing copies of requested records directly to U. S. EPA, the Permittee may assert a claim of confidentiality in accordance with 40 CFR 2, Subpart B.

B.8 Certification [326 IAC 2-7-4(f)][326 IAC 2-7-6(1)][326 IAC 2-7-5(3)(C)]

- (a) A certification required by this permit meets the requirements of 326 IAC 2-7-6(1) if:

DRAFT

- (1) it contains a certification by a "responsible official" as defined by 326 IAC 2-7-1(35), and
- (2) the certification states that, based on information and belief formed after reasonable inquiry, the statements and information in the document are true, accurate, and complete.
- (b) The Permittee may use the attached Certification Form, or its equivalent with each submittal requiring certification. One (1) certification may cover multiple forms in one (1) submittal.
- (c) A "responsible official" is defined at 326 IAC 2-7-1(35).

B.9 Annual Compliance Certification [326 IAC 2-7-6(5)]

- (a) The Permittee shall annually submit a compliance certification report which addresses the status of the source's compliance with the terms and conditions contained in this permit, including emission limitations, standards, or work practices. All certifications shall cover the time period from January 1 to December 31 of the previous year, and shall be submitted no later than July 1 of each year to:

Indiana Department of Environmental Management
Compliance and Enforcement Branch, Office of Air Quality
100 North Senate Avenue
MC 61-53 IGCN 1003
Indianapolis, Indiana 46204-2251

and

United States Environmental Protection Agency, Region 5
Air and Radiation Division, Air Enforcement Branch - Indiana (AE-17J)
77 West Jackson Boulevard
Chicago, Illinois 60604-3590

- (b) The annual compliance certification report required by this permit shall be considered timely if the date postmarked on the envelope or certified mail receipt, or affixed by the shipper on the private shipping receipt, is on or before the date it is due. If the document is submitted by any other means, it shall be considered timely if received by IDEM, OAQ on or before the date it is due.
- (c) The annual compliance certification report shall include the following:
 - (1) The appropriate identification of each term or condition of this permit that is the basis of the certification;
 - (2) The compliance status;
 - (3) Whether compliance was continuous or intermittent;
 - (4) The methods used for determining the compliance status of the source, currently and over the reporting period consistent with 326 IAC 2-7-5(3); and
 - (5) Such other facts, as specified in Sections D of this permit, as IDEM, OAQ may require to determine the compliance status of the source.

The submittal by the Permittee does require a certification that meets the requirements of 326 IAC 2-7-6(1) by a "responsible official" as defined by 326 IAC 2-7-1(35).

DRAFT

B.10 Preventive Maintenance Plan [326 IAC 2-7-5(12)][326 IAC 1-6-3]

- (a) A Preventive Maintenance Plan meets the requirements of 326 IAC 1-6-3 if it includes, at a minimum:

- (1) Identification of the individual(s) responsible for inspecting, maintaining, and repairing emission control devices;
- (2) A description of the items or conditions that will be inspected and the inspection schedule for said items or conditions; and
- (3) Identification and quantification of the replacement parts that will be maintained in inventory for quick replacement.

The Permittee shall implement the PMPs.

- (b) If required by specific condition(s) in Section D of this permit where no PMP was previously required, the Permittee shall prepare and maintain Preventive Maintenance Plans (PMPs) no later than ninety (90) days after issuance of this permit or ninety (90) days after initial start-up, whichever is later, including the following information on each facility:

- (1) Identification of the individual(s) responsible for inspecting, maintaining, and repairing emission control devices;
- (2) A description of the items or conditions that will be inspected and the inspection schedule for said items or conditions; and
- (3) Identification and quantification of the replacement parts that will be maintained in inventory for quick replacement.

If, due to circumstances beyond the Permittee's control, the PMPs cannot be prepared and maintained within the above time frame, the Permittee may extend the date an additional ninety (90) days provided the Permittee notifies:

Indiana Department of Environmental Management
Compliance and Enforcement Branch, Office of Air Quality
100 North Senate Avenue
MC 61-53 IGCN 1003
Indianapolis, Indiana 46204-2251

The PMP extension notification does not require a certification that meets the requirements of 326 IAC 2-7-6(1) by a "responsible official" as defined by 326 IAC 2-7-1(35).

The Permittee shall implement the PMPs.

- (c) A copy of the PMPs shall be submitted to IDEM, OAQ upon request and within a reasonable time, and shall be subject to review and approval by IDEM, OAQ. IDEM, OAQ may require the Permittee to revise its PMPs whenever lack of proper maintenance causes or is the primary contributor to an exceedance of any limitation on emissions. The PMPs and their submittal do not require a certification that meets the requirements of 326 IAC 2-7-6(1) by a "responsible official" as defined by 326 IAC 2-7-1(35).
- (d) To the extent the Permittee is required by 40 CFR Part 60/63 to have an Operation Maintenance, and Monitoring (OMM) Plan for a unit, such Plan is deemed to satisfy the PMP requirements of 326 IAC 1-6-3 for that unit.

DRAFT

B.11 Emergency Provisions [326 IAC 2-7-16]

(a) An emergency, as defined in 326 IAC 2-7-1(12), is not an affirmative defense for an action brought for noncompliance with a federal or state health-based emission limitation.

(b) An emergency, as defined in 326 IAC 2-7-1(12), constitutes an affirmative defense to an action brought for noncompliance with a technology-based emission limitation if the affirmative defense of an emergency is demonstrated through properly signed, contemporaneous operating logs or other relevant evidence that describe the following:

- (1) An emergency occurred and the Permittee can, to the extent possible, identify the causes of the emergency;
- (2) The permitted facility was at the time being properly operated;
- (3) During the period of an emergency, the Permittee took all reasonable steps to minimize levels of emissions that exceeded the emission standards or other requirements in this permit;
- (4) For each emergency lasting one (1) hour or more, the Permittee notified IDEM, OAQ or Northern Regional Office within four (4) daytime business hours after the beginning of the emergency, or after the emergency was discovered or reasonably should have been discovered;

Telephone Number: 1-800-451-6027 (ask for Office of Air Quality, Compliance and Enforcement Branch), or

Telephone Number: 317-233-0178 (ask for Office of Air Quality, Compliance and Enforcement Branch)

Facsimile Number: 317-233-6865

Northern Regional Office phone: (574) 245-4870; fax: (574) 245-4877.

- (5) For each emergency lasting one (1) hour or more, the Permittee submitted the attached Emergency Occurrence Report Form or its equivalent, either by mail or facsimile to:

Indiana Department of Environmental Management
Compliance and Enforcement Branch, Office of Air Quality
100 North Senate Avenue
MC 61-53 IGCN 1003
Indianapolis, Indiana 46204-2251

within two (2) working days of the time when emission limitations were exceeded due to the emergency.

The notice fulfills the requirement of 326 IAC 2-7-5(3)(C)(ii) and must contain the following:

- (A) A description of the emergency;
- (B) Any steps taken to mitigate the emissions; and
- (C) Corrective actions taken.

The notification which shall be submitted by the Permittee does not require a certification that meets the requirements of 326 IAC 2-7-6(1) by a "responsible official" as defined by 326 IAC 2-7-1(35).

- (6) The Permittee immediately took all reasonable steps to correct the emergency.

DRAFT

- (c) In any enforcement proceeding, the Permittee seeking to establish the occurrence of an emergency has the burden of proof.
- (d) This emergency provision supersedes 326 IAC 1-6 (Malfunctions). This permit condition is in addition to any emergency or upset provision contained in any applicable requirement.
- (e) The Permittee seeking to establish the occurrence of an emergency shall make records available upon request to ensure that failure to implement a PMP did not cause or contribute to an exceedance of any limitations on emissions. However, IDEM, OAQ may require that the Preventive Maintenance Plans required under 326 IAC 2-7-4(c)(8) be revised in response to an emergency.
- (f) Failure to notify IDEM, OAQ by telephone or facsimile of an emergency lasting more than one (1) hour in accordance with (b)(4) and (5) of this condition shall constitute a violation of 326 IAC 2-7 and any other applicable rules.
- (g) If the emergency situation causes a deviation from a technology-based limit, the Permittee may continue to operate the affected emitting facilities during the emergency provided the Permittee immediately takes all reasonable steps to correct the emergency and minimize emissions.

B.12 Permit Shield [326 IAC 2-7-15][326 IAC 2-7-20][326 IAC 2-7-12]

- (a) Pursuant to 326 IAC 2-7-15, the Permittee has been granted a permit shield. The permit shield provides that compliance with the conditions of this permit shall be deemed compliance with any applicable requirements as of the date of permit issuance, provided that either the applicable requirements are included and specifically identified in this permit or the permit contains an explicit determination or concise summary of a determination that other specifically identified requirements are not applicable. The Indiana statutes from IC 13 and rules from 326 IAC, referenced in conditions in this permit, are those applicable at the time the permit was issued. The issuance or possession of this permit shall not alone constitute a defense against an alleged violation of any law, regulation or standard, except for the requirement to obtain a Part 70 permit under 326 IAC 2-7 or for applicable requirements for which a permit shield has been granted.

This permit shield does not extend to applicable requirements which are promulgated after the date of issuance of this permit unless this permit has been modified to reflect such new requirements.
- (b) If, after issuance of this permit, it is determined that the permit is in nonconformance with an applicable requirement that applied to the source on the date of permit issuance, IDEM, OAQ shall immediately take steps to reopen and revise this permit and issue a compliance order to the Permittee to ensure expeditious compliance with the applicable requirement until the permit is reissued. The permit shield shall continue in effect so long as the Permittee is in compliance with the compliance order.
- (c) No permit shield shall apply to any permit term or condition that is determined after issuance of this permit to have been based on erroneous information supplied in the permit application. Erroneous information means information that the Permittee knew to be false, or in the exercise of reasonable care should have been known to be false, at the time the information was submitted.
- (d) Nothing in 326 IAC 2-7-15 or in this permit shall alter or affect the following:

DRAFT

- (1) The provisions of Section 303 of the Clean Air Act (emergency orders), including the authority of the U.S. EPA under Section 303 of the Clean Air Act;
- (2) The liability of the Permittee for any violation of applicable requirements prior to or at the time of this permit's issuance;
- (3) The applicable requirements of the acid rain program, consistent with Section 408(a) of the Clean Air Act; and
- (4) The ability of U.S. EPA to obtain information from the Permittee under Section 114 of the Clean Air Act.
- (e) This permit shield is not applicable to any change made under 326 IAC 2-7-20(b)(2) (Sections 502(b)(10) of the Clean Air Act changes) and 326 IAC 2-7-20(c)(2) (trading based on State Implementation Plan (SIP) provisions).
- (f) This permit shield is not applicable to modifications eligible for group processing until after IDEM, OAQ, has issued the modifications. [326 IAC 2-7-12(c)(7)]
- (g) This permit shield is not applicable to minor Part 70 permit modifications until after IDEM, OAQ, has issued the modification. [326 IAC 2-7-12(b)(8)]

B.13 Prior Permits Superseded [326 IAC 2-1.1-9.5][326 IAC 2-7-10.5]

- (a) All terms and conditions of permits established prior to T085-36889-00102 and issued pursuant to permitting programs approved into the state implementation plan have been either:
 - (1) incorporated as originally stated,
 - (2) revised under 326 IAC 2-7-10.5, or
 - (3) deleted under 326 IAC 2-7-10.5.
- (b) Provided that all terms and conditions are accurately reflected in this permit, all previous registrations and permits are superseded by this Part 70 operating permit.

B.14 Termination of Right to Operate [326 IAC 2-7-10][326 IAC 2-7-4(a)]

The Permittee's right to operate this source terminates with the expiration of this permit unless a timely and complete renewal application is submitted at least nine (9) months prior to the date of expiration of the source's existing permit, consistent with 326 IAC 2-7-3 and 326 IAC 2-7-4(a).

B.15 Permit Modification, Reopening, Revocation and Reissuance, or Termination [326 IAC 2-7-5(6)(C)][326 IAC 2-7-8(a)][326 IAC 2-7-9]

- (a) This permit may be modified, reopened, revoked and reissued, or terminated for cause. The filing of a request by the Permittee for a Part 70 Operating Permit modification, revocation and reissuance, or termination, or of a notification of planned changes or anticipated noncompliance does not stay any condition of this permit. [326 IAC 2-7-5(6)(C)] The notification by the Permittee does require a certification that meets the requirements of 326 IAC 2-7-6(1) by a "responsible official" as defined by 326 IAC 2-7-1(35).
- (b) This permit shall be reopened and revised under any of the circumstances listed in IC 13-15-7-2 or if IDEM, OAQ determines any of the following:
 - (1) That this permit contains a material mistake.

DRAFT

- (2) That inaccurate statements were made in establishing the emissions standards or other terms or conditions.
- (3) That this permit must be revised or revoked to assure compliance with an applicable requirement. [326 IAC 2-7-9(a)(3)]
- (c) Proceedings by IDEM, OAQ to reopen and revise this permit shall follow the same procedures as apply to initial permit issuance and shall affect only those parts of this permit for which cause to reopen exists. Such reopening and revision shall be made as expeditiously as practicable. [326 IAC 2-7-9(b)]
- (d) The reopening and revision of this permit, under 326 IAC 2-7-9(a), shall not be initiated before notice of such intent is provided to the Permittee by IDEM, OAQ at least thirty (30) days in advance of the date this permit is to be reopened, except that IDEM, OAQ may provide a shorter time period in the case of an emergency. [326 IAC 2-7-9(c)]

B.16 Permit Renewal [326 IAC 2-7-3][326 IAC 2-7-4][326 IAC 2-7-8(e)]

- (a) The application for renewal shall be submitted using the application form or forms prescribed by IDEM, OAQ and shall include the information specified in 326 IAC 2-7-4. Such information shall be included in the application for each emission unit at this source, except those emission units included on the trivial or insignificant activities list contained in 326 IAC 2-7-1(21) and 326 IAC 2-7-1(42). The renewal application does require a certification that meets the requirements of 326 IAC 2-7-6(1) by a "responsible official" as defined by 326 IAC 2-7-1(35).

Request for renewal shall be submitted to:

Indiana Department of Environmental Management
Permit Administration and Support Section, Office of Air Quality
100 North Senate Avenue
MC 61-53 IGCN 1003
Indianapolis, Indiana 46204-2251

- (b) A timely renewal application is one that is:
 - (1) Submitted at least nine (9) months prior to the date of the expiration of this permit; and
 - (2) If the date postmarked on the envelope or certified mail receipt, or affixed by the shipper on the private shipping receipt, is on or before the date it is due. If the document is submitted by any other means, it shall be considered timely if received by IDEM, OAQ on or before the date it is due.
- (c) If the Permittee submits a timely and complete application for renewal of this permit, the source's failure to have a permit is not a violation of 326 IAC 2-7 until IDEM, OAQ takes final action on the renewal application, except that this protection shall cease to apply if, subsequent to the completeness determination, the Permittee fails to submit by the deadline specified, pursuant to 326 IAC 2-7-4(a)(2)(D), in writing by IDEM, OAQ any additional information identified as being needed to process the application.

B.17 Permit Amendment or Modification [326 IAC 2-7-11][326 IAC 2-7-12]

- (a) Permit amendments and modifications are governed by the requirements of 326 IAC 2-7-11 or 326 IAC 2-7-12 whenever the Permittee seeks to amend or modify this permit.
- (b) Any application requesting an amendment or modification of this permit shall be submitted to:

DRAFT

Indiana Department of Environmental Management
Permit Administration and Support Section, Office of Air Quality
100 North Senate Avenue
MC 61-53 IGCN 1003
Indianapolis, Indiana 46204-2251

Any such application does require a certification that meets the requirements of 326 IAC 2-7-6(1) by a "responsible official" as defined by 326 IAC 2-7-1(35).

- (c) The Permittee may implement administrative amendment changes addressed in the request for an administrative amendment immediately upon submittal of the request. [326 IAC 2-7-11(c)(3)]

B.18 Permit Revision Under Economic Incentives and Other Programs [326 IAC 2-7-5(8)][326 IAC 2-7-12(b)(2)]

- (a) No Part 70 permit revision or notice shall be required under any approved economic incentives, marketable Part 70 permits, emissions trading, and other similar programs or processes for changes that are provided for in a Part 70 permit.
- (b) Notwithstanding 326 IAC 2-7-12(b)(1) and 326 IAC 2-7-12(c)(1), minor Part 70 permit modification procedures may be used for Part 70 modifications involving the use of economic incentives, marketable Part 70 permits, emissions trading, and other similar approaches to the extent that such minor Part 70 permit modification procedures are explicitly provided for in the applicable State Implementation Plan (SIP) or in applicable requirements promulgated or approved by the U.S. EPA.

B.19 Operational Flexibility [326 IAC 2-7-20][326 IAC 2-7-10.5]

- (a) The Permittee may make any change or changes at the source that are described in 326 IAC 2-7-20(b) or (c) without a prior permit revision, if each of the following conditions is met:

- (1) The changes are not modifications under any provision of Title I of the Clean Air Act;
- (2) Any preconstruction approval required by 326 IAC 2-7-10.5 has been obtained;
- (3) The changes do not result in emissions which exceed the limitations provided in this permit (whether expressed herein as a rate of emissions or in terms of total emissions);
- (4) The Permittee notifies the:

Indiana Department of Environmental Management
Permit Administration and Support Section, Office of Air Quality
100 North Senate Avenue
MC 61-53 IGCN 1003
Indianapolis, Indiana 46204-2251

and

United States Environmental Protection Agency, Region 5
Air and Radiation Division, Regulation Development Branch - Indiana (AR-18J)
77 West Jackson Boulevard
Chicago, Illinois 60604-3590

DRAFT

in advance of the change by written notification at least ten (10) days in advance of the proposed change. The Permittee shall attach every such notice to the Permittee's copy of this permit; and

- (5) The Permittee maintains records on-site, on a rolling five (5) year basis, which document all such changes and emission trades that are subject to 326 IAC 2-7-20(b)(1) and (c)(1). The Permittee shall make such records available, upon reasonable request, for public review.

Such records shall consist of all information required to be submitted to IDEM, OAQ in the notices specified in 326 IAC 2-7-20(b)(1) and (c)(1).

- (b) The Permittee may make Section 502(b)(10) of the Clean Air Act changes (this term is defined at 326 IAC 2-7-1(37)) without a permit revision, subject to the constraint of 326 IAC 2-7-20(a). For each such Section 502(b)(10) of the Clean Air Act change, the required written notification shall include the following:

- (1) A brief description of the change within the source;
- (2) The date on which the change will occur;
- (3) Any change in emissions; and
- (4) Any permit term or condition that is no longer applicable as a result of the change.

The notification which shall be submitted is not considered an application form, report or compliance certification. Therefore, the notification by the Permittee does not require a certification that meets the requirements of 326 IAC 2-7-6(1) by a "responsible official" as defined by 326 IAC 2-7-1(35).

- (c) Emission Trades [326 IAC 2-7-20(c)]
The Permittee may trade emissions increases and decreases at the source, where the applicable SIP provides for such emission trades without requiring a permit revision, subject to the constraints of Section (a) of this condition and those in 326 IAC 2-7-20(c).
- (d) Alternative Operating Scenarios [326 IAC 2-7-20(d)]
The Permittee may make changes at the source within the range of alternative operating scenarios that are described in the terms and conditions of this permit in accordance with 326 IAC 2-7-5(9). No prior notification of IDEM, OAQ or U.S. EPA is required.
- (e) Backup fuel switches specifically addressed in, and limited under, Section D of this permit shall not be considered alternative operating scenarios. Therefore, the notification requirements of part (a) of this condition do not apply.

B.20 Source Modification Requirement [326 IAC 2-7-10.5]

A modification, construction, or reconstruction is governed by the requirements of 326 IAC 2.

B.21 Inspection and Entry [326 IAC 2-7-6][IC 13-14-2-2][IC 13-30-3-1][IC 13-17-3-2]

Upon presentation of proper identification cards, credentials, and other documents as may be required by law, and subject to the Permittee's right under all applicable laws and regulations to assert that the information collected by the agency is confidential and entitled to be treated as such, the Permittee shall allow IDEM, OAQ, U.S. EPA, or an authorized representative to perform the following:

DRAFT

- (a) Enter upon the Permittee's premises where a Part 70 source is located, or emissions related activity is conducted, or where records must be kept under the conditions of this permit;
- (b) As authorized by the Clean Air Act, IC 13-14-2-2, IC 13-17-3-2, and IC 13-30-3-1, have access to and copy any records that must be kept under the conditions of this permit;
- (c) As authorized by the Clean Air Act, IC 13-14-2-2, IC 13-17-3-2, and IC 13-30-3-1, inspect any facilities, equipment (including monitoring and air pollution control equipment), practices, or operations regulated or required under this permit;
- (d) As authorized by the Clean Air Act, IC 13-14-2-2, IC 13-17-3-2, and IC 13-30-3-1, sample or monitor substances or parameters for the purpose of assuring compliance with this permit or applicable requirements; and
- (e) As authorized by the Clean Air Act, IC 13-14-2-2, IC 13-17-3-2, and IC 13-30-3-1, utilize any photographic, recording, testing, monitoring, or other equipment for the purpose of assuring compliance with this permit or applicable requirements.

B.22 Transfer of Ownership or Operational Control [326 IAC 2-7-11]

- (a) The Permittee must comply with the requirements of 326 IAC 2-7-11 whenever the Permittee seeks to change the ownership or operational control of the source and no other change in the permit is necessary.
- (b) Any application requesting a change in the ownership or operational control of the source shall contain a written agreement containing a specific date for transfer of permit responsibility, coverage and liability between the current and new Permittee. The application shall be submitted to:

Indiana Department of Environmental Management
Permit Administration and Support Section, Office of Air Quality
100 North Senate Avenue
MC 61-53 IGCN 1003
Indianapolis, Indiana 46204-2251

Any such application does require a certification that meets the requirements of 326 IAC 2-7-6(1) by a "responsible official" as defined by 326 IAC 2-7-1(35).
- (c) The Permittee may implement administrative amendment changes addressed in the request for an administrative amendment immediately upon submittal of the request. [326 IAC 2-7-11(c)(3)]

B.23 Annual Fee Payment [326 IAC 2-7-19][326 IAC 2-7-5(7)][326 IAC 2-1.1-7]

- (a) The Permittee shall pay annual fees to IDEM, OAQ within thirty (30) calendar days of receipt of a billing. Pursuant to 326 IAC 2-7-19(b), if the Permittee does not receive a bill from IDEM, OAQ the applicable fee is due April 1 of each year.
- (b) Except as provided in 326 IAC 2-7-19(e), failure to pay may result in administrative enforcement action or revocation of this permit.
- (c) The Permittee may call the following telephone numbers: 1-800-451-6027 or 317-233-4230 (ask for OAQ, Billing, Licensing, and Training Section), to determine the appropriate permit fee.

DRAFT

B.24 Credible Evidence [326 IAC 2-7-5(3)][326 IAC 2-7-6][62 FR 8314][326 IAC 1-1-6]

For the purpose of submitting compliance certifications or establishing whether or not the Permittee has violated or is in violation of any condition of this permit, nothing in this permit shall preclude the use, including the exclusive use, of any credible evidence or information relevant to whether the Permittee would have been in compliance with the condition of this permit if the appropriate performance or compliance test or procedure had been performed.

DRAFT

SECTION C

SOURCE OPERATION CONDITIONS

Entire Source

Emission Limitations and Standards [326 IAC 2-7-5(1)]

C.1 Particulate Emission Limitations For Processes with Process Weight Rates Less Than One Hundred (100) Pounds per Hour [326 IAC 6-3-2]

Pursuant to 326 IAC 6-3-2(e)(2), particulate emissions from any process not exempt under 326 IAC 6-3-1(b) or (c) which has a maximum process weight rate less than 100 pounds per hour and the methods in 326 IAC 6-3-2(b) through (d) do not apply shall not exceed 0.551 pounds per hour.

C.2 Opacity [326 IAC 5-1]

Pursuant to 326 IAC 5-1-2 (Opacity Limitations), except as provided in 326 IAC 5-1-1 (Applicability) and 326 IAC 5-1-3 (Temporary Alternative Opacity Limitations), opacity shall meet the following, unless otherwise stated in this permit:

- (a) Opacity shall not exceed an average of forty percent (40%) in any one (1) six (6) minute averaging period as determined in 326 IAC 5-1-4.
- (b) Opacity shall not exceed sixty percent (60%) for more than a cumulative total of fifteen (15) minutes (sixty (60) readings as measured according to 40 CFR 60, Appendix A, Method 9 or fifteen (15) one (1) minute nonoverlapping integrated averages for a continuous opacity monitor) in a six (6) hour period.

C.3 Open Burning [326 IAC 4-1][IC 13-17-9]

The Permittee shall not open burn any material except as provided in 326 IAC 4-1-3, 326 IAC 4-1-4 or 326 IAC 4-1-6. The previous sentence notwithstanding, the Permittee may open burn in accordance with an open burning approval issued by the Commissioner under 326 IAC 4-1-4.1.

C.4 Incineration [326 IAC 4-2][326 IAC 9-1-2]

The Permittee shall not operate an incinerator except as provided in 326 IAC 4-2 or in this permit. The Permittee shall not operate a refuse incinerator or refuse burning equipment except as provided in 326 IAC 9-1-2 or in this permit.

C.5 Fugitive Dust Emissions [326 IAC 6-4]

The Permittee shall not allow fugitive dust to escape beyond the property line or boundaries of the property, right-of-way, or easement on which the source is located, in a manner that would violate 326 IAC 6-4 (Fugitive Dust Emissions). 326 IAC 6-4-2(4) is not federally enforceable.

C.6 Stack Height [326 IAC 1-7]

The Permittee shall comply with the applicable provisions of 326 IAC 1-7 (Stack Height Provisions), for all exhaust stacks through which a potential (before controls) of twenty-five (25) tons per year or more of particulate matter or sulfur dioxide is emitted by using ambient air quality modeling pursuant to 326 IAC 1-7-4. The provisions of 326 IAC 1-7-1(3), 326 IAC 1-7-2, 326 IAC 1-7-3(c) and (d), 326 IAC 1-7-4, and 326 IAC 1-7-5(a), (b), and (d) are not federally enforceable.

C.7 Asbestos Abatement Projects [326 IAC 14-10][326 IAC 18][40 CFR 61, Subpart M]

- (a) Notification requirements apply to each owner or operator. If the combined amount of regulated asbestos containing material (RACM) to be stripped, removed or disturbed is at least 260 linear feet on pipes or 160 square feet on other facility components, or at least thirty-five (35) cubic feet on all facility components, then the notification requirements of 326 IAC 14-10-3 are mandatory. All demolition projects require notification whether or not asbestos is present.

DRAFT

- (b) The Permittee shall ensure that a written notification is sent on a form provided by the Commissioner at least ten (10) working days before asbestos stripping or removal work or before demolition begins, per 326 IAC 14-10-3, and shall update such notice as necessary, including, but not limited to the following:
- (1) When the amount of affected asbestos containing material increases or decreases by at least twenty percent (20%); or
 - (2) If there is a change in the following:
 - (A) Asbestos removal or demolition start date;
 - (B) Removal or demolition contractor; or
 - (C) Waste disposal site.
- (c) The Permittee shall ensure that the notice is postmarked or delivered according to the guidelines set forth in 326 IAC 14-10-3(2).
- (d) The notice to be submitted shall include the information enumerated in 326 IAC 14-10-3(3).

All required notifications shall be submitted to:

Indiana Department of Environmental Management
Compliance and Enforcement Branch, Office of Air Quality
100 North Senate Avenue
MC 61-53 IGCN 1003
Indianapolis, Indiana 46204-2251

The notice shall include a signed certification from the owner or operator that the information provided in this notification is correct and that only Indiana licensed workers and project supervisors will be used to implement the asbestos removal project. The notifications do not require a certification that meets the requirements of 326 IAC 2-7-6(1) by a "responsible official" as defined by 326 IAC 2-7-1(35).

- (e) **Procedures for Asbestos Emission Control**
The Permittee shall comply with the applicable emission control procedures in 326 IAC 14-10-4 and 40 CFR 61.145(c). Per 326 IAC 14-10-1, emission control requirements are applicable for any removal or disturbance of RACM greater than three (3) linear feet on pipes or three (3) square feet on any other facility components or a total of at least 0.75 cubic feet on all facility components.
- (f) **Demolition and Renovation**
The Permittee shall thoroughly inspect the affected facility or part of the facility where the demolition or renovation will occur for the presence of asbestos pursuant to 40 CFR 61.145(a).
- (g) **Indiana Licensed Asbestos Inspector**
The Permittee shall comply with 326 IAC 14-10-1(a) that requires the owner or operator, prior to a renovation/demolition, to use an Indiana Licensed Asbestos Inspector to thoroughly inspect the affected portion of the facility for the presence of asbestos. The requirement to use an Indiana Licensed Asbestos inspector is not federally enforceable.

DRAFT

Testing Requirements [326 IAC 2-7-6(1)]

C.8 Performance Testing [326 IAC 3-6]

- (a) For performance testing required by this permit, a test protocol, except as provided elsewhere in this permit, shall be submitted to:

Indiana Department of Environmental Management
Compliance and Enforcement Branch, Office of Air Quality
100 North Senate Avenue
MC 61-53 IGCN 1003
Indianapolis, Indiana 46204-2251

no later than thirty-five (35) days prior to the intended test date. The protocol submitted by the Permittee does not require a certification that meets the requirements of 326 IAC 2-7-6(1) by a "responsible official" as defined by 326 IAC 2-7-1(35).

- (b) The Permittee shall notify IDEM, OAQ of the actual test date at least fourteen (14) days prior to the actual test date. The notification submitted by the Permittee does not require a certification that meets the requirements of 326 IAC 2-7-6(1) by a "responsible official" as defined by 326 IAC 2-7-1(35).
- (c) Pursuant to 326 IAC 3-6-4(b), all test reports must be received by IDEM, OAQ not later than forty-five (45) days after the completion of the testing. An extension may be granted by IDEM, OAQ if the Permittee submits to IDEM, OAQ a reasonable written explanation not later than five (5) days prior to the end of the initial forty-five (45) day period.

Compliance Requirements [326 IAC 2-1.1-11]

C.9 Compliance Requirements [326 IAC 2-1.1-11]

The commissioner may require stack testing, monitoring, or reporting at any time to assure compliance with all applicable requirements by issuing an order under 326 IAC 2-1.1-11. Any monitoring or testing shall be performed in accordance with 326 IAC 3 or other methods approved by the commissioner or the U. S. EPA.

Compliance Monitoring Requirements [326 IAC 2-7-5(1)][326 IAC 2-7-6(1)]

C.10 Compliance Monitoring [326 IAC 2-7-5(3)][326 IAC 2-7-6(1)][40 CFR 64][326 IAC 3-8]

- (a) For new units:
Unless otherwise specified in the approval for the new emission unit(s), compliance monitoring for new emission units shall be implemented on and after the date of initial start-up.
- (b) For existing units:
Unless otherwise specified in this permit, for all monitoring requirements not already legally required, the Permittee shall be allowed up to ninety (90) days from the date of permit issuance to begin such monitoring. If, due to circumstances beyond the Permittee's control, any monitoring equipment required by this permit cannot be installed and operated no later than ninety (90) days after permit issuance, the Permittee may extend the compliance schedule related to the equipment for an additional ninety (90) days provided the Permittee notifies:

DRAFT

Indiana Department of Environmental Management
Compliance and Enforcement Branch, Office of Air Quality
100 North Senate Avenue
MC 61-53 IGCN 1003
Indianapolis, Indiana 46204-2251

in writing, prior to the end of the initial ninety (90) day compliance schedule, with full justification of the reasons for the inability to meet this date.

The notification which shall be submitted by the Permittee does require a certification that meets the requirements of 326 IAC 2-7-6(1) by a "responsible official" as defined by 326 IAC 2-7-1(35).

- (c) For monitoring required by CAM, at all times, the Permittee shall maintain the monitoring, including but not limited to, maintaining necessary parts for routine repairs of the monitoring equipment.
- (d) For monitoring required by CAM, except for, as applicable, monitoring malfunctions, associated repairs, and required quality assurance or control activities (including, as applicable, calibration checks and required zero and span adjustments), the Permittee shall conduct all monitoring in continuous operation (or shall collect data at all required intervals) at all times that the pollutant-specific emissions unit is operating. Data recorded during monitoring malfunctions, associated repairs, and required quality assurance or control activities shall not be used for purposes of this part, including data averages and calculations, or fulfilling a minimum data availability requirement, if applicable. The owner or operator shall use all the data collected during all other periods in assessing the operation of the control device and associated control system. A monitoring malfunction is any sudden, infrequent, not reasonably preventable failure of the monitoring to provide valid data. Monitoring failures that are caused in part by poor maintenance or careless operation are not malfunctions.

C.11 Instrument Specifications [326 IAC 2-1.1-11][326 IAC 2-7-5(3)][326 IAC 2-7-6(1)]

- (a) When required by any condition of this permit, an analog instrument used to measure a parameter related to the operation of an air pollution control device shall have a scale such that the expected maximum reading for the normal range shall be no less than twenty percent (20%) of full scale. The analog instrument shall be capable of measuring values outside of the normal range.
- (b) The Permittee may request that the IDEM, OAQ approve the use of an instrument that does not meet the above specifications provided the Permittee can demonstrate that an alternative instrument specification will adequately ensure compliance with permit conditions requiring the measurement of the parameters.

Corrective Actions and Response Steps [326 IAC 2-7-5][326 IAC 2-7-6]

C.12 Emergency Reduction Plans [326 IAC 1-5-2][326 IAC 1-5-3]

Pursuant to 326 IAC 1-5-2 (Emergency Reduction Plans; Submission):

- (a) The Permittee shall maintain the most recently submitted written emergency reduction plans (ERPs) consistent with safe operating procedures.
- (b) Upon direct notification by IDEM, OAQ that a specific air pollution episode level is in effect, the Permittee shall immediately put into effect the actions stipulated in the approved ERP for the appropriate episode level. [326 IAC 1-5-3]

DRAFT

C.13 Risk Management Plan [326 IAC 2-7-5(11)][40 CFR 68]

If a regulated substance, as defined in 40 CFR 68, is present at a source in more than a threshold quantity, the Permittee must comply with the applicable requirements of 40 CFR 68.

C.14 Response to Excursions or Exceedances [40 CFR 64][326 IAC 3-8][326 IAC 2-7-5][326 IAC 2-7-6]

-
- (I) Upon detecting an excursion where a response step is required by the D Section, or an exceedance of a limitation, not subject to CAM, in this permit:
- (a) The Permittee shall take reasonable response steps to restore operation of the emissions unit (including any control device and associated capture system) to its normal or usual manner of operation as expeditiously as practicable in accordance with good air pollution control practices for minimizing excess emissions.
 - (b) The response shall include minimizing the period of any startup, shutdown or malfunction. The response may include, but is not limited to, the following:
 - (1) initial inspection and evaluation;
 - (2) recording that operations returned or are returning to normal without operator action (such as through response by a computerized distribution control system); or
 - (3) any necessary follow-up actions to return operation to normal or usual manner of operation.
 - (c) A determination of whether the Permittee has used acceptable procedures in response to an excursion or exceedance will be based on information available, which may include, but is not limited to, the following:
 - (1) monitoring results;
 - (2) review of operation and maintenance procedures and records; and/or
 - (3) inspection of the control device, associated capture system, and the process.
 - (d) Failure to take reasonable response steps shall be considered a deviation from the permit.
 - (e) The Permittee shall record the reasonable response steps taken.
- (II)
- (a) *CAM Response to excursions or exceedances.*
 - (1) Upon detecting an excursion or exceedance, subject to CAM, the Permittee shall restore operation of the pollutant-specific emissions unit (including the control device and associated capture system) to its normal or usual manner of operation as expeditiously as practicable in accordance with good air pollution control practices for minimizing emissions. The response shall include minimizing the period of any startup, shutdown or malfunction and taking any necessary corrective actions to restore normal operation and prevent the likely recurrence of the cause of an excursion or exceedance (other than those caused by excused startup or shutdown conditions). Such actions may include initial inspection and evaluation, recording that operations returned to normal without operator action (such as through response by a computerized

DRAFT

distribution control system), or any necessary follow-up actions to return operation to within the indicator range, designated condition, or below the applicable emission limitation or standard, as applicable.

- (2) Determination of whether the Permittee has used acceptable procedures in response to an excursion or exceedance will be based on information available, which may include but is not limited to, monitoring results, review of operation and maintenance procedures and records, and inspection of the control device, associated capture system, and the process.
- (b) If the Permittee identifies a failure to achieve compliance with an emission limitation, subject to CAM, or standard, subject to CAM, for which the approved monitoring did not provide an indication of an excursion or exceedance while providing valid data, or the results of compliance or performance testing document a need to modify the existing indicator ranges or designated conditions, the Permittee shall promptly notify the IDEM, OAQ and, if necessary, submit a proposed significant permit modification to this permit to address the necessary monitoring changes. Such a modification may include, but is not limited to, reestablishing indicator ranges or designated conditions, modifying the frequency of conducting monitoring and collecting data, or the monitoring of additional parameters.
- (c) Based on the results of a determination made under paragraph (II)(a)(2) of this condition, the EPA or IDEM, OAQ may require the Permittee to develop and implement a Quality Improvement Plan (QIP). The Permittee shall develop and implement a QIP if notified to in writing by the EPA or IDEM, OAQ.
- (d) Elements of a QIP:
The Permittee shall maintain a written QIP, if required, and have it available for inspection. The plan shall conform to 40 CFR 64.8 b (2).
- (e) If a QIP is required, the Permittee shall develop and implement a QIP as expeditiously as practicable and shall notify the IDEM, OAQ if the period for completing the improvements contained in the QIP exceeds 180 days from the date on which the need to implement the QIP was determined.
- (f) Following implementation of a QIP, upon any subsequent determination pursuant to paragraph (II)(c) of this condition the EPA or the IDEM, OAQ may require that the Permittee make reasonable changes to the QIP if the QIP is found to have:
 - (1) Failed to address the cause of the control device performance problems;
or
 - (2) Failed to provide adequate procedures for correcting control device performance problems as expeditiously as practicable in accordance with good air pollution control practices for minimizing emissions.
- (g) Implementation of a QIP shall not excuse the Permittee from compliance with any existing emission limitation or standard, or any existing monitoring, testing, reporting, or recordkeeping requirement that may apply under federal, state, or local law, or any other applicable requirements under the Act.
- (h) *CAM recordkeeping requirements.*
 - (1) The Permittee shall maintain records of monitoring data, monitor performance data, corrective actions taken, any written quality improvement plan required pursuant to paragraph (II)(c) of this condition and any activities undertaken to implement a quality improvement plan,

DRAFT

and other supporting information required to be maintained under this condition (such as data used to document the adequacy of monitoring, or records of monitoring maintenance or corrective actions). Section C - General Record Keeping Requirements of this permit contains the Permittee's obligations with regard to the records required by this condition.

- (2) Instead of paper records, the owner or operator may maintain records on alternative media, such as microfilm, computer files, magnetic tape disks, or microfiche, provided that the use of such alternative media allows for expeditious inspection and review, and does not conflict with other applicable recordkeeping requirements

C.15 Actions Related to Noncompliance Demonstrated by a Stack Test [326 IAC 2-7-5][326 IAC 2-7-6]

- (a) When the results of a stack test performed in conformance with Section C - Performance Testing, of this permit exceed the level specified in any condition of this permit, the Permittee shall submit a description of its response actions to IDEM, OAQ no later than seventy-five (75) days after the date of the test.
- (b) A retest to demonstrate compliance shall be performed no later than one hundred eighty (180) days after the date of the test. Should the Permittee demonstrate to IDEM, OAQ that retesting in one hundred eighty (180) days is not practicable, IDEM, OAQ may extend the retesting deadline.
- (c) IDEM, OAQ reserves the authority to take any actions allowed under law in response to noncompliant stack tests.

The response action documents submitted pursuant to this condition do require a certification that meets the requirements of 326 IAC 2-7-6(1) by a "responsible official" as defined by 326 IAC 2-7-1(35).

Record Keeping and Reporting Requirements [326 IAC 2-7-5(3)][326 IAC 2-7-19]

C.16 Emission Statement [326 IAC 2-7-5(3)(C)(iii)][326 IAC 2-7-5(7)][326 IAC 2-7-19(c)][326 IAC 2-6]

In accordance with the compliance schedule specified in 326 IAC 2-6-3(b)(1), starting in 2004 and every three (3) years thereafter, the Permittee shall submit by July 1 an emission statement covering the previous calendar year. The emission statement shall contain, at a minimum, the information specified in 326 IAC 2-6-4(c) and shall meet the following requirements:

- (1) Indicate estimated actual emissions of all pollutants listed in 326 IAC 2-6-4(a);
- (2) Indicate estimated actual emissions of regulated pollutants as defined by 326 IAC 2-7-1(33) ("Regulated pollutant, which is used only for purposes of Section 19 of this rule") from the source, for purpose of fee assessment.

The statement must be submitted to:

Indiana Department of Environmental Management
Technical Support and Modeling Section, Office of Air Quality
100 North Senate Avenue
MC 61-50 IGCN 1003
Indianapolis, Indiana 46204-2251

The emission statement does require a certification that meets the requirements of 326 IAC 2-7-6(1) by a "responsible official" as defined by 326 IAC 2-7-1(35).

DRAFT

C.17 General Record Keeping Requirements [326 IAC 2-7-5(3)][326 IAC 2-7-6]

- (a) Records of all required monitoring data, reports, and support information required by this permit shall be retained for a period of at least five (5) years from the date of monitoring sample, measurement, report, or application. Support information includes the following, where applicable:

- (AA) All calibration and maintenance records.
- (BB) All original strip chart recordings for continuous monitoring instrumentation.
- (CC) Copies of all reports required by the Part 70 permit.

Records of required monitoring information include the following, where applicable:

- (AA) The date, place, as defined in this permit, and time of sampling or measurements.
- (BB) The dates analyses were performed.
- (CC) The company or entity that performed the analyses.
- (DD) The analytical techniques or methods used.
- (EE) The results of such analyses.
- (FF) The operating conditions as existing at the time of sampling or measurement.

These records shall be physically present or electronically accessible at the source location for a minimum of three (3) years. The records may be stored elsewhere for the remaining two (2) years as long as they are available upon request. If the Commissioner makes a request for records to the Permittee, the Permittee shall furnish the records to the Commissioner within a reasonable time.

- (b) Unless otherwise specified in this permit, for all record keeping requirements not already legally required, the Permittee shall be allowed up to ninety (90) days from the date of permit issuance or the date of initial start-up, whichever is later, to begin such record keeping.

C.18 General Reporting Requirements [326 IAC 2-7-5(3)(C)][326 IAC 2-1.1-11] [40 CFR 64][326 IAC 3-8]

- (a) The Permittee shall submit the attached Quarterly Deviation and Compliance Monitoring Report or its equivalent. Proper notice submittal under Section B -Emergency Provisions satisfies the reporting requirements of this paragraph. Any deviation from permit requirements, the date(s) of each deviation, the cause of the deviation, and the response steps taken must be reported except that a deviation required to be reported pursuant to an applicable requirement that exists independent of this permit, shall be reported according to the schedule stated in the applicable requirement and does not need to be included in this report. This report shall be submitted not later than thirty (30) days after the end of the reporting period. The Quarterly Deviation and Compliance Monitoring Report shall include a certification that meets the requirements of 326 IAC 2-7-6(1) by a "responsible official" as defined by 326 IAC 2-7-1(35). A deviation is an exceedance of a permit limitation or a failure to comply with a requirement of the permit.

On and after the date by which the Permittee must use monitoring that meets the requirements of 40 CFR Part 64 and 326 IAC 3-8, the Permittee shall submit CAM reports to the IDEM, OAQ.

A report for monitoring under 40 CFR Part 64 and 326 IAC 3-8 shall include, at a minimum, the information required under paragraph (a) of this condition and the following information, as applicable:

- (1) Summary information on the number, duration and cause (including unknown cause, if applicable) of excursions or exceedances, as applicable, and the corrective actions taken;
- (2) Summary information on the number, duration and cause (including unknown cause, if applicable) for monitor downtime incidents (other than downtime

DRAFT

associated with zero and span or other daily calibration checks, if applicable);
and

- (3) A description of the actions taken to implement a QIP during the reporting period as specified in Section C-Response to Excursions or Exceedances. Upon completion of a QIP, the owner or operator shall include in the next summary report documentation that the implementation of the plan has been completed and reduced the likelihood of similar levels of excursions or exceedances occurring.

The Permittee may combine the Quarterly Deviation and Compliance Monitoring Report and a report pursuant to 40 CFR 64 and 326 IAC 3-8.

- (b) The address for report submittal is:

Indiana Department of Environmental Management
Compliance and Enforcement Branch, Office of Air Quality
100 North Senate Avenue
MC 61-53 IGCN 1003
Indianapolis, Indiana 46204-2251

- (c) Unless otherwise specified in this permit, any notice, report, or other submission required by this permit shall be considered timely if the date postmarked on the envelope or certified mail receipt, or affixed by the shipper on the private shipping receipt, is on or before the date it is due. If the document is submitted by any other means, it shall be considered timely if received by IDEM, OAQ on or before the date it is due.
- (d) Reporting periods are based on calendar years, unless otherwise specified in this permit. For the purpose of this permit "calendar year" means the twelve (12) month period from January 1 to December 31 inclusive.

Stratospheric Ozone Protection

C.19 Compliance with 40 CFR 82 and 326 IAC 22-1

Pursuant to 40 CFR 82 (Protection of Stratospheric Ozone), Subpart F, except as provided for motor vehicle air conditioners in Subpart B, the Permittee shall comply with applicable standards for recycling and emissions reduction.

DRAFT

SECTION D.1

EMISSIONS UNIT OPERATION CONDITIONS

Emissions Unit Description:

(a)

Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
A030000	Truck Dump No. 1 *2006	600	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A020000	Truck Dump No. 2 *2006	600	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
Truck Dump No. 3 (with doors)	Truck Dump No. 3 *2013	360	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A030100	Discharge Conveyor No. 1 *2006	600	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A020100	Discharge Conveyor No. 2 *2006	600	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A040000	Bean Receiving Leg No. 1 *2006	600	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A050000	Bean Receiving Leg No. 2 *2006	600	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A010000	Rail Dump and Rail Collection Conveyor *2006	600	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A010100	Rail Scale Discharge Conveyor *2006	360	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A150100	Cross Bin No 1 thru 3 *2006	600	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A120100	Cross Bin No 4 thru 6 *2006	600	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A151000	Discharge Bin No 1 thru 3 *2006	360	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A121000	Discharge Bin No 4 thru 6 *2006	360	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD

DRAFT

A153000	Day Bin Leg *2006	360	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
AF-2 A200000	Grain Receiving/Meal Loadout Baghouse *2006, **2010	38,000 acfm @ 0.005 grain/acf outlet gr loading		Stack AF-2	Yes under NSPS DD
A152000	West Bin Cross Conveyor 1-3 *2006, **2010 and 2011	360	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A122000	East Bin Cross Conveyor 4-6 *2006, **2010	360	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A130100	West Bin Feed Conveyor *2006, **2010	600	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A100100	East Bin Feed Conveyor *2006, **2010	600	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
G020500	Meal Storage Feed Conveyor *2006, **2010, **2011, ** 2015	200	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
G070300	Truck Meal Loadout Feed Conveyor *2006, **2010, ** 2015	300	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
G290000	Truck Collection Conveyor *2010, ** 2015	300	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
G280000	Truck Loader No.1 *2010, ** 2015	330	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
G270000	Truck Loader No.2 *2010, ** 2015	330	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
G080000	Truck Pelleted Hull Loadout Bin *2006, **2010, ** 2015	148	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
G070000	Truck Meal Loadout Bin *2006, **2010, 2012, ** 2015	300	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
G180000	Rail Pelleted Hull Loadout Bin *2010, ** 2015	148	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
G130000	Rail Meal Loadout Bin *2006, **2010, 2012, **2015	300	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG

DRAFT

G160000	Pellet Hulls Conveyor to Loadout *2006, **2010, 2012, ** 2015	17.0	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
G170000	Rail Car Collection Conveyor *2006, **2010, ** 2015	300	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
G220000	Rail Car Loadout *2010, ** 2015	330	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
G010100	Meal Reclaim Conveyor *2006, **2010, ** 2015	200	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
G010200	Meal Reclaim Leg *2010, ** 2015	200	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
AF-1	Meal Loadout Baghouse *2015	22,125 cfm @ 0.005 grain/acf outlet gr loading		Stack AF-1	
Rail Receiving Leg	Rail Receiving Leg *2013	360	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
Truck Dump No. 3 Receiving Conveyor	Truck Dump No. 3 Receiving Conveyor *2013	360	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
Truck Dump No. 3 Receiving Leg	Truck Dump No. 3 Receiving Leg *2013	360	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
Scalperator Leg	Scalperator Leg *2013	210	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
Scalperator Feed Conveyor	Scalperator Feed Conveyor *2013	210	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
Scalperator Jack Leg	Scalperator Jack Leg *2013	210	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
Scalperator	Scalperator *2013	210	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A060000	Screener *2006, **2011 and 2012	264	Prep Exhaust Baghouse	Stack AF-3	Yes under NESHAP GGGG

DRAFT

A160300	VSC Leg Feed Conveyor *2006, **2010 and 2012	264	Prep Exhaust Baghouse	Stack AF-3	Yes under NESHAP GGGG
A170000	Screenings Tank *2006	5	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
A170300	Screenings Recycle Leg *2006	5	Prep exhaust baghouse	Stack AF-3	Yes under NSPS DD and NESHAP GGGG
B011300	Bean Weigh Scale *2006, **2012	264	Prep exhaust baghouse	Stack AF-3	Yes under NSPS DD and NESHAP GGGG
B310000	Screenings Weight Belt *2006	5	Prep exhaust baghouse	Stack AF-3	Yes under NSPS DD and NESHAP GGGG
AF-7	Pod Grinder/Screeners Baghouse *2011	5,000 acfm		Stack AF-7	Yes under NESHAP GGGG
B310200	Pod Grinder/Destoner *2006, **2010	5	Pod Grinder/Screeners Baghouse	Stack AF-7	Yes under NESHAP GGGG
B011200	VSC Feed Leg *2006, **2012	264	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
A060400	Screeners Feed Conveyor *2010, **2012	264	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
B010100	Whole Bean Aspirator No 1 *2006, **2012	264	Prep exhaust baghouse	Stack AF-3	Yes under NSPS DD and NESHAP GGGG
B020100	Whole Bean Aspirator No 2 *2006, **2012	264	Prep exhaust baghouse	Stack AF-3	Yes under NSPS DD and NESHAP GGGG
B010900	Whole Bean Aspirator Cyclone *2006, **2012	264	Prep exhaust baghouse	Stack AF-3	Yes under NSPS DD and NESHAP GGGG
B030800	Conditioned Bean Feed Conveyor *2006, **2012	264	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
B030900	Hull Collection Conveyor *2006, **2012	17.0	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
E130000	Hull Screeners No.1 *2006	9.6	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
E150000	Hull Screeners No.2 *2006	9.6	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
B440000	Secondary Hull Collection L-Path *2010, **2012	17.0	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
B430000	Secondary Hull Collection Conveyor *2010, **2012	17.0	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG

DRAFT

E070300	4 Hour Hull Tank *2006, **2012	17.0	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
E080000	Pellet Cooler *2006, **2012	17.0	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
E090000	Pellet Cooler Cyclone *2006, **2012	17.0	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
E050200	Hull Hammer Mill Feeder *2006, **2012	17.0	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
E050000	Hull Hammer Mill *2006, **2012	17.0	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
E050100	Hull Hammer Mill Plenum *2006, **2012	17.0	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
G050100	Pelleted Hulls Leg *2006, **2012	17.0	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
G050300	Pelleted Hulls Storage Conveyor *2006, **2010, 2012, **2015	17.0	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
E050400	Hulls Addition Screw *2011, **2012	17.0	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
B010300	Conditioner Bean Loop Path *2006, **2012	264	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
AF-3 G100000	Prep exhaust baghouse *2006	28,900 acfm @ 0.005 grain/acf outlet grain loading		Stack AF-3	Yes under NESHAP GGGG
	Bean Storage Bins #2, #3, #6, and #7 *2006	600 tons/hr and each Bin has a maximum storage capacity of 500,000 bushels	None	None	Yes under NSPS DD
	Bean Storage Bins #4 and #8 *2013	600 tons/hr and each Bin has a maximum storage capacity of 500,000 bushels	None	None	Yes under NSPS DD

DRAFT

	Bean Storage Silos #1 and #5 *2008	600 tons/hr and each Bin has a maximum storage capacity of 500,000 bushels	None	None	Yes under NSPS DD
--	---------------------------------------	--	------	------	-------------------

Note *Approved in the year indicated above for construction.

Note **Approved in the year indicated above for modification.

(b)

Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
Piles #1 and #2 *2008	Two (2) covered seasonal grain storage piles	each with a maximum storage capacity of 1,000,000 bushels of soybeans	None	None	Yes under NSPS DD

Note *Approved in the year indicated above for construction.

(c)

G150000	Meal Conveyor to Loadout *2006, **2012, ** 2015	198	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHA P GGGG
---------	--	-----	----------------------------	------------	------------------------

(The information describing the process contained in this emissions unit description box is descriptive information and does not constitute enforceable conditions.)

Emission Limitations and Standards [326 IAC 2-7-5(1)]

D.1.1 PSD Minor Limit for Particulate [326 IAC 2-2]

In order to render the requirements of 326 IAC 2-2 (Prevention of Significant Deterioration (PSD)) not applicable, the Permittee shall comply with the following PM, PM₁₀, and PM_{2.5} limits:

- (a) The amount of soybeans processed shall be less than 2,251,836 tons per twelve (12) consecutive month period, with compliance determined at the end of each month.
- (b) The PM, PM₁₀, and PM_{2.5} emissions from the following Processes shall be less than the emission limits listed in the table below:

Process	Control	PM Limit (lbs/hour)	PM ₁₀ Limit (lbs/hour)	PM _{2.5} Limit (lbs/hour)
Meal Loadout	Baghouse AF-1	0.95	0.95	0.95
Grain Receiving/Meal Loadout	Baghouse AF-2	1.64	1.64	1.64
Prep Area	Baghouse AF-3	1.26	1.26	1.26

- (c) The PM, PM₁₀, and PM_{2.5} emissions from the following Process shall be less than the emission limits listed in the table below:

DRAFT

Process	Control	PM Limit (lbs/hour)	PM ₁₀ Limit (lbs/hour)	PM _{2.5} Limit (lbs/hour)
Pod Grinder/Destoner	Pod Grinder/ Screener Baghouse AF-7	1.5	1.5	1.5

Compliance with the soybean usage limit in Condition D.1.1(a) in combination with the PM, PM₁₀, and PM_{2.5} emission limits in Conditions D.1.1(b), D.1.1(c), D.2.1, D.3.1, D.5.1, D.6.1 and D.7.1 and with the potential to emit PM, PM₁₀, and PM_{2.5} from all other emission units at the source, shall limit the PM, PM₁₀, and PM_{2.5} emissions from the entire source to less than 250 tons per twelve (12) consecutive month period and render the requirements of 326 IAC 2-2 (Prevention of Significant Deterioration (PSD)) not applicable.

D.1.2 Particulate Emissions Limitations [326 IAC 6-3-2]

- (a) Pursuant to 326 IAC 6-3-2, the particulate emissions from each of the following processes shall not exceed the pound per hour limitations specified in the following table:

Emission unit ID	Emissions Units	¹ Baghouse ID	Maximum Process Weight (tons/hour) for each unit	326 IAC 6-3 Limit (lbs/hr) for each unit
G160000	Pellet Hulls Conveyor to Loadout	AF-1	17.0	27.36
B030900	Hull Collection Conveyor	AF-3	17.0	27.36
E130000 and E150000	Hull Screener No. 1 and No. 2	AF-3	9.6	18.66
B430000	Secondary Hull Collection Conveyor	AF-3	17.0	27.36
B440000	Secondary Hull Collection L-Path	AF-3	17.0	27.36
E080000	Pellet Cooler	AF-3	17.0	27.36
E050000, E050200, and E050100	Hull Hammer Mill, Hull Hammer Mill Feeder, and Hull Hammer Mill Plenum	AF-3	17.0	27.36
G050100	Pelleted Hulls Leg	AF-3	17.0	27.36
G050300	Pelleted Hulls Storage Conveyor	AF-1	17.0	27.36
E050400	Hulls Addition Screw	AF-3	17.0	27.36
B310200	Pod Grinder/Destoner	AF-7	5.0	12.05

Note 1: For emission units that exhaust through the same stack, the source will need to demonstrate compliance with 326 IAC 6-3-2 during normal operations using the most stringent limit (e.g. calculated from the emission unit operating at the lowest process weight in ton/hr).

The particulate emissions limitations from the above table shall be calculated using the following equation:

Interpolation of the data for the process weight rate up to sixty thousand (60,000) pounds per hour shall be accomplished by use of the equation:

$$E = 4.10 P^{0.67} \quad \text{where } E = \text{rate of emission in pounds per hour; and} \\ P = \text{process weight rate in tons per hour}$$

- (b) Pursuant to 326 IAC 6-3-2, the particulate emissions from each of the following processes shall not exceed the pound per hour limitations specified in the following table:

DRAFT

Emission unit ID	Emissions Units	¹ Baghouse ID	Maximum Process Weight (tons/hour) for each unit	326 IAC 6-3 Limit (lbs/hr) for each unit
A030000 and A020000	Truck Dumps No. 1 and No. 2	AF-2	600	71.16
Truck Dump No. 3	Truck Dump No. 3	AF-2	360	65.09
A030100, A020100, A040000, A050000, A130100, and A100100	Discharge Conveyors No. 1 and No. 2, Bean Receiving Legs No. 1 and No. 2, and East and West Bin Feed Conveyors	AF-2	600	71.16
A010000	Rail Dump and Rail Collection Conveyor	AF-2	600	71.16
A150100 and A120100	Cross Bins No 1 thru 6	AF-2	600	71.16
A153000, A010100, A151000, A121000, A152000, and A122000	Day Bin Leg, Rail Scale Discharge Conveyor, Discharge Bin No 1 thru 6, West Bin Cross Conveyor 1-3, and East Bin Cross Conveyor 4-6	AF-2	360.0	65.09
G280000 and G270000	Truck Loader No.1 and No. 2	AF-1	330	64.09
G220000	Rail Car Loadout (Pellets/Hulls)	AF-1	330	64.09
G130000 and G070000	Rail Meal Loadout Bin and Truck Meal Loadout Bin	AF-1	300	63.00
G150000	Meal Conveyor to Loadout	AF-1	198	58.40
G020500	Meal Storage Feed Conveyor	AF-2	200	58.51
G070300, G170000 and G290000	Truck Meal Loadout Feed Conveyor, Rail Car Collection Conveyor and Truck Collection Conveyor	AF-1	300	63.00
G010100 and G010200	Meal Reclaim Conveyor and Meal Reclaim Leg	AF-1	200	58.51
Piles #1 and #2	Two (2) covered seasonal grain storage piles, identified as Piles #1 and #2	N/A	360	65.09
A060000	Screener	AF-3	264	61.56
B011300	Bean Weigh Scale	AF-3	264	61.56
B011200, A160300, B060400 and B030800	VSC Feed Leg, VSC Leg Feed Conveyor, Screener Feed Conveyor and Conditioned Bean Feed Conveyor	AF-3	264	61.56
B010100 and B020100	Whole Bean Aspiration No. 1 and No. 2	AF-3	264	61.56
B010300	Conditioner Bean Loop Path	AF-3	264	61.56
	Bean Storage Bins #2, #3, #4, #6, #7, and #8	N/A	600	71.16
	Bean Storage Silos #1 and #5	N/A	600	71.16
Rail Receiving Leg, Truck Dump No. 3 Receiving Conveyor & Truck Dump No. 3 Receiving Leg	Rail Receiving Leg, Truck Dump No. 3 Receiving Conveyor & Truck Dump No. 3 Receiving Leg	AF-2	360	65.09
Scalperator Leg, Scalperator Feed Conveyor, Scalperator Jack Leg, Scalperator	Scalperator Leg, Scalperator Feed Conveyor, Scalperator Jack Leg, Scalperator	AF-2	210	59.03

Note 1: For emission units that exhaust through the same stack, the source will need to demonstrate compliance with 326 IAC 6-3-2 during normal operations using the most stringent limit (e.g. calculated from the emission unit operating at the lowest process weight in ton/hr).

DRAFT

The particulate emissions limitations from the above table shall be calculated using the following equation:

Interpolation and extrapolation of the data for the process weight rate in excess of sixty thousand (60,000) pounds per hour shall be accomplished by use of the equation:

$$E = 55.0 P^{0.11} - 40 \quad \text{where } E = \text{rate of emission in pounds per hour; and} \\ P = \text{process weight rate in tons per hour}$$

- (c) Pursuant to 326 IAC 6-3-2(e)(3), when the process weight exceeds 200 tons per hour, the maximum allowable emissions may exceed the emission limits shown in paragraph (a), provided the concentration of particulate matter in the gas discharged to the atmosphere is less than 0.10 pounds per 1,000 pounds of gases.

D.1.3 Significant Source Modification Avoidance Limit [326 IAC 2-7-10.5(f)]

Pursuant to Minor Source Modification (MSM) No. 085-24676-00102, issued on April 28, 2008, in order to render the requirements of 326 IAC 2-7-10.5(f) not applicable, the Permittee shall comply with the following PM, PM₁₀, and PM_{2.5} limits:

- (a) The emissions for PM shall not exceed 0.061 pound/Ton of material for the two (2) Storage Bean Piles #1 and #2;
- (b) The emissions for PM₁₀ shall not exceed 0.034 pound/Ton of material for the two (2) Storage Bean Piles #1 and #2;
- (c) The emissions for PM_{2.5} shall not exceed 0.0058 pound/Ton of material for the two (2) Storage Bean Piles #1 and #2; and
- (d) The soybean throughput to the two (2) Storage Bean Piles #1 and #2 shall be less than 8,000,000 bushels per twelve (12) consecutive month period with compliance determined at the end of each month.

D.1.4 Preventive Maintenance Plan [326 IAC 2-7-5(12)]

A Preventive Maintenance Plan is required for these facilities and their control devices. Section B - Preventive Maintenance Plan contains the Permittee's obligation with regard to the preventive maintenance plan required by this condition.

Compliance Determination Requirements [326 IAC 2-7-5(1)]

D.1.5 Particulate Control

- (a) In order to assure compliance with Conditions D.1.1(b) and D.1.1(c), baghouses AF-1, AF-2, AF-3, and AF-7 shall be in operation and control emissions from all emission units exhausting to stacks AF-1, AF-2, AF-3, and AF-7 at all times when an emission unit that the baghouses control are in operation.
- (b) In order to assure compliance with Condition D.1.2, Baghouse AF-1 shall be in operation and control emissions from all emission units exhausting to baghouse AF-1 at all times an emission unit that the baghouse controls is in operation.
- (c) In order to assure compliance with Condition D.1.2, Baghouse AF-2 shall be in operation and control emissions from Truck Dumps No. 1 and No. 2 at all times the Truck Dumps No. 1 and No.2 are in operation.
- (d) In order to assure compliance with Condition D.1.2, baghouse AF-3 shall be in operation and control emissions from the Hull Hammer Mill, Hull Hammer Mill Feeder, and Hull

DRAFT

Hammer Mill Plenum at all times the Hull Hammer Mill, Hull Hammer Mill Feeder, and Hull Hammer Mill Plenum are in operation.

- (e) In the event that bag failure is observed in a multi-compartment baghouse, if operations will continue for ten (10) days or more after the failure is observed before the failed units will be repaired or replaced, the Permittee shall promptly notify the IDEM, OAQ of the expected date the failed units will be repaired or replaced. The notification shall also include the status of the applicable compliance monitoring parameters with respect to normal, and the results of any response actions taken up to the time of notification.

D.1.6 Testing Requirements [326 IAC 2-1.1-11]

- (a) In order to demonstrate compliance with Conditions D.1.1(b) and D.1.2, the Permittee shall perform PM, PM10 and PM2.5 testing for the baghouse AF-1, utilizing methods as approved by the Commissioner at least once every five (5) years from the date of the most recent valid compliance demonstration.
- (b) In order to demonstrate compliance with Conditions D.1.1(b), D.1.2 and E.2.2 (40 CFR 60.302(b)(1)), the Permittee shall perform PM, PM10 and PM2.5 testing for the baghouse AF-2 (associated with the grain receiving/meal loadout system), utilizing methods as approved by the Commissioner at least once every five (5) years from the date of the most recent valid compliance demonstration.
- (c) In order to demonstrate compliance with Conditions D.1.1(b) and D.1.2 and E.2.2 (40 CFR 60.302(b)(1)), the Permittee shall perform PM, PM10 and PM2.5 testing for the baghouse AF-3 (associated with the prep system), utilizing methods as approved by the Commissioner at least once every five (5) years from the date of the most recent valid compliance demonstration.

Testing shall be conducted in accordance with the provisions of 326 IAC 3-6 (Source Sampling Procedures). Section C - Performance Testing contains the Permittee's obligation with regard to the performance testing required by this condition. PM10 and PM2.5 include filterable and condensable particulate matter.

Compliance Monitoring Requirements [326 IAC 2-7-5(1)][326 IAC 2-7-6(1)]

D.1.7 Visible Emissions Notations [40 CFR 64]

- (a) Pursuant to 40 CFR 64 (CAM), visible emission notations of the stacks AF-1, AF-2, AF-3, and AF-7 exhausts shall be performed once per day during normal daylight operations. A trained employee shall record whether emissions are normal or abnormal.
- (b) For processes operated continuously, "normal" means those conditions prevailing, or expected to prevail, eighty percent (80%) of the time the process is in operation, not counting startup or shut down time.
- (c) In the case of batch or discontinuous operations, readings shall be taken during that part of the operation that would normally be expected to cause the greatest emissions.
- (d) A trained employee is an employee who has worked at the plant at least one (1) month and has been trained in the appearance and characteristics of normal visible emissions for that specific process.
- (e) If abnormal emissions are observed, the Permittee shall take a reasonable response. Section C - Response to Excursions or Exceedances contains the Permittee's obligation with regard to the reasonable response steps required by this condition. Failure to take response shall be considered a deviation from this permit.

DRAFT

D.1.8 Parametric Monitoring [40 CFR 64]

- (a) The Permittee shall record pressure drop across baghouses AF-2, and AF-3, used in conjunction with the grain receiving/meal loadout system and prep system, at least once per day when the grain receiving/meal loadout system and prep system are in operation. When for any one reading, the pressure drop across Baghouses AF-2 and AF-3 is outside the normal range presented in the table below or established during the latest stack test, the Permittee shall take a reasonable response. Section C - Response to Excursions or Exceedances contains the Permittee's obligation with regard to the reasonable response steps required by this condition. A pressure reading that is outside the above mentioned range is not a deviation from this permit. Failure to take response steps shall be considered a deviation from this permit.

Baghouse	Lower Limit (inches of water)	Upper Limit (inches of water)
AF-2	2.0	8.0
AF-3	6.0	12.0

- (b) The instrument used for determining the pressure shall comply with Section C - Instrument Specifications, of this permit, shall be subject to approval by IDEM, OAQ, and shall be calibrated or replaced at least once every twelve (12) months or more frequently if recommended by the instrument manufacture's specifications.

D.1.9 Broken or Failed Bag Detection [40 CFR 64]

- (a) For a single compartment baghouse controlling emissions from a process operated continuously, a failed unit and the associated process shall be shut down immediately until the failed unit has been repaired or replaced. Operations may continue only if the event qualifies as an emergency and the Permittee satisfies the requirements of the emergency provisions of this permit (Section B - Emergency Provisions).
- (b) For a single compartment baghouse controlling emissions from a batch process, the feed to the process shall be shut down immediately until the failed unit has been repaired or replaced. The emissions unit shall be shut down no later than the completion of the processing of the material in the emissions unit. Operations may continue only if the event qualifies as an emergency and the Permittee satisfies the requirements of the emergency provisions of this permit (Section B - Emergency Provisions).

Bag failure can be indicated by a significant drop in the baghouse's pressure reading with abnormal visible emissions, by an opacity violation, or by other means such as gas temperature, flow rate, air infiltration, leaks, dust traces, or triboflows.

Record Keeping and Reporting Requirements [326 IAC 2-7-5(3)][326 IAC 2-7-19]**D.1.10 Record Keeping Requirements**

- (a) To document the compliance status with Condition D.1.1(a), the Permittee shall maintain records of the quantity of soybeans processed.
- (b) To document the compliance status with Condition D.1.3(d), the Permittee shall maintain monthly records of the soybean throughput in the two (2) Storage Bean Piles #1 and #2. Records necessary to demonstrate compliance shall be available no later than 30 days of the end of each compliance period.
- (c) To document the compliance status with Condition D.1.7 the Permittee shall maintain a daily record of visible emission notations of the stack exhaust from Stacks AF-1, AF-2, AF-3, and AF-7. The Permittee shall include in its daily record when a visible emission notation is not taken and the reason for the lack of visible emission notation (e.g. the process did not operate that day).

DRAFT

- (d) To document the compliance status with Condition D.1.8 the Permittee shall maintain a daily record of the pressure drop across baghouses AF-2 and AF-3, used to control the grain receiving and prep system. The Permittee shall include in its daily record when a pressure drop reading is not taken and the reason for the lack of a pressure drop reading (e.g. the process did not operate that day).
- (e) Section C - General Record Keeping Requirements of this permit contains the Permittee's obligations with regard to the records required by this condition.

D.1.11 Reporting Requirements

A quarterly summary of the information to document the compliance status with Conditions D.1.1(a) and Condition D.1.3(d) shall be submitted using the reporting forms located at the end of this permit, or their equivalent, not later than thirty (30) days after the end of the quarter being reported. Section C - General Reporting contains the Permittee's obligation with regard to the reporting required by this condition. The reports submitted by the Permittee do require a certification that meets the requirements of 326 IAC 2-7-6(1) by a "responsible official," as defined by 326 IAC 2-7-1(35).

DRAFT

SECTION D.2

EMISSIONS UNIT OPERATION CONDITIONS

Emissions Unit Description:

(c)

Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
C200100	Flaker Feed Loop Conveyor *2010, **2012	247	Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
C010600	Flake Collection Conveyor (12 flakers) *2006, **2010 and 2012	247	Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
C010000	Flaking Roll No. 1 *2013	22.9	***Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
C020000	Flaking Roll No. 2 *2006	22.9	Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
C030000	Flaking Roll No. 3 *2006	22.9	Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
C040000	Flaking Roll No. 4 *2012	22.9	***Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
C050000	Flaking Roll No. 5 *2006	22.9	Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
C060000	Flaking Roll No. 6 *2006	22.9	Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
C070000	Flaking Roll No. 7 *2012	22.9	***Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
C080000	Flaking Roll No. 8 *2006	22.9	Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
C090000	Flaking Roll No. 9 *2006	22.9	Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
C100000	Flaking Roll No. 10 *2013	22.9	***Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
C0110000	Flaking Roll No. 11 *2009	22.9	***Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
C0120000	Flaking Roll No. 12 *2009	22.9	***Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
AF-4 C110000	Flaker aspiration baghouse *2006	24,000 acfm @ 0.005 grain/acf outlet grain loading		Stack AF-4	Yes under NESHAP GGGG
B040000	Hulloosenator No. 1 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B080100	Hulloosenator No. 2 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B130000	Hulloosenator No. 3 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG

DRAFT

B170000	Hulloosensor No. 4 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B050000	Cascade Dryer No. 1 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B090000	Cascade Dryer No. 2 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B140000	Cascade Dryer No. 3 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B180000	Cascade Dryer No. 4 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B210000	CCD Cyclone *2006, **2010	42,000 cfm	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B060000	Cracking Roll No.1 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B100000	Cracking Roll No.2 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B150000	Cracking Roll No.3 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B190000	Cracking Roll No.4 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B070000	Cascade Conditioner No. 1 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B110000	Cascade Conditioner No. 2 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B160000	Cascade Conditioner No. 3 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B200000	Cascade Conditioner No. 4 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B230000	CCC Cyclone *2006, **2010	42,000 cfm	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
E130100	Secondary Aspirator No 1 *2006, **2010	9.6	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
E150100	Secondary Aspirator No 2 *2006, **2010	9.6	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
E160000	Secondary Aspirator Cyclone *2006, **2010	9.6	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
A160100	Feed Day Tank Conveyor *2006	600	Hot dehulling baghouse	Stack AF-5	Yes under NSPS DD
A160000 (Day Tank) A160500 (Aspirator) B420000 (Cyclone)	Day Tank (with aspirator and cyclone) *2006, **2010, **2012, and **2014	264	Hot dehulling baghouse	Stack AF-5	Yes under NSPS DD
AF-5 B260000	Hot dehulling baghouse *2006	43,000 acfm @ 0.005 grain/acf outlet grain loading		Stack AF-5	Yes under NESHAP GGGG

DRAFT

E020300	Grinding Discharge Conveyor *2011, **2012	198	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E020400	Hammer Mill Mixing Conveyor *2006, **2011 and 2012	198	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E010100	Meal L-Path Conveyor *2006, **2012	198	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E010300	Meal Hammer Mill Feed Conveyor *2006, **2012	198	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E020200	Meal Hammer Mill Feeder No. 1 *2006	74.0	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E030200	Meal Hammer Mill Feeder No. 2 *2006	74.0	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E040200	Meal Hammer Mill Feeder No. 3 (switch) *2006	74.0	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E020000	Meal Hammer Mill No. 1 *2006	74.0	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E030000	Meal Hammer Mill No. 2 *2006	74.0	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E040000	Meal Hammer Mill No. 3 (switch) *2006	74.0	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E020100	Meal Hammer Mill Bin No. 1 *2006	74.0	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E030100	Meal Hammer Mill Bin No. 2 *2006	74.0	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E040100	Meal Hammer Mill Bin No. 3 (switch) *2006	74.0	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E230200	Meal Hammer Mill Feeder No. 5 *2012	74.0	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E230000	Meal Hammer Mill No. 5 *2012	74.0	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E230100	Meal Hammer Mill Bin No. 5 *2012	74.0	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
G010300	Meal Leg *2006, **2010 and 2012	198	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
AF-6 E110000	Mill Grinding Baghouse *2006	18,000 acfm @ 0.005 grain/acf outlet grain loading		Stack AF-6	Yes under NESHAP GGGG
B010000	VSC No. 1 *2006, **2012	132	VSC Cyclone No. 1	Stack S-1	Yes under NESHAP GGGG

DRAFT

B020000	VSC No. 2 *2006, **2012	132	VSC Cyclone No. 2	Stack S-6	Yes under NESHAP GGGG
B010500	VSC Air Heater No. 1 *2006, **2012	264	VSC Cyclone No. 1	Stack S-1	Yes under NESHAP GGGG
B020500	VSC Air Heater No. 2 *2015	264	VSC Cyclone No. 2	Stack S-6	Yes under NESHAP GGGG
B010700	VSC Cyclone No. 1 *2006, **2010	42,000 cfm		Stack S-1	Yes under NESHAP GGGG
B020700	VSC Cyclone No. 2 *2015	42,000 cfm		Stack S-6	Yes under NESHAP GGGG
B120000	Jet Dryer No. 1 *2006, **2010 and 2012	132	Jet Dryer Baghouse AF-8	Stack S-1	Yes under NESHAP GGGG
B030000	Jet Dryer No. 2 *2006, **2012	132	Jet Dryer Baghouse AF-9	Stack S-1	Yes under NESHAP GGGG
B120100A	Jet Dryer Baghouse AF-8	74,000 acfm		Stack S-1	Yes under NESHAP GGGG
B120100B	Jet Dryer Baghouse AF-9	74,000 acfm		Stack S-1	Yes under NESHAP GGGG

Note *Approved in the year indicated above for construction.
 Note **Approved in the year indicated above for modification.
 Note ***The Flaker aspiration baghouse has been determined to be integral to the process for this unit.

(The information describing the process contained in this emissions unit description box is descriptive information and does not constitute enforceable conditions.)

Emission Limitations and Standards [326 IAC 2-7-5(1)]

D.2.1 PSD Minor Limit for Particulate [326 IAC 2-2]

In order to render the requirements of 326 IAC 2-2 (Prevention of Significant Deterioration (PSD)) not applicable, the Permittee shall comply with the following PM, PM₁₀, and PM_{2.5} limits:

The PM, PM₁₀, and PM_{2.5} emissions from the following Processes shall be less than the emission limits listed in the table below:

Process	Control	PM Limit (lbs/hour)	PM ₁₀ Limit (lbs/hour)	PM _{2.5} Limit (lbs/hour)
Jet Dryer/VSC	VSC cyclones and Jet Dryer Baghouses AF-8 and AF-9	4.93	3.35	3.35
Hot Dehulling	Baghouse AF-5	2.56	2.56	2.56
Flaker Aspiration	Baghouse AF-4	1.03	1.03	1.03
Meal Grinding	Baghouse AF-6	0.945	0.945	0.945

Compliance with the above limits, in combination with the soybean usage limit in Condition D.1.1(a) and with the PM, PM₁₀, and PM_{2.5} emission limits in Conditions D.1.1(b), D.1.1(c), D.3.1, D.5.1, D.6.1, and D.7.1, and with the potential to emit PM, PM₁₀, and PM_{2.5} from all other emission units at the source, shall limit the PM, PM₁₀, and PM_{2.5} emissions from the entire source to less than 250 tons per twelve (12) consecutive month period and render the requirements of 326 IAC 2-2 (Prevention of Significant Deterioration (PSD)) not applicable.

D.2.2 Particulate Emissions Limitations [326 IAC 6-3-2]

(a) Pursuant to 326 IAC 6-3-2, the particulate emissions from each of the following processes shall not exceed the pound per hour limitations specified in the following table:

DRAFT

Emission unit ID	Emissions Units	¹ Baghouse ID	Maximum Process Weight (tons/hour) for each unit	326 IAC 6-3 Limit (lbs/hr) for each unit
E130100 and E150100	Secondary Aspirator No 1, and No. 2	AF-5	9.6	18.66
C020000, C030000, C050000, C060000, C080000, and C090000	Flaking Rolls No. 2, 3, 5, 6, 8, and 9	AF-4	22.9	33.41

Note 1: For emission units that exhaust through the same stack, the source will need to demonstrate compliance with 326 IAC 6-3-2 during normal operations using the most stringent limit (e.g. calculated from the emission unit operating at the lowest process weight in ton/hr).

The particulate emissions limitations from the above table shall be calculated using the following equation:

Interpolation of the data for the process weight rate up to sixty thousand (60,000) pounds per hour shall be accomplished by use of the equation:

$$E = 4.10 P^{0.67} \quad \text{where } E = \text{rate of emission in pounds per hour; and} \\ P = \text{process weight rate in tons per hour}$$

- (b) Pursuant to 326 IAC 6-3-2, the particulate emissions from each of the following processes shall not exceed the pound per hour limitations specified in the following table:

Emission unit ID	Emissions Units	¹ Baghouse ID	Maximum Process Weight (tons/hour) for each unit	326 IAC 6-3 Limit (lbs/hr) for each unit
B120000 and B030000	Jet Dryer No. 1 and No. 2	Jet Dryer Baghouses AF-8 and AF-9	132	54.11
B010500	VSC Air Heater No. 1	VSC Cyclone No. 1	264	61.56
B020500	VSC Air Heater No. 2	VSC Cyclone No. 2	264	61.56
B010000 and B020000	Vertical Seed Conditioner (VSC) No. 1 and No. 2	VSC Cyclones	132	54.11
B040000, B080100, B130000, and B170000	Hulloosinator No. 1, No. 2, No. 3, and No. 4	AF-5	66.0	47.20
B050000, B090000, B140000, and B180000	Cascade Dryer No. 1, No. 2, No. 3 and No. 4	AF-5	66.0	47.20
B060000, B100000, B150000, and B190000	Cracking Roll No.1, No. 2, No. 3 and No. 4	AF-5	66.0	47.20
B070000, B110000, B160000, and B200000	Cascade Conditioner No. 1, No. 2, No. 3 and No. 4	AF-5	66.0	47.20
C200100 and C010600	Flaker Feed Loop Conveyor and Flake Collection Conveyor	AF-4	247	60.82

DRAFT

Emission unit ID	Emissions Units	¹ Baghouse ID	Maximum Process Weight (tons/hour) for each unit	326 IAC 6-3 Limit (lbs/hr) for each unit
E020300, E020400, E010100, and E010300	Grinding Discharge Conveyor, Hammer Mill Mixing Conveyor, Meal L-Path Conveyor, and Meal Hammer Mill Feed Conveyor	AF-6	198	58.40
E020200, E030200, E040200, E020000, E030000, and E040000	Meal Hammer Mill Feeders No. 1, No. 2 and No. 3, Meal Hammer Mills No. 1, No. 2 and No. 3	AF-6	74	48.30
E230200 and E230000	Meal Hammer Mill Feeder No. 5 and Meal Hammer Mill No. 5	AF-6	74.0	48.30
G010300	Meal Leg	AF-6	198	58.40
E020100, E030100, and E040100	Meal Hammer Mill Bins No. 1, No. 2 and No. 3	AF-6	74	48.30
E230100	Meal Hammer Mill Bin No. 5	AF-6	74.0	48.30
A160100	Feed Day Tank Conveyor	AF-5	600	71.16
A160000 A160500 B420000	Day Tank (with aspirator and cyclone)	AF-5	264	61.56

Note 1: For emission units that exhaust through the same stack, the source will need to demonstrate compliance with 326 IAC 6-3-2 during normal operations using the most stringent limit (e.g. calculated from the emission unit operating at the lowest process weight in ton/hr).

The particulate emissions limitations from the above table shall be calculated using the following equation:

Interpolation and extrapolation of the data for the process weight rate in excess of sixty thousand (60,000) pounds per hour shall be accomplished by use of the equation:

$$E = 55.0 P^{0.11} - 40 \quad \text{where } E = \text{rate of emission in pounds per hour; and} \\ P = \text{process weight rate in tons per hour}$$

D.2.3 Preventive Maintenance Plan [326 IAC 2-7-5(12)]

A Preventive Maintenance Plan is required for these facilities and their control devices. Section B - Preventive Maintenance Plan contains the Permittee's obligation with regard to the preventive maintenance plan required by this condition.

Compliance Determination Requirements [326 IAC 2-7-5(1)]

D.2.4 Particulate Control

- In order to assure compliance with Conditions D.2.1, baghouses AF-4, AF-5, AF-6, the VSC cyclones, and jet dryers baghouses AF-8 and AF-9 shall be in operation and control emissions from all emission units exhausting to stacks, AF-4, AF-5, AF-6, AF-8, AF-9, S-1 and S-6 at all times when an emission unit that the baghouses or the cyclones control is in operation.
- In order to assure compliance with Conditions D.2.2, baghouse AF-4 shall be in operation and control emissions from the Flaker Feed Loop Conveyor and Flake Collection Conveyor at all times the Flaker Feed Loop Conveyor and Flake Collection Conveyor are in operation.
- In order to assure compliance with Conditions D.2.2, baghouse AF-5 shall be in operation and control emissions from the Hulloosensors No. 1, No. 2, No. 3, and No. 4 and Cracking Rolls No.1, No. 2, No. 3 and No. 4 at all times the Hulloosensors No. 1, No. 2, No. 3, and No. 4, Cracking Rolls No.1, No. 2, No. 3 and No. 4 are in operation.

DRAFT

- (d) In order to assure compliance with Conditions D.2.2, baghouse AF-6 shall be in operation and control emissions from the Meal Hammer Mill Feeders No. 1, No. 2, No. 3, and No. 5 and Meal Hammer Mills No. 1, No. 2, No. 3, and No. 5 at all times the Meal Hammer Mill Feeders No. 1, No. 2, No. 3, and No. 5 and Meal Hammer Mills No. 1, No. 2, No. 3, and No. 5 are in operation.
- (e) In the event that bag failure is observed in a multi-compartment baghouse, if operations will continue for ten (10) days or more after the failure is observed before the failed units will be repaired or replaced, the Permittee shall promptly notify the IDEM, OAQ of the expected date the failed units will be repaired or replaced. The notification shall also include the status of the applicable compliance monitoring parameters with respect to normal, and the results of any response actions taken up to the time of notification.

D.2.5 Testing Requirements [326 IAC 2-1.1-11]

- (a) In order to demonstrate compliance with Conditions D.2.1 and D.2.2, the Permittee shall conduct PM, PM₁₀, and PM_{2.5} testing of the baghouses AF-4, AF-5, AF-6, AF-8, and AF-9 and VSC cyclones (associated with the flaking system, dehulling system, meal grinding/conveying, and VSC system), utilizing methods as approved by the Commissioner at least once every five (5) years from the date of the most recent valid compliance demonstration.
- (b) In order to demonstrate compliance with Conditions D.2.1 and D.2.2, the Permittee shall conduct PM, PM₁₀, and PM_{2.5} testing of the baghouses AF-8 and AF-9 (associated with the jet drying), utilizing methods as approved by the Commissioner at least once every five (5) years from the date of the most recent valid compliance demonstration.
- (c) In order to demonstrate compliance with Conditions D.2.1 and D.2.2(b), the Permittee shall conduct PM, PM₁₀, and PM_{2.5} testing of the VSC Cyclone No. 2, utilizing methods as approved by the Commissioner at least once every five (5) years from the date of the most recent valid compliance demonstration.

These tests shall be repeated at least once every five (5) years from the date of the most recent valid compliance demonstration. Testing shall be conducted in accordance with the provisions of 326 IAC 3-6 (Source Sampling Procedures). Section C - Performance Testing contains the Permittee's obligation with regard to the performance testing required by these conditions. PM₁₀ and PM_{2.5} include filterable and condensable particulate matter.

Compliance Monitoring Requirements [326 IAC 2-7-5(1)][326 IAC 2-7-6(1)]

D.2.6 Visible Emissions Notations [40 CFR 64]

- (a) Visible emission notations of the stacks AF-4, AF-5, AF-6, S-1, and S-6 exhausts shall be performed once per day during normal daylight operations. A trained employee shall record whether emissions are normal or abnormal.
- (b) For processes operated continuously, "normal" means those conditions prevailing, or expected to prevail, eighty percent (80%) of the time the process is in operation, not counting startup or shut down time.
- (c) In the case of batch or discontinuous operations, readings shall be taken during that part of the operation that would normally be expected to cause the greatest emissions.
- (d) A trained employee is an employee who has worked at the plant at least one (1) month and has been trained in the appearance and characteristics of normal visible emissions for that specific process.

DRAFT

- (e) If abnormal emissions are observed, the Permittee shall take a reasonable response. Section C - Response to Excursions or Exceedances contains the Permittee's obligation with regard to the reasonable response steps required by this condition. Failure to take response shall be considered a deviation from this permit.

D.2.7 Parametric Monitoring [40 CFR 64]

The Permittee shall record the pressure drop across baghouses AF-4, AF-5, AF-6, AF-8, and AF-9 used in conjunction with the flaking system, dehulling system, and meal grinding system, at least once per day when the flaking system, dehulling system, and meal grinding system are in operation. When for any one reading, the pressure drop across Baghouses AF-4, AF-5 and AF-6, AF-8 and AF-9 is outside the normal range presented in the table below or established during the latest stack test, the Permittee shall take reasonable response steps. Section C - Response to Excursions or Exceedances contains the Permittee's obligation with regard to the reasonable response steps required by this condition. A pressure reading that is outside the above mentioned range is not a deviation from this permit. Failure to take response steps shall be considered a deviation from this permit.

Baghouse	Lower Limit (inches of water)	Upper Limit (inches of water)
AF-4	0.5	11.0
AF-5	0.5	13.0
AF-6	0.5	18.0
AF-8	0.5	8.0
AF-9	0.5	8.0

The instrument used for determining the pressure shall comply with Section C - Instrument Specifications, of this permit, shall be subject to approval by IDEM, OAQ and shall be calibrated or replaced at least once every twelve (12) months or more frequently if recommended by the instrument manufacture's specifications.

D.2.8 Broken or Failed Bag Detection [40 CFR 64]

- (a) For a single compartment baghouse controlling emissions from a process operated continuously, a failed unit and the associated process shall be shutdown immediately until the failed unit has been repaired or replaced. Operations may continue only if the event qualifies as an emergency and the Permittee satisfies the requirements of the emergency provisions of this permit (Section B - Emergency Provisions).
- (b) For a single compartment baghouse controlling emissions from a batch process, the feed to the process shall be shut down immediately until the failed unit has been repaired or replaced. The emissions unit shall be shut down no later than the completion of the processing of the material in the emissions unit. Operations may continue only if the event qualifies as an emergency and the Permittee satisfies the requirements of the emergency provisions of this permit (Section B - Emergency Provisions).

Bag failure can be indicated by a significant drop in the baghouse's pressure reading with abnormal visible emissions, by an opacity violation, or by other means such as gas temperature, flow rate, air infiltration, leaks, dust traces or triboflows.

D.2.9 Cyclone Failure Detection [326 IAC 2-7-6(1)][326 IAC 2-7-5(1)]

In the event that cyclone failure has been observed:

The feed to the process shall be shut down immediately until the failed units have been repaired or replaced. The emission units shall be shut down no later than the completion of the processing of the material in the emission unit. Operations may continue only if the event qualifies as an emergency and the Permittee satisfies the requirements of the emergency provisions of this permit (Section B - Emergency Provisions).

DRAFT

Record Keeping and Reporting Requirements [326 IAC 2-7-5(3)][326 IAC 2-7-19]**D.2.10 Record Keeping Requirements**

- (a) To document the compliance status with Condition D.2.6, the Permittee shall maintain a daily record of visible emission notations of the stack exhaust from Stacks AF-4, AF-5, AF-6, S-1, and S-6. The Permittee shall include in its daily record when a visible emission notation is not taken and the reason for the lack of visible emission notation (e.g. the process did not operate that day).
- (b) To document the compliance status with Condition D.2.7 the Permittee shall maintain a daily record of the pressure drop across baghouses, AF-4, AF-5, AF-6, AF-8, and AF-9 used to control loadout, flaking, dehulling system, and meal grinding. The Permittee shall include in its daily record when a pressure drop reading is not taken and the reason for the lack of a pressure drop reading (e.g. the process did not operate that day).
- (c) Section C - General Record Keeping Requirements of this permit contains the Permittee's obligations with regard to the records required by this condition.

DRAFT

SECTION D.3

EMISSIONS UNIT OPERATION CONDITIONS

Emissions Unit Description:					
(d)					
Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
D010000	Soybean oil extractor *2006, **2010 and 2012	264	Mineral oil absorber	Stack S-4	Yes under NESHAP GGGG
	One (1) set of evaporators *2006		Mineral oil absorber	Stack S-4	Yes under NESHAP GGGG
D020000	One (1) Desolventizer/toaster *2006, **2010		Mineral oil absorber	Stack S-4	Yes under NESHAP GGGG
	One (1) set of water separators *2006		Mineral oil absorber	Stack S-4	Yes under NESHAP GGGG
D060000	Main Vent Condenser *2006		Mineral oil absorber	Stack S-4	Yes under NESHAP GGGG
	Five (5) hexane storage tank *2006 for original tank, and 2010 for other tanks **2010 for original tank	20,690 gallons each	Mineral oil absorber	Stack S-4	Yes under NESHAP GGGG
I220000	One (1) soybean oil pre-treat Tank *2010	35,170 gallons			Yes under NESHAP GGGG
	Three (3) soybean oil storage tank (Degummed Oil Tanks #1 and #2 and Crude Oil Tank #3) *2006 for original tank and 2010 for other tanks, **2010 for original tank	725,000 gallons each			Yes under NESHAP GGGG
D070000	Mineral oil absorber *2006			Stack S-4	Yes under NESHAP GGGG
D310000-1	DC Deck No. 1 *2006, **2010 and 2012	208	DC Deck Cyclone No. 1	Stack S-2	Yes under NESHAP GGGG
D310000-2	DC Deck No. 2 *2006, **2010, 2011, and 2012	208	DC Deck Cyclone No. 2	Stack S-2	Yes under NESHAP GGGG
D310000-3	DC Deck No. 3 *2006, **2010 and 2012	208	DC Deck Cyclone No. 3	Stack S-2	Yes under NESHAP GGGG
D310000-4	DC Deck No. 4 *2006, **2010 and 2012	208	DC Deck Cyclone No. 4	Stack S-2	Yes under NESHAP GGGG
D310700	DC Deck Cyclone No.1 *2006, **2010 and 2011	18,000 scfm		Stack S-2	Yes under NESHAP GGGG

DRAFT

D310800	DC Deck Cyclone No.2 *2006, **2010 and 2011	18,000 scfm		Stack S-2	Yes under NESHAP GGGG
D310900	DC Deck Cyclone No.3 *2010, ** 2011	18,000 scfm		Stack S-2	Yes under NESHAP GGGG
D311000	DC Deck Cyclone No.4 *2010, **2011	18,000 scfm		Stack S-2	Yes under NESHAP GGGG
Note *Approved in the year indicated above for construction. Note **Approved in the year indicated above for modification. (The information describing the process contained in this emissions unit description box is descriptive information and does not constitute enforceable conditions.)					

Emission Limitations and Standards [326 IAC 2-7-5(1)]**D.3.1 PSD Minor Limits for Particulate [326 IAC 2-2]**

In order to render the requirements of 326 IAC 2-2 (Prevention of Significant Deterioration (PSD)) not applicable, the Permittee shall comply with the following PM, PM₁₀, and PM_{2.5} limits:

The PM, PM₁₀, and PM_{2.5} emissions from the following Processes shall be less than the emission limits listed in the table below:

Process	Control	PM Limit (lbs/hour)	PM ₁₀ Limit (lbs/hour)	PM _{2.5} Limit (lbs/hour)
DC Decks	DC Deck Cyclone No. 1 DC Deck Cyclone No. 2 DC Deck Cyclone No. 3 DC Deck Cyclone No. 4	10.74	7.28	7.28

Compliance with the above limits, in combination with the soybean usage limit in Condition D.1.1(a) and with the PM, PM₁₀, and PM_{2.5} emission limits in Conditions D.1.1(b), D.1.1(c), D.2.1, D.5.1, D.6.1 and D.7.1 and with the potential to emit PM, PM₁₀, and PM_{2.5} from all other emission units at the source, shall limit the PM, PM₁₀, and PM_{2.5} emissions from the entire source to less than 250 tons per twelve (12) consecutive month period and render the requirements of 326 IAC 2-2 (Prevention of Significant Deterioration (PSD)) not applicable.

D.3.2 PSD Minor Limit for VOC [326 IAC 2-2]

In order to render the requirements of 326 IAC 2-2 not applicable, the VOC emissions from the following Processes shall be less than the emission limits in the table below:

Process	Control	VOC Limit (lbs/hour)
Soybean oil extractor system Normal operation	Mineral oil absorber	9.3
DC Decks Normal operation	DC Decks Cyclones	32.8

Compliance with the above VOC emission limits, the VOC emission limits in Condition D.4.1, the VOC emission limit in Condition D.5.2, the VOC emission limits in Condition D.7.2, and the potential to emit VOC from other units at the source shall limit the VOC emissions from the entire source to less than 250 tons per twelve (12) consecutive month period and render the requirements of 326 IAC 2-2 Prevention of Significant Deterioration (PSD) not applicable.

D.3.3 Particulate Emissions Limitations [326 IAC 6-3-2]

Pursuant to 326 IAC 6-3-2, the particulate emissions from each of the following processes shall

DRAFT

not exceed the pound per hour limitations specified in the following table:

Emission unit ID	Emissions Units	Cyclone ID	Maximum Process Weight (tons/hour) for each unit	326 IAC 6-3 Limit (lbs/hr) for each unit
D310000-1, D310000-2, D310000-3, and D310000-4	DC Decks No. 1, No. 2, No. 3, and No. 4	DC Deck Cyclones No. 1 through 4	208	58.93

The particulate emissions limitations from the above table shall be calculated using the following equation:

Interpolation and extrapolation of the data for the process weight rate in excess of sixty thousand (60,000) pounds per hour shall be accomplished by use of the equation:

$$E = 55.0 P^{0.11} - 40 \quad \text{where } E = \text{rate of emission in pounds per hour; and} \\ P = \text{process weight rate in tons per hour}$$

D.3.4 Volatile Organic Compounds (VOC) [326 IAC 8-1-6]

Pursuant to Significant Source Modification No. 085-29971-00102, as revised in Significant Source Modification No. 085-31960-00102 and 326 IAC 8-1-6 (New Facilities; General Reduction Requirements) the Permittee shall comply with the following:

- The VOC emissions from the combined condenser and mineral oil absorber system for the extractor vent system shall not exceed 0.048 pounds per ton of soybean processed and shall not exceed 9.3 pounds per hour.
- The VOC emissions from the meal dryers and meal cooler (DC Decks No. 1, No. 2, No. 3, and No. 4) shall not exceed 0.03 gallons of VOC per ton of soybean processed and shall not exceed 32.8 pounds per hour.
- The overall solvent loss ratio shall not exceed 0.141 gallons per ton of soybean crushed from the whole plant per twelve (12) consecutive month period, with compliance determined at the end of each month. The Permittee shall also follow the leak detection and repair program as part of BACT.
- The maximum annual throughput of soybeans processed shall not exceed 2,251,836 tons per twelve (12) consecutive month period, with compliance determined at the end of each month.
- BACT for the fugitive hexane loss shall include an enhanced inspection, maintenance, and repair program (LDAR Program). No later than 60 days of achieving full production, but in no case later than 180 days after initial startup, the Permittee shall institute the following enhanced inspection, maintenance, and repair program for equipment in VOC service located in the solvent extraction portion of the installation. Equipment in vacuum service is exempt from this monitoring requirement pursuant to 40 CFR Part 60.482-1(d).

Table 1

Equipment	Leak Standard
Pumps	500 ppm
Valves	500 ppm
Pressure relief Devices	500 ppm
Flanges, Connectors, and Seals	10,000 ppm

DRAFT

- (1) The Permittee shall determine compliance with the standards in Table 1 by using the procedures of 40 CFR Part 60, Appendix A, Method 21. The instrument shall be calibrated before each day of its use by the procedures as specified in Method 21. A leak is defined as an instrument reading of 500 ppm above background or greater, except for flanges, and connectors where a leak is defined as 10,000 ppm above background.
- (2) The Permittee shall immediately tag all detected leaks with a weatherproof, and readily visible, identification tag with a distinct number. Once a leaking component is detected, first-attempt repairs must be done no later than five days and be completed no later than 15 days of detecting the leaking components. If the repair cannot be accomplished no later than 15 days, then the Permittee shall send a notice of inability to repair to the OAQ no later than 20 days of detecting the leak. The notice must be received by the Compliance and Enforcement Branch, Office of Air Quality, 100 North Senate Avenue, MC 61-53 IGCN 1003, Indianapolis, Indiana 46204-2251, no later than 20 days after the leak was detected. At a minimum the notice shall include the following:
 - (A) Equipment, operator, and instrument identification number;
 - (B) Date of leak detection;
 - (C) Measured concentration (ppm) and background (ppm);
 - (D) Leak identification number associated with the corresponding tag; and
 - (E) Reason of inability to repair no later than 5 to 15 days of detection.
- (3) The Permittee shall maintain records of the following to verify compliance with the enhanced inspection, maintenance, and repair program:
 - (A) equipment inspected;
 - (B) date of inspection; and
 - (C) determination of whether a leak was detected.
- (4) If a leak is detected, the Permittee shall record the following information to verify compliance with the enhanced inspection, maintenance, and repair program:
 - (A) the equipment, operator, and instrument identification number;
 - (B) measured concentration;
 - (C) leak identification number associated with the corresponding tag;
 - (D) date of repair;
 - (E) reason for non-repair if unable to repair no later than 5 to 15 days of detection; and
 - (F) maintenance recheck if repaired-date, concentration, background.
- (5) Definitions contained in 40 CFR Part 60, Subpart VV shall be utilized where necessary to implement this program.

DRAFT

D.3.5 Preventive Maintenance Plan [326 IAC 2-7-5(12)]

A Preventive Maintenance Plan is required for these facilities and their control devices. Section B - Preventive Maintenance Plan contains the Permittee's obligation with regard to the preventive maintenance plan required by this condition.

Compliance Determination Requirements [326 IAC 2-7-5(1)]**D.3.6 Particulate Control**

In order to assure compliance with Conditions D.3.1, and D.3.3, DC Deck Cyclones No. 1, No. 2, No. 3, and No. 4 shall be in operation and control emissions from the DC Decks No. 1, No. 2, No. 3, and No. 4 at all times the DC Decks No. 1, No. 2, No. 3, and No. 4 are in operation.

D.3.7 Volatile Organic Compounds (VOC)

In order to assure compliance with Conditions D.3.2 and D.3.4(a), the mineral oil absorber system and the soybean oil stripper shall be in operation and control emissions from the oil extractor process at all times the oil extractor process is in operation.

D.3.8 Testing Requirements [326 IAC 2-1.1-11]

- (a) In order to demonstrate compliance with Conditions D.3.1 and D.3.3, the Permittee shall conduct PM, PM₁₀, and PM_{2.5} testing on stack S-2 (associated with the meal dryers and cooler), utilizing methods as approved by the Commissioner at least once every five (5) years from the date of the most recent valid compliance demonstration. PM₁₀ and PM_{2.5} include filterable and condensable particulate matter.
- (b) In order to demonstrate compliance with Conditions D.3.2 and D.3.4(a), the Permittee shall perform VOC testing on the mineral oil absorber stack (stack S-4) and determine the mineral oil absorber's mineral oil flow rate and the temperature of mineral oil to the absorber, utilizing methods as approved by the Commissioner at least once every five (5) years from the date of the most recent valid compliance demonstration.
- (c) In order to demonstrate compliance with Conditions D.3.2 and D.3.4(b), the Permittee shall perform VOC testing on the meal dryers and cooler cyclones stack (stack S-2), utilizing methods as approved by the Commissioner at least once five (5) years from the date of the most recent valid compliance demonstration.

Testing shall be conducted in accordance with the provisions of 326 IAC 3-6 (Source Sampling Procedures). Section C - Performance Testing contains the Permittee's obligation with regard to the performance testing required by these conditions.

D.3.9 Leak Detection and Repair (LDAR) Program [326 IAC 8-1-6]

Pursuant to 326 IAC 8-1-6, the following is required to demonstrate compliance with the requirements of Condition D.3.4(c):

- (a) For pumps
 - (i) For the first year:
 - (A) Weekly visual check for leakage; and
 - (B) Semi-annual organic vapor analyzer inspection (leak definition = 500 ppm above background concentrations).
 - (ii) After the first year:
 - (A) Weekly visual check for leakage;

DRAFT

- (B) Annual organic vapor analyzer inspection (leak definition = 500 ppm above background concentrations).
 - (iii) When a unit has a leak detected during an annual organic vapor analyzer inspection, the frequency of organic vapor analyzer inspections shall become semi-annual;
 - (iv) When that unit has no leak detected for two (2) consecutive semi-annual vapor analyzer inspections, the frequency of the inspections shall return to annual.
- (b) For valves
 - (i) For the first year:
 - (A) Semi-annual organic vapor analyzer inspection (leak definition = 500 ppm above background concentrations).
 - (ii) After the first year:
 - (A) Annual organic vapor analyzer inspection (leak definition = 500 ppm above background concentrations);
 - (B) When a unit has a leak detected during an annual organic vapor analyzer inspection, the frequency of organic vapor analyzer inspections shall become semi-annual; and
 - (C) When that unit has no leak detected for two (2) consecutive semi-annual vapor analyzer inspections, the frequency of the inspections shall return to annual.
- (c) For pressure relief devices:
 - (i) No later than five (5) calendar days after a pressure release, the pressure release device shall be monitored to confirm conditions of no detectable emissions, as indicated by an instrument reading of less than 500 ppm above background concentrations or a calibrated LEL Monitor reading of less than 3%. Any pressure relief device that is equipped with a closed vent system capable of capturing and transporting leakage through the pressure relief device to a control device is exempt from the above requirement.
- (d) For connectors, flanges, and seals, the annual organic vapor analyzer inspections shall be made (leak definition = 10,000 ppm above background concentrations).

Compliance Monitoring Requirements [326 IAC 2-7-5(1)][326 IAC 2-7-6(1)]

D.3.10 Visible Emissions Notations [40 CFR 64]

-
- (a) Visible emission notations of Stack S-2 exhaust shall be performed once per day during normal daylight operations. A trained employee shall record whether emissions are normal or abnormal.
 - (b) For processes operated continuously, "normal" means those conditions prevailing, or expected to prevail, eighty percent (80%) of the time the process is in operation, not counting startup or shut down time.
 - (c) In the case of batch or discontinuous operations, readings shall be taken during that part of the operation that would normally be expected to cause the greatest emissions.

DRAFT

- (d) A trained employee is an employee who has worked at the plant at least one (1) month and has been trained in the appearance and characteristics of normal visible emissions for that specific process.
- (e) If abnormal emissions are observed, the Permittee shall take a reasonable response. Section C - Response to Excursions or Exceedances contains the Permittee's obligation with regard to the reasonable response steps required by this condition. Failure to take response shall be considered a deviation from this permit.

D.3.11 Monitoring for Mineral Oil Absorber and Mineral Oil Stripping Column [40 CFR 64]

- (a) The Permittee shall monitor and record the mineral oil flow rate to the mineral oil absorber at least once per day.
- (b) A continuous monitoring system shall be calibrated, maintained, and operated on the mineral oil absorber for measuring operating temperature. For purposes of this condition continuous shall mean temperature measurement no less than once per minute. The output of this system shall be recorded as a 3-hour block average. The Permittee shall operate the mineral oil absorber at or below the 3-hour block average temperature as determined from the most recent valid stack test.
 - (1) The Permittee shall determine the 3-hour block average temperature from the most recent valid stack test that demonstrates compliance with the limits in conditions D.3.2 and D.3.4(a).
 - (2) On and after the date the stack test results are available, the Permittee shall maintain the temperature of the mineral oil to the absorber at or below the 3-hour block average temperature as observed during the compliant stack test.
- (c) A continuous monitoring system shall be calibrated, maintained, and operated on the mineral oil stripper for measuring the temperature of mineral oil to the stripper. For purposes of this condition continuous shall mean temperature measurement no less than once per minute. The output of this system shall be recorded as a 3-hour block average. The Permittee shall operate the mineral oil stripper at or above the 3-hour block average temperature as determined from the most recent valid stack test.
 - (1) The Permittee shall determine the 3-hour block average temperature from the most recent valid stack test that demonstrates compliance with limits in condition D.3.2.
 - (2) On and after the date the stack test results are available, the Permittee shall operate the mineral oil stripper at or above the 3-hour block average temperature as observed during the compliant stack test.
- (d) If any of the following operating conditions occur, the Permittee shall take a reasonable response. Section C - Response to Excursions or Exceedances contains the Permittee's obligation with regard to the reasonable response steps required by this condition. Failure to take response steps shall be considered a deviation from this permit.
 - (1) When the mineral oil flow rate reading is below the minimum mineral oil flow rate for any one reading. The minimum mineral oil flow rate to the mineral oil absorber will be as recommended by the manufacturer or the minimum flow rate established during the latest stack test.
 - (2) When the 3-hour block average temperature reading of the mineral oil to the absorber is above the temperature for any 3-hour block average. The 3-hour block average temperature of the mineral oil to the absorber will be as

DRAFT

recommended by the manufacturer or the maximum temperature established during the latest stack test.

- (3) When the 3-hour block average temperature reading of the mineral oil to the stripper is below the minimum temperature for any 3-hour block average. The minimum temperature of the mineral oil to the stripper will be as recommended by the manufacturer or the minimum temperature established during the latest stack test.

Operating conditions above or below the values specified in (1) through (3) above shall not be considered a deviation from this permit. Failure to take response steps shall be considered a deviation from this permit.

- (e) The instruments used for determining the flow rate and temperature readings shall be subject to approval by IDEM, OAQ, and shall be calibrated or replaced at least once every six (6) months.
- (f) The gauge employed to take the mineral oil flow to the mineral oil absorber shall have a scale such that the expected normal reading shall be no less than 20 percent of full scale and be accurate within + 10% of full scale reading. The instrument shall be quality assured and maintained as specified by the vendor.

D.3.12 Cyclone Failure Detection [326 IAC 2-7-6(1)][326 IAC 2-7-5(1)][40 CFR 64]

In the event that cyclone failure has been observed:

The feed to the process shall be shut down immediately until the failed units have been repaired or replaced. The emission units shall be shut down no later than the completion of the processing of the material in the emission unit. Operations may continue only if the event qualifies as an emergency and the Permittee satisfies the requirements of the emergency provisions of this permit (Section B - Emergency Provisions).

Record Keeping and Reporting Requirements [326 IAC 2-7-5(3)][326 IAC 2-7-19]

D.3.13 Record Keeping Requirements

- (a) To document the compliance status with Condition D.3.4(c), the Permittee shall maintain monthly records of the source wide solvent loss ratio (SLR).
- (b) To document the compliance status with Condition D.3.4(d), the Permittee shall maintain records of the monthly soybean throughput.
- (c) To document the compliance status with Condition D.3.10, the Permittee shall maintain a daily record of visible emission notations of the stack exhaust from Stack S-2. The Permittee shall include in its daily record when a visible emission notation is not taken and the reason for the lack of visible emission notation (e.g. the process did not operate that day).
- (d) To document the compliance status with Condition D.3.11, the Permittee shall maintain a daily record of the mineral oil flow rate, the 3-hour block average temperatures of the mineral oil to the absorber, and the 3-hour block average temperatures of the mineral oil to the stripping column. The Permittee shall include in its daily record when a parametric notation is not taken and the reason for the lack of parametric notation (e.g. the process did not operate that day).
- (e) Section C - General Record Keeping Requirements of this permit contains the Permittee's obligations with regard to the records required by this condition.

DRAFT

D.3.14 Reporting Requirements

Quarterly summaries of the information to document the compliance status with Conditions D.3.4(c) and D.3.4(d) shall be submitted using the reporting forms located at the end of this permit, or their equivalent, not later than thirty (30) days after the end of the quarter being reported. Section C - General Reporting contains the Permittee's obligation with regard to the reporting required by this condition. The reports submitted by the Permittee do require a certification that meets the requirements of 326 IAC 2-7-6(1) by a "responsible official," as defined by 326 IAC 2-7-1(35).

DRAFT

SECTION D.4

EMISSIONS UNIT OPERATION CONDITIONS

Emissions Unit Description:

(e)

Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
CL-5045	1st Primary Transester Column *2006, **2007 and 2010	12,960 gals/hr	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS NNN, NSPS RRR, NSPS VV, and NESHAP FFFF
CL-5046	1st Secondary Transester Column *2006, **2007 and 2010	12,960 gals/hr	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS NNN, NSPS RRR, NSPS VV, and NESHAP FFFF
CL-5063	2nd Primary Transester Column *2006, **2007 and 2010	12,960 gals/hr	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS NNN, NSPS RRR, NSPS VV, and NESHAP FFFF
CL-5064	2nd Secondary Transester Column *2010	12,960 gals/hr	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS NNN, NSPS RRR, NSPS VV, and NESHAP FFFF
R-8171	Esterification Reactor *2010	12,960 gals/hr	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS NNN, NSPS RRR, NSPS VV, and NESHAP FFFF
VU010000	Vacuum group package *2006, **2007 and 2010	12,960 gals/hr	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS NNN, NSPS VV, and NESHAP FFFF
	Biodiesel Mineral Oil Absorber *2010			Stack S-5	Yes under NSPS VV, and NESHAP FFFF
	Biodiesel Water Absorber *2006, **2007	2.2 gpm		Stack S-5	Yes under NSPS VV, and NESHAP FFFF
Biodiesel Distillation	Biodiesel Distillation *2013	15	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSP VVa, NSPS NNN, NESHAP FFFF
I040000	Biodiesel Storage Tank #4 *2006, **2007	725,000 gals			Yes under NSPS VV, and NESHAP FFFF
I050000	Biodiesel Storage Tank #5 *2006, **2007	725,000 gals			Yes under NSPS VV, and NESHAP FFFF
I060000	Biodiesel Storage Tank #6 *2009	3250,000			Yes under NSPS VV, and NESHAP FFFF
I070000	Biodiesel Storage Tank #7 *2006, **2007	325,000 gals			Yes under NSPS VV, and NESHAP FFFF
I080000	Biodiesel Storage Tank #8 *2006, **2007	325,000 gals			Yes under NSPS VV, and NESHAP FFFF
I090000	Biodiesel Storage Tank #9 *2006, **2007	325,000 gals			Yes under NSPS VV, and NESHAP FFFF

DRAFT

I100000	Biodiesel Storage Tank #10 *2006, **2007	325,000 gals			Yes under NSPS VV, and NESHAP FFFF
I110000	Biodiesel Storage Tank #11 *2007	325,000 gals			Yes under NSPS VV, and NESHAP FFFF
1140000	Biodiesel Storage Tank #0 *2009	735,000			Yes under NSPS VV, and NESHAP FFFF
I120000	Glycerin Tank #12 *2006, **2010	360,000 gals			Yes under NSPS VV, and NESHAP FFFF
I130000	Glycerin Tank #13 *2006, **2010	360,000 gals			Yes under NSPS VV, and NESHAP FFFF
I250000	Methanol Storage Tank #1 *2006, **2007 and **2010	38,850 gallons	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS VV, and NESHAP FFFF
I260000	Methanol Storage Tank #2 *2006, **2007 and 2010	38,850 gallons	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS VV, and NESHAP FFFF
I270000	Methanol Storage Tank #3 *2006, **2007 and 2010	38,850 gallons	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS VV, and NESHAP FFFF
I280000	Methanol Storage Tank #4 *2006, **2007 and 2010	38,850 gallons	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS VV, and NESHAP FFFF
I290000	Methanol Storage Tank #5 *2006, **2007 and 2010	38,850 gallons	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS VV, and NESHAP FFFF
I300000	Methanol Storage Tank #6 *2007	38,850 gallons	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS VV, and NESHAP FFFF
I230000	Sodium Methylate (catalyst) Storage Tank #1 *2006, **2007 and 2010	38,850 gallons	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS VV, and NESHAP FFFF
I240000	Sodium Methylate (catalyst) Storage Tank #2 *2007, **2010	38,850 gallons	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS VV, and NESHAP FFFF
Rail Rack	Loading Rack (Rail) *2006, **2007, **2010 and **2011	500 gallons per minute			Yes under NSPS VV, and NESHAP FFFF
Truck Rack #1	Loading Rack (Truck) *2006, **2007, **2010 and **2011	430 gallons per minute			Yes under NSPS VV, and NESHAP FFFF
Truck Rack #2	Loading Rack (Truck) *2011	430 gallons per minute			Yes under NSPS VV, and NESHAP FFFF

Note *Approved in the year indicated above for construction.

Note **Approved in the year indicated above for modification.

(The information describing the process contained in this emissions unit description box is descriptive information and does not constitute enforceable conditions.)

DRAFT

Emission Limitations and Standards [326 IAC 2-7-5(1)]

D.4.1 PSD Minor Limits for VOC [326 IAC 2-2]

In order to render the requirements of 326 IAC 2-2 (Prevention of Significant Deterioration (PSD)) not applicable, the Permittee shall comply with the following:

- (a) The VOC emissions from the following Processes shall be less than the emission limits listed in the table below:

Process	Control	VOC Limit (lbs/hour)	Operation Limit
Biodiesel manufacturing process Normal operation	Mineral Oil Absorber and Water absorber	0.30	
Biodiesel manufacturing process with methanol tank loading	Mineral Oil Absorber and Water absorber	0.63	1,000 hours per twelve (12) consecutive months.
Biodiesel manufacturing process upset operation	Mineral Oil Absorber and Water absorber	29.4	24 hours per twelve (12) consecutive months.
Glycerin storage tanks	None	0.0011	
Biodiesel wastewater	None	0.77	
Biodiesel fugitive emissions	LDR as required by 40 CFR 60, Subpart VV	0.64	

- (b) The amount of purchased seed oil shall be less than 80 million gallons per twelve (12) consecutive month period, with compliance determined at the end of each month.
- (c) The amount of seed oil processed to manufacture biodiesel shall be less than 110,000,000 gallons per twelve (12) consecutive month period, with compliance determined at the end of each month.
- (d) The VOC emissions from the loading racks shall be less than 0.02 lbs/kgal.
- (e) The maximum biodiesel loadout throughput rate for the loading racks shall be less than 110,000,000 gallons per twelve (12) consecutive month period, with compliance determined at the end of each month.

Compliance with the purchased seed oil limit in Condition D.4.1(b), in combination with the above VOC emission limits in Condition D.4.1 shall limit the potential to emit of VOC from the biodiesel process to less than 100 tons per twelve (12) consecutive month period and render the requirements of 326 IAC 2-2 (Prevention of Significant Deterioration (PSD)) not applicable.

Compliance with the purchased seed oil limit in Condition D.4.1(b), in combination with the above VOC emission limits in Condition D.4.1, the VOC emission limits in Condition D.3.2, the VOC emission limit in Condition D.5.2, the VOC emission limits in Condition D.7.2, and the potential to emit from other units at the source, shall limit the VOC emissions from the entire source to less than 250 tons per twelve (12) consecutive month period and render the requirements of 326 IAC 2-2 (Prevention of Significant Deterioration (PSD)) not applicable.

D.4.2 VOC Best Available Control Technology (BACT) Limits [326 IAC 8-1-6]

Pursuant to 326 IAC 8-1-6 (New Facilities, General Reduction Requirements) the Permittee shall

DRAFT

comply with the following Best Available Control Technology (BACT) requirements:

Process	Control	VOC control efficiency, %	VOC Limit (lbs/hour)	Operation Limit
Biodiesel manufacturing process Normal operation	Mineral Oil Absorber and Water absorber	Overall VOC control efficiency of 99%	0.30	NA
Biodiesel manufacturing process with methanol tank loading	Mineral Oil Absorber and Water absorber	Overall VOC control efficiency of 99%	0.63	1,000 hours per twelve (12) consecutive months
Biodiesel manufacturing process upset operation	Mineral Oil Absorber and Water absorber	Overall VOC control efficiency of 95%	29.4	24 hours per twelve (12) consecutive months
Glycerin storage tanks	None	None	0.0011	NA
Biodiesel wastewater	None	None	0.77	NA
Biodiesel fugitive emissions	None	None	0.64	LDAR as required by 40 CFR 60, Subpart VV

D.4.3 Preventive Maintenance Plan [326 IAC 2-7-5(12)]

A Preventive Maintenance Plan is required for these facilities and their control devices. Section B - Preventive Maintenance Plan contains the Permittee's obligation with regard to the preventive maintenance plan required by this condition.

Compliance Determination Requirements [326 IAC 2-7-5(1)]

D.4.4 Volatile Organic Compounds (VOC)

In order to assure compliance with Condition D.4.1(a), the mineral oil absorber and water absorber shall be in operation and control emissions from the biodiesel manufacturing process and the methanol tank unloading at all times the biodiesel manufacturing process and the methanol tank unloading process are in operation.

D.4.5 Testing Requirements [326 IAC 2-1.1-11]

In order to demonstrate compliance with Conditions D.4.1(a) and D.4.2, the Permittee shall perform VOC testing on the inlet and outlet of the Mineral Oil Absorber with methanol unloading and without methanol unloading; and determine the Mineral Oil Absorber's mineral oil flow rate and water absorber's water flow rate, utilizing methods as approved by the Commissioner at least once every five (5) years from the date of the most recent valid compliance demonstration. Testing shall be conducted in accordance with the provisions of 326 IAC 3-6 (Source Sampling Procedures). Section C - Performance Testing contains the Permittee's obligation with regard to the performance testing required by this condition.

Compliance Monitoring Requirements [326 IAC 2-7-5(1)][326 IAC 2-7-6(1)]

D.4.6 Monitoring for mineral oil absorber and water absorber [40 CFR 64]

- The Permittee shall monitor and record the mineral oil flow rate for the mineral oil absorber at least once per day.
- The Permittee shall monitor and record the water flow rate for the water absorber at least once per day.
- A continuous monitoring system shall be calibrated, maintained, and operated on the mineral oil absorber for measuring the temperature of the mineral oil to the mineral oil absorber. For purposes of this condition continuous shall mean temperature measurement no less than once per minute. The output of this system shall be recorded

DRAFT

as a 3-hour average. The Permittee shall operate the mineral oil absorber at or below the 3-hour average temperature as determined from the most recent valid stack test.

- (1) The Permittee shall determine the 3-hour average temperature from the most recent valid stack test that demonstrates compliance with the limits in condition D.4.1(a) and D.4.2.
 - (2) On and after the date the stack test results are available, the Permittee shall operate the mineral oil absorber at or below the 3-hour average temperature as observed during the compliant stack test.
- (d) A continuous monitoring system shall be calibrated, maintained, and operated for measuring the temperature of the water to the water absorber. For purposes of this condition continuous shall mean temperature measurement no less than once per minute. The output of this system shall be recorded as a 3-hour average. The Permittee shall operate the water absorber at or below the 3-hour average temperature as determined from the most recent valid stack test.
 - (1) The Permittee shall determine the 3-hour average temperature from the most recent valid stack test that demonstrates compliance with limits in condition D.4.1(a) and D.4.2.
 - (2) On and after the date the stack test results are available, the Permittee shall operate the water absorber at or below the 3-hour average temperature as observed during the compliant stack test.
- (e) If any of the following operating conditions occur, the Permittee shall take a reasonable response. Section C - Response to Excursions or Exceedances contains the Permittee's obligation with regard to the reasonable response steps required by this condition. Failure to take response steps shall be considered a deviation from this permit.
 - (1) When the mineral oil flow rate reading is below the minimum flow rate for any one reading. The minimum flow rate for the mineral oil absorber will be 1.5 gpm or the minimum mineral oil flow rate established during the latest stack test.
 - (2) When the water flow rate reading is below the minimum flow rate for any one reading. The minimum flow rate for the water absorber will be 2.2 gpm or the minimum water flow rate established during the latest stack test.
 - (3) When the mineral oil absorber 3-hour average temperature reading is above the temperature for any 3-hour average. The 3-hour average temperature for the mineral oil absorber will be as recommended by the manufacturer or the maximum temperature established during the latest stack test.
 - (4) When the water absorber 3-hour average temperature reading is above the 3-hour average temperature for any one reading. The 3-hour average temperature for the water absorber will be as recommended by the manufacturer or the maximum temperature established during the latest stack test.

Operating conditions above or below the values specified in (1) through (4) above shall not be considered a deviation from this permit. Failure to take response steps shall be considered a deviation from this permit.
- (f) The instruments used for determining the flow rate and temperature reading shall be subject to approval by IDEM, OAQ, and shall be calibrated at least once every six (6) months.

DRAFT

- (g) The gauges employed to take the mineral oil flow and water flow across the mineral oil absorber or water absorber, respectively, shall have a scale such that the expected normal reading shall be no less than 20 percent of full scale and be accurate within + 10% of full scale reading. The instrument shall be quality assured and maintained as specified by the vendor.

Record Keeping and Reporting Requirements [326 IAC 2-7-5(3)][326 IAC 2-7-19]

D.4.7 Record Keeping Requirements

- (a) To document the compliance status with Condition D.4.1 (b), the Permittee shall maintain records of the amount of the purchased seed oil.
- (b) To document the compliance status with Condition D.4.1(c), the Permittee shall maintain records of the amount of the seed oil used to manufacture biodiesel.
- (c) To document the compliance status with Condition D.4.1(a) and D.4.2, the Permittee shall maintain records of the operating hours for the biodiesel manufacturing process during the following operating scenarios:
 - (1) Normal operation with methanol tank loading.
 - (2) Upset conditions.
- (d) To document the compliance status with Condition D.4.1(e), the Permittee shall maintain records of the amount of the biodiesel loaded out through the biodiesel loading racks.
- (e) To document the compliance status with Conditions D.4.5(a) and (b), the Permittee shall maintain a daily record of the mineral oil flow rate of the mineral oil absorber, and the water flow rate of the water absorber. The Permittee shall include in its daily record when a parametric notation is not taken and the reason for the lack of a parametric notation (e.g. the process did not operate that day).
- (f) To document the compliance status with Condition D.4.5(c) and (d), the Permittee shall maintain a daily record of the 3-hour average operating temperatures of the mineral oil absorber and water absorber. The Permittee shall include in its daily record when a parametric notation is not taken and the reason for the lack of parametric notation (e.g. the process did not operate that day).
- (g) Section C - General Record Keeping Requirements of this permit contains the Permittee's obligations with regard to the records required by this condition.

D.4.8 Reporting Requirements

Quarterly summaries of the information to document the compliance status with Conditions D.4.1(b), (c), and (e) shall be submitted using the reporting forms located at the end of this permit, or their equivalent, not later than thirty (30) days after the end of the quarter being reported. Section C - General Reporting contains the Permittee's obligation with regard to the reporting required by this condition. The reports submitted by the Permittee do require a certification that meets the requirements of 326 IAC 2-7-6(1) by a "responsible official," as defined by 326 IAC 2-7-1(35).

DRAFT

SECTION D.5**EMISSIONS UNIT OPERATION CONDITIONS**

Emissions Unit Description:					
(f)					
Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
B-1	Main Boiler, natural gas fired *2006	220 MMBtu/hr	Low NOx burner and Flue gas recirculation	Stack S-3	Yes under NSPS Db and NESHAP DDDDD
B-S	Temporary Mobile Boiler, Firing Natural Gas, *2013	40 MMBtu/hr	None	Stack S-MB	-
Note *Approved in the year indicated above for construction.					
Insignificant Activities					
Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
	Two (2) natural gas-fired space heaters *2013	0.25 MMBtu per hour, each	None		
Note *Approved in the year indicated above for construction.					
(The information describing the process contained in this emissions unit description box is descriptive information and does not constitute enforceable conditions.)					

Emission Limitations and Standards [326 IAC 2-7-5(1)]**D.5.1 Particulate [326 IAC 6-2-4]**

- (a) Pursuant to 326 IAC 6-2-4 (Particulate Emission Limitations for Sources of Indirect Heating), the total particulate emissions from the boiler shall be less than 0.265 pounds per million British thermal units (lb/MMBtu) heat input.
- (b) Pursuant to 326 IAC 6-2-4 (Particulate Emission Limitations for Sources of Indirect Heating), the total particulate emissions from the temporary mobile boiler (B-S) shall be less than 0.255 pounds per million British thermal units (lb/MMBtu) heat input.

D.5.2 Preventive Maintenance Plan [326 IAC 2-7-5(12)]

A Preventive Maintenance Plan is required for these facilities and their control devices. Section B - Preventive Maintenance Plan contains the Permittee's obligation with regard to the preventive maintenance plan required by this condition.

Compliance Monitoring Requirements [326 IAC 2-7-5(1)][326 IAC 2-7-6(1)]**D.5.3 Continuous Emission Monitoring [326 IAC 3-5]**

- (a) Pursuant to 326 IAC 3-5-1(c)(2)(A) (Continuous Monitoring of Emissions), continuous emission monitoring systems (CEMS) and related equipment for the boiler shall be calibrated, maintained, and operated for measuring NO_x, in accordance with applicable federal regulations and 326 IAC 3-5.
- (b) The CEMS shall be operated at all times, except during CEMS malfunctions, reasonable periods of necessary CEMS calibration or CEMS maintenance activities. CEMS

DRAFT

calibration and maintenance activities shall be properly documented and shall be conducted pursuant to the standard operating procedures under 326 IAC 3-5-4(a).

- (c) The Permittee shall keep records in accordance with 326 IAC 3-5-6(b) that includes the following:
 - (1) All documentation relating to:
 - (A) design, installation, and testing of all elements of the monitoring system; and
 - (B) required corrective action or compliance plan activities.
 - (2) All maintenance logs, calibration checks, and other required quality assurance activities.
 - (3) All records of corrective and preventive action.
 - (4) A log of plant operations, including the following:
 - (A) Date of facility downtime.
 - (B) Time of commencement and completion of each downtime.
 - (C) Reason for each downtime.
- (d) In accordance with 326 IAC 3-5-7(5), the Permittee shall submit reports of continuous monitoring system instrument downtime, except for zero (0) and span checks, which shall be reported separately. The reports shall include the following:
 - (1) Date of downtime.
 - (2) Time of commencement.
 - (3) Duration of each downtime.
 - (4) Reasons for each downtime.
 - (5) Nature of system repairs and adjustments.
- (e) Except where permit conditions streamline similar applicable requirements pursuant to 326 IAC 2-7-24, nothing in this permit shall excuse the Permittee from complying with 326 IAC 3-5.

DRAFT

SECTION D.6**EMISSIONS UNIT OPERATION CONDITIONS**

Emissions Unit Description:					
(g)					
Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
G010000	Meal Bin No. 1*** *2006, **2010, 2011, and 2012	198	Meal Bin Filter No. 1	Stack MBF-1	Yes under NESHAP GGGG
G020000	Meal Bin No. 2*** *2006, **2010, 2011, and 2012	198	Meal Bin Filter No. 2	Stack MBF-2	Yes under NESHAP GGGG
G030000	Meal Bin No. 3*** *2006, **2010, 2011, and 2012	198	Meal Bin Filter No. 3	Stack MBF-3	Yes under NESHAP GGGG
G040000	Meal Bin No. 4*** *2006, **2010, 2011, and 2012	198	Meal Bin Filter No. 4	Stack MBF-4	Yes under NESHAP GGGG
G050000	Meal Bin No. 5*** *2010, **2011 and 2012	198	Meal Bin Filter No. 5	Stack MBF-5	Yes under NESHAP GGGG
Note *Approved in the year indicated above for construction. Note **Approved in the year indicated above for modification. Note ***There are five meal bins. However, the plant is only physically capable of loading one meal bin at a time. Thus, the PTE for these units is calculated at a rate of 198 tons/hr for all five meal bins combined.					
(The information describing the process contained in this emissions unit description box is descriptive information and does not constitute enforceable conditions.)					

Emission Limitations and Standards [326 IAC 2-7-5(1)]**D.6.1 PSD Minor Limit for Particulate [326 IAC 2-2]**

In order to render the requirements of 326 IAC 2-2 (Prevention of Significant Deterioration (PSD)) not applicable, the Permittee shall comply with the following:

The PM, PM₁₀ and PM_{2.5} emissions from the following Processes shall be less than the emission limits listed in the table below:

Process	Control	PM Limit (lbs/hour)	PM ₁₀ Limit (lbs/hour)	PM _{2.5} Limit (lbs/hour)
Meal Bin No. 1	Meal Bin Filter No. 1	0.93	0.93	0.93
Meal Bin No. 2	Meal Bin Filter No. 2	0.93	0.93	0.93
Meal Bin No. 3	Meal Bin Filter No. 3	0.93	0.93	0.93
Meal Bin No. 4	Meal Bin Filter No. 4	0.93	0.93	0.93
Meal Bin No. 5	Meal Bin Filter No. 5	0.93	0.93	0.93

Compliance with the above limits, in combination with the soybean usage limit in Condition D.1.1(a) and with the PM, PM₁₀ and PM_{2.5} emission limits in Conditions D.1.1(b), D.1.1(c), D.2.1, D.3.1, D.5.1, and D.7.1, and with the potential to emit PM, PM₁₀ and PM_{2.5} from other emission units at the source, shall limit the PM, PM₁₀ and PM_{2.5} emissions from the entire source to less

DRAFT

than 250 tons per twelve (12) consecutive month period and render the requirements of 326 IAC 2-2 Prevention of Significant Deterioration (PSD) not applicable.

D.6.2 Particulate Emissions Limitations [326 IAC 6-3-2]

Pursuant to 326 IAC 6-3-2, the particulate emissions from each of the following processes shall not exceed the pound per hour limitations specified in the following table:

Emission unit ID	Emissions Units	Control	Maximum Process Weight (tons/hour) for each unit	326 IAC 6-3 Limit (lbs/hr) for each unit
G010000, G020000, G030000, G040000 and G050000	Meal Bins No. 1 thru 5	Meal Bin Filters No. 1 thru No. 5	198	58.40

The particulate emissions limitations from the above table shall be calculated using the following equation:

Interpolation and extrapolation of the data for the process weight rate in excess of sixty thousand (60,000) pounds per hour shall be accomplished by use of the equation:

$$E = 55.0 P^{0.11} - 40 \quad \text{where } E = \text{rate of emission in pounds per hour; and} \\ P = \text{process weight rate in tons per hour}$$

D.6.3 Preventive Maintenance Plan [326 IAC 2-7-5(12)]

A Preventive Maintenance Plan is required for these facilities and their control devices. Section B - Preventive Maintenance Plan contains the Permittee's obligation with regard to the preventive maintenance plan required by this condition.

Compliance Determination Requirements [326 IAC 2-7-5(1)]

D.6.4 Particulate Control

- (a) In order to assure compliance with Condition D.6.1, Meal Bin Filters No. 1 thru No. 5 shall be in operation and control emissions from the Meal Bins No. 1 thru 5 at all times the Meal Bins No. 1 thru 5 are in operation.
- (b) In the event that bag failure is observed in a multi-compartment baghouse, if operations will continue for ten (10) days or more after the failure is observed before the failed units will be repaired or replaced, the Permittee shall promptly notify the IDEM, OAQ of the expected date the failed units will be repaired or replaced. The notification shall also include the status of the applicable compliance monitoring parameters with respect to normal, and the results of any response actions taken up to the time of notification.

Compliance Monitoring Requirements [326 IAC 2-7-5(1)][326 IAC 2-7-6(1)]

D.6.5 Visible Emissions Notations

- (a) Visible emission notations of Stack exhausts MBF-1, MBF-2, MBF-3, MBF-4 and MBF-5 shall be performed once per day during normal daylight operations. A trained employee shall record whether emissions are normal or abnormal.
- (b) For processes operated continuously, "normal" means those conditions prevailing, or expected to prevail, eighty percent (80%) of the time the process is in operation, not counting startup or shut down time.

DRAFT

- (c) In the case of batch or discontinuous operations, readings shall be taken during that part of the operation that would normally be expected to cause the greatest emissions.
- (d) A trained employee is an employee who has worked at the plant at least one (1) month and has been trained in the appearance and characteristics of normal visible emissions for that specific process.
- (e) If abnormal emissions are observed, the Permittee shall take reasonable response steps. Section C - Response to Excursions or Exceedances contains the Permittee's obligation with regard to the reasonable response steps required by this condition. Failure to take response shall be considered a deviation from this permit.

D.6.6 Broken or Failed Bag Detection

- (a) For a single compartment baghouse controlling emissions from a process operated continuously, a failed unit and the associated process shall be shutdown immediately until the failed unit has been repaired or replaced. Operations may continue only if the event qualifies as an emergency and the Permittee satisfies the requirements of the emergency provisions of this permit (Section B - Emergency Provisions).
- (b) For a single compartment baghouse controlling emissions from a batch process, the feed to the process shall be shut down immediately until the failed unit has been repaired or replaced. The emissions unit shall be shut down no later than the completion of the processing of the material in the emissions unit. Operations may continue only if the event qualifies as an emergency and the Permittee satisfies the requirements of the emergency provisions of this permit (Section B - Emergency Provisions).

Bag failure can be indicated by a significant drop in the baghouse's pressure reading with abnormal visible emissions, by an opacity violation, or by other means such as gas temperature, flow rate, air infiltration, leaks, dust traces or triboflows.

Record Keeping and Reporting Requirements [326 IAC 2-7-5(3)][326 IAC 2-7-19]

D.6.7 Record Keeping Requirements

- (a) To document the compliance status with Condition D.6.5 the Permittee shall maintain a daily record of visible emission notations of the stack exhaust from Stacks MBF-1, MBF-2, MBF-3, MBF-4 and MBF-5. The Permittee shall include in its daily record when a visible emission notation is not taken and the reason for the lack of visible emission notation (e.g. the process did not operate that day).
- (b) Section C - General Record Keeping Requirements of this permit contains the Permittee's obligations with regard to the records required by this condition.

DRAFT

SECTION D.7 EMISSIONS UNIT OPERATION CONDITIONS

Emissions Unit Description: Insignificant Activities

Description	Capacity	Control	Affected Facility?
Kaolin Receiving Tank [326 IAC 2-2][326 IAC 6-3-2] *2006, **2010	10,800 and 40 tons per hour	Bin Filter	
Hull Overflow Tank [326 IAC 2-2][326 IAC 6-3-2] *2006, **2010 and 2012	13,900 cu. ft and 17 tons per hour	None	
diesel/#2 fuel oil storage tank *2006, **2011	44,839 gallons	None	
Cooling tower with a maximum drift rate of 0.005% *2006	11,000 gpm	None	
Three (3) Emergency Diesel Fire Pumps *2006	575 BHP each	None	Yes under NSPS III and NESHAP ZZZZ
Two (2) natural gas-fired space heaters *2013	0.25 MMBtu per hour, each	None	
Diatomaceous Earth (DE) Storage Bin [326 IAC 6-3-2] *2009, **2011	767 tons per year	Filter	
Hot Oil Heater *2013	6 MMBtu/hr	None	Yes, Under NESHAP DDDDD
Natural Gas-Fired Glycerin Steam Boiler, *2014	1.0 MMBtu/hr	None	Yes, Under NESHAP DDDDD
Glycerin Refinery *2013, **2014	7.0 tons per hour	None	Yes, Under NSPS VVa, NSPS NNN, and NESHAP FFFF

Note *Approved in the year indicated above for construction.

Note **Approved in the year indicated above for modification.

(The information describing the process contained in this emissions unit description box is descriptive information and does not constitute enforceable conditions.)

Emission Limitations and Standards [326 IAC 2-7-5(1)]

D.7.1 PSD Minor Limits for Particulate [326 IAC 2-2]

In order to render the requirements of 326 IAC 2-2 (Prevention of Significant Deterioration (PSD)) not applicable, the Permittee shall comply with the following PM, PM₁₀, and PM_{2.5} limits:

The PM, PM₁₀, and PM_{2.5} emissions from the following emission units shall be less than the emission limits listed in the table below:

Process	Control	PM Limit (lbs/hour)	PM10 Limit (lbs/hour)	PM2.5 Limit (lbs/hour)
Kaolin Receiving Tank	Bin Filter	1.9	1.9	1.9

Compliance with the above limits, in combination with the soybean usage limit in Condition D.1.1(a) in combination with the PM, PM₁₀, and PM_{2.5} emission limits in Conditions D.1.1(b), D.1.1(c), D.2.1, D.3.1, D.6.1 and D.5.1, and with the potential to emit PM, PM₁₀, and PM_{2.5} from other emission units at the source, shall limit the PM, PM₁₀, and PM_{2.5} emissions from the entire

DRAFT

source to less 250 tons per twelve (12) consecutive month period and render the requirements of 326 IAC 2-2 (Prevention of Significant Deterioration (PSD)) not applicable.

D.7.2 Particulate Emissions Limitations [326 IAC 6-3-2]

- (a) Pursuant to 326 IAC 6-3-2, the particulate emissions from each of the following processes shall not exceed the pound per hour limitations specified in the following table:

Emissions Units	Baghouse ID	Maximum Process Weight (tons/hour) for each unit	326 IAC 6-3 Limit (lbs/hr) for each unit
Diatomaceous Earth (DE) Storage Bin	Filter	0.0875	0.80

The particulate emissions limitations from the above table shall be calculated using the following equation:

Interpolation of the data for the process weight rate up to sixty thousand (60,000) pounds per hour shall be accomplished by use of the equation:

$$E = 4.10 P^{0.67} \quad \text{where } E = \text{rate of emission in pounds per hour; and} \\ P = \text{process weight rate in tons per hour}$$

- (b) Pursuant to 326 IAC 6-3-2, the particulate emissions from each of the following processes shall not exceed the pound per hour limitations specified in the following table:

Emissions Units	Baghouse ID	Maximum Process Weight (tons/hour) for each unit	326 IAC 6-3 Limit (lbs/hr) for each unit
Kaolin Receiving Tank	Bin Filter	40	42.53

The particulate emissions limitations from the above table shall be calculated using the following equation:

Interpolation and extrapolation of the data for the process weight rate in excess of sixty thousand (60,000) pounds per hour shall be accomplished by use of the equation:

$$E = 55.0 P^{0.11} - 40 \quad \text{where } E = \text{rate of emission in pounds per hour; and} \\ P = \text{process weight rate in tons per hour}$$

D.7.3 Particulate [326 IAC 6-2-4]

- (a) Pursuant to 326 IAC 6-2-4 (Particulate Emission Limitations for Sources of Indirect Heating), the particulate emissions from the one (1) 6 MMBtu/hr hot oil heater shall be less than 0.266 pounds per million British thermal units (lb/MMBtu) heat input.
- (b) Pursuant to 326 IAC 6-2-4 (Particulate Emission Limitations for Sources of Indirect Heating), the particulate emissions from one (1) 1.0 MMBtu/hr natural gas-fired glycerin steam boiler shall be less than 0.255 pounds per million British thermal units (lb/MMBtu) heat input.

D.7.4 Preventive Maintenance Plan [326 IAC 2-7-5(12)]

A Preventive Maintenance Plan is required for these facilities and their control devices. Section B - Preventive Maintenance Plan contains the Permittee's obligation with regard to the preventive maintenance plan required by this condition.

DRAFT

Compliance Monitoring Requirements [326 IAC 2-7-5(1)][326 IAC 2-7-6(1)]

D.7.5 Visible Emissions Notations

- (a) Visible emission notations of the Kaolin Receiving Tank exhaust shall be performed once per day during normal daylight operations when exhausting to the atmosphere. A trained employee shall record whether emissions are normal or abnormal.
- (b) For processes operated continuously, "normal" means those conditions prevailing, or expected to prevail, eighty percent (80%) of the time the process is in operation, not counting startup or shut down time.
- (c) In the case of batch or discontinuous operations, readings shall be taken during that part of the operation that would normally be expected to cause the greatest emissions.
- (d) A trained employee is an employee who has worked at the plant at least one (1) month and has been trained in the appearance and characteristics of normal visible emissions for that specific process.
- (e) If abnormal emissions are observed, the Permittee shall take reasonable response steps. Section C - Response to Excursions or Exceedances contains the Permittee's obligation with regard to the reasonable response steps required by this condition. Failure to take response shall be considered a deviation from this permit.

D.7.6 Broken or Failed Bag Detection

- (a) For a single compartment baghouse controlling emissions from a process operated continuously, a failed unit and the associated process shall be shutdown immediately until the failed unit has been repaired or replaced. Operations may continue only if the event qualifies as an emergency and the Permittee satisfies the requirements of the emergency provisions of this permit (Section B - Emergency Provisions).
- (b) For a single compartment baghouse controlling emissions from a batch process, the feed to the process shall be shut down immediately until the failed unit has been repaired or replaced. The emissions unit shall be shut down no later than the completion of the processing of the material in the emissions unit. Operations may continue only if the event qualifies as an emergency and the Permittee satisfies the requirements of the emergency provisions of this permit (Section B - Emergency Provisions).

Bag failure can be indicated by a significant drop in the baghouse's pressure reading with abnormal visible emissions, by an opacity violation, or by other means such as gas temperature, flow rate, air infiltration, leaks, dust traces or triboflows.

Record Keeping and Reporting Requirements [326 IAC 2-7-5(3)][326 IAC 2-7-19]

D.7.7 Record Keeping Requirements

- (a) To document the compliance status with Condition D.7.5 the Permittee shall maintain a daily record of visible emission notations from the Kaolin Receiving Tank exhaust. The Permittee shall include in its daily record when a visible emission notation is not taken and the reason for the lack of visible emission notation (e.g. the process did not operate that day).
- (b) Section C - General Record Keeping Requirements of this permit contains the Permittee's obligations with regard to the records required by this condition.

DRAFT

SECTION D.8

Reserved

DRAFT

SECTION E.1

NSPS

Emissions Unit Description:

(f)

Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
B-1	Main Boiler, natural gas fired *2006	220 MMBtu/hr	Low NOx burner and Flue gas recirculation	Stack S-3	Yes under NSPS Db and NESHAP DDDDD

Note *Approved in the year indicated above for construction.

(The information describing the process contained in this facility description box is descriptive information and does not constitute enforceable conditions.)

New Source Performance Standards (NSPS) Requirements [326 IAC 2-7-5(1)]

E.1.1 General Provisions Relating to New Source Performance Standards [326 IAC 12-1][40 CFR Part 60, Subpart A]

(a) Pursuant to 40 CFR 60.1, the Permittee shall comply with the provisions of 40 CFR Part 60, Subpart A – General Provisions, which are incorporated by reference as 326 IAC 12-1, for the emission unit(s) listed above, except as otherwise specified in 40 CFR Part 60, Subpart Db.

(b) Pursuant to 40 CFR 60.4, the Permittee shall submit all required notifications and reports to:

Indiana Department of Environmental Management
Compliance and Enforcement Branch, Office of Air Quality
100 North Senate Avenue
MC 61-53 IGCN 1003
Indianapolis, Indiana 46204-2251

E.1.2 Industrial-Commercial-Institutional Steam Generating Units NSPS [326 IAC 12][40 CFR Part 60, Subpart Db]

The Permittee shall comply with the following provisions of 40 CFR Part 60, Subpart Db (included as Attachment A to the operating permit), which are incorporated by reference as 326 IAC 12, for the emission unit(s) listed above:

- (1) 40 CFR 60.40b(a), (g), and (j)
- (2) 40 CFR 60.41b
- (3) 40 CFR 60.42b (e), (g), and (k)
- (4) 40 CFR 60.43b (f), (g), and (h)
- (5) 40 CFR 60.44b(a), (h), (i), (l)(1), and (l)(2)
- (6) 40 CFR 60.45b(a), (b) and (j)
- (7) 40 CFR 60.46b(a), (b), (c), (d), (e)(1), and (e)(4)
- (8) 40 CFR 60.47b (f)
- (9) 40 CFR 60.48b(a), (b), (c), (d), (e)(2), (e)(3), (f), and (g)
- (10) 40 CFR 60.49b (a)(1), (b), (d), (f), (g), (h)(2), (h)(4), (i), (j), (o), (r), (v), and (w)

DRAFT

SECTION E.2

NSPS

Emissions Unit Description:

(a)

Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
A030000	Truck Dump No. 1 *2006	600	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A020000	Truck Dump No. 2 *2006	600	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
Truck Dump No. 3 (with doors)	Truck Dump No. 3 *2013	360	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A030100	Discharge Conveyor No. 1 *2006	600	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A020100	Discharge Conveyor No. 2 *2006	600	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A040000	Bean Receiving Leg No. 1 *2006	600	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A050000	Bean Receiving Leg No. 2 *2006	600	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A010000	Rail Dump and Rail Collection Conveyor *2006	600	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A010100	Rail Scale Discharge Conveyor *2006	360	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A150100	Cross Bin No 1 thru 3 *2006	600	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A120100	Cross Bin No 4 thru 6 *2006	600	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A151000	Discharge Bin No 1 thru 3 *2006	360	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A121000	Discharge Bin No 4 thru 6 *2006	360	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A153000	Day Bin Leg *2006	360	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
AF-2 A200000	Grain Receiving/Meal Loadout Baghouse *2006, **2010	38,000 acfm @ 0.005 grain/acf outlet gr loading		Stack AF-2	Yes under NSPS DD
A152000	West Bin Cross Conveyor 1-3 *2006, **2010 and 2011	360	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A122000	East Bin Cross Conveyor 4-6 *2006, **2010	360	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD

DRAFT

A130100	West Bin Feed Conveyor *2006, **2010	600	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A100100	East Bin Feed Conveyor *2006, **2010	600	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
Rail Receiving Leg	Rail Receiving Leg *2013	360	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
Truck Dump No. 3 Receiving Conveyor	Truck Dump No. 3 Receiving Conveyor *2013	360	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
Truck Dump No. 3 Receiving Leg	Truck Dump No. 3 Receiving Leg *2013	360	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
Scalperator Leg	Scalperator Leg *2013	210	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
Scalperator Feed Conveyor	Scalperator Feed Conveyor *2013	210	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
Scalperator Jack Leg	Scalperator Jack Leg *2013	210	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
Scalperator	Scalperator *2013	210	Grain Receiving/Meal Loadout Baghouse	Stack AF-2	Yes under NSPS DD
A170300	Screenings Recycle Leg *2006	5	Prep exhaust baghouse	Stack AF-3	Yes under NSPS DD and NESHAP GGGG
B011300	Bean Weigh Scale *2006, **2012	264	Prep exhaust baghouse	Stack AF-3	Yes under NSPS DD and NESHAP GGGG
B310000	Screenings Weight Belt *2006	5	Prep exhaust baghouse	Stack AF-3	Yes under NSPS DD and NESHAP GGGG
B010100	Whole Bean Aspirator No 1 *2006, **2012	264	Prep exhaust baghouse	Stack AF-3	Yes under NSPS DD and NESHAP GGGG
B020100	Whole Bean Aspirator No 2 *2006, **2012	264	Prep exhaust baghouse	Stack AF-3	Yes under NSPS DD and NESHAP GGGG
B010900	Whole Bean Aspirator Cyclone *2006, **2012	264	Prep exhaust baghouse	Stack AF-3	Yes under NSPS DD and NESHAP GGGG
	Bean Storage Bins #2, #3, #6, and #7 *2006	600 tons/hr and each Bin has a maximum storage capacity of 500,000 bushels	None	None	Yes under NSPS DD

DRAFT

	Bean Storage Bins #4 and #8 *2013	600 tons/hr and each Bin has a maximum storage capacity of 500,000 bushels	None	None	Yes under NSPS DD
	Bean Storage Silos #1 and #5 *2008	600 tons/hr and each Bin has a maximum storage capacity of 500,000 bushels	None	None	Yes under NSPS DD

Note *Approved in the year indicated above for construction.

Note **Approved in the year indicated above for modification.

(b)

Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
Piles #1 and #2 *2008	Two (2) covered seasonal grain storage piles	each with a maximum storage capacity of 1,000,000 bushels of soybeans	None	None	Yes under NSPS DD

Note *Approved in the year indicated above for construction.

(c)

Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
A160100	Feed Day Tank Conveyor *2006	600	Hot dehulling baghouse	Stack AF-5	Yes under NSPS DD
A160000 (Day Tank) A160400 (Aspirator) B420000 (Cyclone)	Day Tank (with Aspirator and cyclone) *2006, **2010 , **2012, and **2014	264	Hot dehulling baghouse	Stack AF-5	Yes under NSPS DD

Note *Approved in the year indicated above for construction.

Note **Approved in the year indicated above for modification.

(The information describing the process contained in this facility description box is descriptive information and does not constitute enforceable conditions.)

New Source Performance Standards (NSPS) Requirements [326 IAC 2-7-5(1)]**E.2.1 General Provisions Relating to New Source Performance Standards [326 IAC 12-1][40 CFR Part 60, Subpart A]**

- (a) Pursuant to 40 CFR 60.1, the Permittee shall comply with the provisions of 40 CFR Part 60, Subpart A - General Provisions, which are incorporated by reference in 326 IAC 12-1, for the emission unit(s) listed above, except as otherwise specified in 40 CFR Part 60, Subpart DD.

DRAFT

- (b) Pursuant to 40 CFR 60.4, the Permittee shall submit all required notifications and reports to:

Indiana Department of Environmental Management
Compliance and Enforcement Branch, Office of Air Quality
100 North Senate Avenue
MC 61-53 IGCN 1003
Indianapolis, Indiana 46204-2251

E.2.2 Grain Elevators NSPS [326 IAC 12][40 CFR Part 60, Subpart DD]

The Permittee shall comply with the following provisions of 40 CFR Part 60, Subpart DD (included as Attachment B to the operating permit), which are incorporated by reference as 326 IAC 12, for the emission units listed above:

- (1) 40 CFR 60.300
- (2) 40 CFR 60.301
- (3) 40 CFR 60.302 (a)(1), (b), (c)(1), (c)(2), and (c)(3)
- (4) 40 CFR 60.303
- (5) 40 CFR 60.304

The stack testing requirements under 40 CFR § 60.303 shall not apply to the Bean Storage Bins #2, #3, #4, #6, #7, and #8, Bean Storage Silos #1 and #5, and the seasonal grain storage Piles #1 and #2.

DRAFT

SECTION E.3

NSPS

Emissions Unit Description:

(e)

Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
CL-5045	1st Primary Transester Column *2006, **2007 and 2010	12,960 gals/hr	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS NNN, NSPS RRR, NSPS VV, and NESHAP FFFF
CL-5046	1st Secondary Transester Column *2006, **2007 and 2010	12,960 gals/hr	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS NNN, NSPS RRR, NSPS VV, and NESHAP FFFF
CL-5063	2nd Primary Transester Column *2006, **2007 and 2010	12,960 gals/hr	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS NNN, NSPS RRR, NSPS VV, and NESHAP FFFF
CL-5064	2nd Secondary Transester Column *2010	12,960 gals/hr	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS NNN, NSPS RRR, NSPS VV, and NESHAP FFFF
R-8171	Esterification Reactor *2010	12,960 gals/hr	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS NNN, NSPS RRR, NSPS VV, and NESHAP FFFF
VU010000	Vacuum group package *2006, **2007 and 2010	12,960 gals/hr	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS NNN, NSPS VV, and NESHAP FFFF
	Biodiesel Mineral Oil Absorber *2010			Stack S-5	Yes under NSPS VV, and NESHAP FFFF
	Biodiesel Water Absorber *2006, **2007	0.448 gpm		Stack S-5	Yes under NSPS VV, and NESHAP FFFF
I040000	Biodiesel Storage Tank #4 *2006, **2007	725,000 gals			Yes under NSPS VV, and NESHAP FFFF
I050000	Biodiesel Storage Tank #5 *2006, **2007	725,000 gals			Yes under NSPS VV, and NESHAP FFFF
I060000	Biodiesel Storage Tank #6 *2009	325,000			Yes under NSPS VV, and NESHAP FFFF
I070000	Biodiesel Storage Tank #7 *2006, **2007	325,000 gals			Yes under NSPS VV, and NESHAP FFFF
I080000	Biodiesel Storage Tank #8 *2006, **2007	325,000 gals			Yes under NSPS VV, and NESHAP FFFF
I090000	Biodiesel Storage Tank #9 *2006, **2007	325,000 gals			Yes under NSPS VV, and NESHAP FFFF

DRAFT

I100000	Biodiesel Storage Tank #10 *2006, **2007	325,000 gals			Yes under NSPS VV, and NESHAP FFFF
I110000	Biodiesel Storage Tank #11 *2007	325,000 gals			Yes under NSPS VV, and NESHAP FFFF
1140000	Biodiesel Storage Tank #0 *2009	735,000			Yes under NSPS VV, and NESHAP FFFF
I120000	Glycerin Tank #12 *2006, **2010	360,000 gals			Yes under NSPS VV, and NESHAP FFFF
I130000	Glycerin Tank #13 *2006, **2010	360,000 gals			Yes under NSPS VV, and NESHAP FFFF
I250000	Methanol Storage Tank #1 *2006, **2007 and **2010	38,850 gallons	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS VV, and NESHAP FFFF
I260000	Methanol Storage Tank #2 *2006, **2007 and 2010	38,850 gallons	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS VV, and NESHAP FFFF
I270000	Methanol Storage Tank #3 *2006, **2007 and 2010	38,850 gallons	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS VV, and NESHAP FFFF
I280000	Methanol Storage Tank #4 *2006, **2007 and 2010	38,850 gallons	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS VV, and NESHAP FFFF
I290000	Methanol Storage Tank #5 *2006, **2007 and 2010	38,850 gallons	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS VV, and NESHAP FFFF
I300000	Methanol Storage Tank #6 *2007	38,850 gallons	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS VV, and NESHAP FFFF
I230000	Sodium Methylate (catalyst) Storage Tank #1 *2006, **2007 and 2010	38,850 gallons	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS VV, and NESHAP FFFF
I240000	Sodium Methylate (catalyst) Storage Tank #2 *2007, **2010	38,850 gallons	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS VV, and NESHAP FFFF
Rail Rack	Loading Rack (Rail) *2006, **2007, **2010 and **2011	500 gallons per minute			Yes under NSPS VV, and NESHAP FFFF
Truck Rack #1	Loading Rack (Truck) *2006, **2007, **2010 and **2011	430 gallons per minute			Yes under NSPS VV, and NESHAP FFFF
Truck Rack #2	Loading Rack (Truck) *2011	430 gallons per minute			Yes under NSPS VV, and NESHAP FFFF

Note *Approved in the year indicated above for construction.

Note **Approved in the year indicated above for modification.

(The information describing the process contained in this facility description box is descriptive information and does not constitute enforceable conditions.)

DRAFT

New Source Performance Standards (NSPS) Requirements [326 IAC 2-7-5(1)]**E.3.1 General Provisions Relating to New Source Performance Standards [326 IAC 12-1][40 CFR Part 60, Subpart A]**

(a) Pursuant to 40 CFR 60.1, the Permittee shall comply with the provisions of 40 CFR Part 60, Subpart A - General Provisions, which are incorporated by reference in 326 IAC 12-1, for the emission unit(s) listed above, except as otherwise specified in 40 CFR Part 60, Subpart VV.

(b) Pursuant to 40 CFR 60.4, the Permittee shall submit all required notifications and reports to:

Indiana Department of Environmental Management
Compliance and Enforcement Branch, Office of Air Quality
100 North Senate Avenue
MC 61-53 IGCN 1003
Indianapolis, Indiana 46204-2251

E.3.2 Equipment Leaks of VOC in the Synthetic Organic Chemicals Manufacturing Industry for which Construction, Reconstruction, or Modification Commenced After January 5, 1981, and on or Before November 7, 2006 NSPS [326 IAC 12][40 CFR Part 60, Subpart VV]

The Permittee shall comply with the following provisions of 40 CFR Part 60, Subpart VV (included as Attachment C to the operating permit), which are incorporated by reference as 326 IAC 12, for the emission units listed above:

- (1) 40 CFR 60.480(a), (b), (c), and (f)
- (2) 40 CFR 60.481
- (3) 40 CFR 60.482-1
- (4) 40 CFR 60.482-2
- (5) 40 CFR 60.482-3
- (6) 40 CFR 60.482-4
- (7) 40 CFR 60.482-5
- (8) 40 CFR 60.482-6
- (9) 40 CFR 60.482-7
- (10) 40 CFR 60.482-8
- (11) 40 CFR 60.482-9
- (12) 40 CFR 60.482-10
- (13) 40 CFR 60.483-1
- (14) 40 CFR 60.483-2
- (15) 40 CFR 60.484
- (16) 40 CFR 60.485
- (17) 40 CFR 60.486
- (18) 40 CFR 60.487
- (19) 40 CFR 60.488
- (20) 40 CFR 60.489

DRAFT

SECTION E.4**NSPS****Emissions Unit Description:**

(e)

Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
CL-5045	1st Primary Transester Column *2006, **2007 and 2010	12,960 gals/hr	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS NNN, NSPS RRR, NSPS VV, and NESHAP FFFF
CL-5046	1st Secondary Transester Column *2006, **2007 and 2010	12,960 gals/hr	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS NNN, NSPS RRR, NSPS VV, and NESHAP FFFF
CL-5063	2nd Primary Transester Column *2006, **2007 and 2010	12,960 gals/hr	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS NNN, NSPS RRR, NSPS VV, and NESHAP FFFF
CL-5064	2nd Secondary Transester Column *2010	12,960 gals/hr	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS NNN, NSPS RRR, NSPS VV, and NESHAP FFFF
R-8171	Esterification Reactor *2010	12,960 gals/hr	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS NNN, NSPS RRR, NSPS VV, and NESHAP FFFF
VU010000	Vacuum group package *2006, **2007 and 2010	12,960 gals/hr	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS NNN, NSPS VV, and NESHAP FFFF
Biodiesel Distillation	Biodiesel Distillation *2013	15	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS VVa, NSPS NNN, and NESHAP FFFF

Note *Approved in the year indicated above for construction.

Note **Approved in the year indicated above for modification.

Insignificant Activities:

Description	Capacity	Control	Affected Facility?
Glycerin Refinery *2013, **2014	7.0 tons per hour	None	Yes, Under NSPS VVa, NSPS NNN, and NESHAP FFFF

Note *Approved in the year indicated above for construction.

(The information describing the process contained in this facility description box is descriptive information and does not constitute enforceable conditions.)

DRAFT

New Source Performance Standards (NSPS) Requirements [326 IAC 2-7-5(1)]**E.4.1 General Provisions Relating to New Source Performance Standards [326 IAC 12-1][40 CFR Part 60, Subpart A]**

(a) Pursuant to 40 CFR 60.1, the Permittee shall comply with the provisions of 40 CFR Part 60, Subpart A - General Provisions, which are incorporated by reference in 326 IAC 12-1, for the emission unit(s) listed above, except as otherwise specified in 40 CFR Part 60, Subpart NNN.

(b) Pursuant to 40 CFR 60.4, the Permittee shall submit all required notifications and reports to:

Indiana Department of Environmental Management
Compliance and Enforcement Branch, Office of Air Quality
100 North Senate Avenue
MC 61-53 IGCN 1003
Indianapolis, Indiana 46204-2251

E.4.2 Volatile Organic Compound (VOC) Emissions From Synthetic Organic Chemical Manufacturing Industry (SOCMI) Distillation Operations NSPS [326 IAC 12][40 CFR Part 60, Subpart NNN]

The Permittee shall comply with the following provisions of 40 CFR Part 60, Subpart NNN (included as Attachment D to the operating permit), which are incorporated by reference as 326 IAC 12, for the emission units listed above:

- (1) 40 CFR 60.660(a), (b)(1), and (c)(4)
- (2) 40 CFR 60.661
- (3) 40 CFR 60.662(c)
- (4) 40 CFR 60.664(e) and (f)
- (5) 40 CFR 60.665(a), (b), (h), (k), (l), (m), and (p)
- (6) 40 CFR 60.666
- (7) 40 CFR 60.667
- (8) 40 CFR 60.668

DRAFT

SECTION E.5

NSPS

Emissions Unit Description:

(e)

Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
CL-5045	1st Primary Transester Column *2006, **2007 and 2010	12,960 gals/hr	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS NNN, NSPS RRR, NSPS VV, and NESHAP FFFF
CL-5046	1st Secondary Transester Column *2006, **2007 and 2010	12,960 gals/hr	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS NNN, NSPS RRR, NSPS VV, and NESHAP FFFF
CL-5063	2nd Primary Transester Column *2006, **2007 and 2010	12,960 gals/hr	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS NNN, NSPS RRR, NSPS VV, and NESHAP FFFF
CL-5064	2nd Secondary Transester Column *2010	12,960 gals/hr	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS NNN, NSPS RRR, NSPS VV, and NESHAP FFFF
R-8171	Esterification Reactor *2010	12,960 gals/hr	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS NNN, NSPS RRR, NSPS VV, and NESHAP FFFF

Note *Approved in the year indicated above for construction.

Note **Approved in the year indicated above for modification.

(The information describing the process contained in this facility description box is descriptive information and does not constitute enforceable conditions.)

New Source Performance Standards (NSPS) Requirements [326 IAC 2-7-5(1)]

E.5.1 General Provisions Relating to New Source Performance Standards [326 IAC 12-1][40 CFR Part 60, Subpart A]

(a) Pursuant to 40 CFR 60.1, the Permittee shall comply with the provisions of 40 CFR Part 60, Subpart A - General Provisions, which are incorporated by reference in 326 IAC 12-1, for the emission unit(s) listed above, except as otherwise specified in 40 CFR Part 60, Subpart RRR.

(b) Pursuant to 40 CFR 60.4, the Permittee shall submit all required notifications and reports to:

Indiana Department of Environmental Management
 Compliance and Enforcement Branch, Office of Air Quality
 100 North Senate Avenue
 MC 61-53 IGCN 1003
 Indianapolis, Indiana 46204-2251

DRAFT

E.5.2 Volatile Organic Compound Emissions From Synthetic Organic Chemical Manufacturing Industry (SOCMI) Reactor Processes NSPS [326 IAC 12][40 CFR Part 60, Subpart RRR]

The Permittee shall comply with the following provisions of 40 CFR Part 60, Subpart RRR (included as Attachment E to the operating permit), which are incorporated by reference as 326 IAC 12, for the emission units listed above:

- (1) 40 CFR 60.700(c)(4)
- (2) 40 CFR 60.701
- (3) 40 CFR 60.704(g)
- (4) 40 CFR 60.705(h), (l)(4), and (o)
- (5) 40 CFR 60.706
- (6) 40 CFR 60.707
- (7) 40 CFR 60.708

DRAFT

SECTION E.6

NSPS

Emissions Unit Description: Insignificant Activities

Description	Capacity	Control	Affected Facility?
Three (3) Diesel Fire Pumps *2006	575 BHP each	None	Yes under NSPS IIII and NESHAP ZZZZ
Three (3) diesel-fired emergency generators *2016	One (1) 250 kW and Two (2) 150 kW	None	Yes under NSPS IIII and NESHAP ZZZZ

Note *Approved in the year indicated above for construction.

(The information describing the process contained in this facility description box is descriptive information and does not constitute enforceable conditions.)

New Source Performance Standards (NSPS) Requirements [326 IAC 2-7-5(1)]

E.6.1 General Provisions Relating to New Source Performance Standards [326 IAC 12-1][40 CFR Part 60, Subpart A]

- (a) Pursuant to 40 CFR 60.1, the Permittee shall comply with the provisions of 40 CFR Part 60, Subpart A - General Provisions, which are incorporated by reference in 326 IAC 12-1, for the emission unit(s) listed above, except as otherwise specified in 40 CFR Part 60, Subpart IIII.
- (b) Pursuant to 40 CFR 60.4, the Permittee shall submit all required notifications and reports to:

Indiana Department of Environmental Management
Compliance and Enforcement Branch, Office of Air Quality
100 North Senate Avenue
MC 61-53 IGCN 1003
Indianapolis, Indiana 46204-2251

E.6.2 Stationary Compression Ignition Internal Combustion Engines NSPS [326 IAC 12][40 CFR Part 60, Subpart IIII]

The Permittee shall comply with the following provisions of 40 CFR Part 60, Subpart IIII (included as Attachment F to the operating permit), which are incorporated by reference as 326 IAC 12:

- (a) Three (3) Diesel Fire Pumps
- (1) 40 CFR 60.4200(a)(2)(ii)
 - (2) 40 CFR 60.4205(c)
 - (3) 40 CFR 60.4206
 - (4) 40 CFR 60.4207(a) and (b)
 - (5) 40 CFR 60.4209
 - (6) 40 CFR 60.4211(a), (b) and (f)
 - (7) 40 CFR 60.4214(b) and (c)
 - (8) 40 CFR 60.4218
 - (9) 40 CFR 60.4219
 - (10) Table 3 to Subpart IIII
 - (11) Table 4 to Subpart IIII
 - (12) Table 8 to Subpart IIII

DRAFT

(b) Three (3) diesel-fired emergency generators

- (1) 40 CFR 60.4200(a)(2)(i), (a)(4)
- (2) 40 CFR 60.4205(b) and (e)
- (3) 40 CFR 60.4206
- (4) 40 CFR 60.4207(b)
- (5) 40 CFR 60.4208
- (6) 40 CFR 60.4209
- (7) 40 CFR 60.4211(a), (c) and (f)
- (8) 40 CFR 60.4212
- (9) 40 CFR 60.4214(b)
- (10) 40 CFR 60.4218
- (11) 40 CFR 60.4219
- (12) Table 5 to Subpart IIII
- (13) Table 8 to Subpart IIII

DRAFT

SECTION E.7

NESHAP

Emissions Unit Description:

(e)

Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
CL-5045	1st Primary Transester Column *2006, **2007 and 2010	12,960 gals/hr	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS NNN, NSPS RRR, NSPS VV, and NESHAP FFFF
CL-5046	1st Secondary Transester Column *2006, **2007 and 2010	12,960 gals/hr	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS NNN, NSPS RRR, NSPS VV, and NESHAP FFFF
CL-5063	2nd Primary Transester Column *2006, **2007 and 2010	12,960 gals/hr	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS NNN, NSPS RRR, NSPS VV, and NESHAP FFFF
CL-5064	2nd Secondary Transester Column *2010	12,960 gals/hr	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS NNN, NSPS RRR, NSPS VV, and NESHAP FFFF
R-8171	Esterification Reactor *2010	12,960 gals/hr	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS NNN, NSPS RRR, NSPS VV, and NESHAP FFFF
VU010000	Vacuum group package *2006, **2007 and 2010	12,960 gals/hr	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS NNN, NSPS VV, and NESHAP FFFF
	Biodiesel Mineral Oil Absorber *2010			Stack S-5	Yes under NSPS VV, and NESHAP FFFF
	Biodiesel Water Absorber *2006, **2007	0.448 gpm		Stack S-5	Yes under NSPS VV, and NESHAP FFFF
Biodiesel Distillation	Biodiesel Distillation *2013	15	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS VVa, NSPS NNN& NESHAP FFFF
I040000	Biodiesel Storage Tank #4 *2006, **2007	725,000 gals			Yes under NSPS VV, and NESHAP FFFF
I050000	Biodiesel Storage Tank #5 *2006, **2007	725,000 gals			Yes under NSPS VV, and NESHAP FFFF
1060000	Biodiesel Storage Tank #6 *2009	325,000			Yes under NSPS VV, and NESHAP FFFF
I070000	Biodiesel Storage Tank #7 *2006, **2007	325,000 gals			Yes under NSPS VV, and NESHAP FFFF

DRAFT

I080000	Biodiesel Storage Tank #8 *2006, **2007	325,000 gals			Yes under NSPS VV, and NESHAP FFFF
I090000	Biodiesel Storage Tank #9 *2006, **2007	325,000 gals			Yes under NSPS VV, and NESHAP FFFF
I100000	Biodiesel Storage Tank #10 *2006, **2007	325,000 gals			Yes under NSPS VV, and NESHAP FFFF
I110000	Biodiesel Storage Tank #11 *2007	325,000 gals			Yes under NSPS VV, and NESHAP FFFF
I140000	Biodiesel Storage Tank #0 *2009	735,000			Yes under NSPS VV, and NESHAP FFFF
I120000	Glycerin Tank #12 *2006, **2010	360,000 gals			Yes under NSPS VV, and NESHAP FFFF
I130000	Glycerin Tank #13 *2006, **2010	360,000 gals			Yes under NSPS VV, and NESHAP FFFF
I250000	Methanol Storage Tank #1 *2006, **2007 and **2010	38,850 gallons	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS VV, and NESHAP FFFF
I260000	Methanol Storage Tank #2 *2006, **2007 and 2010	38,850 gallons	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS VV, and NESHAP FFFF
I270000	Methanol Storage Tank #3 *2006, **2007 and 2010	38,850 gallons	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS VV, and NESHAP FFFF
I280000	Methanol Storage Tank #4 *2006, **2007 and 2010	38,850 gallons	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS VV, and NESHAP FFFF
I290000	Methanol Storage Tank #5 *2006, **2007 and 2010	38,850 gallons	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS VV, and NESHAP FFFF
I300000	Methanol Storage Tank #6 *2007	38,850 gallons	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS VV, and NESHAP FFFF
I230000	Sodium Methylate (catalyst) Storage Tank #1 *2006, **2007 and 2010	38,850 gallons	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS VV, and NESHAP FFFF
I240000	Sodium Methylate (catalyst) Storage Tank #2 *2007, **2010	38,850 gallons	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS VV, and NESHAP FFFF
Rail Rack	Loading Rack (Rail) *2006, **2007, **2010 and **2011	500 gallons per minute			Yes under NSPS VV, and NESHAP FFFF
Truck Rack #1	Loading Rack (Truck) *2006, **2007, **2010 and **2011	430 gallons per minute			Yes under NSPS VV, and NESHAP FFFF
Truck Rack #2	Loading Rack (Truck) *2011	430 gallons per minute			Yes under NSPS VV, and NESHAP FFFF

Note *Approved in the year indicated above for construction.

Note **Approved in the year indicated above for modification.

Insignificant Activities:

DRAFT

Description	Capacity	Control	Affected Facility?
Glycerin Refinery *2013, **2014	7.0 tons per hour	None	Yes, Under NSPS VVa, NSPS NNN, and NESHAP FFFF
Glycerin Truck/ Rail Loadout	7.0 tons per hour	None	Yes, Under NSPS VVa and NESHAP FFFF
Glycerin Yellow Tank	8,000 gallons	None	Yes, Under NSPS VVa and NESHAP FFFF
Glycerin Yellow Tank 1	50,000 gallons	None	Yes, Under NSPS VVa and NESHAP FFFF
Glycerin Yellow Tank 2	50,000 gallons	None	Yes, Under NSPS VVa and NESHAP FFFF
Glycerin Yellow Tank 3	50,000 gallons	None	Yes, Under NSPS VVa and NESHAP FFFF
Glycerin Yellow Tank 4	50,000 gallons	None	Yes, Under NSPS VVa and NESHAP FFFF

Note *Approved in the year indicated above for construction.

(The information describing the process contained in this facility description box is descriptive information and does not constitute enforceable conditions.)

National Emission Standards for Hazardous Air Pollutants (NESHAP) Requirements [326 IAC 2-7-5(1)]

E.7.1 General Provisions Relating to National Emission Standards for Hazardous Air Pollutants under 40 CFR 63 [326 IAC 20-1][40 CFR 63, Subpart A]

- (a) Pursuant to 40 CFR 63.1, the Permittee shall comply with the provisions of 40 CFR Part 63, Subpart A - General Provisions, which are incorporated by reference as 326 IAC 20-1, for the emission units listed above, except as otherwise specified in 40 CFR Part 63, Subpart FFFF.
- (b) Pursuant to 40 CFR 63.10, the Permittee shall submit all required notifications and reports to:

Indiana Department of Environmental Management
 Compliance and Enforcement Branch, Office of Air Quality
 100 North Senate Avenue
 MC 61-53 IGCN 1003
 Indianapolis, Indiana 46204-2251

E.7.2 Miscellaneous Organic Chemical Manufacturing NESHAP [40 CFR Part 63, Subpart FFFF][326 IAC 20-84]

The Permittee shall comply with the following provisions of 40 CFR Part 63, Subpart FFFF (included as Attachment G to the operating permit), which are incorporated by reference as 326 IAC 20-84, for the emission units listed above:

DRAFT

- (1) 40 CFR 63.2430
- (2) 40 CFR 63.2435(a), (b), (d), and (e)
- (3) 40 CFR 63.2440
- (4) 40 CFR 63.2445(a)(2), (c), (d), and (f)
- (5) 40 CFR 63.2450(a), (c)(1), (c)(2), (e)(1), (e)(2), (g), (h), (k)(5), (l), (m), (p), and (r)
- (6) 40 CFR 63.2460
- (7) 40 CFR 63.2470
- (8) 40 CFR 63.2475
- (9) 40 CFR 63.2480(a), (b), and (d)
- (10) 40 CFR 63.2500(b) through (f)
- (11) 40 CFR 63.2505
- (12) 40 CFR 63.2515
- (13) 40 CFR 63.2520
- (14) 40 CFR 63.2525(b), (c), (d), (f), and (g)
- (15) 40 CFR 63.2540
- (16) 40 CFR 63.2545
- (17) 40 CFR 63.2550
- (18) Table 2 to Subpart FFFF
- (19) Table 4 to Subpart FFFF
- (20) Table 5 to Subpart FFFF
- (21) Table 6 to Subpart FFFF
- (22) Table 9 to Subpart FFFF
- (23) Table 11 to Subpart FFFF
- (24) Table 12 to Subpart FFFF

DRAFT

SECTION E.8

NESHAP

Emissions Unit Description:

(a)

Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
G020500	Meal Storage Feed Conveyor *2006, **2010, **2011, ** 2015	200	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
G070300	Truck Meal Loadout Feed Conveyor *2006, **2010, ** 2015	300	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
G290000	Truck Collection Conveyor *2010, ** 2015	300	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
G280000	Truck Loader No.1 *2010, ** 2015	330	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
G270000	Truck Loader No.2 *2010, ** 2015	330	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
G080000	Truck Pelleted Hull Loadout Bin *2006, **2010, ** 2015	148	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
G070000	Truck Meal Loadout Bin *2006, **2010, 2012, ** 2015	300	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
G180000	Rail Pelleted Hull Loadout Bin *2010, ** 2015	148	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
G130000	Rail Meal Loadout Bin *2006, **2010, 2012, **2015	300	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
G160000	Pellet Hulls Conveyor to Loadout *2006, **2010, 2012, ** 2015	17.0	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
G170000	Rail Car Collection Conveyor *2006, **2010, ** 2015	300	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
G220000	Rail Car Loadout *2010, ** 2015	330	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
G010100	Meal Reclaim Conveyor *2006, **2010, ** 2015	200	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
G010200	Meal Reclaim Leg *2010, ** 2015	200	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG

DRAFT

A060000	Screener *2006, **2011 and 2012	264	Prep Exhaust Baghouse	Stack AF-3	Yes under NESHAP GGGG
A160300	VSC Leg Feed Conveyor *2006, **2010 and 2012	264	Prep Exhaust Baghouse	Stack AF-3	Yes under NESHAP GGGG
A170000	Screenings Tank *2006	5	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
A170300	Screenings Recycle Leg *2006	5	Prep exhaust baghouse	Stack AF-3	Yes under NSPS DD and NESHAP GGGG
B011300	Bean Weigh Scale *2006, **2012	264	Prep exhaust baghouse	Stack AF-3	Yes under NSPS DD and NESHAP GGGG
B310000	Screenings Weight Belt *2006	5	Prep exhaust baghouse	Stack AF-3	Yes under NSPS DD and NESHAP GGGG
AF-7	Pod Grinder/Screener Baghouse *2011	5,000 acfm		Stack AF-7	Yes under NESHAP GGGG
B310200	Pod Grinder/Destoner *2006, **2010	5	Pod Grinder/ Screener Baghouse	Stack AF-7	Yes under NESHAP GGGG
B011200	VSC Feed Leg *2006, **2012	264	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
A060400	Screener Feed Conveyor *2010, **2012	264	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
B010100	Whole Bean Aspirator No 1 *2006, **2012	264	Prep exhaust baghouse	Stack AF-3	Yes under NSPS DD and NESHAP GGGG
B020100	Whole Bean Aspirator No 2 *2006, **2012	264	Prep exhaust baghouse	Stack AF-3	Yes under NSPS DD and NESHAP GGGG
B010900	Whole Bean Aspirator Cyclone *2006, **2012	264	Prep exhaust baghouse	Stack AF-3	Yes under NSPS DD and NESHAP GGGG
B030800	Conditioned Bean Feed Conveyor *2006, **2012	264	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
B030900	Hull Collection Conveyor *2006, **2012	17.0	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
E130000	Hull Screener No.1 *2006	9.6	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
E150000	Hull Screener No.2 *2006	9.6	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
B440000	Secondary Hull Collection L-Path *2010, **2012	17.0	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
B430000	Secondary Hull Collection Conveyor *2010, **2012	17.0	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
E070300	4 Hour Hull Tank *2006, **2012	17.0	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
E080000	Pellet Cooler *2006, **2012	17.0	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG

DRAFT

E090000	Pellet Cooler Cyclone *2006, **2012	17.0	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
E050200	Hull Hammer Mill Feeder *2006, **2012	17.0	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
E050000	Hull Hammer Mill *2006, **2012	17.0	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
E050100	Hull Hammer Mill Plenum *2006, **2012	17.0	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
G050100	Pelleted Hulls Leg *2006, **2012	17.0	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
G050300	Pelleted Hulls Storage Conveyor *2006, **2010, 2012, **2015	17.0	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG
E050400	Hulls Addition Screw *2011, **2012	17.0	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
B010300	Conditioner Bean Loop Path *2006, **2012	264	Prep exhaust baghouse	Stack AF-3	Yes under NESHAP GGGG
AF-3 G100000	Prep exhaust baghouse *2006	28,900 acfm @ 0.005 grain/acf outlet grain loading		Stack AF-3	Yes under NESHAP GGGG

Note *Approved in the year indicated above for construction.

Note **Approved in the year indicated above for modification.

(c)

Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
C200100	Flaker Feed Loop Conveyor *2010, **2012	247	Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
C010600	Flake Collection Conveyor (12 flakers) *2006, **2010 and 2012	247	Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
C010000	Flaking Roll No. 1 *2013	22.9	***Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
C020000	Flaking Roll No. 2 *2006	22.9	Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
C030000	Flaking Roll No. 3 *2006	22.9	Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
C040000	Flaking Roll No. 4 *2012	22.9	***Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
C050000	Flaking Roll No. 5 *2006	22.9	Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
C060000	Flaking Roll No. 6 *2006	22.9	Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
C070000	Flaking Roll No. 7 *2012	22.9	***Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
C080000	Flaking Roll No. 8 *2006	22.9	Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
C090000	Flaking Roll No. 9 *2006	22.9	Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
C100000	Flaking Roll No. 10 *2013	22.9	***Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG

DRAFT

C0110000	Flaking Roll No. 11 *2009	22.9	***Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
C0120000	Flaking Roll No. 12 *2009	22.9	***Flaker aspiration baghouse	Stack AF-4	Yes under NESHAP GGGG
AF-4 C110000	Flaker aspiration baghouse *2006	24,000 acfm @ 0.005 grain/acf outlet grain loading		Stack AF-4	Yes under NESHAP GGGG
B040000	Hulloosinator No. 1 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B080100	Hulloosinator No. 2 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B130000	Hulloosinator No. 3 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B170000	Hulloosinator No. 4 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B050000	Cascade Dryer No. 1 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B090000	Cascade Dryer No. 2 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B140000	Cascade Dryer No. 3 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B180000	Cascade Dryer No. 4 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B210000	CCD Cyclone *2006, **2010	42,000 cfm	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B060000	Cracking Roll No.1 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B100000	Cracking Roll No.2 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B150000	Cracking Roll No.3 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B190000	Cracking Roll No.4 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B070000	Cascade Conditioner No. 1 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B110000	Cascade Conditioner No. 2 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B160000	Cascade Conditioner No. 3 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B200000	Cascade Conditioner No. 4 *2006, **2012	66.0	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
B230000	CCC Cyclone *2006, **2010	42,000 cfm	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
E130100	Secondary Aspirator No 1 *2006, **2010	9.6	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
E150100	Secondary Aspirator No 2 *2006, **2010	9.6	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG
E160000	Secondary Aspirator Cyclone *2006, **2010	9.6	Hot dehulling baghouse	Stack AF-5	Yes under NESHAP GGGG

DRAFT

AF-5 B260000	Hot dehulling baghouse *2006	43,000 acfm @ 0.005 grain/acf outlet grain loading		Stack AF-5	Yes under NESHAP GGGG
E020300	Grinding Discharge Conveyor *2011, **2012	198	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E020400	Hammer Mill Mixing Conveyor *2006, **2011 and 2012	198	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E010100	Meal L-Path Conveyor *2006, **2012	198	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E010300	Meal Hammer Mill Feed Conveyor *2006, **2012	198	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E020200	Meal Hammer Mill Feeder No. 1 *2006	74.0	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E030200	Meal Hammer Mill Feeder No. 2 *2006	74.0	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E040200	Meal Hammer Mill Feeder No. 3 (switch) *2006	74.0	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E020000	Meal Hammer Mill No. 1 *2006	74.0	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E030000	Meal Hammer Mill No. 2 *2006	74.0	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E040000	Meal Hammer Mill No. 3 (switch) *2006	74.0	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E020100	Meal Hammer Mill Bin No. 1 *2006	74.0	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E030100	Meal Hammer Mill Bin No. 2 *2006	74.0	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E040100	Meal Hammer Mill Bin No. 3 (switch) *2006	74.0	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E230200	Meal Hammer Mill Feeder No. 5 *2012	74.0	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E230000	Meal Hammer Mill No. 5 *2012	74.0	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
E230100	Meal Hammer Mill Bin No. 5 *2012	74.0	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
G010300	Meal Leg *2006, **2010 and 2012	198	Meal Grinding Baghouse	Stack AF-6	Yes under NESHAP GGGG
G150000	Meal Conveyor to Loadout *2006, **2012, ** 2015	198	Meal Loadout Baghouse AF-1	Stack AF-1	Yes under NESHAP GGGG

DRAFT

AF-6 E110000	Mill Grinding Baghouse *2006	18,000 acfm @ 0.005 grain/acf outlet grain loading		Stack AF-6	Yes under NESHAP GGGG
B010000	VSC No. 1 *2006, **2012	132	VSC Cyclone No. 1	Stack S-1	Yes under NESHAP GGGG
B020000	VSC No. 2 *2006, **2012	132	VSC Cyclone No. 2	Stack S-6	Yes under NESHAP GGGG
B010500	VSC Air Heater No. 1 *2006, **2012	264	VSC Cyclone No. 1	Stack S-1	Yes under NESHAP GGGG
B020500	VSC Air Heater No. 2 *2015	264	VSC Cyclone No. 2	Stack S-6	Yes under NESHAP GGGG
B010700	VSC Cyclone No. 1 *2006, **2010	42,000 cfm		Stack S-1	Yes under NESHAP GGGG
B020700	VSC Cyclone No. 2 *2015	42,000 cfm		Stack S-6	Yes under NESHAP GGGG
B120000	Jet Dryer No. 1 *2006, **2010 and 2012	132	Jet Dryer Baghouse AF-8	Stack S-1	Yes under NESHAP GGGG
B030000	Jet Dryer No. 2 *2006, **2012	132	Jet Dryer Baghouse AF-9	Stack S-1	Yes under NESHAP GGGG
B120100A	Jet Dryer Baghouse AF- 8	74,000 acfm		Stack S-1	Yes under NESHAP GGGG
B120100B	Jet Dryer Baghouse AF- 9	74,000 acfm		Stack S-1	Yes under NESHAP GGGG

Note *Approved in the year indicated above for construction.

Note **Approved in the year indicated above for modification.

Note ***The Flaker aspiration baghouse has been determined to be integral to the process for this unit.

(d)

Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
D010000	Soybean oil extractor *2006, **2010 and 2012	264	Mineral oil absorber	Stack S-4	Yes under NESHAP GGGG
	One (1) set of evaporators *2006		Mineral oil absorber	Stack S-4	Yes under NESHAP GGGG
D020000	One (1) Desolventizer/toaster *2006, **2010		Mineral oil absorber	Stack S-4	Yes under NESHAP GGGG
	One (1) set of water separators *2006		Mineral oil absorber	Stack S-4	Yes under NESHAP GGGG
D060000	Main Vent Condenser *2006		Mineral oil absorber	Stack S-4	Yes under NESHAP GGGG
	Five (5) hexane storage tank *2006 for original tank, and 2010 for other tanks **2010 for original tank	20,690 gallons each	Mineral oil absorber	Stack S-4	Yes under NESHAP GGGG
I220000	One (1) soybean oil pre- treat Tank *2010	35,170 gallons			Yes under NESHAP GGGG

DRAFT

	Three (3) soybean oil storage tank (Degummed Oil Tanks #1 and #2 and Crude Oil Tank #3) *2006 for original tank and 2010 for other tanks, **2010 for original tank	725,000 gallons each			Yes under NESHAP GGGG
D070000	Mineral oil absorber *2006			Stack S-4	Yes under NESHAP GGGG
D310000-1	DC Deck No. 1 *2006, **2010 and 2012	208	DC Deck Cyclone No. 1	Stack S-2	Yes under NESHAP GGGG
D310000-2	DC Deck No. 2 *2006, **2010, 2011, and 2012	208	DC Deck Cyclone No. 2	Stack S-2	Yes under NESHAP GGGG
D310000-3	DC Deck No. 3 *2006, **2010 and 2012	208	DC Deck Cyclone No. 3	Stack S-2	Yes under NESHAP GGGG
D310000-4	DC Deck No. 4 *2006, **2010 and 2012	208	DC Deck Cyclone No. 4	Stack S-2	Yes under NESHAP GGGG
D310700	DC Deck Cyclone No. 1 *2006, **2010 and 2011	18,000 scfm		Stack S-2	Yes under NESHAP GGGG
D310800	DC Deck Cyclone No. 2 *2006, **2010 and 2011	18,000 scfm		Stack S-2	Yes under NESHAP GGGG
D310900	DC Deck Cyclone No. 3 *2010, ** 2011	18,000 scfm		Stack S-2	Yes under NESHAP GGGG
D311000	DC Deck Cyclone No. 4 *2010, **2011	18,000 scfm		Stack S-2	Yes under NESHAP GGGG

Note *Approved in the year indicated above for construction.

Note **Approved in the year indicated above for modification.

(g)

Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
G010000	Meal Bin No. 1*** *2006, **2010, 2011, and 2012	198	Meal Bin Filter No. 1	Stack MBF-1	Yes under NESHAP GGGG
G020000	Meal Bin No. 2*** *2006, **2010, 2011, and 2012	198	Meal Bin Filter No. 2	Stack MBF-2	Yes under NESHAP GGGG
G030000	Meal Bin No. 3*** *2006, **2010, 2011, and 2012	198	Meal Bin Filter No. 3	Stack MBF-3	Yes under NESHAP GGGG
G040000	Meal Bin No. 4*** *2006, **2010, 2011, and 2012	198	Meal Bin Filter No. 4	Stack MBF-4	Yes under NESHAP GGGG
G050000	Meal Bin No. 5*** *2010, **2011 and 2012	198	Meal Bin Filter No. 5	Stack MBF-5	Yes under NESHAP GGGG

Note *Approved in the year indicated above for construction.

Note **Approved in the year indicated above for modification.

Note ***There are five meal bins. However, the plant is only physically capable of loading one meal bin at a time. Thus, the PTE for these units is calculated at a rate of 198 tons/hr for all five meal bins combined.

(The information describing the process contained in this facility description box is descriptive information and does not constitute enforceable conditions.)

DRAFT

National Emission Standards for Hazardous Air Pollutants (NESHAP) Requirements [326 IAC 2-7-5(1)]

E.8.1 General Provisions Relating to National Emission Standards for Hazardous Air Pollutants under 40 CFR 63 [326 IAC 20-1][40 CFR 63, Subpart A]

- (a) Pursuant to 40 CFR 63.1, the Permittee shall comply with the provisions of 40 CFR Part 63, Subpart A - General Provisions, which are incorporated by reference as 326 IAC 20-1, for the emission units listed above, except as otherwise specified in 40 CFR Part 63, Subpart GGGG.
- (b) Pursuant to 40 CFR 63.10, the Permittee shall submit all required notifications and reports to:

Indiana Department of Environmental Management
Compliance and Enforcement Branch, Office of Air Quality
100 North Senate Avenue
MC 61-53 IGCN 1003
Indianapolis, Indiana 46204-2251

E.8.2 Solvent Extraction for Vegetable Oil Production NESHAP [40 CFR Part 63, Subpart GGGG][326 IAC 20-60]

The Permittee shall comply with the following provisions of 40 CFR Part 63, Subpart GGGG (included as Attachment H to the operating permit), which are incorporated by reference as 326 IAC 20-60, for the emission units listed above:

- (1) 40 CFR 63.2830
- (2) 40 CFR 63.2831
- (3) 40 CFR 63.2832(a)
- (4) 40 CFR 63.2833
- (5) Table 1 to 63.2833(6)
- (6) 40 CFR 63.2834
- (7) Table 1 of 63.2834(c)
- (8) 40 CFR 63.2840(a), (b), (c), (d), and (f)
- (9) Table 1 of 63.2840(ix)
- (10) 40 CFR 63.2850(a), (c), (d), and (e)
- (11) Table 1 of 63.2850
- (12) Table 2 of 63.2850(b), and (c)
- (13) 40 CFR 63.2851
- (14) 40 CFR 63.2852
- (15) 40 CFR 63.2853
- (16) Table 1 of 63.2853
- (17) 40 CFR 63.2854
- (18) 40 CFR 63.2855
- (19) 40 CFR 63.2860(b), (c), and (d)
- (20) 40 CFR 63.2861
- (21) 40 CFR 63.2862
- (22) 40 CFR 63.2863
- (23) 40 CFR 63.2870
- (24) Table 1 of 63.2870
- (25) 40 CFR 63.2871
- (26) 40 CFR 63.2872

DRAFT

SECTION E.9

NESHAP

Emissions Unit Description: Insignificant Activities

Description	Capacity	Control	Affected Facility?
Three (3) Emergency Diesel Fire Pumps *2006	575 BHP each	None	Yes under NSPS III and NESHAP ZZZZ
Three (3) diesel-fired emergency generators *2016	One (1) 250 kW and Two (2) 150 kW	None	Yes under NSPS III and NESHAP ZZZZ

Note *Approved in the year indicated above for construction.

(The information describing the process contained in this facility description box is descriptive information and does not constitute enforceable conditions.)

National Emission Standards for Hazardous Air Pollutants (NESHAP) Requirements [326 IAC 2-7-5(1)]

E.9.1 General Provisions Relating to National Emission Standards for Hazardous Air Pollutants under 40 CFR 63 [326 IAC 20-1][40 CFR 63, Subpart A]

(a) Pursuant to 40 CFR 63.1, the Permittee shall comply with the provisions of 40 CFR Part 63, Subpart A - General Provisions, which are incorporated by reference as 326 IAC 20-1, for the emission units listed above, except as otherwise specified in 40 CFR Part 63, Subpart ZZZZ.

(b) Pursuant to 40 CFR 63.10, the Permittee shall submit all required notifications and reports to:

Indiana Department of Environmental Management
Compliance and Enforcement Branch, Office of Air Quality
100 North Senate Avenue
MC 61-53 IGCN 1003
Indianapolis, Indiana 46204-2251

E.9.2 Stationary Reciprocating Internal Combustion Engines NESHAP [40 CFR Part 63, Subpart ZZZZ][326 IAC 20-82]

The Permittee shall comply with the following provisions of 40 CFR Part 63, Subpart ZZZZ (included as Attachment I to the operating permit), which are incorporated by reference as 326 IAC 20-82, for the emission units listed above:

- (1) 40 CFR 63.6580
- (2) 40 CFR 63.6585(a) and (b)
- (3) 40 CFR 63.6590(a)(2)(i)
- (4) 40 CFR 63.6590(b)(1)(i)
- (5) 40 CFR 63.6645(f)
- (6) 40 CFR 63.6665
- (7) 40 CFR 63.6670(a)
- (8) 40 CFR 63.6675

DRAFT

SECTION E.10

NESHAP

Emissions Unit Description:

(f)

Unit ID	Description	Capacity	Control	Discharging to Stack	Affected Facility?
B-1	Main Boiler, natural gas fired and #2 fuel oil as back up fuel *2006	220 MMBtu/hr	Low NOx burner and Flue gas recirculation	Stack S-3	Yes under NSPS Db and NESHAP DDDDD

Note *Approved in the year indicated above for construction.

Insignificant Activities:

Description	Capacity	Control	Affected Facility?
Hot Oil Heater *2013	6 MMBtu/hr	None	Yes, Under NESHAP DDDDD
Natural Gas-Fired Glycerin Steam Boiler, *2014	1.0 MMBtu/hr	None	Yes, Under NESHAP DDDDD

Note *Approved in the year indicated above for construction.

(The information describing the process contained in this facility description box is descriptive information and does not constitute enforceable conditions.)

National Emission Standards for Hazardous Air Pollutants (NESHAP) Requirements [326 IAC 2-7-5(1)]

E.10.1 General Provisions Relating to National Emission Standards for Hazardous Air Pollutants under 40 CFR 63 [326 IAC 20-1][40 CFR 63, Subpart A]

(a) Pursuant to 40 CFR 63.1, the Permittee shall comply with the provisions of 40 CFR Part 63, Subpart A - General Provisions, which are incorporated by reference as 326 IAC 20-1, for the emission units listed above, except as otherwise specified in 40 CFR Part 63, Subpart DDDDD.

(b) Pursuant to 40 CFR 63.10, the Permittee shall submit all required notifications and reports to:

Indiana Department of Environmental Management
Compliance and Enforcement Branch, Office of Air Quality
100 North Senate Avenue
MC 61-53 IGCN 1003
Indianapolis, Indiana 46204-2251

E.10.2 Industrial, Commercial, and Institutional Boilers and Process Heaters NESHAP [40 CFR Part 63, Subpart DDDDD][326 IAC 20-95]

The Permittee shall comply with the following provisions of 40 CFR Part 63, Subpart DDDDD (included as Attachment J to the operating permit), which are incorporated by reference as 326 IAC 20-95, for the emission units listed above:

DRAFT

Main Boiler (B-1)

- (1) 40 CFR 63.7480;
- (2) 40 CFR 63.7485;
- (3) 40 CFR 63.7490(a);
- (4) 40 CFR 63.7495(b), (d);
- (5) 40 CFR 63.7499;
- (6) 40 CFR 63.7500(a), (b), (f);
- (7) 40 CFR 63.7501;
- (8) 40 CFR 63.7505(a);
- (9) 40 CFR 63.7510(g);
- (10) 40 CFR 63.7515;
- (11) 40 CFR 63.7520;
- (12) 40 CFR 63.7521;
- (13) 40 CFR 63.7525(a), (c);
- (14) 40 CFR 63.7530;
- (15) 40 CFR 63.7533;
- (16) 40 CFR 63.7535;
- (17) 40 CFR 63.7540(a), (b), (d);
- (18) 40 CFR 63.7545;
- (19) 40 CFR 63.7550;
- (20) 40 CFR 63.7555;
- (21) 40 CFR 63.7560;
- (22) 40 CFR 63.7565;
- (23) 40 CFR 63.7570;
- (24) 40 CFR 63.7575;
- (25) Table 2 to 40 CFR 63 Subpart DDDDD;
- (26) Table 3 to 40 CFR 63 Subpart DDDDD;
- (27) Table 4 to 40 CFR 63 Subpart DDDDD;
- (28) Table 5 to 40 CFR 63 Subpart DDDDD;
- (29) Table 6 to 40 CFR 63 Subpart DDDDD;
- (30) Table 7 to 40 CFR 63 Subpart DDDDD;
- (31) Table 8 to 40 CFR 63 Subpart DDDDD;
- (32) Table 9 to 40 CFR 63 Subpart DDDDD;
- (33) Table 10 to 40 CFR 63 Subpart DDDDD.

Glycerin Steam Boiler

- (1) 40 CFR 63.74780;
- (2) 40 CFR 63.7485;
- (3) 40 CFR 63.7490(a), (b);
- (4) 40 CFR 63.7495(a), (d);
- (5) 40 CFR 63.7499;
- (6) 40 CFR 63.7500(e);
- (7) 40 CFR 63.7501;
- (8) 40 CFR 63.7505(a);
- (9) 40 CFR 63.7510(g);
- (10) 40 CFR 63.7515;
- (11) 40 CFR 63.7520;
- (12) 40 CFR 63.7521(f)(1);
- (13) 40 CFR 63.7530;
- (14) 40 CFR 63.7540;
- (15) 40 CFR 63.7545(c);
- (16) 40 CFR 63.7550;
- (17) 40 CFR 63.7555;
- (18) 40 CFR 63.7560;
- (19) 40 CFR 63.7565;
- (20) 40 CFR 63.7570;
- (21) 40 CFR 63.7575;

DRAFT

- (22) Table 3 to 40 CFR 63 Subpart DDDDD;
- (23) Table 9 to 40 CFR 63 Subpart DDDDD;
- (24) Table 10 to 40 CFR 63 Subpart DDDDD.

Hot Oil Heater

- (1) 40 CFR 63.74780;
- (2) 40 CFR 63.7485;
- (3) 40 CFR 63.7490(a), (b);
- (4) 40 CFR 63.7495(a), (d);
- (5) 40 CFR 63.7499;
- (6) 40 CFR 63.7500(e);
- (7) 40 CFR 63.7501;
- (8) 40 CFR 63.7505(a);
- (9) 40 CFR 63.7510(g);
- (10) 40 CFR 63.7515;
- (11) 40 CFR 63.7521(f)(1);
- (12) 40 CFR 63.7530;
- (13) 40 CFR 63.7540;
- (14) 40 CFR 63.7545(c);
- (15) 40 CFR 63.7550;
- (16) 40 CFR 63.7555;
- (17) 40 CFR 63.7560;
- (18) 40 CFR 63.7565;
- (19) 40 CFR 63.7570;
- (20) 40 CFR 63.7575;
- (21) Table 9 to 40 CFR 63 Subpart DDDDD;
- (22) Table 10 to 40 CFR 63 Subpart DDDDD.

DRAFT

SECTION E.11

NSPS

Emissions Unit Description:

(e)

Unit ID	Description	Capacity (tons/hr)	Control	Discharging to Stack	Affected Facility?
Biodiesel Distillation	Biodiesel Distillation *2013	15	Mineral Oil Absorber and water absorber	Stack S-5	Yes under NSPS VVa, NSPS NNN& NESHAP FFFF

Note *Approved in the year indicated above for construction.

Insignificant Activities:

Description	Capacity	Control	Affected Facility?
Glycerin Refinery *2013, **2014	7.0 tons per hour	None	Yes, Under NSPS VVa, NSPS NNN, and NESHAP FFFF
Glycerin Truck/ Rail Loadout	7.0 tons per hour	None	Yes, Under NSPS VVa and NESHAP FFFF
Glycerin Yellow Tank	8,000 gallons	None	Yes, Under NSPS VVa and NESHAP FFFF
Glycerin Yellow Tank 1	50,000 gallons	None	Yes, Under NSPS VVa and NESHAP FFFF
Glycerin Yellow Tank 2	50,000 gallons	None	Yes, Under NSPS VVa and NESHAP FFFF
Glycerin Yellow Tank 3	50,000 gallons	None	Yes, Under NSPS VVa and NESHAP FFFF
Glycerin Yellow Tank 4	50,000 gallons	None	Yes, Under NSPS VVa and NESHAP FFFF

Note *Approved in the year indicated above for construction.

(The information describing the process contained in this facility description box is descriptive information and does not constitute enforceable conditions.)

New Source Performance Standards (NSPS) Requirements [326 IAC 2-7-5(1)]

E.11.1 General Provisions Relating to New Source Performance Standards [326 IAC 12-1][40 CFR Part 60, Subpart A]

- (a) Pursuant to 40 CFR 60.1, the Permittee shall comply with the provisions of 40 CFR Part 60, Subpart A - General Provisions, which are incorporated by reference in 326 IAC 12-1, for the emission units listed above, except as otherwise specified in 40 CFR Part 60, Subpart VVa.
- (b) Pursuant to 40 CFR 60.4, the Permittee shall submit all required notifications and reports to:

Indiana Department of Environmental Management
 Compliance and Enforcement Branch, Office of Air Quality

DRAFT

100 North Senate Avenue
MC 61-53 IGCN 1003
Indianapolis, Indiana 46204-2251

E.11.2 Equipment Leaks for VOC in the Synthetic Organic Chemicals Manufacturing Industry for which Construction, Reconstruction or Modification Commenced after November 7, 2006 NSPS [326 IAC 12][40 CFR Part 60, Subpart VVa]

The Permittee shall comply with the following provisions of 40 CFR Part 60, Subpart VVa (included as Attachment K to the operating permit), which are incorporated by reference as 326 IAC 12, for the emission units listed above:

- (1) 40 CFR Part 60.480a
- (2) 40 CFR Part 60.481a
- (3) 40 CFR Part 60.482-1a
- (4) 40 CFR Part 60.482-2a
- (5) 40 CFR Part 60.482-3a
- (6) 40 CFR Part 60.482-4a
- (7) 40 CFR Part 60.482-5a
- (8) 40 CFR Part 60.482-6a
- (9) 40 CFR Part 60.482-7a
- (10) 40 CFR Part 60.482-8a
- (11) 40 CFR Part 60.482-9a
- (12) 40 CFR Part 60.482-10a
- (13) 40 CFR Part 60.482-11a
- (14) 40 CFR Part 60.483-1a
- (15) 40 CFR Part 60.483-2a
- (16) 40 CFR Part 60.484a
- (17) 40 CFR Part 60.485a
- (18) 40 CFR Part 60.486a
- (19) 40 CFR Part 60.487a
- (20) 40 CFR Part 60.488a
- (21) 40 CFR Part 60.489a

DRAFT

SECTION E.12

NSPS

Emissions Unit Description:

(h)

Unit ID	Description	Output Power Rating	Control	Discharging to Stack	Affected Facility?
GT-1	Natural Gas-fired power generator *2016 (Originally constructed in 1999)	4.6 MW	none	outside	Yes, under NSPS GG and NESHAP YYYY

Note *Approved in the year indicated above for construction.

(The information describing the process contained in this emissions unit description box is descriptive information and does not constitute enforceable conditions.)

New Source Performance Standards (NSPS) Requirements [326 IAC 2-7-5(1)]

E.12.1 General Provisions Relating to New Source Performance Standards [326 IAC 12-1][40 CFR Part 60, Subpart A]

(a) Pursuant to 40 CFR 60.1, the Permittee shall comply with the provisions of 40 CFR Part 60, Subpart A – General Provisions, which are incorporated by reference as 326 IAC 12-1, for the emission unit listed above, except as otherwise specified in 40 CFR Part 60, Subpart GG.

(b) Pursuant to 40 CFR 60.4, the Permittee shall submit all required notifications and reports to:

Indiana Department of Environmental Management
Compliance and Enforcement Branch, Office of Air Quality
100 North Senate Avenue
MC 61-53 IGCN 1003
Indianapolis, Indiana 46204-2251

E.12.2 Stationary Gas Turbines NSPS [326 IAC 12][40 CFR Part 60, Subpart GG]

The Permittee shall comply with the following provisions of 40 CFR Part 60, Subpart GG (included as Attachment L to the operating permit), which are incorporated by reference as 326 IAC 12, for the emission unit listed above:

- (1) 40 CFR 60.330
- (2) 40 CFR 60.331
- (3) 40 CFR 60.332
- (4) 40 CFR 60.333
- (5) 40 CFR 60.334
- (6) 40 CFR 60.335
- (7) 40 CFR 60.336

DRAFT

SECTION E.13

NESHAP

Emissions Unit Description:

(h)

Unit ID	Description	Output Power Rating	Control	Discharging to Stack	Affected Facility?
GT-1	Natural Gas-fired power generator *2016 (Originally constructed in 1999)	4.6 MW	none	outside	Yes, under NSPS GG and NESHAP YYYY

Note *Approved in the year indicated above for construction.

(The information describing the process contained in this emissions unit description box is descriptive information and does not constitute enforceable conditions.)

National Emission Standards for Hazardous Air Pollutants (NESHAP) Requirements [326 IAC 2-7-5(1)]

E.13.1 General Provisions Relating to National Emission Standards for Hazardous Air Pollutants under 40 CFR Part 63 [326 IAC 20-1][40 CFR Part 63, Subpart A]

- (a) Pursuant to 40 CFR 63.1 the Permittee shall comply with the provisions of 40 CFR Part 63, Subpart A – General Provisions, which are incorporated by reference as 326 IAC 20-1, for the emission unit listed above, except as otherwise specified in 40 CFR Part 63, Subpart YYYY.
- (b) Pursuant to 40 CFR 63.10, the Permittee shall submit all required notifications and reports to:

Indiana Department of Environmental Management
Compliance and Enforcement Branch, Office of Air Quality
100 North Senate Avenue
MC 61-53 IGCN 1003
Indianapolis, Indiana 46204-2251

E.13.2 Stationary Combustion Turbines NESHAP [40 CFR Part 63, Subpart YYYY][326 IAC 20-90]

The Permittee shall comply with the following provisions of 40 CFR Part 63, Subpart YYYY (included as Attachment M to the operating permit), which are incorporated by reference as 326 IAC 20-90, for the emission unit listed above:

- (1) 40 CFR 63.6080
- (2) 40 CFR 63.6085(a), (b)
- (3) 40 CFR 63.6090 (a)(1) and (b)(4)
- (4) 40 CFR 63.6170
- (5) 40 CFR 63.6175

E.13.3 Preventive Maintenance Plan [326 IAC 2-7-5(12)]

A Preventive Maintenance Plan is required for this facility and any associated control devices. Section B - Preventive Maintenance Plan contains the Permittee's obligation with regard to the preventive maintenance plan required by this condition.

DRAFT

**INDIANA DEPARTMENT OF ENVIRONMENTAL MANAGEMENT
OFFICE OF AIR QUALITY
COMPLIANCE AND ENFORCEMENT BRANCH
PART 70 OPERATING PERMIT
CERTIFICATION**

Source Name: Louis Dreyfus Company Agricultural Industries, LLC
Source Address: 7344 State Road 15 South, Claypool, Indiana 46510
Part 70 Permit No.: T085-36889-00102

This certification shall be included when submitting monitoring, testing reports/results or other documents as required by this permit.

Please check what document is being certified:

- ☐ Annual Compliance Certification Letter
- ☐ Test Result (specify)
- ☐ Report (specify)
- ☐ Notification (specify)
- ☐ Affidavit (specify)
- ☐ Other (specify)

I certify that, based on information and belief formed after reasonable inquiry, the statements and information in the document are true, accurate, and complete.

Signature:

Printed Name:

Title/Position:

Phone:

Date:

DRAFT

INDIANA DEPARTMENT OF ENVIRONMENTAL MANAGEMENT
OFFICE OF AIR QUALITY
COMPLIANCE AND ENFORCEMENT BRANCH
100 North Senate Avenue
MC 61-53 IGCN 1003
Indianapolis, Indiana 46204-2251
Phone: (317) 233-0178
Fax: (317) 233-6865

PART 70 OPERATING PERMIT
EMERGENCY OCCURRENCE REPORT

Source Name: Louis Dreyfus Company Agricultural Industries, LLC
Source Address: 7344 State Road 15 South, Claypool, Indiana 46510
Part 70 Permit No.: T085-36889-00102

This form consists of 2 pages

Page 1 of 2

- ☐ This is an emergency as defined in 326 IAC 2-7-1(12)
- The Permittee must notify the Office of Air Quality (OAQ), within four (4) daytime business hours (1-800-451-6027 or 317-233-0178, ask for Compliance Section); and
 - The Permittee must submit notice in writing or by facsimile within two (2) working days (Facsimile Number: 317-233-6865), and follow the other requirements of 326 IAC 2-7-16.

If any of the following are not applicable, mark N/A

Facility/Equipment/Operation:
Control Equipment:
Permit Condition or Operation Limitation in Permit:
Description of the Emergency:
Describe the cause of the Emergency:

DRAFT

If any of the following are not applicable, mark N/A

Page 2 of 2

Date/Time Emergency started:
Date/Time Emergency was corrected:
Was the facility being properly operated at the time of the emergency? Y N
Type of Pollutants Emitted: TSP, PM-10, SO ₂ , VOC, NO _x , CO, Pb, other:
Estimated amount of pollutant(s) emitted during emergency:
Describe the steps taken to mitigate the problem:
Describe the corrective actions/response steps taken:
Describe the measures taken to minimize emissions:
If applicable, describe the reasons why continued operation of the facilities are necessary to prevent imminent injury to persons, severe damage to equipment, substantial loss of capital investment, or loss of product or raw materials of substantial economic value:

Form Completed by: _____

Title / Position: _____

Date: _____

Phone: _____

DRAFT

**INDIANA DEPARTMENT OF ENVIRONMENTAL MANAGEMENT
OFFICE OF AIR QUALITY
COMPLIANCE AND ENFORCEMENT BRANCH**

Part 70 Quarterly Report

Source Name: Louis Dreyfus Company Agricultural Industries, LLC
 Source Address: 7344 State Road 15 South, Claypool, Indiana 46510-9746
 Part 70 Permit No.: T085-29197-00102
 Facility: Grain Receiving/Meal Loadout and Prep Area
 Parameter: Soybeans processed
 Limit: shall be less than 2,251,836 tons per twelve (12) consecutive month period, with compliance determined at the end of each month.

QUARTER: _____ YEAR: _____

Month	Column 1	Column 2	Column 1 + Column 2
	This Month	Previous 11 Months	12 Month Total

☐ No deviation occurred in this quarter.

☐ Deviation/s occurred in this quarter.

Deviation has been reported on:

Submitted by: _____

Title / Position: _____

Signature: _____

Date: _____

Phone: _____

DRAFT

**INDIANA DEPARTMENT OF ENVIRONMENTAL MANAGEMENT
OFFICE OF AIR QUALITY
COMPLIANCE AND ENFORCEMENT BRANCH**

Part 70 Quarterly Report

Source Name: Louis Dreyfus Company Agricultural Industries, LLC
Source Address: 7344 State Road 15 South, Claypool, Indiana 46510-9746
Part 70 Permit No.: T085-29197-00102
Facility: Biodiesel Manufacturing
Parameter: Purchased seed oil
Limit: shall be less than 80 million gallons per twelve (12) consecutive month period,
with compliance determined at the end of each month.

QUARTER: _____ YEAR: _____

Month	Column 1	Column 2	Column 1 + Column 2
	This Month	Previous 11 Months	12 Month Total

☐ No deviation occurred in this quarter.

☐ Deviation/s occurred in this quarter.

Deviation has been reported on:

Submitted by: _____

Title / Position: _____

Signature: _____

Date: _____

Phone: _____

DRAFT

**INDIANA DEPARTMENT OF ENVIRONMENTAL MANAGEMENT
OFFICE OF AIR QUALITY
COMPLIANCE AND ENFORCEMENT BRANCH**

Part 70 Quarterly Report

Source Name: Louis Dreyfus Company Agricultural Industries, LLC
Source Address: 7344 State Road 15 South, Claypool, Indiana 46510-9746
Part 70 Permit No.: T085-29197-00102
Facility: Biodiesel Reactor
Parameter: Seed oil processed to manufacture biodiesel
Limit: shall be less than 110,000,000 gallons per twelve (12) consecutive month period,
with compliance determined at the end of each month.

QUARTER: _____ YEAR: _____

Month	Column 1	Column 2	Column 1 + Column 2
	This Month	Previous 11 Months	12 Month Total

☐ No deviation occurred in this quarter.

☐ Deviation/s occurred in this quarter.

Deviation has been reported on:

Submitted by: _____

Title / Position: _____

Signature: _____

Date: _____

Phone: _____

DRAFT

**INDIANA DEPARTMENT OF ENVIRONMENTAL MANAGEMENT
OFFICE OF AIR QUALITY
COMPLIANCE AND ENFORCEMENT BRANCH**

Part 70 Quarterly Report

Source Name: Louis Dreyfus Company Agricultural Industries, LLC
Source Address: 7344 State Road 15 South, Claypool, Indiana 46510-9746
Part 70 Permit No.: T085-29197-00102
Facility: Biodiesel Manufacturing Process with Methanol Tank Loading
Parameter: Operating Hours
Limit: shall be less than 1,000 hours per twelve (12) consecutive month period with compliance determined at the end of each month.

QUARTER: _____ YEAR: _____

Month	Column 1	Column 2	Column 1 + Column 2
	This Month	Previous 11 Months	12 Month Total

☐ No deviation occurred in this quarter.

☐ Deviation/s occurred in this quarter.

Deviation has been reported on:

Submitted by: _____

Title / Position: _____

Signature: _____

Date: _____

Phone: _____

DRAFT

**INDIANA DEPARTMENT OF ENVIRONMENTAL MANAGEMENT
OFFICE OF AIR QUALITY
COMPLIANCE AND ENFORCEMENT BRANCH**

Part 70 Quarterly Report

Source Name: Louis Dreyfus Company Agricultural Industries, LLC
 Source Address: 7344 State Road 15 South, Claypool, Indiana 46510-9746
 Part 70 Permit No.: T085-29197-00102
 Facility: Biodiesel Manufacturing Process upset operation
 Parameter: Operating Hours
 Limit: shall be less than 24 hours per twelve (12) consecutive month period with compliance determined at the end of each month.

QUARTER: _____ YEAR: _____

Month	Column 1	Column 2	Column 1 + Column 2
	This Month	Previous 11 Months	12 Month Total

☐ No deviation occurred in this quarter.

☐ Deviation/s occurred in this quarter.
 Deviation has been reported on:

Submitted by: _____
 Title / Position: _____
 Signature: _____
 Date: _____
 Phone: _____

DRAFT

**INDIANA DEPARTMENT OF ENVIRONMENTAL MANAGEMENT
OFFICE OF AIR QUALITY
COMPLIANCE AND ENFORCEMENT BRANCH**

Part 70 Quarterly Report

Source Name: Louis Dreyfus Company Agricultural Industries, LLC
Source Address: 7344 State Road 15 South, Claypool, Indiana 46510-9746
Part 70 Permit No.: T085-29197-00102
Facility: Biodiesel Loading Racks
Parameter: Throughout Rate
Limit: shall be less than 110,000,000 gallons per twelve (12) consecutive month period with compliance determined at the end of each month.

QUARTER: _____ YEAR: _____

Month	Column 1	Column 2	Column 1 + Column 2
	This Month	Previous 11 Months	12 Month Total

☐ No deviation occurred in this quarter.

☐ Deviation/s occurred in this quarter.

Deviation has been reported on:

Submitted by: _____

Title / Position: _____

Signature: _____

Date: _____

Phone: _____

DRAFT

**INDIANA DEPARTMENT OF ENVIRONMENTAL MANAGEMENT
OFFICE OF AIR QUALITY
COMPLIANCE AND ENFORCEMENT BRANCH**

Part 70 Quarterly Report

Source Name: Louis Dreyfus Company Agricultural Industries, LLC
Source Address: 7344 State Road 15 South, Claypool, Indiana 46510-9746
Part 70 Permit No.: T085-29197-00102
Facility: Storage Bean Piles #1 and #2
Parameter: Soybean throughput
Limit: shall be less than 8,000,000 bushels per twelve (12) consecutive month period with compliance determined at the end of each month.

QUARTER: _____ YEAR: _____

Month	Column 1	Column 2	Column 1 + Column 2
	This Month	Previous 11 Months	12 Month Total

☐ No deviation occurred in this quarter.

☐ Deviation/s occurred in this quarter.

Deviation has been reported on:

Submitted by: _____

Title / Position: _____

Signature: _____

Date: _____

Phone: _____

DRAFT

**INDIANA DEPARTMENT OF ENVIRONMENTAL MANAGEMENT
OFFICE OF AIR QUALITY
COMPLIANCE AND ENFORCEMENT BRANCH**

Part 70 Quarterly Report

Source Name: Louis Dreyfus Company Agricultural Industries, LLC
Source Address: 7344 State Road 15 South, Claypool, Indiana 46510-9746
Part 70 Permit No.: T085-29197-00102
Facility: Overall source
Parameter: Solvent Loss Ratio
Limit: The overall solvent loss ratio shall not exceed 0.141 gallons per ton of soybean crushed from the whole plant per twelve (12) consecutive month period, with compliance determined at the end of each month. The Permittee shall also follow the leak detection and repair program as part of BACT.

QUARTER: _____ **YEAR:** _____

Month	Hexane Usage This Month (gal)	Total Grain Processed This Month (tons)	Solvent Loss Ratio (gal/ton)

Solvent Loss Ratio (gal/ton) = Hexane Usage for This Month (gal) / Total Grain Processed for This Month (tons)

- ☐ No deviation occurred in this quarter.
- ☐ Deviations occurred in this quarter.
Deviation has been reported on: _____

Submitted By: _____

Title/Position: _____

Signature: _____

Date: _____

Phone: _____

DRAFT

**INDIANA DEPARTMENT OF ENVIRONMENTAL MANAGEMENT
OFFICE OF AIR QUALITY
COMPLIANCE AND ENFORCEMENT BRANCH
PART 70 OPERATING PERMIT
QUARTERLY DEVIATION AND COMPLIANCE MONITORING REPORT**

Source Name: Louis Dreyfus Company Agricultural Industries, LLC
Source Address: 7344 State Road 15 South, Claypool, Indiana 46510
Part 70 Permit No.: T085-36889-00102

Months: _____ to Year: _____

Page 1 of 2

This report shall be submitted quarterly based on a calendar year. Proper notice submittal under Section B -Emergency Provisions satisfies the reporting requirements of paragraph (a) of Section C- General Reporting. Any deviation from the requirements of this permit, the date(s) of each deviation, the probable cause of the deviation, and the response steps taken must be reported. A deviation required to be reported pursuant to an applicable requirement that exists independent of the permit, shall be reported according to the schedule stated in the applicable requirement and does not need to be included in this report. Additional pages may be attached if necessary. If no deviations occurred, please specify in the box marked "No deviations occurred this reporting period".

☐ NO DEVIATIONS OCCURRED THIS REPORTING PERIOD.

☐ THE FOLLOWING DEVIATIONS OCCURRED THIS REPORTING PERIOD

Permit Requirement (specify permit condition #)

Date of Deviation:

Duration of Deviation:

Number of Deviations:

Probable Cause of Deviation:

Response Steps Taken:

Permit Requirement (specify permit condition #)

Date of Deviation:

Duration of Deviation:

Number of Deviations:

Probable Cause of Deviation:

Response Steps Taken:

DRAFT

Page 2 of 2

Permit Requirement (specify permit condition #)	
Date of Deviation:	Duration of Deviation:
Number of Deviations:	
Probable Cause of Deviation:	
Response Steps Taken:	
Permit Requirement (specify permit condition #)	
Date of Deviation:	Duration of Deviation:
Number of Deviations:	
Probable Cause of Deviation:	
Response Steps Taken:	
Permit Requirement (specify permit condition #)	
Date of Deviation:	Duration of Deviation:
Number of Deviations:	
Probable Cause of Deviation:	
Response Steps Taken:	

Form Completed by: _____

Title / Position: _____

Date: _____

Phone: _____

**Indiana Department of Environmental Management
Office of Air Quality**

**Technical Support Document (TSD) for a Part 70 Significant Source
Modification and Significant Permit Modification**

Source Description and Location

Source Name:	Louis Dreyfus Company Agricultural Industries LLC
Source Location:	7344 State Road 15 South, Claypool, Indiana 46510
County:	Kosciusko
SIC Code:	2075 (Soybean Oil Mills)
	2079 (Shortening, Table Oils, Margarine, and Other Edible Fats and Oils, Not Elsewhere Classified)
	2869 (Industrial Organic Chemicals, Not Elsewhere Classified)
Operation Permit No.:	T085-36889-00102
Operation Permit Issuance Date:	January 5, 2017
Significant Source Modification No.:	085-39532-00102
Significant Permit Modification No.:	085-39729-00102
Permit Reviewer:	Daniel W. Pell

Existing Approvals

The source was issued Part 70 Operating Permit Renewal No. T085-36889-00102 on January 5, 2017. The source has since received the following approvals:

- (a) Administrative Amendment No.: 085-38449-00102, issued on June 16, 2017; and
- (b) Significant Permit Modification No.: 085-39843-00102, issued September 5, 2018.

County Attainment Status

The source is located in Kosciusko County.

Pollutant	Designation
SO ₂	Better than national standards.
CO	Unclassifiable or attainment effective November 15, 1990.
O ₃	Unclassifiable or attainment effective July 20, 2012, for the 2008 8-hour ozone standard. ¹
PM _{2.5}	Unclassifiable or attainment effective April 5, 2005, for the annual PM _{2.5} standard.
PM _{2.5}	Unclassifiable or attainment effective December 13, 2009, for the 24-hour PM _{2.5} standard.
PM ₁₀	Unclassifiable effective November 15, 1990.
NO ₂	Cannot be classified or better than national standards.
Pb	Unclassifiable or attainment effective December 31, 2011.
¹ Unclassifiable or attainment effective October 18, 2000, for the 1-hour ozone standard which was revoked effective June 15, 2005.	

- (a) Ozone Standards
Volatile organic compounds (VOC) and Nitrogen Oxides (NO_x) are regulated under the Clean Air Act (CAA) for the purposes of attaining and maintaining the National Ambient Air Quality Standards (NAAQS) for ozone. Therefore, VOC and NO_x emissions are considered when evaluating the rule applicability relating to ozone. Kosciusko County has been designated as attainment or unclassifiable for ozone. Therefore, VOC and NO_x emissions were reviewed pursuant to the requirements for Prevention of Significant Deterioration (PSD), 326 IAC 2-2.

- (b) **PM_{2.5}**
Kosciusko County has been classified as attainment for PM_{2.5}. Therefore, direct PM_{2.5}, SO₂, and NO_x emissions were reviewed pursuant to the requirements for Prevention of Significant Deterioration (PSD), 326 IAC 2-2.
- (c) **Other Criteria Pollutants**
Kosciusko County has been classified as attainment or unclassifiable in Indiana for all the other criteria pollutants. Therefore, these emissions were reviewed pursuant to the requirements for Prevention of Significant Deterioration (PSD), 326 IAC 2-2.

Fugitive Emissions

- (a) Louis Dreyfus is a nested source, with a biodiesel production plant (one of the twenty-eight (28) listed source categories) and a soybean oil extraction plant (a non-listed source). Therefore, the fugitive emissions from the biodiesel production plant (including the associated paved road emissions) are counted toward the determination of PSD, Emission Offset, and Part 70 applicability. However, the fugitive emissions from the soybean oil extraction plant are not counted toward the determination of PSD, Emission Offset, and Part 70 applicability, except as required in paragraph (b).
- (b) There is an applicable New Source Performance Standard that was in effect on August 7, 1980. Therefore, fugitive emissions from the grain elevator (including the associated paved road emissions) are counted toward the determination of PSD, Emission Offset, and Part 70 Permit applicability.

Greenhouse Gas (GHG) Emissions

On June 23, 2014, in the case of *Utility Air Regulatory Group v. EPA*, cause no. 12-1146, (available at http://www.supremecourt.gov/opinions/13pdf/12-1146_4g18.pdf) the United States Supreme Court ruled that the U.S. EPA does not have the authority to treat greenhouse gases (GHGs) as an air pollutant for the purpose of determining operating permit applicability or PSD Major source status. On July 24, 2014, the U.S. EPA issued a memorandum to the Regional Administrators outlining next steps in permitting decisions in light of the Supreme Court's decision. U.S. EPA's guidance states that U.S. EPA will no longer require PSD or Title V permits for sources "previously classified as 'Major' based solely on greenhouse gas emissions."

The Indiana Environmental Rules Board adopted the GHG regulations required by U.S. EPA at 326 IAC 2-2-1(zz), pursuant to Ind. Code § 13-14-9-8(h) (Section 8 rulemaking). A rule, or part of a rule, adopted under Section 8 is automatically invalidated when the corresponding federal rule, or part of the rule, is invalidated. Due to the United States Supreme Court Ruling, IDEM, OAQ cannot consider GHG emissions to determine operating permit applicability or PSD applicability to a source or modification.

Source Status - Existing Source

The table below summarizes the potential to emit of the entire source, prior to the proposed modification, after consideration of all enforceable limits established in the effective permits:

Process / Emission Unit	Source-Wide Emissions Before Modification (ton/year)							
	PM	PM ₁₀ *	PM _{2.5} **	SO ₂	NO _x	VOC	CO	Combined HAPs
Soybean Oil Extraction Plant								
Stack AF-2 (and uncaptured emissions)	21.65	12.77	8.38	-	-	-	-	-
Stack AF-1 (and uncaptured emissions)	23.91	9.08	8.83	-	-	-	-	-
Stacks MBF-1 through MBF-5	20.37	20.37	20.37	-	-	-	-	-
Piles #1 and #2 (no control)	7.32	4.08	0.70	-	-	-	-	-
Stack AF-3	5.52	5.52	5.52	-	-	-	-	-
Stack AF-7	6.57	6.57	6.57	-	-	-	-	-
Stacks S-1 and S-6	21.59	14.67	14.67	-	-	-	-	-
Stack AF-5	11.21	11.21	11.21	-	-	-	-	-
Stack AF-4	4.51	4.51	4.51	-	-	-	-	-
Stack AF-6	4.14	4.14	4.14	-	-	-	-	-
Stack S-2	47.04	31.89	31.89	-	-	-	-	-
Bin Filter	8.32	8.32	8.32	-	-	-	-	-
Hull Bin Filter	0.84	0.21	0.04	-	-	-	-	-
Bean Storage Bin No. 2 thru 4, 6, 7, and 8 and Bean Storage Silo No. 1 and 5	30.40	7.66	1.34	-	-	-	-	-
Diatomaceous Earth (DE) Storage Bin	0.38	0.06	0.02	-	-	-	-	-
Main Gas Vent (Soybean Oil Extractor System)	-	-	-	-	-	40.73	-	26.07
Desolventized Meal Dryers & Cooler	-	-	-	-	-	143.66	-	91.94
Meal Storage	-	-	-	-	-	fugitive	-	1.21
Bound In Product & Byproduct Desolventized Meal (fugitive)	-	-	-	-	-	fugitive	-	113.00
Plant Startup/Shutdown	-	-	-	-	-	fugitive	-	10.75
General (equipment failure, leaks, etc.)	-	-	-	-	-	fugitive	-	207.21

Process / Emission Unit	Source-Wide Emissions Before Modification (ton/year)							
	PM	PM ₁₀ *	PM _{2.5} **	SO ₂	NO _x	VOC	CO	Combined HAPs
Plant Upsets	-	-	-	-	-	fugitive	-	16.90
Combustion Sources	2.75	8.14	8.14	1.01	145.84	6.29	82.42	1.80
Temporary Mobile Boiler (B-S)	0.33	1.31	1.31	0.10	17.18	0.94	14.43	0.32
GT-1 Turbine	1.53	1.53	1.53	0.79	23.11	16.04	27.97	0.24
Three (3) Diesel Emergency Generators	0.41	0.41	0.41	0.38	5.71	0.46	1.23	5.00E-03
Glycerin Cooling Tower	0.19	0.13	negl.	-	-	-	-	-
Total PTE from Soybean Oil Extraction Plant with Roads	235.16	155.81	138.68	2.28	191.84	208.14	126.04	469.45
Biodiesel Production Plant								
Biodiesel Methanol Absorbers (worse case) ¹	-	-	-	-	-	1.83	-	1.83
Loading Racks ¹	-	-	-	-	-	1.10	-	1.10
Biodiesel Storage Tanks ¹	-	-	-	-	-	4.31	-	4.31
Glycerin Storage Tanks	-	-	-	-	-	negl.	-	4.82E-03
Biodiesel Wastewater (fugitive)	-	-	-	-	-	3.37	-	3.37
Equipment Leaks (fugitive)	-	-	-	-	-	2.80	-	2.80
Diesel/#2 Fuel Oil Storage Tank	-	-	-	-	-	0.01	-	0.01
One (1) Soybean Oil Pre-Treat Tank	-	-	-	-	-	0.01	-	0.01
Six (6) Methanol Tanks	-	-	-	-	-	0.07	-	0.07
Three (3) Soybean Oil Tanks (Degummed Oil Tanks #1 and #2 and Crude Oil Tank #3)	-	-	-	-	-	<= 1.0	-	<= 1.0
Purchased Soybean Oil Unloading	-	-	-	-	-	4.00	-	4.00
Two (2) Sodium Methylate Tanks	-	-	-	-	-	0.01	-	0.01
Five (5) Hexane tanks	-	-	-	-	-	2.01	-	2.01
Road Traffic at Source ²	16.18	3.24	0.79	-	-	-	-	-
Cooling Towers	4.34	4.34	4.34	-	-	-	-	-
Biodiesel Distillation	-	-	-	-	-	1.31	-	1.31
Hot Oil Heater	0.05	0.20	0.20	0.02	2.58	0.14	2.16	0.05
Natural Gas-Fired Glycerin Steam Boiler	0.01	0.03	0.03	negl.	0.43	0.02	0.36	0.01
Glycerin Tanks	-	-	-	-	-	negl.	-	-

Process / Emission Unit	Source-Wide Emissions Before Modification (ton/year)							Combined HAPs
	PM	PM ₁₀ *	PM _{2.5} **	SO ₂	NO _x	VOC	CO	
Glycerin Refinery/ Rail Loadout	-	-	-	-	-	negl.	-	-
Glycerin Refinery	-	-	-	-	-	negl.	-	2.00E-04
Total PTE at Biodiesel Plant	20.58	7.81	5.37	0.02	3.01	22.02	2.52	21.90
Total PTE at Entire Source including Fugitives from Roads	239.56	160.38	143.25	2.29	194.85	230.15	128.57	491.35
PSD Major Source Thresholds	250	250	250	250	250	250	250	10

negl. = negligible
Notes:
¹The VOC emissions from these units are mainly methanol which is also a HAP. Assumed all VOCs are HAPs as the worst case scenario.
² Fugitive road emissions are being counted for the soybean extraction plant because the operations are covered by the source category definition in NSPS DD (a pre-1980 NSPS).
 In addition, fugitive road emissions are being counted because they include roads at the biodiesel production plant which is one of the 28 source categories.
 * Under the Part 70 Permit program (40 CFR 70), PM₁₀ and PM_{2.5}, not particulate matter (PM), are each considered as a regulated air pollutant".
 **PM_{2.5} listed is direct PM_{2.5}.

- (a) Louis Dreyfus is a nested source, with a biodiesel production plant (one of the twenty-eight (28) listed source categories) and a soybean oil extraction plant (a non-listed source). Louis Dreyfus is an existing minor PSD stationary source and will continue to be minor under 326 IAC 2-2 because the emissions of each PSD regulated pollutant will continue to be less than the PSD major source thresholds.
- (b) This existing source is a major source of HAPs, as defined in 40 CFR 63.2, because HAP emissions are equal to or greater than twenty-five (25) tons per year for a combination of HAPs. Therefore, this source is a major source under Section 112 of the Clean Air Act (CAA).
- (c) These emissions are based on the TSD of Administrative Amendment No.: 085-38449-00102, issued on June 16, 2017.

Description of Proposed Modification

The Office of Air Quality (OAQ) has reviewed an application, submitted by Louis Dreyfus Company Agricultural Industries LLC on January 23, 2018, relating to requests for the following:

- (a) to establish a 326 IAC 8-1-6 VOC BACT limit for the Biodiesel Manufacturing Operation which uses soybean oil to produce a biodiesel fuel.

The original Title V permit for the biodiesel process that was established to meet the requirements for Best Available Control Technology (BACT), were based on estimates provided by Louis Dreyfus (before plant was constructed and operational) that uncontrolled VOC emissions were more than 25 tons per year. Initial stack test results for the biodiesel manufacturing process recorded VOC emissions at the inlet to the scrubber system to be less than 25 tons per year. Based on these stack test results, IDEM OAQ removed the mostly duplicative VOC BACT requirements from the permit, although the comparable PSD Minor Limits for VOC (pound per hour) and LDAR requirements were retained to keep allowable VOC emissions below PSD major source thresholds. Louis Dreyfus now believes that this was an inaccurate assessment of the uncontrolled VOC emissions to the scrubber system under various operating conditions and rates, and believes that uncontrolled potential VOC emissions exceed the 25 tons per year applicability threshold of 326 IAC 8-1-6. Therefore, the biodiesel manufacturing process is

subject to the requirements of 326 IAC 8-1-6 (Best Available Control Technology (BACT)). The affected biodiesel manufacturing process equipment are noted as follows:

Affected Biodiesel Manufacturing Process Equipment
1st Primary Transester Column
1st Secondary Transester Column
2nd Primary Transester Column
2nd Secondary Transester Column
Esterification Reactor
Vacuum group package
Biodiesel Distillation
Methanol Storage Tank #1
Methanol Storage Tank #2
Methanol Storage Tank #3
Methanol Storage Tank #4
Methanol Storage Tank #5
Methanol Storage Tank #6
Sodium Methylate (catalyst) Storage Tank #1
Sodium Methylate (catalyst) Storage Tank #2

There are no physical modifications being made to the existing Biodiesel Manufacturing Operation.

- (b) Clarify the applicability of NESHAP Subpart YYYY for the existing gas turbine, identified as GT-1. This unit was originally constructed in 1999 in New Jersey and was then relocated to its current location in 2016. Due to this construction year, it is now classified as an existing stationary combustion turbine under Subpart YYYY.

Note:

- (c) The source requested to change the existing Leak Detection and Repair (LDAR) program language in Condition D.3.4(e) of the permit, which contains VOC BACT requirements for the soybean oil processing operation. The current Leak Detection and Repair (LDAR) program follows the procedures contained in 40 CFR Part 60, Method 21. The source has requested that the current Leak Detection and Repair (LDAR) program be changed to a system that utilizes olfactory, visual, and auditory (OVA) detection to identify leaks. Sufficient justification has not been provided by the source to demonstrate that a change to the existing Leak Detection and Repair (LDAR) program to use an OVA method would continue to demonstrate the same level of leak detection. Therefore, at this time, this change is not being made.

The PTE from the entire source will not change as a result of this permit modification. No emission units are being removed or added to the source.

Enforcement Issues

There are no pending enforcement actions related to this modification.

Emission Calculations

See Appendix A of this Technical Support Document for detailed emission calculations.

Permit Level Determination – PSD

- (a) There is no physical change or change in the method of operation occurring at the source as a result of this modification and there are no PTE increases of regulated NSR pollutants.

- (b) This existing minor PSD stationary source will continue to be minor under 326 IAC 2-2 because the emissions of each PSD regulated pollutant will continue to be less than the PSD major source thresholds. Therefore, pursuant to 326 IAC 2-2, the PSD requirements do not apply.

Federal Rule Applicability Determination

Due to the modification at this source, federal rule applicability has been reviewed as follows:

New Source Performance Standards (NSPS):

- (a) There are no New Source Performance Standards (NSPS) (326 IAC 12 and 40 CFR Part 60) included in the permit for this proposed modification.

National Emission Standards for Hazardous Air Pollutants (NESHAP):

- (b) The natural gas-fired power generator, identified as Turbine GT-1, is subject to the National Emission Standards for Hazardous Air Pollutants for Stationary Combustion Turbines, 40 CFR 63, Subpart YYYY and 326 IAC 20-90, because the unit was originally constructed in 1999, and due to this construction year is now classified an existing stationary combustion turbine under Subpart YYYY located at a major source of HAP emissions. Pursuant to 40 CFR 63.6090(a)(1), a change in ownership of an existing stationary combustion turbine does not make that stationary combustion turbine a new stationary combustion turbine.

GT-1 is subject to the following portions of Subpart YYYY:

- (1) 40 CFR 63.6080
 - (2) 40 CFR 63.6085(a), (b)
 - (3) 40 CFR 63.6090(a)(1) and (b)(4)
 - (4) 40 CFR 63.6170
 - (5) 40 CFR 63.6175
- (c) There are no other National Emission Standards for Hazardous Air Pollutants under 40 CFR 63, 326 IAC 14 and 326 IAC 20 included for this proposed modification.

Compliance Assurance Monitoring (CAM):

- (d) There are no Compliance Assurance Monitoring (CAM) requirements for this proposed modification.

State Rule Applicability Determination

Due to the modification at this source, state rule applicability has been reviewed as follows:

326 IAC 2-2 (PSD)

PSD applicability is discussed under the Permit Level Determination – PSD.

326 IAC 8-1-6 (New facilities; general reduction requirements)

The uncontrolled VOC PTE from the Biodiesel Manufacturing Process are greater than 25 tons per year. Therefore, pursuant to 326 IAC 8-1-6, IDEM has established VOC BACT limits for the Biodiesel Manufacturing Process, as follows:

Process	Control	VOC control efficiency, %	VOC Limit (lbs/hour)	Operation Limit
Biodiesel manufacturing process Normal operation	Mineral Oil Absorber and Water absorber	Overall VOC control efficiency of 99%	0.30	NA

Process	Control	VOC control efficiency, %	VOC Limit (lbs/hour)	Operation Limit
Biodiesel manufacturing process with methanol tank loading	Mineral Oil Absorber and Water absorber	Overall VOC control efficiency of 99%	0.63	1,000 hours per twelve (12) consecutive months
Biodiesel manufacturing process upset operation	Mineral Oil Absorber and Water absorber	Overall VOC control efficiency of 95%	29.4	24 hours per twelve (12) consecutive months
Glycerin storage tanks	None	None	0.0011	NA
Biodiesel wastewater	None	None	0.77	NA
Biodiesel fugitive emissions	None	None	0.64	LDAR as required by 40 CFR 60, Subpart VV

Compliance Determination and Monitoring Requirements

- (a) The testing requirements applicable to this proposed modification are as follows:

Testing Requirements			
Emission Units	Pollutant	Date of Initial Valid Demonstration	Frequency of Testing
Biodiesel Manufacturing Process (Normal Operations)	VOC	June 3, 2015 *	once every five (5) years
Biodiesel Manufacturing Process with Methanol Tank Loading			
Biodiesel Manufacturing Process Upset Operation			

*This test evaluated the inlet and outlet of the mineral oil absorber and provided data to determine the control efficiency; therefore, new testing within 180 days of permit issuance is not required.

- (b) The monitoring requirements applicable to this proposed modification are as follows:

Control Device / Emission Unit	Type of Parametric Monitoring	Frequency	Range or Specification
Mineral Oil Absorber	Mineral Oil Flow Rate	At least once per day	The minimum flow rate for the mineral oil absorber will be 1.5 gpm or the minimum mineral oil flow rate established during the latest stack test
	Temperature	Continuous	At or below the 3-hour average temperature as determined from the most recent valid stack test
Water Absorber	Water Flow Rate	At least once per day	The minimum flow rate for the water absorber will be 2.2 gpm or the minimum water flow rate established during the latest stack test
	Temperature	Continuous	At or below the 3-hour average temperature as determined from the most recent valid stack test

Proposed Changes

The following changes listed below are due to the proposed modification. Deleted language appears as ~~strike through~~ text and new language appears as **bold** text:

1. The construction year was added to the description for the existing gas turbine, identified GT-1, to determine applicability of NESHAP Subpart YYYY. Also, the language describing the power rating units for the power rating for GT-1 were clarified. In Section D.8 and Section E.12, the construction year was added to the description for the existing gas turbine, identified GT-1, to determine applicability of NESHAP Subpart YYYY. The rule citations for the NESHAP Subpart YYYY were updated based on the year of construction of the gas turbine, GT-1.
 2. In Section D.4, a 326 IAC 8-1-6 Best Available Control Technology (BACT) limit for the Biodiesel Manufacturing Process was added. Also, Testing and Record Keeping Requirements were added to Section D.4 as a result of the BACT Limit addition. The Condition numbers were updated as needed.
- A.2 Emission Units and Pollution Control Equipment Summary [326 IAC 2-7-4(c)(3)][326 IAC 2-7-5(14)]

This stationary source consists of the following emission units and pollution control devices:

(a) *****

(h)

Unit ID	Description	Capacity (tons/hr) Output Power Rating	Control	Discharging to Stack	Affected Facility?
GT-1	Natural Gas-fired power generator *2016 (Originally constructed in 1999)	4.6 MW	none	outside	Yes, under NSPS GG and NESHAP YYYY

Note *Approved in 2016 for construction. Original construction of the Natural Gas-fired power generator, GT-1, occurred in New Jersey in 1999.

D.4.2 VOC Best Available Control Technology (BACT) Limits [326 IAC 8-1-6]

Pursuant to 326 IAC 8-1-6 (New Facilities, General Reduction Requirements) the Permittee shall comply with the following Best Available Control Technology (BACT) requirements:

Process	Control	VOC control efficiency, %	VOC Limit (lbs/hour)	Operation Limit
Biodiesel manufacturing process Normal operation	Mineral Oil Absorber and Water absorber	Overall VOC control efficiency of 99%	0.30	NA
Biodiesel manufacturing process with methanol tank loading	Mineral Oil Absorber and Water absorber	Overall VOC control efficiency of 99%	0.63	1,000 hours per twelve (12) consecutive months
Biodiesel manufacturing process upset operation	Mineral Oil Absorber and Water absorber	Overall VOC control efficiency of 95%	29.4	24 hours per twelve (12) consecutive months
Glycerin storage tanks	None	None	0.0011	NA

Process	Control	VOC control efficiency, %	VOC Limit (lbs/hour)	Operation Limit
Biodiesel wastewater	None	None	0.77	NA
Biodiesel fugitive emissions	None	None	0.64	LDAR as required by 40 CFR 60, Subpart VV

D.4.54 Testing Requirements [326 IAC 2-1.1-11]

In order to demonstrate compliance with Conditions D.4.1(a) **and D.4.2**, the Permittee shall perform VOC testing on the **inlet and** outlet of the Mineral Oil Absorber with methanol unloading and without methanol unloading; and determine the Mineral Oil Absorber's mineral oil flow rate and water absorber's water flow rate, utilizing methods as approved by the Commissioner at least once every five (5) years from the date of the most recent valid compliance demonstration. Testing shall be conducted in accordance with the provisions of 326 IAC 3-6 (Source Sampling Procedures). Section C - Performance Testing contains the Permittee's obligation with regard to the performance testing required by this condition.

D.4.6 Monitoring for mineral oil absorber and water absorber [40 CFR 64]

- (a) *****
 (c) *****

- (1) The Permittee shall determine the 3-hour average temperature from the most recent valid stack test that demonstrates compliance with the limits in condition D.4.1(a) **and D.4.2**.

D.4.76 Record Keeping Requirements

- (a) *****

- (c) To document the compliance status with Condition D.4.1(a) **and D.4.2**, the Permittee shall maintain records of the operating hours for the biodiesel manufacturing process during the following operating scenarios:

- *****
 (d) *****

- (1) The Permittee shall determine the 3-hour average temperature from the most recent valid stack test that demonstrates compliance with limits in condition D.4.1(a) **and D.4.2**.

SECTION D.8 EMISSIONS UNIT OPERATION CONDITIONS

***** (h)					
Unit ID	Description	Capacity (tons/hr) Output Power Rating	Control	Discharging to Stack	Affected Facility?
GT-1	Natural Gas-fired power generator *2016 (Originally constructed in 1999)	4.6 MW	none	outside	Yes, under NSPS GG and NESHAP YYYY
Note *Approved in the year indicated above for construction. *****					

(h)

Unit ID	Description	Capacity (tons/hr) Output Power Rating	Control	Discharging to Stack	Affected Facility?
GT-1	Natural Gas-fired power generator *2016 (Originally constructed in 1999)	4.6 MW	none	outside	Yes, under NSPS GG and NESHAP YYYY

Note *Approved in the year indicated above for construction.

SECTION E.13

NESHAP

Emissions Unit Description:

(h)

Unit ID	Description	Capacity (tons/hr) Output Power Rating	Control	Discharging to Stack	Affected Facility?
GT-1	Natural Gas-fired power generator *2016 (Originally constructed in 1999)	4.6 MW	none	outside	Yes, under NSPS GG and NESHAP YYYY

Note *Approved in the year indicated above for construction.

E.13.2 Stationary Combustion Turbines NESHAP [40 CFR Part 63, Subpart YYYY][326 IAC 20-90]

The Permittee shall comply with the following provisions of 40 CFR Part 63, Subpart YYYY (included as Attachment M to the operating permit), which are incorporated by reference as 326 IAC 20-90, for the emission unit listed above:

- (1) 40 CFR 63.6080
- (2) 40 CFR 63.6085(a), (b)
- (3) 40 CFR 63.6090(a)(2) (a)(1); (b)(4)
- (4) ~~40 CFR 63.6092~~
- (5) ~~40 CFR 63.6095(a)(2), (c), (d)~~
- (6) ~~40 CFR 63.6100~~
- (7) ~~40 CFR 63.6105~~
- (8) ~~40 CFR 63.6110~~
- (9) ~~40 CFR 63.6115~~
- (10) ~~40 CFR 63.6120~~
- (11) ~~40 CFR 63.6125~~
- (12) ~~40 CFR 63.6130~~
- (13) ~~40 CFR 63.6135~~
- (14) ~~40 CFR 63.6140~~
- (15) ~~40 CFR 63.6145~~
- (16) ~~40 CFR 63.6150~~
- (17) ~~40 CFR 63.6155~~
- (18) ~~40 CFR 63.6160~~
- (19) ~~40 CFR 63.6165~~

~~(420)~~ 40 CFR 63.6170(a), (b)(1-5)
~~(521)~~ 40 CFR 63.6175

Conclusion and Recommendation

Unless otherwise stated, information used in this review was derived from the application and additional information submitted by the applicant. An application for the purposes of this review was received on January 23, 2018.

The construction of this proposed modification shall be subject to the conditions of the attached proposed Part 70 Significant Source Modification No. 085-39532-00102. The operation of this proposed modification shall be subject to the conditions of the attached Significant Permit Modification No. 085-39729-00102.

The staff recommends to the Commissioner that the Part 70 Significant Source Modification and the Significant Permit Modification be approved.

IDEM Contact

- (a) Questions regarding this proposed permit can be directed to Daniel W. Pell at the Indiana Department Environmental Management, Office of Air Quality, Permits Branch, 100 North Senate Avenue, MC 61-53 IGCN 1003, Indianapolis, Indiana 46204-2251 or by telephone at (317) 234-8532 or toll free at 1-800-451-6027, extension 4-8532.
- (b) A copy of the findings is available on the Internet at: <http://www.in.gov/ai/appfiles/idem-caats/>
- (c) For additional information about air permits and how the public and interested parties can participate, refer to the IDEM Air Permits page on the Internet at: <http://www.in.gov/idem/airquality/2356.htm>; and the Citizens' Guide to IDEM on the Internet at: <http://www.in.gov/idem/6900.htm>.

Appendix A: Emissions Calculations

Source Summary
Potential to Emit Before Control

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

Potential To Emit before Control

Stack ID / Control / Emission Units	PM (tons/yr)	PM ₁₀ (tons/yr)	PM _{2.5} (tons/yr)	SO ₂ (tons/yr)	NO _x (tons/yr)	VOC (tons/yr)	CO (tons/yr)	Total HAPs (tons/yr)
Soybean Oil Extraction Plant								
Stack AF-2 (and uncaptured emissions)	3,571.19	1,718.97	293.92	-	-	-	-	-
Stack AF-1 (and uncaptured emissions)	1,318.23	471.73	244.17	-	-	-	-	-
Stacks MBF-1, MBF-2, MBF-3, MBF-4 and MBF-5	21.68	5.46	0.95	-	-	-	-	-
Piles #1 and #2 (no control)	192.37	107.22	18.29	-	-	-	-	-
Stack AF-3	1,271.29	640.27	364.79	-	-	-	-	-
Stack AF-7	43.80	10.95	10.95	-	-	-	-	-
Stack S-1 and Stack S-6	624.41	156.10	50.65	-	-	-	-	-
Stack AF-5	8,732.13	2,185.46	2,122.69	-	-	-	-	-
Stack AF-4	62.65	83.54	83.54	-	-	-	-	-
Stack AF-6	9,112.98	2,359.61	2,230.59	-	-	-	-	-
Stack S-2	6,559.49	1,639.87	1,639.87	-	-	-	-	-
Bin Filter	3.47	0.56	0.21	-	-	-	-	-
Hull Bin Filter	0.84	0.21	0.04	-	-	-	-	-
Bean Storage Bin No. 2 thru 4, 6, 7, and 8 and Bean Storage Silo No. 1 and 5	30.40	7.66	1.34	-	-	-	-	-
Diatomaceous Earth (DE) Storage Bin	0.38	0.06	0.02	-	-	-	-	-
Main Gas Vent (Soybean Oil Extractor System)	-	-	-	-	-	5,980.80	-	3,827.71
Desolventized Meal Dryers & Cooler	-	-	-	-	-	202.20	-	129.41
Meal Storage	-	-	-	-	-	fugitive	-	1.21
Bound In Product & Byproduct Desolventized Meal (fugitive)	-	-	-	-	-	fugitive	-	113.00
Plant Startup/Shutdown	-	-	-	-	-	fugitive	-	10.75
General (equipment failure, leaks, etc.)	-	-	-	-	-	fugitive	-	207.21
Plant Upsets	-	-	-	-	-	fugitive	-	16.90
Combustion Sources	2.75	8.14	8.14	1.01	145.84	6.29	82.42	1.80
Temporary Mobile Boiler (B-S)	0.33	1.31	1.31	0.10	17.18	0.94	14.43	0.32
GT-1 Turbine	1.53	1.53	1.53	0.79	23.11	16.04	27.97	0.24
Three (3) Diesel Emergency Generators *2016	0.41	0.41	0.41	0.38	5.71	0.46	1.23	5.00E-03
Glycerin Cooling Tower	0.19	0.13	5.95E-04	-	-	-	-	-
Total PTE from Soybean Oil Extraction Plant with Roads	31,566.69	9,402.43	7,074.20	2.28	191.84	6,206.74	126.04	4,308.55
Biodiesel Production Plant								
Biodiesel Methanol Absorbers (worse case) ¹	-	-	-	-	-	154.60	-	154.60
Loading Racks ¹	-	-	-	-	-	10.72	-	10.72
Biodiesel Storage Tanks ¹	-	-	-	-	-	4.31	-	4.31
Glycerine Storage Tanks	-	-	-	-	-	0.00	-	0.00
Biodiesel Wastewater (fugitive)	-	-	-	-	-	3.37	-	3.37
Equipment Leaks (fugitive) ¹	-	-	-	-	-	12.74	-	12.74
Diesel/#2 Fuel Oil Storage Tank	-	-	-	-	-	0.01	-	0.01
One (1) Soybean Oil Pre-Treat Tank	-	-	-	-	-	0.01	-	0.01
Six (6) Methanol Tanks	-	-	-	-	-	4.49	-	4.49
Three (3) Soybean Oil Tanks (Degummed Oil Tanks #1 and #2 and Crude Oil Tank #3)	-	-	-	-	-	<= 1.0	-	<= 1.0
Purchased Soybean Oil Unloading	-	-	-	-	-	4.00	-	4.00
Two (2) Sodium Methylate Tanks	-	-	-	-	-	0.60	-	0.60
Five (5) Hexane tanks	-	-	-	-	-	2.01	-	2.01
Road Traffic at Source ²	16.18	3.24	0.79	-	-	-	-	-
Cooling Towers	4.34	4.34	4.34	-	-	-	-	-
Biodiesel Distillation	-	-	-	-	-	1.31	-	1.31
Hot Oil Heater	0.05	0.20	0.20	0.02	2.58	0.14	2.16	0.05
Natural Gas-Fired Glycerin Steam Boiler	0.01	0.03	0.03	2.58E-03	0.43	0.02	0.36	0.01
Glycerin Tanks	-	-	-	-	-	7.70E-06	-	-
Glycerin Refinery/ Rail Loadout	-	-	-	-	-	3.31E-04	-	-
Glycerin Refinery	-	-	-	-	-	2.00E-04	-	0.0002
Total PTE at Biodiesel Plant	20.58	7.81	5.37	0.02	3.01	199.35	2.52	199.24
Total PTE at Entire Source including Fugitives from Roads	31,571.09	9,407.00	7,078.77	2.29	194.85	6,406.09	128.57	4,507.79

Notes:

¹ The VOC emissions from these units are mainly methanol which is also a HAP. Assume all VOCs are HAPs in the worst case scenario.

² Fugitive road emissions are being counted for the soybean extraction plant because the operations are covered by the source category definition in NSPS DD (a pre-1980 NSPS). In addition, fugitive road emissions are being counted because they include roads at the biodiesel production plant which is one of 28 source categories.

Appendix A: Emissions Calculations

Source Summary
Limited Potential to Emit

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

Limited Potential To Emit

Stack ID / Control / Emission Units	PM (tons/yr)	PM ₁₀ (tons/yr)	PM _{2.5} (tons/yr)	SO ₂ (tons/yr)	NO _x (tons/yr)	VOC (tons/yr)	CO (tons/yr)	Total HAPs (tons/yr)
Soybean Oil Extraction Plant								
Stack AF-2 (and uncaptured emissions)	21.65	12.77	8.38	-	-	-	-	-
Stack AF-1 (and uncaptured emissions)	23.91	9.08	8.83	-	-	-	-	-
Stacks MBF-1, MBF-2, MBF-3, MBF-4 and MBF-5	20.37	20.37	20.37	-	-	-	-	-
Piles #1 and #2 (no control)	7.32	4.08	0.70	-	-	-	-	-
Stack AF-3	5.52	5.52	5.52	-	-	-	-	-
Stack AF-7	6.57	6.57	6.57	-	-	-	-	-
Stack S-1 and Stack S-6	21.59	14.67	14.67	-	-	-	-	-
Stack AF-5	11.21	11.21	11.21	-	-	-	-	-
Stack AF-4	4.51	4.51	4.51	-	-	-	-	-
Stack AF-6	4.14	4.14	4.14	-	-	-	-	-
Stack S-2	47.04	31.89	31.89	-	-	-	-	-
Bin Filter	8.32	8.32	8.32	-	-	-	-	-
Hull Bin Filter	0.84	0.21	0.04	-	-	-	-	-
Bean Storage Bin No. 2 thru 4, 6, 7, and 8 and Bean Storage Silo No. 1 and 5	30.40	7.66	1.34	-	-	-	-	-
Diatomaceous Earth (DE) Storage Bin	0.38	0.06	0.02	-	-	-	-	-
Main Gas Vent (Soybean Oil Extractor System)	-	-	-	-	-	40.73	-	26.07
Desolventized Meal Dryers & Cooler	-	-	-	-	-	143.66	-	91.94
Meal Storage	-	-	-	-	-	fugitive	-	1.21
Bound In Product & Byproduct Desolventized Meal (fugitive)	-	-	-	-	-	fugitive	-	113.00
Plant Startup/Shutdown	-	-	-	-	-	fugitive	-	10.75
General (equipment failure, leaks, etc.)	-	-	-	-	-	fugitive	-	207.21
Plant Upsets	-	-	-	-	-	fugitive	-	16.90
Combustion Sources	2.75	8.14	8.14	1.01	145.84	6.29	82.42	1.80
Temporary Mobile Boiler (B-S)	0.33	1.31	1.31	0.10	17.18	0.94	14.43	0.32
GT-1 Turbine	1.53	1.53	1.53	0.79	23.11	16.04	27.97	0.24
Three (3) Diesel Emergency Generators *2016	0.41	0.41	0.41	0.38	5.71	0.46	1.23	5.00E-03
Glycerin Cooling Tower	0.19	0.13	5.95E-04	-	-	-	-	-
Total Limited PTE from Soybean Oil Extraction Plant with Roads	235.16	155.81	138.68	2.28	191.84	208.14	126.04	469.45
Biodiesel Production Plant								
Biodiesel Methanol Absorbers (worst case) ¹	-	-	-	-	-	1.83	-	1.83
Loading Racks ¹	-	-	-	-	-	1.10	-	1.10
Biodiesel Storage Tanks ¹	-	-	-	-	-	4.31	-	4.31
Glycerine Storage Tanks	-	-	-	-	-	4.82E-03	-	4.82E-03
Biodiesel Wastewater (fugitive)	-	-	-	-	-	3.37	-	3.37
Equipment Leaks (fugitive) ¹	-	-	-	-	-	2.80	-	2.80
Diesel/#2 Fuel Oil Storage Tank	-	-	-	-	-	0.01	-	0.01
One (1) Soybean Oil Pre-Treat Tank	-	-	-	-	-	0.01	-	0.01
Six (6) Methanol Tanks	-	-	-	-	-	0.07	-	0.07
Three (3) Soybean Oil Tanks (Degummed Oil Tanks #1 and #2 and Crude Oil Tank #3)	-	-	-	-	-	<= 1.0	-	<= 1.0
Purchased Soybean Oil Unloading	-	-	-	-	-	4.00	-	4.00
Two (2) Sodium Methylate Tanks	-	-	-	-	-	0.01	-	0.01
Five (5) Hexane tanks	-	-	-	-	-	2.01	-	2.01
Road Traffic at Source ²	16.18	3.24	0.79	-	-	-	-	-
Cooling Towers	4.34	4.34	4.34	-	-	-	-	-
Biodiesel Distillation	-	-	-	-	-	1.31	-	1.31
Hot Oil Heater	0.05	0.20	0.20	0.02	2.58	0.14	2.16	0.05
Natural Gas-Fired Glycerin Steam Boiler	0.01	0.03	0.03	2.58E-03	0.43	0.02	0.36	0.01
Glycerin Tanks	-	-	-	-	-	7.70E-06	-	-
Glycerin Refinery/ Rail Loadout	-	-	-	-	-	3.31E-04	-	-
Glycerin Refinery	-	-	-	-	-	2.00E-04	-	2.00E-04
Total Limited PTE from Biodiesel Production Plant	20.58	7.81	5.37	0.02	3.01	22.02	2.52	21.90
Total Limited PTE at Entire Source including Fugitives from Roads	239.56	160.38	143.25	2.29	194.85	230.15	128.57	491.35

Notes:

¹ The VOC emissions from these units are mainly methanol which is also a HAP. Assume all VOCs are HAPs in the worst case scenario.

² Fugitive road emissions are being counted for the soybean extraction plant because the operations are covered by the source category definition in NSPS DD (a pre-1980 NSPS).
In addition, fugitive road emissions are being counted because they include roads at the biodiesel production plant which is one of 28 source categories.

Appendix A: Emissions Calculations
Solar T60 Power Generator (GT-1)

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

Emission Unit	Unit ID	Fuel Type	MW output	MMBtu/hr	Fuel Heat Content	Heat Content Units
Solar T60 Generator (GT-1)	GT-1	Natural Gas	4.60	52.77	1020	mmBtu/mmcf

Basis for Emission Factors

	Solar T60MPU Technical Data
	AP 42 Factors (Chapter 3, Section 1, Table 3.1-2a)

Emission Factors									
Emission Unit	Fuel Type	Units	NO _x	CO	VOC	SO ₂	PM	PM ₁₀	PM _{2.5}
Solar T60 Generator (GT-1)	Natural Gas	lbs/hr	5.28	6.39	3.66	0.18	0.35	0.35	0.35
		lb/MW-hr	1.08	1.31	0.38				
		ppmvd	25.0	50.0	50.0				
		lb/MMBtu	0.100	0.121	0.069	0.0034	0.0066	0.0066	0.0066

Maximum Potential to Emit - Natural Gas

			Potential to Emit (tons/year)						
Emission Unit	Hrs/ Yr	Fuel Type	NO _x	CO	VOC	SO ₂	PM	PM ₁₀	PM _{2.5}
Solar T60 Generator (natural gas)	8760	Natural Gas	23.11	27.97	16.04	0.79	1.53	1.53	1.53

Methodology

Emission Factors are from AP 42 (Chapter 3, Section 1, Table 3.1-2a)

Potential to Emit (tons/year) = Emission Factors (lbs/hr) * Hrs/Yr

Natural Gas HAPs		HAPs - Organics										
	1,3-Butadiene	Acetaldehyde	Acrolein	Benzene	Ethylbenzene	Formaldehyde	Napthalene	PAH	Propylene Oxide	Toluene	Xylenes	Total - Organics
Emission Factor in lb/MMBtu	4.30E-07	4.0E-05	6.40E-06	1.20E-05	3.20E-05	7.1E-04	1.30E-06	2.20E-06	2.90E-05	1.30E-04	6.4E-05	
Potential Emission in tons/yr	9.94E-05	9.25E-03	1.48E-03	2.77E-03	7.40E-03	0.16	3.00E-04	5.08E-04	6.70E-03	3.00E-02	1.48E-02	0.24
											Worst HAP	0.16
												Formaldehyde

Methodology

Natural Gas Emission Factors are from AP 42 (Chapter 3, Section 1, Table 3.1-3)

Potential Emissions in tons/yr = MMBtu/hr * Emission Factor in lb/MMBtu

Appendix A: Emission Calculations
Reciprocating Internal Combustion Engines - Diesel Fuel
Output Rating (<=600 HP)
Maximum Input Rate (<=4.2 MMBtu/hr)

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

Two (2) diesel powered emergency generators, rated at 150 kW, each.
One (1) diesel powered emergency generator, rated at 250 kW.

Total kW	550
-----------------	-----

Emissions calculated based on output rating (hp)

Output Horsepower Rating (hp)	737.0
Maximum Hours Operated per Year	500
Potential Throughput (hp-hr/yr)	368,500

	Pollutant						
	PM*	PM10*	direct PM2.5*	SO2	NOx	VOC	CO
Emission Factor in lb/hp-hr	0.0022	0.0022	0.0022	0.0021	0.0310	0.0025	0.0067
Potential Emission in tons/yr	0.41	0.41	0.41	0.38	5.71	0.46	1.23

*PM and PM2.5 emission factors are assumed to be equivalent to PM10 emission factors. No information was given regarding which method was used to determine the factor or the fraction of PM10 which is condensable.

Hazardous Air Pollutants (HAPs)

	Pollutant							
	Benzene	Toluene	Xylene	1,3-Butadiene	Formaldehyde	Acetaldehyde	Acrolein	Total PAH HAPs***
Emission Factor in lb/hp-hr****	6.53E-06	2.86E-06	2.00E-06	2.74E-07	8.26E-06	5.37E-06	6.48E-07	1.18E-06
Potential Emission in tons/yr	1.20E-03	5.28E-04	3.68E-04	5.04E-05	1.52E-03	9.89E-04	1.19E-04	2.17E-04
Potential Emission of Total HAPs (tons/yr)								5.00E-03

***PAH = Polyaromatic Hydrocarbon (PAHs are considered HAPs, since they are considered Polycyclic Organic Matter)

****Emission factors in lb/hp-hr were calculated using emission factors in lb/MMBtu and a brake specific fuel consumption of 7,000 Btu / hp-hr (AP-42 Table 3.3-1).

Methodology

Emission Factors are from AP 42 (Supplement B 10/96) Tables 3.4-1 , 3.4-2, 3.4-3, and 3.4-4.

Output Horsepower Rating (hp) = Total kW * 1.34 hp/kW

Potential Throughput (hp-hr/yr) = [Output Horsepower Rating (hp)] * [Maximum Hours Operated per Year]

Potential Emission (tons/yr) = [Potential Throughput (hp-hr/yr)] * [Emission Factor (lb/hp-hr)] / [2,000 lb/ton]

Appendix A: Emission Calculations
Glycerin Cooling Tower
Particulate Emissions

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

Water Flow Rate =	875	gpm
Drift Eliminators =	0.005%	--

Potential to Emit (lb/hr)			Potential to Emit (ton/yr)		
PM	PM ₁₀	PM _{2.5}	PM	PM ₁₀	PM _{2.5}
0.044	0.030	1.36E-04	0.19	0.13	5.95E-04

Methodology

PTE of PM (tpy) = 875 gpm * 60 min/hr * 8.34 lb/gal * 8760 hr * 2000 lb/ton / 1000000 ppm * 0.005%

PTE of PM10 (lb/hr) = 875 gpm * 60 min/hr * 8.34 lb/gal * 8760 hr * 2000 lb/ton / 1000000 ppm * 0.005% * 68.6%

PTE of PM2.5 (tpy) = 875 gpm * 60 min/hr * 8.34 lb/gal * 8760 hr * 2000 lb/ton / 1000000 ppm * 0.005% * 0.31%

INPUT MAXIMUM TOTAL DISSOLVED SOLIDS (TDS) CONCENTRATION

Max TDS = ppmw

EPRI Droplet Diameter (μm)	Droplet Volume (μm ³)	Droplet Mass (μg)	Particle Mass (solids) (μg)	Solid Particle Volume (μm ³)	Solid Particle Diameter (μm)	EPRI % Mass Smaller
10	524	5.24E-04	1.05E-06	0.48	0.969	0.000
			Interpolation --->		2.500	0.310
20	4189	4.19E-03	8.38E-06	3.81	1.937	0.196
30	14137	1.41E-02	2.83E-05	12.85	2.906	0.226
40	33510	3.35E-02	6.70E-05	30.46	3.875	0.514
50	65450	6.54E-02	1.31E-04	59.50	4.844	1.816
60	113097	1.13E-01	2.26E-04	102.82	5.812	5.702
70	179594	1.80E-01	3.59E-04	163.27	6.781	21.348
			Interpolation --->		10.000	68.638
90	381704	3.82E-01	7.63E-04	347.00	8.719	49.812
110	696910	6.97E-01	1.39E-03	633.55	10.656	70.509
130	1150347	1.15E+00	2.30E-03	1045.77	12.593	82.023
150	1767146	1.77E+00	3.53E-03	1606.50	14.531	88.012
180	3053628	3.05E+00	6.11E-03	2776.03	17.437	91.032
210	4849048	4.85E+00	9.70E-03	4408.23	20.343	92.468
240	7238229	7.24E+00	1.45E-02	6580.21	23.250	94.091
270	10305995	1.03E+01	2.06E-02	9369.09	26.156	94.689
300	14137167	1.41E+01	2.83E-02	12851.97	29.062	96.288
350	22449298	2.24E+01	4.49E-02	20408.45	33.906	97.011
400	33510322	3.35E+01	6.70E-02	30463.93	38.749	98.34
450	47712938	4.77E+01	9.54E-02	43375.40	43.593	99.071
500	65449847	6.54E+01	1.31E-01	59499.86	48.436	99.071
600	113097336	1.13E+02	2.26E-01	102815.76	58.124	100.000

0.31 % of PM is PM2.5

68.6 % of PM is PM10

Calculations based on approach presented in: Calculating Realistic PM10 Emissions from Cooling Towers
 Joel Reisman and Gordon Frisbie, Environmental Progress (Vol 21, No 2), July 2002

**Appendix A: Emissions Calculations
Portable Boiler - Natural Gas Combustion (B-S)**

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

Heat Input Capacity MMBtu/hr	HHV mmBtu mmscf	Potential Throughput MMCF/yr
40.0	1020	343.5

	Pollutant						
Emission Factor in lb/MMCF	PM* 1.9	PM10* 7.6	direct PM2.5* 7.6	SO2 0.6	NOx 100 **see below	VOC 5.5	CO 84
Potential Emission in tons/yr	0.33	1.3	1.3	0.10	17.2	0.9	14.4

*PM emission factor is filterable PM only. PM10 emission factor is filterable and condensable PM10 combined.

PM2.5 emission factor is filterable and condensable PM2.5 combined.

**Emission Factors for NOx: Uncontrolled = 100, Low NOx Burner = 50, Low NOx Burners/Flue gas recirculation = 32

Methodology

All emission factors are based on normal firing.

MMBtu = 1,000,000 Btu

MMCF = 1,000,000 Cubic Feet of Gas

Emission Factors are from AP 42, Chapter 1.4, Tables 1.4-1, 1.4-2, 1.4-3, SCC #1-02-006-02, 1-01-006-02, 1-03-006-02, and 1-03-006-03

Potential Throughput (MMCF) = Heat Input Capacity (MMBtu/hr) x 8,760 hrs/yr x 1 MMCF/1,020 MMBtu

Emission (tons/yr) = Throughput (MMCF/yr) x Emission Factor (lb/MMCF)/2,000 lb/ton

HAPS Calculations

	HAPs - Organics					
Emission Factor in lb/MMcf	Benzene 2.1E-03	Dichlorobenzene 1.2E-03	Formaldehyde 7.5E-02	Hexane 1.8E+00	Toluene 3.4E-03	Total - Organics
Potential Emission in tons/yr	3.607E-04	2.061E-04	1.288E-02	3.092E-01	5.840E-04	3.232E-01

	HAPs - Metals					
Emission Factor in lb/MMcf	Lead 5.0E-04	Cadmium 1.1E-03	Chromium 1.4E-03	Manganese 3.8E-04	Nickel 2.1E-03	Total - Metals
Potential Emission in tons/yr	8.588E-05	1.889E-04	2.405E-04	6.527E-05	3.607E-04	9.413E-04
					Total HAPs	3.242E-01
					Worst HAP	3.092E-01

Methodology is the same as above.

The five highest organic and metal HAPs emission factors are provided above.

Additional HAPs emission factors are available in AP-42, Chapter 1.4.

**Appendix A: Emissions Calculations
Glycerin Tanks - VOC Emissions
Potential to Emit Before Control**

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

Unit ID	Equipment Description	Capacity	Tank Diameter (ft)	Tank Height (ft)	Pot'l VOC Working Loss (lbs/yr)	Pot'l VOC Standing Loss (lbs/yr)	Pot'l VOC Max Emissions (lbs/yr)	Pot'l VOC Max Emissions (tons/year)	Notes
Trivial Activity T-0360	Glycerin Yellow Tank	8,000 gallons	10	14	6.00E-04	0.00E+00	6.00E-04	3.00E-07	Assume 1 turnover per week (52 per year)
Trivial Activity T-0361	Glycerin Tank 1	50,000 gallons	14	44	3.60E-03	1.00E-04	3.70E-03	1.85E-06	Assume 1 turnover per week (52 per year)
Trivial Activity T-0362	Glycerin Tank 2	50,000 gallons	14	44	3.60E-03	1.00E-04	3.70E-03	1.85E-06	Assume 1 turnover per week (52 per year)
Trivial Activity T-0363	Glycerin Tank 3	50,000 gallons	14	44	3.60E-03	1.00E-04	3.70E-03	1.85E-06	Assume 1 turnover per week (52 per year)
Trivial Activity T-0364	Glycerin Tank 4	50,000 gallons	14	44	3.60E-03	1.00E-04	3.70E-03	1.85E-06	Assume 1 turnover per week (52 per year)
Total								7.70E-06	tons/year

Appendix A: Emission Calculations Bean Storage Bins No. 4 and 8

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

PTE for Bean Storage Bin No. 4 and 8

PM Emission Factor	0.025	lb/ton	Source:	SCC 3-02-005-40
PM10/PM2.5 Emission Factor	0.006	lb/ton		SCC 3-02-005-40
Hourly Loading Rate	260.5	tons/hr		
Aspiration Rate	116,000	cfm		58,000 cfm, each
% Control PM	N/A			
% Control PM10/PM2.5	N/A			

Hourly throughput is based on maximum transport system capacity.

Potential PM emissions Bean Storage Bin No. 4 and 8

		Uncontrolled total		Controlled total	
Max Hourly	=	6.51	lbs/hr	6.51	lbs/hr
Max Yearly	=	28.52	tons/yr	28.52	tons/yr

Potential PM10/PM2.5 emissions Bean Storage Bin No. 4 and 8

		Uncontrolled total		Controlled total	
Max Hourly	=	1.64	lbs/hr	1.64	lbs/hr
Max Yearly	=	7.19	tons/yr	7.19	tons/yr

Methodology

Emission factors are from AP 42 Table 9.9.1-1 Particulate Emission Factors for Grain Elevators (3/03)

Hourly Loading Rate (tons/hr) = [2,251,836 (tons/yr) + 30,0000 (tons storage capacity for two bins)] / 8,760 (hr/yr)

The hourly loading rate assumes the two (2) new storage bins process all of the soybeans for the entire plant.

Uncontrolled Emissions (lb/hr) = Hourly Loading Rate (tons/hr) * Emission Factor (lb/ton)

Controlled Emissions (lb/hr) = Uncontrolled Emissions (lb/hr) * (1-Control Efficiency)

Uncontrolled Emissions (ton/yr) = Uncontrolled Emissions (lb/hr) * 8,760 (hr/yr) / 2000 (lbs/ton)

Controlled Potential Emissions (ton/yr) = Uncontrolled Emissions (ton/yr) * (1-Control Efficiency).

Appendix A: Emissions Calculations
Glycerin Steam Boiler - Natural Gas Combustion

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

Heat Input Capacity MMBtu/hr	HHV mmBtu mmscf	Potential Throughput MMCF/yr
1.0	1020	8.6

	Pollutant						
Emission Factor in lb/MMCF	PM* 1.9	PM10* 7.6	direct PM2.5* 7.6	SO2 0.6	NOx 100 **see below	VOC 5.5	CO 84
Potential Emission in tons/yr	0.01	0.03	0.03	2.58E-03	0.4	0.02	0.36

*PM emission factor is filterable PM only. PM10 emission factor is filterable and condensable PM10 combined.

PM2.5 emission factor is filterable and condensable PM2.5 combined.

**Emission Factors for NOx: Uncontrolled = 100, Low NOx Burner = 50, Low NOx Burners/Flue gas recirculation = 32

Methodology

All emission factors are based on normal firing.

MMBtu = 1,000,000 Btu

MMCF = 1,000,000 Cubic Feet of Gas

Emission Factors are from AP 42, Chapter 1.4, Tables 1.4-1, 1.4-2, 1.4-3, SCC #1-02-006-02, 1-01-006-02, 1-03-006-02, and 1-03-006-03

Potential Throughput (MMCF) = Heat Input Capacity (MMBtu/hr) x 8,760 hrs/yr x 1 MMCF/1,020 MMBtu

HAPS Calculations

	HAPs - Organics					
Emission Factor in lb/MMcf	Benzene 2.1E-03	Dichlorobenzene 1.2E-03	Formaldehyde 7.5E-02	Hexane 1.8E+00	Toluene 3.4E-03	Total - Organics
Potential Emission in tons/yr	9.018E-06	5.153E-06	3.221E-04	7.729E-03	1.460E-05	8.080E-03

	HAPs - Metals					
Emission Factor in lb/MMcf	Lead 5.0E-04	Cadmium 1.1E-03	Chromium 1.4E-03	Manganese 3.8E-04	Nickel 2.1E-03	Total - Metals
Potential Emission in tons/yr	2.147E-06	4.724E-06	6.012E-06	1.632E-06	9.018E-06	2.353E-05
					Total HAPs	8.104E-03
					Worst HAP	7.729E-03

Methodology is the same as above.

Hexane

The five highest organic and metal HAPs emission factors are provided above.

Additional HAPs emission factors are available in AP-42, Chapter 1.4.

**Appendix A: Emissions Calculations
Glycerin Refinery**

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

The biodiesel production process produces biodiesel and glycerin. All products are shipped offsite by truck or rail resulting loadout uncontrolled emissions. According to Ap-42, chapter 5.2 - Transportation and Marketing of Petroleum Liquids (6/08), the VOC emission factor from the truck and rail loading rack are:

$$L = 12.46 * (SPM/T)$$

where:

L = loading loss (lbs/kgal)
 S = a saturation factor (see AP-42, Table 5.2-1)
 P = true vapor pressure of the liquid loaded (psia) (see AP-42, Table 7.1-2)
 M = molecular weight of vapors
 T = temperature of the bulk liquid loaded (degree R)

	S	P (psia)	M (lbs/mole lbs)	T (degree R)	L (lbs/kgal)
Glycerin	0.6	0.00005	92.1	582	0.00006

Glycerin Throughput 7.0 ton/hour

	Max. Throughput (kgals/yr)	PTE of VOC (lbs/yr)	PTE of VOC (tons/yr)
Glycerin	11,570	0.66	0.0003

Note:

Weight of one gallon of glycerin is equal to 10.6 lbs.
 All VOC is considered to be HAP.

**Appendix A: Emissions Calculations
Grain Dryer - Natural Gas Combustion**

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

Heat Input Capacity MMBtu/hr	HHV mmBtu mmscf	Potential Throughput MMCF/yr
6.0	1020	51.5

	Pollutant						
	PM*	PM10*	direct PM2.5*	SO2	NOx	VOC	CO
Emission Factor in lb/MMCF	1.9	7.6	7.6	0.6	100 **see below	5.5	84
Potential Emission in tor	0.05	0.2	0.2	0.02	2.6	0.1	2.2

*PM emission factor is filterable PM only. PM10 emission factor is filterable and condensable PM10 combined.

PM2.5 emission factor is filterable and condensable PM2.5 combined.

**Emission Factors for NOx: Uncontrolled = 100, Low NOx Burner = 50, Low NOx Burners/Flue gas recirculation = 32

Methodology

All emission factors are based on normal firing.

MMBtu = 1,000,000 Btu

MMCF = 1,000,000 Cubic Feet of Gas

Emission Factors are from AP 42, Chapter 1.4, Tables 1.4-1, 1.4-2, 1.4-3, SCC #1-02-006-02, 1-01-006-02, 1-03-006-02, and 1-03-006-03

Potential Throughput (MMCF) = Heat Input Capacity (MMBtu/hr) x 8,760 hrs/yr x 1 MMCF/1,020 MMBtu

Emission (tons/yr) = Throughput (MMCF/yr) x Emission Factor (lb/MMCF)/2,000 lb/ton

HAPS Calculations

	HAPs - Organics					
	Benzene	Dichlorobenzene	Formaldehyde	Hexane	Toluene	Total - Organics
Emission Factor in lb/Mt	2.1E-03	1.2E-03	7.5E-02	1.8E+00	3.4E-03	
Potential Emission in tor	5.411E-05	3.092E-05	1.932E-03	4.638E-02	8.760E-05	4.848E-02

	HAPs - Metals					
	Lead	Cadmium	Chromium	Manganese	Nickel	Total - Metals
Emission Factor in lb/Mt	5.0E-04	1.1E-03	1.4E-03	3.8E-04	2.1E-03	
Potential Emission in tor	1.288E-05	2.834E-05	3.607E-05	9.791E-06	5.411E-05	1.412E-04
					Total HAPs	4.862E-02
					Worst HAP	4.638E-02

Methodology is the same as above.

The five highest organic and metal HAPs emission factors are provided above.

Additional HAPs emission factors are available in AP-42, Chapter 1.4.

Appendix A: Emission Calculations
Flaking Roll No. 1 and No. 10

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

PTE for emission units C100000 and C010600
Flaking Roll No. 1 and No. 10

Controlled PM/PM10/PM2.5 Emission Factor	0.037	lb/ton	Source: SCC 3-02-007-88
% Control Cyclone	90.00%		
Uncontrolled PM/PM10/PM2.5 Emission Factor	0.37	lb/ton	
Hourly Throughput Rate	22.9	tons/hr	
Annual Throughput Rate	200,604	tons/yr	
% Control Flaker Aspiration Baghouse	98.70%		

Potential PM/PM10/PM2.5 Emissions
Flaking Roll No. 1 and No. 10

		Uncontrolled for each unit		Controlled for each unit	
Max Hourly	=	8.47	lbs/hr	0.11	lbs/hr
Max Yearly	=	37.11	tons/yr	0.48	tons/yr
		Total Uncontrolled		Total Controlled	
Total Hourly	=	16.95	lbs/hr	0.22	lbs/hr
Total Yearly	=	74.22	tons/yr	0.96	tons/yr

Methodology

Emission factor from AP 42 Table 9.11.1-1 Total Particulate Emission Factors for Soybean Milling (11/1995)

*Values in Table 9.11.1-1 for flaking rolls include cyclone as control device. Therefore, IDEM has calculated the uncontrolled emission factors by removing the control efficiency for the cyclone (90% control efficiency). There is no emission factor for PM10 and PM2.5. Therefore, IDEM has assumed PM10 and PM2.5 = PM.

Uncontrolled Emission Factor (lb/ton) = Controlled Emission Factor (lb/ton)/(1 - 90% Control Efficiency)

Uncontrolled Potential Emissions (lb/hr) = Throughput (ton/hr) * Uncontrolled Emission factor (lb/ton)

Uncontrolled Potential Emissions (ton/yr) = Throughput (ton/hr) * Uncontrolled Emission factor (lb/ton) * 8760 (hours/year) / 2000 (lbs/ton)

Controlled Potential Emissions (lb/hr) = Throughput (ton/hr) * Uncontrolled Emission factor (lb/ton) * (1-Control Efficiency)

Controlled Potential Emissions (ton/yr) = Throughput (ton/yr) * Uncontrolled Emission factor (lb/ton) / 2000 (lbs/ton)* (1-Control Efficiency).

Total Uncontrolled/Controlled = Uncontrolled/Controlled for each unit * 2

The flaker aspiration baghouse is integral to the process, so the potential to emit will be determined after control.

Appendix A: Emissions Calculations
326 IAC 6-3-2 Compliance
Units Less than 30 Tons per Hour Maximum Process Weight

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

Number of Units	Emission unit ID	Emissions Units	Baghouse ID	¹ Uncontrolled PM PTE (lb/hr) for each unit	¹ Controlled PM PTE (lb/hr) for each unit	¹ Maximum Process Weight (tons/hr) for each unit	326 IAC 6-3 Limit (lb/hr) for each unit
1	G160000	Pellet Hulls Conveyor to Loadout	AF-1	1.04	0.01	17.0	27.36
1	B030900	Hull Collection Conveyor	AF-3	1.04	0.01	17.0	27.36
1	B430000	Secondary Hull Collection Conveyor	AF-3	1.04	0.01	17.0	27.36
1	E070300	4 Hour Hull Tank	AF-3	0.43	0.003	17.0	N/A ²
1	B440000	Secondary Hull Collection L-Path	AF-3	1.04	0.01	17.0	27.36
1	E080000	Pellet Cooler	AF-3	25.50	0.15	17.0	27.36
3	E050000, E050200, and E050100	³ Hull Hammer Mill, Hull Hammer Mill Feeder, and Hull Hammer Mill Plenum	AF-3	34.00	0.20	17.0	27.36
1	G050100	Pelleted Hulls Leg	AF-3	1.04	0.01	17.0	27.36
1	G050300	Pelleted Hulls Storage Conveyor	AF-1	1.04	0.01	17.0	27.36
1	E050400	Hulls Addition Screw	AF-3	1.04	0.01	17.0	27.36
2	C040000 and C070000	⁴ Flaking Rolls No. 4 and 7	AF-4	0.13	0.13	22.9	N/A ²
2	C100000 and C010600	⁴ Flaking Rolls No. 1 and 10	AF-4	0.22	0.22	22.9	N/A ³
1		Hull Overflow Tank	Hull Bin Filter	0.19	0.19	17.0	N/A ²

Notes:

- ¹ Process Weight Rates and PTEs are found on the PM summary page.
- ² The PM PTE from these units is less than 0.551 lbs/hr; therefore, they are not subject to 326 IAC 6-3-2.
- ³ The control devices shall be in operation and control emissions from these emission units, at all times they are in operation in order to comply with 326 IAC 6-3-2.
- ⁴ The control devices are integral and shall be in operation and control emissions from these emission units, at all times they are in operation in order to comply with 326 IAC 6-3-2.

Pursuant to 326 IAC 6-3-2, the particulate emissions limitations from the above table shall be calculated using the following equations:

Interpolation of the data for the process weight rate up to sixty thousand (60,000) pounds per hour shall be accomplished by use of the equation:

$$E = 4.10 P^{0.67}$$

Where:

E = Rate of emission in pounds per hour.

P = Process weight rate in tons per hour.

Appendix A: Emissions Calculations
326 IAC 6-3-2 Compliance
Units Greater than 30 Tons per Hour Maximum Process Weight

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

Number of Units	Emission unit ID	Emissions Units	Baghouse ID	¹ Uncontrolled PM PTE (lb/hr) for each unit	¹ Controlled PM PTE (lb/hr) for each unit	¹ Maximum Process Weight (tons/hr) for each unit	326 IAC 6-3 Limit (lb/hr) for each unit
2	G080000 and G180000	Truck and Rail Pelleted Hull Loadout Bins	AF-1	0.51	0.03	148	N/A ³
2	G130000 and G070000	² Rail Meal Loadout Bin and Truck Meal Loadout Bin	AF-1	7.50	0.41	300	63.00
5	G010000, G020000, G030000, G040000, and G050000	Meal Bin No. 1 thru 5 Vent Filters	MBF-1, MBF-2, MBF-3, MBF-4, and MBF-5	4.95	0.01	198	58.40
1	G220000	Rail Car Loadout (Pellets/Hulls)	AF-1	1.03	0.01	330.00	64.09
1	G020500	Meal Storage Feed Conveyor	AF-1	12.20	0.07	200.00	58.51
3	G070300, G170000 and G290000	Truck Meal Loadout Feed Conveyor, Rail Car Collection Conveyor and Truck Collection Conveyor	AF-1	54.90	0.31	300.00	63.00
1	G280000 and G270000	Truck Loader No.1	AF-1	169.29	0.95	330.00	64.09
2	G010100 and G010200	Meal Reclaim Conveyor and Meal Reclaim Leg	AF-1	24.40	0.14	200.00	58.51
1	A060000	Screener	AF-3	16.10	0.10	264	61.56
1	B011300	Bean Weigh Scale	AF-3	16.10	0.10	264	61.56
4	B011200, A160300, A060400 and B030800	VSC Feed Leg, VSC Leg Feed Conveyor, Screener Feed Conveyor and Conditioned Bean Feed Conveyor	AF-3	16.10	0.10	264	61.56
2	B010100 and B020100	Whole Bean Aspiration No. 1 and No. 2	AF-3	16.10	0.10	264	61.56
1	B010300	Conditioner Bean Loop Path	AF-3	26.40	0.16	264	61.56
2	B120000 and B030000	Jet Dryer No. 1 and No. 2	Jet Dryer Cyclones No. 1A, 1B, 2A, and 2B	29.04	0.29	132	54.11
2	B010500 and B020500	VSC Air Heater No. 1 and No. 2	VSC Cyclones	29.04	1.41	264	61.56
2	B010000 and B020000	Vertical Seed Conditioner (VSC) No. 1 and No. 2	VSC Cyclones	13.20	0.64	132	54.11
4	B040000, B080100, B130000, and B170000	² Hulloosensor No. 1, No. 2, No. 3, and No. 4	AF-5	237.60	1.95	66	47.20
4	B050000, B090000, B140000, and B180000	Cascade Dryer No. 1, No. 2, No. 3 and No. 4	AF-5	14.52	0.12	66	47.20
4	B060000, B100000, B150000, and B190000	² Cracking Roll No.1, No. 2, No. 3 and No. 4	AF-5	237.60	1.95	66	47.20
4	B070000, B110000, B160000, and B200000	Cascade Conditioner No. 1, No. 2, No. 3 and No. 4	AF-5	6.60	0.05	66	47.20
1	A160000, A160500 and B420000	Day Tank (with Aspirator and Cyclone)	AF-5	6.60	0.05	264	61.56
2	C200100 and C010600	² Flaker Feed Loop Conveyor and Flake Collection Conveyor	AF-4	14.30	0.13	247	60.82
4	E020300, E020400, E010100, and E010300	Grinding Discharge Conveyor, Hammer Mill Mixing Conveyor, Meal L-Path Conveyor, and Meal Hammer Mill Feed Conveyor	AF-6	12.08	0.06	198	58.40
2	E230200 and E230000	² Meal Hammer Mill Feeder No. 5, Meal Hammer Mill No. 5	AF-6	503.20	2.52	74	48.30
1	G010300	Meal Leg	AF-6	12.08	0.06	198	58.40
1	G150000	Meal Conveyor to Loadout	AF-1	12.08	0.07	198	58.40
1	E230100	Meal Hammer Mill Bin No. 5	AF-6	1.85	0.01	74	48.30
4	D310000-1, D310000-2, D310000-3, and D310000-4	² DC Decks No. 1, No. 2, No. 3, and No. 4	DC Deck Cyclones No. 1 through 4	374.40	7.11	208	58.93
2		Bean Storage Bins No. 4 and No. 8	None	6.51	6.51	600	71.16
1	Truck Dump No. 3	Truck Dump No. 3	AF-2	61.56	0.34	360	65.09
3	Rail Receiving Leg, Truck Dump No. 3 Receiving Conveyor & Truck Dump No. 3 Receiving Leg	Rail Receiving Leg, Truck Dump No. 3 Receiving Conveyor & Truck Dump No. 3 Receiving Leg	AF-2	21.96	0.12	360	65.09
4	Scalperator Leg, Scalperator Feed Conveyor, Scalperator Jack Leg, Scalperator	Scalperator Leg, Scalperator Feed Conveyor, Scalperator Jack Leg, Scalperator	AF-2	12.81	0.07	210	59.04

Notes:¹ Process Weight Rates and PTEs are found on the PM summary page.² The control devices shall be in operation and control emissions from these emission units, at all times they are in operation in order to comply with 326 IAC 6-3-2.³ The PM PTE from these units is less than 0.551 lbs/hr; therefore, they are not subject to 326 IAC 6-3-2.

Pursuant to 326 IAC 6-3-2, the particulate emissions limitations from the above table shall be calculated using the following equation:

Interpolation and extrapolation of the data for the process weight rate in excess of sixty thousand (60,000) pounds per hour shall be accomplished by use of the equation:

$$E = 55.0 P^{0.11} - 40$$

Where:

E = Rate of emission in pounds per hour.

P = Process weight rate in tons per hour.

Pursuant to 326 IAC 6-3-2(e) (Particulate Emission Limitations for Manufacturing Processes), when the process weight rate exceeds two hundred (200) tons per hour, the allowable emissions may exceed that shown in the table in 326 IAC 6-3-2(e) provided the concentration of particulate in the discharge gases to the atmosphere is less than one tenth (0.10) pound per one thousand (1,000) pounds of gases.

Notes:

- 1) Emission factors based SCC codes that start with 3-2-0-0-0 were taken from Table 9.9.1 in AP-42 section 9.1 (02/2003).
- 2) Emission factors based SCC codes that start with 3-2-0-0-7 were taken from Table 11.1.1 in AP-42 section 11.1 (11/1999).
- 3) Emission factors based SCC codes that start with 3-2-0-0-8 were taken from Table 9.9.1 in AP-42 section 9.1 (02/2003).
- 4) Emission factors based SCC codes that start with 3-2-0-0-9 were taken from Table 11.1.2 in AP-42 section 11.2 (02/2004).

2) To obtain the sourcewise uncorrected PTE for a process, the number of units of that process is multiplied by uncorrected PTE for a single unit. The uncorrected PTE for a single unit of that process is shown in more detail on subsequent pages.

3) Capture efficiency at incinerator(s) is 99%, therefore, it is estimated that 5% of uncorrected emissions are uncaptured.

4) The source area of this process is shown in the table.

5) Uncaptured captured emissions from Recycling and Landfill were reduced to 95% due to assumption that they are uncaptured.

6) The uncorrected PTE for the process is calculated by multiplying the uncorrected PTE for a single unit by the number of units of that process.

7) PM_{10} emission factor not available. Pursuant to AP-42, Table 9.1-3, PM_{10} emission factor is scaled to total particulate using the ratio of 25 percent.

8) PM_{10} emission factor not available. Pursuant to AP-42, Table 9.1-3, PM_{10} emission factor can be estimated by taking 50 percent of the $PM_{2.5}$ emission factor.

9) The uncorrected PTE for the process is calculated by multiplying the uncorrected PTE for a single unit by the number of units of that process.

10) The maximum annual limited rate of these units would be the permitted credit rate of 2,555,836 tons per year plus 150,000 tons per year for the standing storage capacity of the plant (for storage silos and temporary storage). This equivalent to the maximum annual limited rate for material moved and limited to the permitted credit rate of 2,555,836 tons per year $\times$ 88% $\times$ 1,541,031 tons/year.

11) The maximum annual limited rate for gasses is the permitted credit rate of 2,555,836 tons per year $\times$ 106.47 tons/year.

12) This emission unit is aspired and controlled via an aspired device down stream.

13) All captured emissions exhaust through stack A-2 and are limited to 1.64 tbs/hr at PM_{10} . All uncaptured limits are shown above.

14) The uncorrected PTE for the process is calculated by multiplying the uncorrected PTE for a single unit by the number of units of that process. Therefore, assuming efficiency is reduced to 99% tons per year but gives the same as a rate.

15) There are five feed lines. However, the plant is only physically capable of loading one feed line at a time. Thus, the PTE for these units is calculated at a rate of 198 tons/hr for all five bins combined. These emissions units do not have a rate.

16) Pursuant to Minor Source Modification (MSM) No. 08-2487-0-0-0, in order to render the requirements of 326 (21.42-1.5) not applicable, the Permittee shall comply with an soybean throughput to the two (2) Storage Bean Piles #1 and #2.

17) The uncorrected fugitive emissions from Truck Load and 2nd Air Fall Capture (Pellets/Meal) and Rail Fall Pelleted (No Load) bins, Rail and Truck Meal Load bins and the units are reacting to Stack A-2 and are not regulated.

Appendix A: Emission Calculations
PM Summary
Stacks AF-3, AF-7, and S-1
Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-30728-00102
Permit Reviewer: Daniel W. Peil

Filter / Stack ID	Control / Stack Description	Emission unit ID	Number of Units	Process	Notes	Unlimited Rate		Limited Rate		Emission Factors (lb/ton)			Aspiration Rate (cfm)	Control Efficiency		Before Control				After Control				Emissions Based on Limited Soybeans Processed			Emissions Based on All Limits			Emission Factor Source			
						tons/hr	tons/yr	tons/yr	PM	PM ₁₀	PM _{2.5}	PM		PM ₁₀	lb/hr	tons/yr	lb/hr	tons/yr	lb/hr	tons/yr	lb/hr	tons/yr	PM	PM ₁₀	PM _{2.5}	PM	PM ₁₀	PM _{2.5}	PM		PM ₁₀	PM _{2.5}	
AF-3	Prep Exhaust Filter, Baghouse AF-3, 26,300 acfm 99.41% PM10/PM2.5 control	A060000	1	Screener		264	2,312,640	2,251.836	0.061	0.034	0.0058	5,000	99.41%	99.41%	16.10	70.54	8.98	39.31	6.71	0.10	0.42	0.05	0.23	0.04	68.68	38.28	6.53			SCC 3-02-005-30			
		A170000	1	Screening Tank		5.0	43,800	N/A	0.028	0.0051	0.0011	500	99.41%	99.41%	0.13	0.55	0.03	0.14	0.02	0.00	0.00	0.00	0.00	0.00	0.55	0.14	0.02			SCC 3-02-005-40			
		A370000	1	Screening Device Lids		5.0	43,800	N/A	0.061	0.034	0.0058	500	99.41%	99.41%	0.31	1.34	0.17	0.74	0.13	0.06	0.01	0.00	0.00	0.00	1.34	0.74	0.13			SCC 3-02-005-30			
		B011300	1	Bean Weight Scale	3	264	2,312,640	2,251.836	0.061	0.034	0.0058	500	99.41%	99.41%	16.10	70.54	8.98	39.31	6.71	0.10	0.42	0.05	0.23	0.04	68.68	38.28	6.53			SCC 3-02-005-30			
		B012000, A160000, A090400 and B020000	4	VSC Feed Leg, VSC Leg Feed Conveyor, Screener Feed Conveyor and Conditioned Bean Feed Conveyors		264	2,312,640	2,251.836	0.061	0.034	0.0058	500	99.41%	99.41%	64.42	282.14	35.90	157.26	26.83	0.38	1.66	0.21	0.93	0.16	68.68	38.28	6.53			SCC 3-02-005-30			
		B010100 and B020100	2	Whole Bean Aspiration No. 1 and No. 2		264	2,312,640	2,251.836	0.061	0.034	0.0058	2,000	99.41%	99.41%	32.21	141.07	17.95	78.63	13.41	0.19	0.83	0.11	0.46	0.08	68.68	38.28	6.53			SCC 3-02-005-30			
		B203000	1	Hull Collection Conveyor		17.0	148,920	N/A	0.061	0.034	0.0058	500	99.41%	99.41%	1.04	4.54	0.58	2.53	0.43	0.03	0.03	0.00	0.03	0.00	4.54	2.53	0.43			SCC 3-02-005-30			
		E070000 and E140000	2	Hull Screener No. 1 and No. 2	10	9.6	84,096	N/A	0.061	0.034	0.0058	N/A	99.41%	99.41%	1.17	5.13	0.65	2.86	0.49	0.01	0.03	0.00	0.03	0.00	5.13	2.86	0.49			SCC 3-02-005-30			
		B430000	1	Secondary Hull Collection Conveyor		17.0	148,920	N/A	0.061	0.034	0.0058	3,000	99.41%	99.41%	1.04	4.54	0.58	2.53	0.43	0.03	0.03	0.00	0.03	0.00	4.54	2.53	0.43			SCC 3-02-005-30			
		E070000	1	4 Hour Hull Tank		17.0	148,920	N/A	0.028	0.0051	0.0011	1,500	99.41%	99.41%	0.43	1.89	0.11	0.47	0.08	0.00	0.01	0.00	0.00	0.00	1.89	0.47	0.08			SCC 3-02-005-40			
		B440000	1	Secondary Hull Collection L-Path		17.0	148,920	N/A	0.061	0.034	0.0058	N/A	99.41%	99.41%	1.04	4.54	0.58	2.53	0.43	0.01	0.03	0.00	0.01	0.00	4.54	2.53	0.43			SCC 3-02-005-30			
		E060000	4	Roller Cooler	6, 7	17.0	148,920	N/A	1.9	0.75	0.75	6,500	99.41%	99.41%	25.50	111.69	12.75	55.85	95.85	0.18	0.69	0.08	0.33	0.35	111.69	55.85	95.85			SCC 3-02-009-16			
		C050000, C060000, and E050100	3	Hull Hammer Mill, Hull Hammer Mill Feeder, and Hull Hammer Mill Plenum	5, 7	17.0	148,920	N/A	2	1.0	1.0	8,000	99.41%	99.41%	102.00	446.76	51.00	223.38	223.38	0.65	2.64	0.30	1.32	1.32	446.76	223.38	223.38			SCC 3-02-007-86			
		E050100	1	Hammer Mills Lids		17.0	148,920	N/A	0.061	0.034	0.0058	1,000	99.41%	99.41%	1.04	4.54	0.58	2.53	0.43	0.03	0.03	0.00	0.03	0.00	4.54	2.53	0.43			SCC 3-02-005-30			
		E060400	1	Hulls Addition Screw		17.0	148,920	N/A	0.061	0.034	0.0058	N/A	99.41%	99.41%	1.04	4.54	0.58	2.53	0.43	0.01	0.03	0.00	0.01	0.00	4.54	2.53	0.43			SCC 3-02-005-30			
		B130000	1	Screening Weighit Belt		5.0	43,800	N/A	0.061	0.034	0.0058	500	99.41%	99.41%	0.31	1.34	0.17	0.74	0.13	0.06	0.01	0.00	0.00	0.00	1.34	0.74	0.13			SCC 3-02-005-30			
		B010000	1	Conditioner Bean Loop Path	6, 11	264	2,312,640	no limit	0.10	0.025	0.0025	42,000	99.41%	99.41%	26.40	115.63	6.60	28.91	28.91	0.16	0.68	0.04	0.17	0.17	N/A	N/A	N/A			SCC 3-02-007-87			
AF-3 Total																296.25	1,271.29	146.18	646.37	347.14	1.11	7.50	0.16	3.74	2.13	866.09	448.96	308.35	5.52	5.52	5.52		
AF-7		Pod Grinder/Screener Baghouse, AF-7, 5000 acfm, 99.00% PM10/PM2.5 control	B310200	1	Pod Grinder/Desolator	6	5.0	43,800	N/A	2.000	0.500	0.500	5,000	99.00%	99.00%	10.00	43.80	2.50	10.95	10.95	0.10	0.44	0.03	0.11	0.11	43.80	10.95	10.95	1.5 lb/hr (6.57 tons/yr)	1.5 lb/hr (6.57 tons/yr)	1.5 lb/hr (6.57 tons/yr)	SCC 3-02-007-86	
	AF-7 Total																10.00	43.80	2.50	10.95	10.95	0.10	0.44	0.03	0.11	0.11	43.80	10.95	10.95	6.57	6.57	6.57	
S-1	Jet Dryer Baghouses 99.00% PM10/PM2.5 control	B120000 and B030000	2	Jet Dryer No. 1 and No. 2		11	132	1,156,320	no limit	0.22	0.055	0.0094	36,000	99.00%	99.00%	58.08	254.39	14.52	63.60	10.87	0.58	2.54	0.15	0.64	0.11	N/A	N/A	N/A			SCC 3-02-005-27		
		S-1, S-6 Total																10.00	43.80	2.50	10.95	10.95	0.10	0.44	0.03	0.11	0.11	43.80	10.95	10.95	6.57	6.57	6.57
S-1, S-6	VSC Cyclones No. 1 st & No. 2 95.14% PM10 control, 80.57% PM2.5 control	B010500 and B020500	2	VSC Air Heater No. 1 and No. 2		11,13	264	2,312,640	no limit	0.22	0.055	0.0094	42,000	95.14%	80.57%	58.08	254.39	14.52	63.60	10.87	2.82	12.36	2.82	12.36	2.11	N/A	N/A	N/A	4.93 lb/hr (21.59 tons/yr)	3.35 lb/hr (14.87 tons/yr)	3.35 lb/hr (14.87 tons/yr)	SCC 3-02-005-27	
		S-1, S-6 Total																10.00	43.80	2.50	10.95	10.95	0.10	0.44	0.03	0.11	0.11	43.80	10.95	10.95	6.57	6.57	6.57
S-1, S-6	Vertical Seed Conditioner (VSC) No. 1 and No. 2	B010000 and B020000	2	Vertical Seed Conditioner (VSC) No. 1 and No. 2		6,11	132	1,156,320	no limit	0.10	0.025	0.0025	42,000	95.14%	80.57%	26.40	115.63	6.60	28.91	28.91	1.28	5.62	1.28	5.62	5.62	N/A	N/A	N/A			SCC 3-02-007-87		
		S-1 and S-6 Total																132.56	524.41	35.64	156.19	156.19	4.98	20.53	4.25	18.61	7.84	5.00	5.00	5.00	21.68	14.62	14.62
Pair 2 Subtotal (tons/yr)																442.81	1,895.70	191.82	796.37	415.44	6.40	28.03	5.11	22.39	9.99	856.09	448.96	308.35	31.65	25.26	25.26		

Notes:
Emission factors based SCC Codes that start with 3-02-005 were taken from Table 9.9.1-1 in AP-42 section 9.9.1 (03/2003).
Emission factors based SCC Codes that start with 3-02-007 were taken from Table 9.11.1-1 in AP-42 section 9.11.1 (11/1995).
Emission factors based SCC Codes that start with 3-02-008 were taken from Table 9.9.2-1 in AP-42 section 9.9.1 (03/2003).
Emission factors based SCC Codes that start with 3-05-038 were taken from Table 11.19.2-4 in AP-42 section 11.19.2 (08/2004).
1) To obtain the sourcewide uncontrolled PTE for a process, the number of units of that process is multiplied by uncontrolled PTE for a single unit. The Uncontrolled PTE for a single unit of that process is shown in more detail on subsequent pages.
2) Capture efficiency at receiving/loadout is 95%; therefore, it is estimated that 5% of uncontrolled emissions are uncaptured.
3) The source utilized oil suppression.
4) Uncaptured emissions from Receiving and Loadout were reduced to 95% due to assumption that 5% are uncaptured.
5) Emission factor adjusted to uncontrolled back calculating from a control efficiency of 90% for cyclones and 60% for fabric filters.
6) PM_{2.5} emission factor not available. Pursuant to AP-42, Table 9.9.1-1 Note 1, PM_{2.5} emission factor scaled to total particulate using the ratio of 25 percent.
7) PM₁₀ emission factor not available. Pursuant to AP-42, Table 9.9.1-2 Note 1, PM₁₀ emission factors can be estimated by taking 50 percent of the filterable PM emission factor.
8) PM_{2.5} emission factor provided when available in AP-42. For all others, the assumption is that PM_{2.5} = PM₁₀. Control Efficiency for PM_{2.5} was assumed to be equal to control efficiency for PM₁₀ unless a control efficiency for PM_{2.5} was not provided.
9) The amount of soybeans processed shall be limited to less than 2,251,836 tons per twelve (12) consecutive month period with compliance determined at the end of each month. If the maximum throughput is less than the limited rate, then "N/A" was listed.
10) This emission unit is aspirated and controlled via an aspirated device down stream.
11) These emission units do not have a limited throughput.
12) The outlet PM emission rate was 0.88 lb/hr and the outlet PM_{2.5} emission rate was 1.05 lb/hr from stack testing of the VSC Cyclone No. 1.
13) Maximum Throughput design rate for each heater is 264 tons per hour. However, the throughput rate is limited by the maximum design capacity of the milling operations (264 tons/hr total). Therefore the PTE calculations are based on a combined throughput of 264tons/hr through both VSC Air Heater No. 1 and No. 2.

Notes:

- 1) Emission factors based on SCC Codes that start with 3-02-005 were taken from Table 9.9.1.1 in AP-42 section 9.1.1 (10/2005).
- 2) Emission factors based on SCC Codes that start with 3-02-007 were taken from Table 9.1.1.1 in AP-42 section 9.1.1 (10/2005).
- 3) Emission factors based on SCC Codes that start with 3-02-008 were taken from Table 9.9.1.2 in AP-42 section 9.1.1 (10/2005).
- 4) Emission factors based on SCC Codes that start with 3-05-008 were taken from Table 11.2.2.4 in AP-42 section 11.2 (8/2004).

1) To obtain the sourcewide uncontrolled PM for a process, the number of rates is multiplied by uncontrolled PM for a single unit. The Uncontrolled PMTE for a single unit of that process is shown more in detail on subsequent pages.

2) Cadmium emission at nonferrous metal refineries is given. It is estimated that 5% of cadmium is uncontrolled.

3) This source utilizes suppression.

4) Emission based on emissions from Recycling and Leadacid were reduced to 95% due to compliance that 5% are uncontrolled.

5) Emission factor adjusted to nonferrous metal calculating from lead content of 95% of cyclones and 90% for fabric filters.

6) PM_{10} emission factor based on $PM_{2.5}$ emission factor scaled to total particulates using the ratio of 25 percent.

7) PM_{10} emission factor not available. Pursuant to AP-42, Table 9.1-2.1 Note 1, PM_{10} emission factors can be estimated by taking 50 percent of the 25 percent $PM_{2.5}$ emission factor.

8) PM_{10} emission factor provided when available in AP-42, Table 9.1-2.1 Note 1, otherwise, the assumption is the PM_{10} = $PM_{2.5}$ Control Efficiency for PM_{10} was assumed to be equal to control efficiency for $PM_{2.5}$ as a control efficiency for PM_{10} was not provided.

9) Emission factor of cyclones that range from 2,255.388 to 2,255.388 lb per million Btu was assumed to be 100 percent compliance determined at the end of the compliance period.

10) Pursuant to Administrative Memorandum #E-2794-02, baghouse #F-4 was determined to be an integral part of the flaring process for Flaring Rule 10, and Line 12.

11) Pursuant to Administrative Memorandum #E-2794-02, baghouse #F-5 was determined to be an integral part of the flaring process for Flaring process, and Line 13.

12) Pursuant to Significant Source Modification (OSR-31260-02), baghouse #F-4 was determined to be an integral part of the flaring process for Flaring Rule 3, and Line 7.

13) Pursuant to Significant Source Modification (OSR-32048-02), baghouse #F-5 was determined to be an integral part of the flaring process for Flaring Rule 3, and Line 7.

14) This emission unit is segregated and controlled as an segregated device under stream.

Appendix A: Emission Calculations
PM Summary
Stack S-2, Insignificant Activities, and Fugitives
Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-30728-00102
Permit Reviewer: Daniel W. Pell

Filter / Stack ID	Control / Stack Description	Emission unit ID	Number of Units	Process	Notes	Unlimited Rate		Limited Rate	Emission Factors (lb/ton)			Aspiration Rate (cm)	Control Efficiency		Before Control					After Control					Emissions Based on Limited Soybeans Processed			Limits			Emission Factor Source	
						tons/hr	tons/yr	tons/yr	PM	PM ₁₀	PM _{2.5}		PM	PM ₁₀	PM _{2.5}	PM		PM ₁₀		PM _{2.5}		PM	PM ₁₀	PM _{2.5}	PM	PM ₁₀	PM _{2.5}					
																lb/hr	tons/yr	lb/hr	tons/yr	lb/hr	tons/yr							lb/hr	tons/yr	lb/hr		tons/yr
S-2	DYDC Stack, DC Deck Cyclones, 18,000 acfm, 99.18% PM control, and 99.66% PM10 control	D310000-1, D310000-2, D310000-3, and D310000-4	4	DC Decks No. 1, No. 2, No. 3, and No. 4	5, 6, 10	208	1,822,080	no limit	1.8	0.5	0.5	63,900	98.10%	97.89%	1497.60	6559.49	374.40	1639.87	1639.87	28.45	124.63	7.90	34.60	34.60	N/A	N/A	N/A	10.74 lb/hr (47.04 tons/yr)	7.28 lb/hr (31.89 tons/yr)	7.28 lb/hr (31.89 tons/yr)	SCC 3-02-007-89	
S-2 Total																1,497.60	6,559.49	374.40	1,639.87	1,639.87	28.45	124.63	7.90	34.60	34.60	0.00	0.00	0.00	47.04	31.89	31.89	
	Bin Filter		1	Kaolin Receiving Tank	5, 10, 11	40	7,000	no limit	0.99	0.16	0.06	750	99.50%	99.50%	39.60	3.47	6.40	0.56	0.21	0.20	0.02	0.03	0.00	0.00	N/A	N/A	N/A	1.9 lb/hr (8.32 tons/yr)	1.9 lb/hr (8.32 tons/yr)	1.9 lb/hr (8.32 tons/yr)	SCC 3-05-038-13	
	Hull Bin Filter		1	Hull Overflow Tank	10, 12	17	67,263	no limit	0.025	0.0063	0.0011	1,000	0.00%	0.00%	0.19	0.84	0.05	0.21	0.04	0.19	0.84	0.05	0.21	0.04	0.84	0.21	0.04	0.84	0.21	0.04	SCC 3-02-005-40	
			1	Bean Storage Bin No. 2, 3, 4, 6, 7, and 8 and Bean Storage Silo No. 1 and 5	13	600	2,431,836	no limit	0.025	0.0063	0.0011	348,000	N/A	N/A	15.00	30.40	3.78	7.66	1.34	N/A	N/A	N/A	N/A	30.40	7.66	1.34	30.40	7.66	1.34	SCC 3-02-005-40		
	Filter		1	Diatomaceous Earth (DE) Storage Bin	10, 14	0.0875	767	no limit	0.99	0.16	0.06	Not Provided	99.50%	99.50%	0.09	0.38	0.01	0.06	0.02	0.00	0.00	0.00	0.00	N/A	N/A	N/A	0.38	0.06	0.02	SCC 3-05-038-13		
Insignificant Activities Total																54.88	38.08	10.24	8.49	3.61	0.39	0.86	0.08	0.23	0.64	31.24	7.87	1.37	39.44	16.26	9.72	
Road Traffic Cooling Towers																7.69	18.35	0.14	5.24	0.16	3.89	16.16	0.14	5.24	0.16	N/A	N/A	N/A	16.16	5.24	0.16	AP 42 Table 13.2.1
Fugitives Total																4.69	20.52	1.73	7.58	5.14	4.69	20.52	1.73	7.58	5.14	N/A	N/A	N/A	20.52	7.58	5.14	
Pages 3 Subtotal (tons/yr) Including Fugitives																1,567.36	6,615.10	386.37	1,695.94	1,646.62	33.53	146.01	9.71	42.39	39.78	31.24	7.87	1.37	107.51	86.72	46.74	
Pages 3 through 4 Total (tons/yr) including Fugitives																7,248.73	31,950.35	2,149.59	9,378.85	7,065.25	82.73	361.51	26.99	118.07	92.02	2,269.87	1,007.88	472.57	234.30	148.64	131.65	

Notes:

Emission factors based SCC Codes that start with 3-02-005 were taken from Table 9.9.1.1 in AP-42 section 9.9.1 (03/2003).

Emission factors based SCC Codes that start with 3-02-007 were taken from Table 9.11.1.1 in AP-42 section 9.11.1 (11/1999).

Emission factors based SCC Codes that start with 3-02-008 were taken from Table 9.9.1.2 in AP-42 section 9.9.1 (03/2003).

Emission factors based SCC Codes that start with 3-05-038 were taken from Table 11.19.2.4 in AP-42 section 11.19.2 (08/2004).

1) To obtain the sourcewide uncontrolled PTE for a process, the number of units of that process is multiplied by uncontrolled PTE for a single unit. The Uncontrolled PTE for a single unit of that process is shown in more detail on subsequent pages.

2) Capture efficiency at receiving/loadout is 99%; therefore, it is estimated that 5% of uncontrolled emission are uncaptured.

3) This source utilizes oil suppression.

4) Uncontrolled captured emissions from Receiving and Loadout were reduced to 95% due to assumption that 5% are uncaptured.

5) Emission factor adjusted to uncontrolled back calculating from a control efficiency of 90% for cyclones and 99% for fabric filters.

6) PM₁₀ emission factor not available. Pursuant to AP-42, Table 9.9.1.1 note b, PM₁₀ emission factor scaled to total particulate using the ratio of 25 percent.

7) PM_{2.5} emission factor not available. Pursuant to AP-42, Table 9.9.1.2 note b, PM_{2.5} emission factor can be estimated by taking 50 percent of the filterable PM emission factor.

8) PM_{2.5} emission factor provided when available in AP-42. For all others, the assumption is that PM_{2.5} = PM₁₀. Control Efficiency for PM_{2.5} was assumed to be equal to control efficiency for PM₁₀ since a control efficiency for PM_{2.5} was not provided.

9) Road Traffic provided by the source in T085-21297-00102, issued on January 24, 2006.

10) These emission units do not have a limited throughput.

11) The hourly capacity of 40 tons/hr was not used to calculate the maximum PTE for the Kaolin Receiving Tank. The quantity of kaolin used to calculate the PTE (that would correspond to the permitted crush rate of 2,251,836 tons per year) would be 7,000 tons.

12) The hourly capacity of 17 tons/hr was not used to calculate the maximum PTE for the Hull Overflow Tank. The quantity of hull used to calculate the PTE (that would correspond to the permitted crush rate of 2,251,836 tons per year) would be 67,263 tons.

13) The hourly capacity of 600 tons/hr of the Bean Storage Bins was not used to calculate the maximum PTE. The maximum annual throughput of the bean storage bins would be the permitted crush rate of 2,251,836 tons per year plus 180,000 tons per year for the standing storage capacity of the plant (all storage silos and temporary storage). This equates to a maximum throughput of 2,431,836 tons per year which is used in calculating the PTE.

14) Diatomaceous Earth (DE) Storage Bin has an hourly usage rate of 175 lb/hr and a storage capacity of 40 tons.

Appendix A: Emission Calculations
Truck Dumps No. 1 and No. 2 and Rail Dump

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

PTE for emission units A030000 and A020000
Truck Dumps No. 1 and No. 2

PM Emission Factor	0.180	lb/ton	Source: SCC 3-02-005-51
PM10/PM2.5 Emission Factor	0.059	lb/ton	SCC 3-02-005-51
Hourly Loading Rate	600	tons/hr	
Limited Loading Rate	2,401,836	tons/yr	
Aspiration Rate	6000	cfm	
% Control PM	99.44%		
% Control PM10/PM2.5	99.44%		

Hourly throughput is based on maximum capacity of transfer system, limited throughput is based on the limited amount of soybeans processed pursuant to T085-21297-00102 issued on January 24, 2006.

Potential PM emissions

Truck Dumps No. 1 and No. 2 (captured)	Uncontrolled for each unit		Controlled for each unit	
Max Hourly	102.60	lbs/hr	0.57	lbs/hr
Max Yearly	449.39	tons/yr	2.52	tons/yr
Limited Yearly	205.36	tons/yr	1.15	tons/yr

Potential PM10/PM2.5 emissions

Truck Dumps No. 1 and No. 2 (captured)	Uncontrolled for each unit		Controlled for each unit	
Max Hourly	33.63	lbs/hr	0.19	lbs/hr
Max Yearly	147.30	tons/yr	0.82	tons/yr
Limited Yearly	67.31	tons/yr	0.38	tons/yr

Potential PM emissions

Truck Dumps No. 1 and No. 2 (uncaptured)	Uncontrolled for each unit	
Max Hourly	5.40	lbs/hr
Max Yearly	23.65	tons/yr
Limited Yearly	10.81	tons/yr

Potential PM10/PM2.5 emissions

Truck Dumps No. 1 and No. 2 (uncaptured)	Uncontrolled for each unit	
Max Hourly	1.77	lbs/hr
Max Yearly	7.75	tons/yr
Limited Yearly	3.54	tons/yr

Notes:

The emission factors are from AP 42, Table 9.9.1-1 Particulate Emission Factors for Grain Elevators (3/03).

The emission factors are from worst case which is straight truck although the source receives about 90% delivery by hopper trucks and about 10% delivery by straight trucks.

Uncontrolled captured emissions were reduced to 95% due to assumption that 5% are uncaptured.

Control efficiency for PM is 99.44% and for PM10 is 99.44% according to the renewal application.

Methodology:

Uncontrolled Emissions (tons/yr) = Throughput (tons/yr) * Emission factor (lb/ton) ÷ 2000 lbs/ton

Limited Emissions (tons/yr) = Limited Throughput (tons/yr) * Emission factor (lb/ton) ÷ 2000 lbs/ton

Controlled Potential Emissions (tons/yr) = Throughput (tons/yr) * Emission factor (lb/ton) / 2000 lbs/ton * (1- Control Efficiency)

Appendix A: Emission Calculations
Discharge Conveyors, Bean Receiving Legs, and Bin Feed Conveyors

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

PTE for emission units A030100, A020100, A040000, A050000, A130100 and A100100

Discharge Conveyors No. 1 and No. 2, Bean Receiving Legs No. 1 and No. 2, and East and West Bin Feed Conveyors

PM Emission Factor	0.061	lb/ton	Source: SCC 3-02-005-30
PM10/PM2.5 Emission Factor	0.034	lb/ton	SCC 3-02-005-30
Hourly Loading Rate	600.0	tons/hr	
Limited Loading Rate	1,686,300	tons/yr	
Aspiration Rate	5000	cfm	
% Control PM	99.44%		
% Control PM10/PM2.5	99.44%		

Hourly throughput is based on maximum capacity of transfer system, limited throughput is based on the limited amount of soybeans processed pursuant to T085-21297-00102 issued on January 24, 2006.

Potential PM emissions

Discharge Conveyors No. 1 and No. 2, Bean Receiving Legs No. 1 and No. 2, and East and West Bin Feed Conveyors

	Uncontrolled for each unit		Controlled for each unit	
Max Hourly	36.60	lbs/hr	0.20	lbs/hr
Max Yearly	160.31	tons/yr	0.90	tons/yr
Limited Yearly	51.43	tons/yr	0.29	tons/yr

Potential PM10/PM2.5 emissions

Discharge Conveyors No. 1 and No. 2, Bean Receiving Legs No. 1 and No. 2, and East and West Bin Feed Conveyors

	Uncontrolled for each unit		Controlled for each unit	
Max Hourly	20.40	lbs/hr	0.11	lbs/hr
Max Yearly	89.35	tons/yr	0.50	tons/yr
Limited Yearly	28.67	tons/yr	0.16	tons/yr

Notes:

Emission factors are from AP 42 Table 9.9.1-1 Particulate Emission Factors for Grain Elevators (3/03)

Control efficiency for PM is 99.44% and for PM10 is 99.44% according to the renewal application.

Methodology:

Uncontrolled Emissions (ton/yr) = Throughput (ton/yr)* Emission factor (lb/ton) / 2000 (lbs/ton)

Limited Emissions (ton/yr) = Limited throughput (ton/yr)* Emission factor (lb/ton) / 2000 (lbs/ton)

Controlled Potential Emissions (ton/yr) = Throughput (ton/yr) * Emission factor (lb/ton) / 2000 (lbs/ton)* (1-Control Efficiency).

Appendix A: Emission Calculations**Cross Bins, Day Bin Leg, Rail Collection and Discharge Conveyors, Discharge Bins, and Bin Cross Conveyors**

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

PTE for emission unit A150100 and A120100
Cross Bins No 1 thru 6

PM Emission Factor	0.061	lb/ton	Source: SCC 3-02-005-30
PM10 /PM2.5Emission Factor	0.034	lb/ton	SCC 3-02-005-30
Hourly Loading Rate	600	tons/hr	Notes:
Limited Loading Rate	1,686,300	tons/yr	* This source utilizes oil suppression
Aspiration Rate	1000	cfm	
% Control PM	99.44%		
% Control PM10/PM2.5	99.44%		

Hourly throughput is based on maximum capacity of transfer system, limited throughput is based on the limited amount of soybeans processed pursuant to T085-21297-00102 issued on January 24, 2006.

Potential PM emissions			
Cross Bins No 1 thru 6	Uncontrolled for each unit		Controlled for each unit
Max Hourly	36.60	lbs/hr	0.20 lbs/hr
Max Yearly	160.31	tons/yr	0.90 tons/yr
Limited Yearly	51.43	tons/yr	0.29 tons/yr

Potential PM10/PM2.5 emissions			
Cross Bins No 1 thru 6	Uncontrolled for each unit		Controlled for each unit
Max Hourly	20.40	lbs/hr	0.11 lbs/hr
Max Yearly	89.35	tons/yr	0.50 tons/yr
Limited Yearly	28.67	tons/yr	0.16 tons/yr

PTE for emission units A153000, A010100, A151000, A121000, A152000 and A122000

Day Bin Leg, Rail Scale Discharge Conveyor, Discharge Bin No 1 thru 6, West Bin Cross Conveyor 1-3, and East Bin Cross Conveyor 4-6

PM Emission Factor	0.061	lb/ton	Source: SCC 3-02-005-30
PM10/PM2.5 Emission Factor	0.034	lb/ton	SCC 3-02-005-30
Hourly Loading Rate	360	tons/hr	* This source utilizes oil suppression
Limited Loading Rate	1,686,300	tons/yr	
Aspiration Rate	2,500	cfm	
% Control PM	99.44%		
% Control PM10/PM2.5	99.44%		

Hourly throughput is based on maximum capacity of transfer system, limited throughput is based on the limited amount of soybeans processed pursuant to T085-21297-00102 issued on January 24, 2006.

Potential PM emissions			
Day Bin Leg, Rail Scale Discharge Conveyor, Discharge Bin No 1 thru 6, West Bin Cross Conveyor 1-3, and East Bin Cross Conveyor 4-6	Uncontrolled for each unit		Controlled for each unit
Max Hourly	21.96	lbs/hr	0.12 lbs/hr
Max Yearly	96.18	tons/yr	0.54 tons/yr
Limited Yearly	51.43	tons/yr	0.29 tons/yr

Potential PM10/PM2.5 emissions			
Day Bin Leg, Rail Scale Discharge Conveyor, Discharge Bin No 1 thru 6, West Bin Cross Conveyor 1-3, and East Bin Cross Conveyor 4-6	Uncontrolled for each unit		Controlled for each unit
Max Hourly	12.24	lbs/hr	0.07 lbs/hr
Max Yearly	53.61	tons/yr	0.30 tons/yr
Limited Yearly	28.67	tons/yr	0.16 tons/yr

Notes:

Emission factors are from AP 42 Table 9.9.1-1 Particulate Emission Factors for Grain Elevators (3/03)
Control efficiency for PM is 99.44% and for PM10 is 99.44% according to the renewal application.

Methodology:

Uncontrolled Emissions (ton/yr) = Throughput (ton/yr)* Emission factor (lb/ton) / 2000 (lbs/ton)
Limited Emissions (ton/yr) = Limited throughput (ton/yr)* Emission factor (lb/ton) / 2000 (lbs/ton)
Controlled Potential Emissions (ton/yr) = Throughput (ton/yr) * Emission factor (lb/ton) / 2000 (lbs/ton)* (1-Control Efficiency).

Appendix A: Emission Calculations
Truck Loader No.1 and No. 2 (captured)

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

PTE for emission units G280000 and G270000
Truck Loader No. 1 and No. 2 (captured)

PM Emission Factor	0.2700	lb/ton	Source: SCC 3-02-007-91
PM10/PM2.5 Emission Factor	0.0675	lb/ton	SCC 3-02-007-91
Hourly Loading Rate	330	tons/hr	
Aspiration Rate	3000	cfm	
Limited Loading Rate	1,686,300	tons/yr	
% Control PM	99.44%		
% Control PM10/PM2.5	99.44%		

Hourly throughput is based on maximum capacity of transfer system, limited throughput is based on the limited amount of soybeans processed pursuant to T085-21297-00102 issued on January 24, 2006.

Potential PM emissions

Truck Loadout No. 1 and No. 2 (captured)

	Uncontrolled for each unit		Controlled for each unit	
Max Hourly	84.65	lbs/hr	0.47	lbs/hr
Max Yearly	370.75	tons/yr	2.08	tons/yr
Limited Yearly	216.27	tons/yr	1.21	tons/yr

Potential PM10/PM2.5 emissions

Truck Loadout No. 1 and No. 2 (captured)

	Uncontrolled for each unit		Controlled for each unit	
Max Hourly	21.16	lbs/hr	0.12	lbs/hr
Max Yearly	92.69	tons/yr	0.52	tons/yr
Limited Yearly	54.07	tons/yr	0.30	tons/yr

Potential PM emissions from

Truck Loadout No. 1 and No. 2 (uncaptured)

	Uncontrolled for each unit	
Max Hourly	4.46	lbs/hr
Max Yearly	19.51	tons/yr
Limited Yearly	11.38	tons/yr

Potential PM10/PM2.5 emissions

Truck Loadout No. 1 and No. 2 (uncaptured)

	Uncontrolled for each unit	
Max Hourly	1.11	lbs/hr
Max Yearly	4.88	tons/yr
Limited Yearly	2.85	tons/yr

Notes

60% of the Pellets/Hulls are shipped via rail.

Emission factors are from AP 42 Table 9.11.1-1, Total Particulate Emission Factors for Soybean Milling (11-95)

Capture efficiency at receiving/loadout is 95%. Therefore 5% of the emissions from receiving/loadout are fugitive emissions.

Control efficiency for PM is 99.44% and for PM10 is 99.44% according to the renewal application.

Methodology

Uncontrolled Emissions (ton/yr) = Throughput (ton/yr)* Emission factor (lb/ton) / 2000 (lbs/ton)

Controlled Potential Emissions (ton/yr) = Throughput (ton/yr) * Emission factor (lb/ton) / 2000 (lbs/ton)* (1-Control Efficiency).

Limited Emissions (ton/yr) = Limited throughput (ton/yr)* Emission factor (lb/ton) / 2000 (lbs/ton)

Appendix A: Emission Calculations
Rail Car Loadout (Pellets/Hulls)

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

PTE for emission units G220000

Rail Car Loadout (Pellets/Hulls)

PM Emission Factor	0.0033	lb/ton	Source: SCC-3-02-008-03
PM10/PM2.5 Emission Factor	0.0008	lb/ton	SCC-3-02-008-03
Hourly Loading Rate	330	tons/hr	
Aspiration Rate	10000	cfm	
Limited Loading Rate	1,686,300	tons/yr	
% Control PM	99.44%		
% Control PM10/PM2.5	99.44%		

Hourly throughput is based on maximum capacity of transfer system, limited throughput is based on the limited amount of soybeans processed pursuant to T085-21297-00102 issued on January 24, 2006.

Potential PM emissions			
Rail Car Loadout (Pellets/Hulls) (captured)	Uncontrolled for each unit		Controlled for each unit
Max Hourly	1.03	lbs/hr	0.01 lbs/hr
Max Yearly	4.53	tons/yr	0.03 tons/yr
Limited Yearly	2.64	tons/yr	0.01 tons/yr

Potential PM10/PM2.5 emissions			
Rail Car Loadout (Pellets/Hulls) (captured)	Uncontrolled for each unit		Controlled for each unit
Max Hourly	0.25	lbs/hr	0.00 lbs/hr
Max Yearly	1.10	tons/yr	0.01 tons/yr
Limited Yearly	0.64	tons/yr	0.00 tons/yr

Potential PM emissions			
Rail Car Loadout (Pellets/Hulls) (uncaptured)	Uncontrolled for each unit		
Max Hourly	0.05	lbs/hr	
Max Yearly	0.24	tons/yr	
Limited Yearly	0.14	tons/yr	

Potential PM10/PM2.5 emissions			
Rail Car Loadout (Pellets/Hulls) (uncaptured)	Uncontrolled for each unit		
Max Hourly	0.01	lbs/hr	
Max Yearly	0.06	tons/yr	
Limited Yearly	0.03	tons/yr	

Appendix A: Emission Calculations
Truck and Rail Meal Loadout Bins

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

PTE for emission unit G130000 and G070000

Rail Meal Loadout Bin and Truck Meal Loadout Bin

PM Emission Factor	0.0250	lb/ton	Source: SCC 3-02-005-40		
PM10 Emission Factor	0.0063	lb/ton	SCC 3-02-005-40		
PM2.5 Emission Factor	0.0011	lb/ton	SCC 3-02-005-40		
Maximum Loading Rate	300	tons/hr			
Limited Loading Rate	1,914,061	tons/yr			
Outlet Grain Loading	0.005	gr/acf	% Control PM	99.44%	
Aspiration Rate	1000	cfm	% Control PM10/PM2.5	99.44%	

Hourly throughput is based on maximum capacity of transfer system, limited throughput is based on the limited amount of soybeans processed pursuant to T085-21297-00102 issued on January 24, 2006.

Potential PM emissions

Rail Meal Loadout Bin (captured) and Truck Meal Loadout Bin (captured)			Uncontrolled for each unit			Controlled for each unit		
	Maximum Potential	7.13	lbs/hr			0.04	lbs/hr	
	Maximum Potential	31.21	tons/yr			0.17	tons/yr	
	Limited Potential	22.73	tons/yr			0.13	tons/yr	

Potential PM10 emissions

Rail Meal Loadout Bin (captured) and Truck Meal Loadout Bin (captured)			Uncontrolled for each unit			Controlled for each unit		
	Maximum Potential	1.78	lbs/hr			0.010	lbs/hr	
	Maximum Potential	7.80	tons/yr			0.044	tons/yr	
	Limited Potential	5.68	tons/yr			0.032	tons/yr	

Potential PM2.5 emissions

Rail Meal Loadout Bin (captured) and Truck Meal Loadout Bin (captured)			Uncontrolled for each unit			Controlled for each unit		
	Maximum Potential	0.30	lbs/hr			0.30	lbs/hr	
	Maximum Potential	1.33	tons/yr			0.0074	tons/yr	
	Limited Potential	0.97	tons/yr			0.0054	tons/yr	

Potential PM emissions

Rail Meal Loadout Bin (uncaptured) and Truck Meal Loadout Bin (uncaptured)			Uncontrolled for each unit			Controlled for each unit		
	Maximum Potential	0.38	lbs/hr					
	Maximum Potential	1.64	tons/yr					
	Limited Potential	1.20	tons/yr					

Potential PM10 emissions

Rail Meal Loadout Bin (uncaptured) and Truck Meal Loadout Bin (uncaptured)			Uncontrolled for each unit			Controlled for each unit		
	Maximum Potential	0.09	lbs/hr					
	Maximum Potential	0.41	tons/yr					
	Limited Potential	0.30	tons/yr					

Potential PM2.5 emissions

Rail Meal Loadout Bin (uncaptured) and Truck Meal Loadout Bin (uncaptured)			Uncontrolled for each unit			Controlled for each unit		
	Maximum Potential	0.02	lbs/hr					
	Maximum Potential	0.07	tons/yr					
	Limited Potential	0.05	tons/yr					

Limited Emissions

Rail Meal Loadout Bin (captured) and Truck Meal Loadout Bin (captured)			Uncontrolled for each unit			Controlled for each unit		
	Limited Controlled	1.430	lb/hr					
	Limited Controlled	6.26	tons/yr					

Notes

40% of the Pellets/Hulls are shipped via rail.
Emission factors are from AP 42 Table 9.9.1-1 Particulate Emission Factors for Grain Elevators (3/03)
Capture efficiency at receiving/loadout is 95%. Therefore 5% of the emissions from receiving/loadout are fugitive emissions.
Control efficiency for PM is 99.44% and for PM10 is 99.44% according to the renewal application.

Methodology

Uncontrolled Emissions (tons/yr) = Throughput (tons/yr) * Emission factor (lb/ton) ÷ 2000 lbs/ton
Controlled Potential Emissions (tons/yr) = Throughput (tons/yr) * Emission factor (lb/ton) ÷ 2000 lbs/ton * (1- Control Efficiency)
Limited Potential (tons/yr) = Limited Throughput (tons/yr) * Emission Factor (lb/ton) ÷ 2000 lbs/ton

Appendix A: Emission Calculations
Truck and Rail Pelleted Hull Loadout Bins, Meal Storage Feed Conveyor,
Truck Meal Loadout Feed Conveyor, and Truck Collection Conveyor

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

PTE for emission units G080000 and G180000
Truck and Rail Pelleted Hull Loadout Bins

PM Emission Factor	0.003	lb/ton	Source: SCC-3-02-008-03
PM10/PM2.5 Emission Factor	0.001	lb/ton	SCC-3-02-008-03
Hourly Loading Rate	148.0	tons/hr	
Aspiration Rate	1000	cfm	
Limited Loading Rate	1,686,300	tons/yr	
% Control PM	99.44%		
% Control PM10/PM2.5	99.44%		

Hourly throughput is based on maximum capacity of transfer system, limited throughput is based on the limited amount of soybeans processed pursuant to T085-21297-00102 issued on January 24, 2006.

Potential PM emissions					
Truck and Rail Pelleted Hull Loadout Bins		Uncontrolled for each unit		Controlled for each unit	
Max Hourly	=	0.49	lbs/hr	0.00	lbs/hr
Max Yearly	=	2.14	tons/yr	0.01	tons/yr
Limited Yearly	=	2.78	tons/yr	0.02	tons/yr

Potential PM10/PM2.5 emissions					
Truck and Rail Pelleted Hull Loadout Bins		Uncontrolled for each unit		Controlled for each unit	
Max Hourly	=	0.12	lbs/hr	0.00	lbs/hr
Max Yearly	=	0.52	tons/yr	0.00	tons/yr
Limited Yearly	=	0.67	tons/yr	0.00	tons/yr

PTE for emission unit G020500
Meal Storage Feed Conveyor

PM Emission Factor	0.061	lb/ton	Source: SCC 3-02-005-30
PM10/PM2.5 Emission Factor	0.034	lb/ton	SCC 3-02-005-30
Hourly Loading Rate	200	tons/hr	
Aspiration Rate	500	cfm	
Limited Loading Rate	1,686,300	tons/yr	
% Control PM	99.44%		
% Control PM10/PM2.5	99.44%		

Hourly throughput is based on maximum capacity of transfer system, limited throughput is based on the limited amount of soybeans processed pursuant to T085-21297-00102 issued on January 24, 2006.

Potential PM emissions					
Meal Storage Feed Conveyor		Uncontrolled for each unit		Controlled for each unit	
Max Hourly	=	12.20	lbs/hr	0.07	lbs/hr
Max Yearly	=	53.44	tons/yr	0.30	tons/yr
Limited Yearly	=	51.43	tons/yr	0.29	tons/yr

Potential PM10/PM2.5 emissions					
Meal Storage Feed Conveyor		Uncontrolled for each unit		Controlled for each unit	
Max Hourly	=	6.80	lbs/hr	0.04	lbs/hr
Max Yearly	=	29.78	tons/yr	0.17	tons/yr
Limited Yearly	=	28.67	tons/yr	0.16	tons/yr

PTE for emission units G070300, G170000 and G290000
Truck Meal Loadout Feed Conveyor, and Truck Collection Conveyor

PM Emission Factor	0.061	lb/ton	Source: SCC 3-02-005-30
PM10/PM2.5 Emission Factor	0.034	lb/ton	SCC 3-02-005-30
Hourly Loading Rate	300	tons/hr	
Aspiration Rate	1500	cfm	
Limited Loading Rate	1,686,300	tons/yr	
% Control PM	99.44%		
% Control PM10/PM2.5	99.44%		

Hourly throughput is based on maximum capacity of transfer system, limited throughput is based on the limited amount of soybeans processed pursuant to T085-21297-00102 issued on January 24, 2006.

Potential PM emissions					
Truck Meal Loadout Feed Conveyor, and Truck Collection Conveyor		Uncontrolled for each unit		Controlled for each unit	
Max Hourly	=	18.30	lbs/hr	0.10	lbs/hr
Max Yearly	=	80.15	tons/yr	0.45	tons/yr
Limited Yearly	=	51.43	tons/yr	0.29	tons/yr

Potential PM10/PM2.5 emissions					
Truck Meal Loadout Feed Conveyor, and Truck Collection Conveyor		Uncontrolled for each unit		Controlled for each unit	
Max Hourly	=	10.20	lbs/hr	0.06	lbs/hr
Max Yearly	=	44.68	tons/yr	0.25	tons/yr
Limited Yearly	=	28.67	tons/yr	0.16	tons/yr

Methodology

Emission factors are from AP 42 Table 9.9.1-1 Particulate Emission Factors for Grain Elevators (3/03)
Potential Emissions (ton/yr) = Throughput (ton/yr)* Emission factor (lb/ton) / 2000 (lbs/ton)
Controlled Potential Emissions (ton/yr) = Throughput (ton/yr) * Emission factor (lb/ton) / 2000 (lbs/ton)* (1-Control Efficiency).
Limited Emissions (ton/yr) = Limited throughput (ton/yr)* Emission factor (lb/ton) / 2000 (lbs/ton)
Control efficiency for PM is 99.44% and for PM10 is 99.44% according to the renewal application.

Appendix A: Emission Calculations
Pellet Hulls Conveyor to Loadout, Meal Reclaim Conveyor and Meal Reclaim Leg

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

PTE for emission unit G160000

Pellet Hulls Conveyor to Loadout

PM Emission Factor	0.061	lb/ton	Source: SCC 3-02-005-30
PM10/PM2.5 Emission Factor	0.034	lb/ton	SCC 3-02-005-30
Hourly Loading Rate	17.0	tons/hr	
Aspiration Rate	500	cfm	
Limited Loading Rate	1,686,300	tons/yr	
% Control PM	99.44%		
% Control PM10/PM2.5	99.44%		

Hourly throughput is based on maximum capacity of transfer system.

Potential PM emissions					
Pellet Hulls Conveyor to Loadout		Uncontrolled for each unit		Controlled for each unit	
Max Hourly	=	1.04	lbs/hr	0.01	lbs/hr
Max Yearly	=	4.54	tons/yr	0.03	tons/yr

Potential PM10/PM2.5 emissions					
Pellet Hulls Conveyor to Loadout		Uncontrolled for each unit		Controlled for each unit	
Max Hourly	=	0.58	lbs/hr	0.00	lbs/hr
Max Yearly	=	2.53	tons/yr	0.01	tons/yr

PTE for emission units G010100 and G010200
Meal Reclaim Conveyor and Meal Reclaim Leg

PM Emission Factor	0.061	lb/ton	Source: SCC 3-02-005-30
PM10/PM2.5 Emission Factor	0.034	lb/ton	SCC 3-02-005-30
Hourly Loading Rate	200	tons/hr	
Aspiration Rate	1000	cfm	
Limited Loading Rate	1,686,300	tons/yr	
% Control PM	99.44%		
% Control PM10/PM2.5	99.44%		

Hourly throughput is based on maximum capacity of transfer system, limited throughput is based on the limited amount of soybeans processed pursuant to T085-21297-00102 issued on January 24, 2006.

Potential PM emissions					
Meal Reclaim Conveyor and Meal Reclaim Leg		Uncontrolled for each unit		Controlled for each unit	
Max Hourly	=	12.20	lbs/hr	0.07	lbs/hr
Max Yearly	=	53.44	tons/yr	0.30	tons/yr
Limited Yearly	=	51.43	tons/yr	0.29	tons/yr

Potential PM10/PM2.5 emissions					
Meal Reclaim Conveyor and Meal Reclaim Leg		Uncontrolled for each unit		Controlled for each unit	
Max Hourly					
Max Yearly	=	6.80	lbs/hr	0.04	lbs/hr
Limited Yearly	=	29.78	tons/yr	0.17	tons/yr
Limited Yearly	=	28.67	tons/yr	0.16	tons/yr

Notes:

Emission factors are from AP-42, Table 9.9.1-1 (Particulate Emission Factors for Grain Elevators (3/03)).
Control efficiency for PM is 99.44% and for PM10 is 99.44% according to Operating Permit Renewal No. T085-29197-00102.

Methodology:

Uncontrolled Emissions (tons/yr) = Throughput (tons/yr) * Emission factor (lb/ton) ÷ 2000 lbs/ton
Controlled Potential Emissions (tons/yr) = Throughput (tons/yr) * Emission factor (lb/ton) ÷ 2000 lbs/ton * (1 - Control Efficiency)
Limited Emissions (tons/yr) = Limited Throughput (tons/yr) * Emission factor (lb/ton) ÷ 2000 lbs/ton

Appendix A: Emission Calculations
Meal Bin No. 1 thru 5 Vent Filters

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

PTE for emission units G010000, G020000, G030000, G040000, G050000

¹Meal Bin No. 1 thru 5 Vent Filters

PM Emission Factor	0.025	lb/ton	Source: SCC 3-02-005-40
PM10/PM2.5 Emission Factor	0.006	lb/ton	SCC 3-02-005-40
Hourly Loading Rate	198	tons/hr	
Aspiration Rate	1000	cfm	
% Control PM	99.82%		
% Control PM10/PM2.5	99.82%		

Hourly throughput is based on maximum capacity of transfer system.

Potential PM emissions					
Meal Bin No. 1 thru 5 Vent Filters		Uncontrolled for each unit		Controlled for each unit	
Max Hourly	=	4.95	lbs/hr	0.01	lbs/hr
Max Yearly	=	21.68	tons/yr	0.04	tons/yr

Potential PM10/PM2.5 emissions					
Meal Bin No. 1 thru 5 Vent Filters		Uncontrolled for each unit		Controlled for each unit	
Max Hourly	=	1.25	lbs/hr	0.00	lbs/hr
Max Yearly	=	5.46	tons/yr	0.01	tons/yr

Notes:

¹ There are five meal bins. However, the plant is only physically capable of loading one meal bin at a time. Therefore, the PTE for these units is calculated at a rate of 198 tons/hr for all five meal bins combined. Emission factors are from AP-42, Table 9.9.1-1 (Particulate Emission Factors for Grain Elevators (3/03)). Control efficiency for PM is 99.82% and for PM10 is 99.82% according to Operating Permit Renewal No. T085-29197-00102.

Methodology:

Uncontrolled Emissions (tons/yr) = Throughput (tons/yr) * Emission factor (lb/ton) ÷ 2000 lbs/ton

Controlled Potential Emissions (tons/yr) = Throughput (tons/yr) * Emission factor (lb/ton) ÷ 2000 lbs/ton * (1 - Control Efficiency)

PTE for emission unit G050300

Pelleted Hulls Storage Conveyor

PM Emission Factor	0.061	lb/ton	Source: SCC 3-02-005-40
PM10/PM2.5 Emission Factor	0.034	lb/ton	SCC 3-02-005-40
Hourly Loading Rate	17	tons/hr	
Aspiration Rate	500	cfm	
% Control PM	99.44%		
% Control PM10/PM2.5	99.44%		

Potential PM emissions					
Pelleted Hulls Storage Conveyor		Uncontrolled for each unit		Controlled for each unit	
Max Hourly	=	1.04	lbs/hr	0.01	lbs/hr
Max Yearly	=	4.54	tons/yr	0.03	tons/yr

Potential PM10/PM2.5 emissions					
Pelleted Hulls Storage Conveyor		Uncontrolled for each unit		Controlled for each unit	
Max Hourly	=	0.58	lbs/hr	3.24E-03	lbs/hr
Max Yearly	=	2.53	tons/yr	0.01	tons/yr

PTE for emission units G150000

Meal Conveyor to Loadout

PM Emission Factor	0.061	lb/ton	Source: SCC 3-02-005-27
PM10/PM2.5 Emission Factor	0.034	lb/ton	SCC 3-02-005-27
Hourly Loading Rate	198	tons/hr	
Aspiration Rate	500	cfm	
% Control PM	99.44%		
% Control PM10/PM2.5	99.44%		

Potential PM emissions					
Meal Conveyor to Loadout		Uncontrolled for each unit		Controlled for each unit	
Max Hourly	=	12.08	lbs/hr	0.07	lbs/hr
Max Yearly	=	52.90	tons/yr	0.30	tons/yr

Potential PM10/PM2.5 emissions					
Meal Conveyor to Loadout		Uncontrolled for each unit		Controlled for each unit	
Max Hourly	=	6.73	lbs/hr	0.04	lbs/hr
Max Yearly	=	29.49	tons/yr	0.17	tons/yr

**Appendix A: Emission Calculations
Storage Piles #1 and #2**

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

PTE for Piles #1 and #2

Covered Seasonal Grain Storage Piles

PM Emission Factor	0.061	lb/ton	Source: SCC 3-02-005-30
PM10/PM2.5 Emission	0.034	lb/ton	SCC 3-02-005-30
Hourly Loading Rate	360	tons/hr	
Limited Rate	8,000,000	bushels per year	

Hourly throughput is based on maximum capacity of transfer system, limited throughput is based on the limited amount of soybeans processed pursuant to 085-24676-00102, issued on April 28, 2008.

Potential PM emissions

Piles #1 and #2		Uncontrolled for each unit
Max Hourly	=	21.96 lbs/hr
Max Yearly	=	96.18 tons/yr
Limited Yearly	=	7.32 tons/yr

Potential PM10/PM2.5 emissions

Piles #1 and #2		Uncontrolled for each unit
Max Hourly	=	12.24 lbs/hr
Max Yearly	=	53.61 tons/yr
Limited Yearly	=	4.08 tons/yr

Notes:

Emission factors are from AP 42, Table 9.9.1-1, Particulate Emission Factors for Grain Elevators (3/03).

Conveyors used with storage piles #1 and #2 have already been accounted for.

See PM Summary tab for complete list of emission units and particulate emissions.

Methodology:

Uncontrolled Emissions (tons/yr) = Throughput (tons/yr) * Emission factor (lb/ton) ÷ 2000 lb/ton

Limited Emissions (tons/yr) = Limited Throughput (bushels/yr) * 60 lbs/bushel ÷ 2,000 lb/ton * Emission factor (lb/ton) ÷ 2000 lb/ton

Controlled Potential Emissions (tons/yr) = Throughput (tons/yr) * Emission factor (lb/ton) ÷ 2000 lb/ton * (1 - Control Efficiency)

**Appendix A: Emission Calculations
Screener and Screenings Tank**

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

**PTE for emission unit A060000
Screener**

PM Emission Factor	0.061	lb/ton	Source: SCC 3-02-005-30
PM10/PM2.5 Emission Factor	0.034	lb/ton	SCC 3-02-005-30
Hourly Loading Rate	264	tons/hr	
Limited Loading Rate	1,686,300	tons/yr	
Aspiration Rate	4400	cfm	
% Control PM	99.41%		
% Control PM10/PM2.5	99.41%		

Hourly throughput is based on maximum capacity of transfer system, limited throughput is based on the limited amount of soybeans processed pursuant to T085-21297-00102 issued on January 24, 2006.

Potential PM emissions

Screener			Uncontrolled for each unit		Controlled for each unit
	Max Hourly	=	16.10	lbs/hr	0.10 lbs/hr
	Max Yearly	=	70.54	tons/yr	0.42 tons/yr
	Limited Yearly	=	51.43	tons/yr	0.30 tons/yr

Potential PM10/PM2.5 emissions

Screener			Uncontrolled for each unit		Controlled for each unit
	Max Hourly	=	8.98	lbs/hr	0.05 lbs/hr
	Max Yearly	=	39.31	tons/yr	0.23 tons/yr
	Limited Yearly	=	28.67	tons/yr	0.17 tons/yr

PTE for emission unit A170000

Screenings Tank

PM Emission Factor	0.025	lb/ton	Source: SCC 3-02-005-40
PM10/PM2.5 Emission Factor	0.0063	lb/ton	SCC 3-02-005-40
Hourly Loading Rate	5.0	tons/hr	Loadings are based on assumption of 2.5% of raw beans will be lost
Aspiration Rate	500	cfm	
% Control PM	99.41%		
% Control PM10/PM2.5	99.41%		

Potential PM emissions

Screenings Tank			Uncontrolled for each unit		Controlled for each unit
	Max Hourly	=	0.13	lbs/hr	0.00 lbs/hr
	Max Yearly	=	0.55	tons/yr	0.00 tons/yr

Potential PM10/PM2.5 emissions

Screenings Tank			Uncontrolled for each unit		Controlled for each unit
	Max Hourly	=	0.03	lbs/hr	0.00 lbs/hr
	Max Yearly	=	0.14	tons/yr	0.00 tons/yr

Notes:

Emission factors are from AP-42, Table 9.9.1-1 (Particulate Emission Factors for Grain Elevators (3/03)).
Control efficiency for PM is 99.41% and for PM10 is 99.41% according to Operating Permit Renewal No. T085-29197-00102.

Methodology:

Uncontrolled Emissions (tons/yr) = Throughput (tons/yr) * Emission factor (lb/ton) ÷ 2000 lbs/ton
Controlled Potential Emissions (tons/yr) = Throughput (tons/yr) * Emission factor (lb/ton) ÷ 2000 (lbs/ton) * (1 - Control Efficiency)

Appendix A: Emission Calculations
Screenings Recycle Leg and Bean Weigh Scale

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell
#REF! #REF!

PTE for emission unit A170300
Screenings Recycle Leg

PM Emission Factor	0.061	lb/ton	Source: SCC 3-02-005-30
PM10/PM2.5 Emission Factor	0.0340	lb/ton	SCC 3-02-005-30
Hourly Loading Rate	5.0	tons/hr	Loadings are based on assumption of 2.5% of raw
Aspiration Rate	500	cfm	beans will be lost
% Control PM	99.41%		
% Control PM10/PM2.5	99.41%		

Potential PM emissions					
Screenings Recycle Leg			Uncontrolled for each unit		Controlled for each unit
	Max Hourly	=	0.31	lbs/hr	0.00 lbs/hr
	Max Yearly	=	1.34	tons/yr	0.01 tons/yr

Potential PM10/PM2.5 emissions					
Screenings Recycle Leg			Uncontrolled for each unit		Controlled for each unit
	Max Hourly	=	0.17	lbs/hr	0.00 lbs/hr
	Max Yearly	=	0.74	tons/yr	0.00 tons/yr

PTE for emission unit B011300
Bean Weigh Scale

PM Emission Factor	0.061	lb/ton	Source: SCC 3-02-005-30
PM10/PM2.5 Emission Factor	0.0340	lb/ton	SCC 3-02-005-30
Hourly Loading Rate	264	tons/hr	* This source utilizes oil suppression
Aspiration Rate	500	cfm	
% Control PM	99.41%		
% Control PM10/PM2.5	99.41%		

Hourly throughput is based on maximum design capacity of milling operations, limited throughput is based on the limited amount of soybeans processed pursuant to T085-21297-00102 issued on January 24, 2006.

Potential PM emissions					
Bean Weigh Scale			Uncontrolled for each unit		Controlled for each unit
	Max Hourly	=	16.10	lbs/hr	0.10 lbs/hr
	Max Yearly	=	70.54	tons/yr	0.42 tons/yr

Potential PM10/PM2.5 emissions					
Bean Weigh Scale			Uncontrolled for each unit		Controlled for each unit
	Max Hourly	=	8.98	lbs/hr	0.05 lbs/hr
	Max Yearly	=	39.31	tons/yr	0.23 tons/yr

Notes:

Emission factors are from AP-42, Table 9.9.1-1 (Particulate Emission Factors for Grain Elevators (3/03)).
Control efficiency for PM is 99.41% and for PM10 is 99.41% according to Operating Permit Renewal No. T085-29197-00102.

Methodology:

Uncontrolled Emissions (tons/yr) = Throughput (tons/yr)* Emission factor (lb/ton) ÷ 2000 lbs/ton
Controlled Potential Emissions (tons/yr) = Throughput (tons/yr) * Emission factor (lb/ton) ÷ 2000 (lbs/ton) * (1 - Control Efficiency)

Appendix A: Emission Calculations**VSC Feed Leg, VSC Leg Feed Conveyor, Screener Feed Conveyor, Conditioned Bean Feed Conveyor, and Whole Bean Aspiration No. 1 & No. 2**

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
 Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
 Significant Source Modification No.: 085-39532-00102
 Significant Permit Modification No.: 085-39729-00102
 Permit Reviewer: Daniel W. Pell
 #REF! #REF!

PTE for emission units B011200, A160300, A060400, and B030800**VSC Feed Leg, VSC Leg Feed Conveyor, Screener Feed Conveyor and Conditioned Bean Feed Conveyor**

PM Emission Factor	0.061	lb/ton	Source: SCC 3-02-005-30
PM10/PM2.5 Emission Factor	0.034	lb/ton	SCC 3-02-005-30
Hourly Loading Rate	264	tons/hr	* This source utilizes oil suppression
Aspiration Rate	500	cfm	
% Control PM	99.41%		
% Control PM10/PM2.5	99.41%		

Hourly throughput is based on maximum design capacity of milling operations, limited throughput is based on the limited amount of soybeans processed pursuant to T085-21297-00102 issued on January 24, 2006.

Potential PM emissions**VSC Feed Leg, VSC Leg Feed Conveyor, Screener Feed Conveyor and Conditioned Bean Feed Conveyor**

		Uncontrolled for each unit		Controlled for each unit	
Max Hourly	=	16.10	lbs/hr	0.10	lbs/hr
Max Yearly	=	70.54	tons/yr	0.42	tons/yr

Potential PM10/PM2.5 emissions**VSC Feed Leg, VSC Leg Feed Conveyor, Screener Feed Conveyor and Conditioned Bean Feed Conveyor**

		Uncontrolled for each unit		Controlled for each unit	
Max Hourly	=	8.98	lbs/hr	0.05	lbs/hr
Max Yearly	=	39.31	tons/yr	0.23	tons/yr

PTE for emission units B010100 and B020100**Whole Bean Aspiration No. 1 and No. 2**

PM Emission Factor	0.061	lb/ton	Source: SCC 3-02-005-30
PM10/PM2.5 Emission Factor	0.034	lb/ton	SCC 3-02-005-30
Hourly Loading Rate	264	tons/hr	* This source utilizes oil suppression
Aspiration Rate	2000	cfm	
% Control PM	99.41%		
% Control PM10/PM2.5	99.41%		

Hourly throughput is based on maximum design capacity of milling operations, limited throughput is based on the limited amount of soybeans processed pursuant to T085-21297-00102 issued on January 24, 2006.

Potential PM emissions**Whole Bean Aspiration No. 1 and No. 2**

		Uncontrolled for each unit		Controlled for each unit	
Max Hourly	=	16.10	lbs/hr	0.10	lbs/hr
Max Yearly	=	70.54	tons/yr	0.42	tons/yr

Potential PM10/PM2.5 emissions**Whole Bean Aspiration No. 1 and No. 2**

		Uncontrolled for each unit		Controlled for each unit	
Max Hourly	=	8.98	lbs/hr	0.05	lbs/hr
Max Yearly	=	39.31	tons/yr	0.23	tons/yr

Notes:

Emission factors are from AP-42, Table 9.9.1-1 (Particulate Emission Factors for Grain Elevators (3/03)).

Control efficiency for PM is 99.41% and for PM10 is 99.41% according to Operating Permit Renewal No. T085-29197-00102.

Methodology:

Uncontrolled Emissions (tons/yr) = Throughput (tons/yr)* Emission factor (lb/ton) ÷ 2000 lbs/ton

Controlled Potential Emissions (tons/yr) = Throughput (tons/yr) * Emission factor (lb/ton) ÷ 2000 (lbs/ton) * (1 - Control Efficiency)

Appendix A: Emission Calculations
Hull Collection Conveyor and Hull Screener No. 1 and No. 2

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell
#REF! #REF!

PTE for emission unit B030900

Hull Collection Conveyor

PM Emission Factor	0.061	lb/ton	Source: SCC 3-02-005-30
PM10/PM2.5 Emission Factor	0.034	lb/ton	SCC 3-02-005-30
Hourly Loading Rate	17.0	tons/hr	* This source utilizes oil suppression.
Aspiration Rate	500	cfm	
% Control PM	99.41%		
% Control PM10/PM2.5	99.41%		

Potential PM emissions

Hull Collection Conveyor		Uncontrolled for each unit		Controlled for each unit
Max Hourly	=	1.037	lbs/hr	0.006 lbs/hr
Max Yearly	=	4.542	tons/yr	0.027 tons/yr

Potential PM10/PM2.5 emissions

Hull Collection Conveyor		Uncontrolled for each unit		Controlled for each unit
Max Hourly	=	0.578	lbs/hr	0.003 lbs/hr
Max Yearly	=	2.532	tons/yr	0.015 tons/yr

PTE for emission units E130000 and E150000

Hull Screener No. 1 and No. 2

PM Emission Factor	0.061	lb/ton	Source: SCC 3-02-005-30
PM10/PM2.5 Emission Factor	0.034	lb/ton	SCC 3-02-005-30
Hourly Loading Rate	9.6	tons/hr	* The hull screeners are aspirated through the
Aspiration Rate	NA	cfm	Secondary Aspiration.
% Control PM	99.41%		
% Control PM10/PM2.5	99.41%		

Hourly throughput is based on assumption that 5% of the raw beans are hulls.

Potential PM emissions

Hull Screener No. 1 and No. 2		Uncontrolled for each unit		Controlled for each unit
Max Hourly	=	0.59	lbs/hr	0.00 lbs/hr
Max Yearly	=	2.56	tons/yr	0.02 tons/yr

Potential PM10/PM2.5 emissions

Hull Screener No. 1 and No. 2		Uncontrolled for each unit		Controlled for each unit
Max Hourly	=	0.33	lbs/hr	0.00 lbs/hr
Max Yearly	=	1.43	tons/yr	0.01 tons/yr

Notes:

Emission factors are from AP-42, Table 9.9.1-1 (Particulate Emission Factors for Grain Elevators (3/03)).

Control efficiency for PM is 99.41% and for PM10 is 99.41% according to Operating Permit Renewal No. T085-29197-00102.

Methodology:

Uncontrolled Emissions (tons/yr) = Throughput (tons/yr)* Emission factor (lb/ton) ÷ 2000 lbs/ton

Controlled Potential Emissions (tons/yr) = Throughput (tons/yr) * Emission factor (lb/ton) ÷ 2000 (lbs/ton) * (1 - Control Efficiency)

Appendix A: Emission Calculations
Secondary Hull Collection Conveyor and 4 Hour Hull Tank

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell
#REF! #REF!

PTE for emission unit B430000
Secondary Hull Collection Conveyor

PM Emission Factor	0.061	lb/ton	Source: SCC 3-02-005-30
PM10/PM2.5 Emission Factor	0.034	lb/ton	SCC 3-02-005-30
Hourly Loading Rate	17.0	tons/hr	
Aspiration Rate	3000	cfm	
% Control PM	99.41%		
% Control PM10/PM2.5	99.41%		

Potential PM emissions

Secondary Hull Collection Conveyor

		Uncontrolled for each unit		Controlled for each unit	
Max Hourly	=	1.037	lbs/hr	0.006	lbs/hr
Max Yearly	=	4.542	tons/yr	0.027	tons/yr

Potential PM10/PM2.5 emissions

Secondary Hull Collection Conveyor

		Uncontrolled for each unit		Controlled for each unit	
Max Hourly	=	0.578	lbs/hr	0.003	lbs/hr
Max Yearly	=	2.532	tons/yr	0.015	tons/yr

PTE for emission unit E070300
4 Hour Hull Tank

PM Emission Factor	0.025	lb/ton	Source: SCC 3-02-005-40
PM10/PM2.5 Emission Factor	0.0063	lb/ton	SCC 3-02-005-40
Hourly Loading Rate	17.0	tons/hr	
Aspiration Rate	1500	cfm	
% Control PM	99.41%		
% Control PM10/PM2.5	99.41%		

Potential PM emissions

4 Hour Hull Tank

		Uncontrolled for each unit		Controlled for each unit	
Max Hourly	=	0.425	lbs/hr	0.003	lbs/hr
Max Yearly	=	1.862	tons/yr	0.011	tons/yr

Potential PM10/PM2.5 emissions

4 Hour Hull Tank

		Uncontrolled for each unit		Controlled for each unit	
Max Hourly	=	0.107	lbs/hr	0.001	lbs/hr
Max Yearly	=	0.469	tons/yr	0.003	tons/yr

Notes:

Emission factors are from AP-42, Table 9.9.1-1 (Particulate Emission Factors for Grain Elevators (3/03)).

Control efficiency for PM is 99.41% and for PM10 is 99.41% according to Operating Permit Renewal No. T085-29197-00102.

Methodology:

Uncontrolled Emissions (tons/yr) = Throughput (tons/yr) * Emission factor (lb/ton) ÷ 2000 lbs/ton

Controlled Potential Emissions (tons/yr) = Throughput (tons/yr) * Emission factor (lb/ton) ÷ 2000 (lbs/ton) * (1 - Control Efficiency)

Appendix A: Emission Calculations
Secondary Hull Collection L-Path and Pellet Cooler

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell
#REF! #REF!

PTE for emission unit B440000
Secondary Hull Collection L-Path

PM Emission Factor	0.061	lb/ton	Source: SCC 3-02-005-30
PM10/PM2.5 Emission Factor	0.034	lb/ton	SCC 3-02-005-30
Hourly Loading Rate	17.0	tons/hr	
Aspiration Rate	Not Provided	cfm	
% Control PM	99.41%		
% Control PM10/PM2.5	99.41%		

Potential PM emissions
Secondary Hull Collection L-Path

		Uncontrolled for each unit		Controlled for each unit	
Max Hourly	=	1.037	lbs/hr	0.006	lbs/hr
Max Yearly	=	4.542	tons/yr	0.027	tons/yr

Potential PM10/PM2.5 emissions
Secondary Hull Collection L-Path

		Uncontrolled for each unit		Controlled for each unit	
Max Hourly	=	0.578	lbs/hr	0.003	lbs/hr
Max Yearly	=	2.532	tons/yr	0.015	tons/yr

PTE for emission unit E080000
Pellet Cooler

PM Emission Factor	1.5	lb/ton	Source: SCC 3-02-008-16
PM10/PM2.5 Emission Factor	0.75	lb/ton	SCC 3-02-008-16
Hourly Loading Rate	17.0	tons/hr	* Emission Factor was adjusted to uncontrolled
Aspiration Rate	6500	cfm	based on assumption that cyclones provide 90%
% Control PM	99.41%		control.
% Control PM10/PM2.5	99.41%		

Potential PM emissions
Pellet Cooler

		Uncontrolled for each unit		Controlled for each unit	
Max Hourly	=	25.500	lbs/hr	0.150	lbs/hr
Max Yearly	=	111.690	tons/yr	0.659	tons/yr

Potential PM10/PM2.5 emissions
Pellet Cooler

		Uncontrolled for each unit		Controlled for each unit	
Max Hourly	=	12.750	lbs/hr	0.075	lbs/hr
Max Yearly	=	55.845	tons/yr	0.329	tons/yr

Notes:

Emission factors are from AP-42, Table 9.9.1-1 (Particulate Emission Factors for Grain Elevators (3/03)).
Control efficiency for PM is 99.41% and for PM10 is 99.41% according to Operating Permit Renewal No. T085-29197-00102.

Methodology:

Uncontrolled Emissions (tons/yr) = Throughput (tons/yr) * Emission factor (lb/ton) ÷ 2000 lbs/ton
Controlled Potential Emissions (tons/yr) = Throughput (tons/yr) * Emission factor (lb/ton) ÷ 2000 (lbs/ton) * (1 - Control Efficiency)

Appendix A: Emission Calculations
Hull Hammer Mill, Hull Hammer Mill Feeder, Hull Hammer Mill Plenum, and Pelleted
Hulls Leg

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell
#REF! #REF!

PTE for emission units E050000, E050200, and E050100

Hull Hammer Mill, Hull Hammer Mill Feeder, and Hull Hammer Mill Plenum

PM Emission Factor	2.0	lb/ton	Source: SCC 3-02-007-86
PM10/PM2.5 Emission Factor	1.0	lb/ton	SCC 3-02-007-86
Hourly Loading Rate	17.0	tons/hr	* Emission Factor was adjusted to uncontrolled
Aspiration Rate	8000	cfm	based on assumption that cyclones provide 90%
% Control PM	99.41%		control.
% Control PM10/PM2.5	99.41%		

Potential PM emissions

Hull Hammer Mill, Hull Hammer Mill Feeder, and Hull Hammer Mill Plenum

			Uncontrolled for each unit		Controlled for each unit
Max Hourly	=	34.000	lbs/hr	0.201	lbs/hr
Max Yearly	=	148.920	tons/yr	0.879	tons/yr

Potential PM10/PM2.5 emissions

Hull Hammer Mill, Hull Hammer Mill Feeder, and Hull Hammer Mill Plenum

			Uncontrolled for each unit		Controlled for each unit
Max Hourly	=	17.000	lbs/hr	0.100	lbs/hr
Max Yearly	=	74.460	tons/yr	0.439	tons/yr

PTE for emission unit G050100

Pelleted Hulls Leg

PM Emission Factor	0.061	lb/ton	Source: SCC 3-02-005-30
PM10/PM2.5 Emission Factor	0.034	lb/ton	SCC 3-02-005-30
Hourly Loading Rate	17.0	tons/hr	
Aspiration Rate	1000	cfm	
% Control PM	99.41%		
% Control PM10/PM2.5	99.41%		

Potential PM emissions

Pelleted Hulls Leg

			Uncontrolled for each unit		Controlled for each unit
Max Hourly	=	1.037	lbs/hr	0.006	lbs/hr
Max Yearly	=	4.542	tons/yr	0.027	tons/yr

Potential PM10/PM2.5 emissions

Pelleted Hulls Leg

			Uncontrolled for each unit		Controlled for each unit
Max Hourly	=	0.578	lbs/hr	0.003	lbs/hr
Max Yearly	=	2.532	tons/yr	0.015	tons/yr

Notes:

Emission factors are from AP-42, Table 9.9.1-1 (Particulate Emission Factors for Grain Elevators (3/03)).

Control efficiency for PM is 99.41% and for PM10 is 99.41% according to Operating Permit Renewal No. T085-29197-00102.

Methodology:

Uncontrolled Emissions (tons/yr) = Throughput (tons/yr)* Emission factor (lb/ton) ÷ 2000 lbs/ton

Controlled Potential Emissions (tons/yr) = Throughput (tons/yr) * Emission factor (lb/ton) ÷ 2000 (lbs/ton) * (1 - Control Efficiency)

Appendix A: Emission Calculations
Pelleted Hulls Storage Conveyor, Screenings Weight Belt and Conditioner Bean Loop Path

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell
#REF! #REF!

PTE for emission unit E050400

Hulls Addition Screw

PM Emission Factor	0.061	lb/ton	Source: SCC 3-02-005-30
PM10/PM2.5 Emission Factor	0.034	lb/ton	SCC 3-02-005-30
Hourly Loading Rate	17.0	tons/hr	
Aspiration Rate	Not Provided	cfm	
% Control PM	99.41%		
% Control PM10/PM2.5	99.41%		

Potential PM emissions

Hulls Addition Screw

		Uncontrolled for each unit		Controlled for each unit
Max Hourly	=	1.037	lbs/hr	0.006 lbs/hr
Max Yearly	=	4.542	tons/yr	0.027 tons/yr

Potential PM10/PM2.5 emissions

Hulls Addition Screw

		Uncontrolled for each unit		Controlled for each unit
Max Hourly	=	0.578	lbs/hr	0.003 lbs/hr
Max Yearly	=	2.532	tons/yr	0.015 tons/yr

PTE for emission unit B310000

Screenings Weight Belt

PM Emission Factor	0.061	lb/ton	Source: SCC 3-02-007-86
PM10/PM2.5 Emission Factor	0.034	lb/ton	SCC 3-02-007-86
Hourly Loading Rate	5.0	tons/hr	
Aspiration Rate	500	cfm	
% Control PM	99.41%		
% Control PM10/PM2.5	99.41%		

Potential PM emissions

Screenings Weight Belt

		Uncontrolled for each unit		Controlled for each unit
Max Hourly	=	0.305	lbs/hr	0.002 lbs/hr
Max Yearly	=	1.336	tons/yr	0.008 tons/yr

Potential PM10/PM2.5 emissions

Screenings Weight Belt

		Uncontrolled for each unit		Controlled for each unit
Max Hourly	=	0.170	lbs/hr	0.001 lbs/hr
Max Yearly	=	0.745	tons/yr	0.004 tons/yr

PTE for emission unit B010300

Conditioner Bean Loop Path

PM Emission Factor	0.10	lb/ton	Source: SCC 3-02-007-87
PM10/PM2.5 Emission Factor	0.025	lb/ton	SCC 3-02-007-87
Hourly Loading Rate	264	tons/hr	
Aspiration Rate	42,000	cfm	
% Control PM	99.41%		
% Control PM10/PM2.5	99.41%		

Potential PM emissions

Conditioner Bean Loop Path

		Uncontrolled for each unit		Controlled for each unit
Max Hourly	=	26.400	lbs/hr	0.156 lbs/hr
Max Yearly	=	115.632	tons/yr	0.682 tons/yr

Potential PM10/PM2.5 emissions

Conditioner Bean Loop Path

		Uncontrolled for each unit		Controlled for each unit
Max Hourly	=	6.600	lbs/hr	0.039 lbs/hr
Max Yearly	=	28.908	tons/yr	0.171 tons/yr

Notes:

Emission factors are from AP-42, Table 9.9.1-1 (Particulate Emission Factors for Grain Elevators (3/03)).

Control efficiency for PM is 99.41% and for PM10 is 99.41% according to Operating Permit Renewal No. T085-29197-00102.

Methodology:

Uncontrolled Emissions (tons/yr) = Throughput (tons/yr)* Emission factor (lb/ton) ÷ 2000 lbs/ton

Controlled Potential Emissions (tons/yr) = Throughput (tons/yr) * Emission factor (lb/ton) ÷ 2000 (lbs/ton) * (1 - Control Efficiency)

Appendix A: Emission Calculations Pod Grinder/Destoner

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

PTE for emission unit B310200 Pod Grinder/Destoner

PM Emission Factor	2.00	lb/ton	Source: SCC 3-02-007-86
PM10/PM2.5 Emission Factor	0.50	lb/ton	SCC 3-02-007-86
Hourly Loading Rate	5.0	tons/hr	
Aspiration Rate	5000	cfm	
% Control PM	99.00%		
% Control PM10/PM2.5	99.00%		

Potential PM emissions Pod Grinder/Destoner

	Uncontrolled for each unit	Controlled for each unit
Max Hourly	= 10.000 lbs/hr	0.100 lbs/hr
Max Yearly	= 43.800 tons/yr	0.438 tons/yr

Potential PM10/PM2.5 emissions Pod Grinder/Destoner

	Uncontrolled for each unit	Controlled for each unit
Max Hourly	= 2.500 lbs/hr	0.025 lbs/hr
Max Yearly	= 10.950 tons/yr	0.110 tons/yr

Methodology

Emission factors are from AP 42 Table 9.9.1-1 Particulate Emission Factors for Grain Elevators (3/03)

Uncontrolled Emissions (ton/yr) = Throughput (ton/yr)* Emission factor (lb/ton) / 2000 (lbs/ton)

Controlled Potential Emissions (ton/yr) = Throughput (ton/yr) * Emission factor (lb/ton) / 2000 (lbs/ton)* (1-Control Efficiency).

Control efficiency for PM is 99.00% and for PM10 is 99.00% according to the renewal application.

**Appendix A: Emission Calculations
Jet Dryer No. 1 and No. 2**

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

**PTE for emission units B120000 and B030000
Jet Dryer No. 1 and No. 2**

PM Emission Factor	0.22	lb/ton	Source: SCC 3-02-005-27
PM10/PM2.5 Emission Factor	0.055	lb/ton	SCC 3-02-005-27
Hourly Loading Rate	132	tons/hr	
Aspiration Rate	36,000	cfm	
% Control PM	99.00%		
% Control PM10/PM2.5	99.00%		

Hourly throughput is based on maximum design capacity of milling operations.

Potential PM emissions

Jet Dryer No. 1 and No. 2	Uncontrolled for each unit		Controlled for each unit	
Max Hourly	=	29.04 lbs/hr	0.29	lbs/hr
Max Yearly	=	127.20 tons/yr	1.27	tons/yr

Potential PM10/PM2.5 emissions

Jet Dryer No. 1 and No. 2	Uncontrolled for each unit		Controlled for each unit	
Max Hourly	=	7.26 lbs/hr	0.07	lbs/hr
Max Yearly	=	31.80 tons/yr	0.32	tons/yr

Notes:

Emission factors are from AP-42, Table 9.9.1-1 (Particulate Emission Factors for Grain Elevators (3/03)).

Control efficiency for PM is 99.00% and for PM10 is 99.00% according to Operating Permit Renewal No. T085-29197-00102.

Methodology:

Uncontrolled Emissions (tons/yr) = Throughput (tons/yr) * Emission factor (lb/ton) ÷ 2000 lbs/ton

Controlled Potential Emissions (tons/yr) = Throughput (tons/yr) * Emission factor (lb/ton) ÷ 2000 (lbs/ton) * (1 - Control Efficiency)

**Appendix A: Emission Calculations
VSC Air Heaters No. 1 and No. 2**

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

**PTE for emission unit B010500
VSC Air Heaters No. 1 and No. 2**

PM Emission Factor	0.22	lb/ton	Source: SCC 3-02-005-27
PM10/PM2.5 Emission Factor	0.055	lb/ton	SCC 3-02-005-27
*Hourly Loading Rate	264	tons/hr	
Aspiration Rate	42,000	cfm	
% Control PM	95.14%		
% Control PM10/PM2.5	80.57%		

* Maximum rated throughput capacity of each heater is 264 tons/hr. However, hourly throughput is based on maximum design capacity of milling operations.

**Potential PM emissions
VSC Air Heaters No. 1 and No. 2**

	Uncontrolled for each unit		Controlled for each unit	
Max Hourly	=	58.08 lbs/hr	2.82	lbs/hr
Max Yearly	=	254.39 tons/yr	12.36	tons/yr

**Potential PM10/PM2.5 emissions
VSC Air Heaters No. 1 and No. 2**

	Uncontrolled for each unit		Controlled for each unit	
Max Hourly	=	14.52 lbs/hr	2.82	lbs/hr
Max Yearly	=	63.60 tons/yr	12.36	tons/yr

Notes:

Emission factors are from AP-42, Table 9.9.1-1 (Particulate Emission Factors for Grain Elevators (3/03)).
 Control efficiency for PM is 99.00% and for PM10 is 99.00% according to Operating Permit Renewal No. T085-29197-00102.

Methodology:

Uncontrolled Emissions (tons/yr) = Throughput (tons/yr) * Emission factor (lb/ton) ÷ 2000 lbs/ton

Controlled Potential Emissions (tons/yr) = Throughput (tons/yr) * Emission factor (lb/ton) ÷ 2000 (lbs/ton) * (1 - Control Efficiency)

Appendix A: Emission Calculations
Vertical Seed Conditioner (VSC) No. 1 and No. 2

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

PTE for emission units B010000 and B020000
Vertical Seed Conditioner (VSC) No. 1 and No. 2

PM Emission Factor	0.10	lb/ton	Source: SCC 3-02-007-87
PM10/PM2.5 Emission Factor	0.025	lb/ton	SCC 3-02-007-87
Hourly Loading Rate	132	tons/hr	
Aspiration Rate	42,000	cfm	
% Control PM	95.14%		
% Control PM10/PM2.5	80.57%		

Potential PM emissions

Vertical Seed Conditioner (VSC) No. 1 and No. 2

		Uncontrolled for each unit		Controlled for each unit	
Max Hourly	=	13.200	lbs/hr	0.642	lbs/hr
Max Yearly	=	57.816	tons/yr	2.810	tons/yr

Potential PM10/PM2.5 emissions

Vertical Seed Conditioner (VSC) No. 1 and No. 2

		Uncontrolled for each unit		Controlled for each unit	
Max Hourly	=	3.300	lbs/hr	0.641	lbs/hr
Max Yearly	=	14.454	tons/yr	2.808	tons/yr

Notes:

Emission factors are from AP-42, Table 9.9.1-1 (Particulate Emission Factors for Grain Elevators (3/03)).

Control efficiency for PM is 95.14% and for PM10 is 80.57% according to Operating Permit Renewal No. T085-29197-00102.

Methodology:

Uncontrolled Emissions (tons/yr) = Throughput (tons/yr) * Emission factor (lb/ton) ÷ 2000 lbs/ton

Controlled Potential Emissions (tons/yr) = Throughput (tons/yr) * Emission factor (lb/ton) ÷ 2000 (lbs/ton) * (1 - Control Efficiency)

Appendix A: Emission Calculations
Hulloosenators No. 1, No. 2, No. 3, and No. 4 and Cascade
Dryers No. 1, No. 2, No. 3 and No. 4

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

PTE for emission units B040000, B080100, B130000, and B170000
Hulloosenator No. 1, No. 2, No. 3, and No. 4

PM Emission Factor	3.60	lb/ton	Source: SCC 3-02-007-85
PM10/PM2.5 Emission Factor	0.90	lb/ton	SCC 3-02-007-85
Hourly Loading Rate	66	tons/hr	Note: Source is aspirated via the Cascade Dryers
Aspiration Rate	NA	cfm	
% Control PM	99.18%		
% Control PM10/PM2.5	99.18%		

Hourly throughput is based on maximum design capacity of milling operations.

Potential PM emissions

Hulloosenator No. 1, No. 2, No. 3, and No. 4

		Uncontrolled for each unit	Controlled for each unit
Max Hourly	=	237.60 lbs/hr	1.95 lbs/hr
Max Yearly	=	1040.69 tons/yr	8.53 tons/yr

Potential PM10/PM2.5 emissions

Hulloosenator No. 1, No. 2, No. 3, and No. 4

		Uncontrolled for each unit	Controlled for each unit
Max Hourly	=	59.40 lbs/hr	0.49 lbs/hr
Max Yearly	=	260.17 tons/yr	2.13 tons/yr

PTE for emission units B050000, B090000, B140000, and B180000
Cascade Dryer No. 1, No. 2, No. 3, and No. 4

PM Emission Factor	0.22	lb/ton	Source: SCC 3-02-005-27
PM10/PM2.5 Emission Factor	0.055	lb/ton	SCC 3-02-005-27
Hourly Loading Rate	66	tons/hr	
Aspiration Rate	30,000	cfm	
% Control PM	99.18%		
% Control PM10/PM2.5	99.18%		

Potential PM emissions

Cascade Dryer No. 1, No. 2, No. 3, and No. 4

		Uncontrolled for each unit	Controlled for each unit
Max Hourly	=	14.520 lbs/hr	0.119 lbs/hr
Max Yearly	=	63.598 tons/yr	0.522 tons/yr

Potential PM10/PM2.5 emissions

Cascade Dryer No. 1, No. 2, No. 3, and No. 4

		Uncontrolled for each unit	Controlled for each unit
Max Hourly	=	3.630 lbs/hr	0.030 lbs/hr
Max Yearly	=	15.899 tons/yr	0.130 tons/yr

Notes:
Emission factors are from AP-42, Table 9.9.1-1 (Particulate Emission Factors for Grain Elevators (3/03)).
Control efficiency for PM is 99.18% and for PM10 is 99.18% according to Operating Permit Renewal No. T085-29197-00102.

Methodology:

Uncontrolled Emissions (tons/yr) = Throughput (tons/yr) * Emission factor (lb/ton) ÷ 2000 lbs/ton
Controlled Potential Emissions (tons/yr) = Throughput (tons/yr) * Emission factor (lb/ton) ÷ 2000 (lbs/ton) * (1 - Control Efficiency)

Appendix A: Emission Calculations
Cracking Rolls No.1, No. 2, No. 3 and No. 4 and Cascade
Conditioners No. 1, No. 2, No. 3 and No. 4

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

PTE for emission units B060000, B100000, B150000, and B190000

Cracking Roll No. 1, No. 2, No. 3, and No. 4

PM Emission Factor	3.60	lb/ton	Source: SCC 3-02-007-85
PM10/PM2.5 Emission Factor	0.90	lb/ton	SCC 3-02-007-85
Hourly Loading Rate	66	tons/hr	Note: This source is aspirated via the
Aspiration Rate	NA	cfm	Cascade Conditioners
% Control PM	99.18%		
% Control PM10/PM2.5	99.18%		

Potential PM emissions

Cracking Roll No. 1, No. 2, No. 3, and No. 4

		Uncontrolled for each unit		Controlled for each unit
Max Hourly	=	237.600	lbs/hr	1.948
Max Yearly	=	1040.688	tons/yr	8.534

Potential PM10/PM2.5 emissions

Cracking Roll No. 1, No. 2, No. 3, and No. 4

		Uncontrolled for each unit		Controlled for each unit
Max Hourly	=	59.400	lbs/hr	0.487
Max Yearly	=	260.172	tons/yr	2.133

PTE for emission units B070000, B110000, B160000, and B200000

Cascade Conditioner No. 1, No. 2, No. 3, and No. 4

PM Emission Factor	0.100	lb/ton	Source: SCC 3-02-007-87
PM10/PM2.5 Emission Factor	0.025	lb/ton	SCC 3-02-007-87
Hourly Loading Rate	66	tons/hr	
Aspiration Rate	42,600	cfm	
% Control PM	99.18%		
% Control PM10/PM2.5	99.18%		

Potential PM emissions

Cascade Conditioner No. 1, No. 2, No. 3, and No. 4

		Uncontrolled for each unit		Controlled for each unit
Max Hourly	=	6.600	lbs/hr	0.054
Max Yearly	=	28.908	tons/yr	0.237

Potential PM10/PM2.5 emissions

Cascade Conditioner No. 1, No. 2, No. 3, and No. 4

		Uncontrolled for each unit		Controlled for each unit
Max Hourly	=	1.650	lbs/hr	0.014
Max Yearly	=	7.227	tons/yr	0.059

Notes:
Emission factors are from AP-42, Table 9.9.1-1 (Particulate Emission Factors for Grain Elevators (3/03)).
Control efficiency for PM is 99.18% and for PM10 is 99.18% according to Operating Permit Renewal No. T085-29197-00102.

Methodology:

Uncontrolled Emissions (tons/yr) = Throughput (tons/yr) * Emission factor (lb/ton) ÷ 2000 lbs/ton
Controlled Potential Emissions (tons/yr) = Throughput (tons/yr) * Emission factor (lb/ton) ÷ 2000 (lbs/ton) * (1 - Control Efficiency)

Appendix A: Emission Calculations
Secondary Aspirator No 1 and No. 2, Feed Day Tank Conveyor,
and Day Bin Vent

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

PTE for emission units E130100, E150100, and A160100

Secondary Aspirator No. 1 and No. 2, and Feed Day Tank Conveyor

PM Emission Factor	0.061	lb/ton	Source: SCC 3-02-005-30
PM10/PM2.5 Emission Factor	0.034	lb/ton	SCC 3-02-005-30
Hourly Loading Rate	9.6	tons/hr	
Aspiration Rate	Not Provided	cfm	
% Control PM	99.18%		
% Control PM10/PM2.5	99.18%		

Potential PM emissions

Secondary Aspirator No. 1 and No. 2, and Feed Day Tank Conveyor

		Uncontrolled for each unit		Controlled for each unit	
Max Hourly	=	0.586	lbs/hr	0.005	lbs/hr
Max Yearly	=	2.565	tons/yr	0.021	tons/yr

Potential PM10/PM2.5 emissions

Secondary Aspirator No. 1 and No. 2, and Feed Day Tank Conveyor

		Uncontrolled for each unit		Controlled for each unit	
Max Hourly	=	0.326	lbs/hr	0.003	lbs/hr
Max Yearly	=	1.430	tons/yr	0.012	tons/yr

PTE for emission units A160000, A160500 and B420000

Day Tank (with Aspirator and Cyclone)

PM Emission Factor	0.025	lb/ton	Source: SCC 3-02-005-40
PM10/PM2.5 Emission Factor	0.0063	lb/ton	SCC 3-02-005-40
Hourly Loading Rate	264	tons/hr	* This source utilizes oil suppression.
Aspiration Rate	6,000	cfm	
% Control PM	99.18%		
% Control PM10/PM2.5	99.18%		

Hourly throughput is based on maximum design capacity of milling operations, limited throughput is based on the limited amount of soybeans processed pursuant to T085-21297-00102 issued on January 24, 2006.

Potential PM emissions

Day Bin Vent

		Uncontrolled for each unit		Controlled for each unit	
Max Hourly	=	6.60	lbs/hr	0.05	lbs/hr
Max Yearly	=	28.91	tons/yr	0.24	tons/yr

Potential PM10/PM2.5 emissions

Day Bin Vent

		Uncontrolled for each unit		Controlled for each unit	
Max Hourly	=	1.66	lbs/hr	0.01	lbs/hr
Max Yearly	=	7.28	tons/yr	0.06	tons/yr

Notes:

Emission factors are from AP-42, Table 9.9.1-1 (Particulate Emission Factors for Grain Elevators (3/03)).

Control efficiency for PM is 99.18% and for PM10 is 99.18% according to Operating Permit Renewal No. T085-29197-00102.

Methodology:

Uncontrolled Emissions (tons/yr) = Throughput (tons/yr) * Emission factor (lb/ton) ÷ 2000 lbs/ton

Controlled Potential Emissions (tons/yr) = Throughput (tons/yr) * Emission factor (lb/ton) ÷ 2000 (lbs/ton) * (1 - Control Efficiency)

Appendix A: Emission Calculations
Flaking Rolls No. 1 through No. 12 and Conveyors

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

PTE for emission units C010000, C020000, C030000, C040000, C050000, C060000, C070000, C080000, C090000, C100000, C0110000, C0110000, C0120000, C200100, and C010600
Flaking Rolls No. 1 through No. 12 and Conveyors

Stack Test done on April 15-17, 2008:
Air Flow Rate = 16,592 dry standard cubic feet/minute
PM Outlet Grain Loading = 0.0009 grain/dry standard cubic feet
PM₁₀ Outlet Grain Loading = 0.0012 grain/dry standard cubic feet
PM/PM₁₀ Control Efficiency = 99.07%

Controlled PTE					
Hourly PM Emissions	0.13	lb/hr	Hourly PM ₁₀ /PM _{2.5} Emissions	0.17	lb/hr
Yearly PM Emissions	0.56	tons/yr	Yearly PM ₁₀ /PM _{2.5} Emissions	0.75	tons/yr

Uncontrolled PTE (Original Capacity)					
Hourly PM Emissions	13.76	lb/hr	Hourly PM ₁₀ /PM _{2.5} Emissions	18.35	lb/hr
Yearly PM Emissions	60.28	tons/yr	Yearly PM ₁₀ /PM _{2.5} Emissions	80.38	tons/yr

Original Capacity					
8	units *	22.9	tons/hr *	8760	hr/yr = 1,604,832 tons/yr
4	units *	20.3	tons/hr *	8760	hr/yr = 711,312 tons/yr
				TOTAL	2,316,144 tons/yr

Current Capacity (085-31979-00102)					
10	units *	22.90	tons/hr *	8760	hr/yr = 2,006,040 tons/yr
2	units *	20.30	tons/hr *	8760	hr/yr = 355,656 tons/yr
				TOTAL	2,361,696 tons/yr

Uncontrolled PTE (After Increase in Capacity)					
Hourly PM Emissions	14.03	lb/hr	Hourly PM ₁₀ /PM _{2.5} Emissions	18.71	lb/hr
Yearly PM Emissions	61.47	tons/yr	Yearly PM ₁₀ /PM _{2.5} Emissions	81.96	tons/yr

Increase in Capacity (085-32885-00102)					
12	units *	22.90	tons/hr *	8760	hr/yr = 2,407,248 tons/yr
				TOTAL	2,407,248 tons/yr

Uncontrolled PTE (After Increase in Capacity)					
Hourly PM Emissions	14.30	lb/hr	Hourly PM ₁₀ /PM _{2.5} Emissions	19.07	lb/hr
Yearly PM Emissions	62.65	tons/yr	Yearly PM ₁₀ /PM _{2.5} Emissions	83.54	tons/yr

Notes:
The stack test result was based on the following capacities: 186 tons/hr (4/15/2008), 185.8 tons/hr (4/16/2008), and 193.8 tons/hr (4/17/2008).

Methodology:
Controlled PTE (lb/hr) = Grain Loading (gr/acf) * Air Flow Rate (acf/min) * 60 min/hr ÷ 7000 grains/lb
Controlled PTE (tons/yr) = PTE (lb/hr) * 8760 hrs/yr ÷ 2000 lb/ton
Uncontrolled PTE Before Increase (lb/hr) = Controlled PTE (lb/hr) ÷ (1 - Control Efficiency)
Uncontrolled PTE Before increase (tons/yr) = Controlled PTE (tons/yr) ÷ (1 - Control Efficiency)
Uncontrolled PTE After Increase (lb/hr) = Uncontrolled PTE Before Increase (lb/hr) * Increase in Capacity ÷ Original Capacity
Uncontrolled PTE After Increase (tons/yr) = Uncontrolled PTE Before Increase (tons/yr) * Increase in Capacity ÷ Original Capacity

Appendix A: Emission Calculations
Hammer Mill Mixing Conveyor, Meal L-Path Conveyor, Meal Hammer Mill Feed Conveyor, Meal Hammer Mill Feeders No. 1, No. 2, No. 3, and No. 5, and Meal Hammer Mills No. 1, No. 2, No. 3, and No. 5

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

PTE for emission units E020300, E020400, E010100, and E010300

Grinding Discharge Conveyor, Hammer Mill Mixing Conveyor, Meal L-Path Conveyor, and Meal Hammer Mill Feed Conveyor

PM Emission Factor	0.061	lb/ton	Source: SCC 3-02-005-30
PM10/PM2.5 Emission Factor	0.034	lb/ton	SCC 3-02-005-30
Hourly Loading Rate	198	tons/hr	
Aspiration Rate	1000	cfm	
% Control PM	99.50%		
% Control PM10/PM2.5	99.50%		

Hourly throughput is based on maximum design capacity of extraction process.

Potential PM emissions

Grinding Discharge Conveyor, Hammer Mill Mixing Conveyor, Meal L-Path Conveyor, and Meal Hammer Mill Feed Conveyor

	Uncontrolled for each unit	Controlled for each unit
Max Hourly	= 12.08 lbs/hr	0.06 lbs/hr
Max Yearly	= 52.90 tons/yr	0.26 tons/yr

Potential PM10/PM2.5 emissions

Grinding Discharge Conveyor, Hammer Mill Mixing Conveyor, Meal L-Path Conveyor, and Meal Hammer Mill Feed Conveyor

	Uncontrolled for each unit	Controlled for each unit
Max Hourly	= 6.73 lbs/hr	0.03 lbs/hr
Max Yearly	= 29.49 tons/yr	0.15 tons/yr

PTE for emission units E020200, E030200, E040200, E230200, E020000, E030000, E040000, and E230000

Meal Hammer Mill Feeders No. 1, No. 2 and No. 3, Meal Hammer Mills No. 1, No. 2 and No. 3

Meal Hammer Mill Feeder No. 5, Meal Hammer Mill No. 5

PM Emission Factor	3.4	lb/ton	Source: SCC 3-02-007-93
PM10/PM2.5 Emission Factor	0.9	lb/ton	SCC 3-02-007-93
Hourly Loading Rate	74	tons/hr	* Emission Factor was adjusted to uncontrolled based on assumption that cyclones provide 90% control
Aspiration Rate	3000	cfm	
% Control PM	99.50%		
% Control PM10/PM2.5	99.50%		

Potential PM emissions

Meal Hammer Mill Feeders No. 1, No. 2 and No. 3, Meal Hammer Mills No. 1, No. 2 and No. 3

Meal Hammer Mill Feeder No. 5, Meal Hammer Mill No. 5

	Uncontrolled for each unit	Controlled for each unit
Max Hourly	= 251.600 lbs/hr	1.258 lbs/hr
Max Yearly	= 1102.008 tons/yr	5.510 tons/yr

Potential PM10/PM2.5 emissions

Meal Hammer Mill Feeders No. 1, No. 2 and No. 3, Meal Hammer Mills No. 1, No. 2 and No. 3

Meal Hammer Mill Feeder No. 5, Meal Hammer Mill No. 5

	Uncontrolled for each unit	Controlled for each unit
Max Hourly	= 62.900 lbs/hr	0.315 lbs/hr
Max Yearly	= 275.502 tons/yr	1.378 tons/yr

Notes:

Emission factors are from AP-42, Table 9.9.1-1 (Particulate Emission Factors for Grain Elevators (3/03)).

Control efficiency for PM is 99.50% and for PM10 is 99.50% according to Operating Permit Renewal No. T085-29197-00102.

Methodology:

Uncontrolled Emissions (tons/yr) = Throughput (tons/yr) * Emission factor (lb/ton) ÷ 2000 lbs/ton

Controlled Potential Emissions (tons/yr) = Throughput (tons/yr) * Emission factor (lb/ton) ÷ 2000 (lbs/ton) * (1 - Control Efficiency)

Appendix A: Emission Calculations
Meal Leg, Meal Conveyor to Loadout, and Meal Hammer Mill Bins No. 1, No. 2,
No. 3, and No. 5

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

PTE for emission units G010300**Meal Leg**

PM Emission Factor	0.061	lb/ton	Source: SCC 3-02-005-30
PM10/PM2.5 Emission Factor	0.034	lb/ton	SCC 3-02-005-30
Hourly Loading Rate	198	tons/hr	
Aspiration Rate	1,000	cfm	
% Control PM	99.50%		
% Control PM10/PM2.5	99.50%		

Potential PM emissions**Meal Leg**

	Uncontrolled for each unit	Controlled for each unit
Max Hourly	= 12.078 lbs/hr	0.060 lbs/hr
Max Yearly	= 52.902 tons/yr	0.265 tons/yr

Potential PM10/PM2.5 emissions**Meal Leg**

	Uncontrolled for each unit	Controlled for each unit
Max Hourly	= 6.732 lbs/hr	0.034 lbs/hr
Max Yearly	= 29.486 tons/yr	0.147 tons/yr

PTE for emission units E020100, E030100, E040100, and E230100**Meal Hammer Mill Bins No. 1, No. 2 and No. 3****Meal Hammer Mill Bin No. 5**

PM Emission Factor	0.025	lb/ton	Source: SCC 3-02-005-40
PM10/PM2.5 Emission Factor	0.0063	lb/ton	SCC 3-02-005-40
Hourly Loading Rate	74	tons/hr	
Aspiration Rate	6,000	cfm	
% Control PM	99.50%		
% Control PM10/PM2.5	99.50%		

Potential PM emissions**Meal Hammer Mill Bins No. 1, No. 2 and No. 3****Meal Hammer Mill Bin No. 5**

	Uncontrolled for each unit	Controlled for each unit
Max Hourly	= 1.850 lbs/hr	0.009 lbs/hr
Max Yearly	= 8.103 tons/yr	0.041 tons/yr

Potential PM10/PM2.5 emissions**Meal Hammer Mill Bins No. 1, No. 2 and No. 3****Meal Hammer Mill Bin No. 5**

	Uncontrolled for each unit	Controlled for each unit
Max Hourly	= 0.466 lbs/hr	0.002 lbs/hr
Max Yearly	= 2.042 tons/yr	0.010 tons/yr

Notes:
Emission factors are from AP-42, Table 9.9.1-1 (Particulate Emission Factors for Grain Elevators (3/03)).
Control efficiency for PM is 99.50% and for PM10 is 99.50% according to Operating Permit Renewal No. T085-29197-00102.

Methodology:

Uncontrolled Emissions (tons/yr) = Throughput (tons/yr) * Emission factor (lb/ton) ÷ 2000 lbs/ton
Controlled Potential Emissions (tons/yr) = Throughput (tons/yr) * Emission factor (lb/ton) ÷ 2000 (lbs/ton) * (1 - Control Efficiency)

Appendix A: Emission Calculations
DC Decks No. 1, No. 2, No. 3, and No. 4

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

PTE for emission units D310000-1, D310000-2, D310000-3, and D310000-4
DC Decks No. 1, No. 2, No. 3, and No. 4

PM Emission Factor	1.80	lb/ton	Source:	SCC 3-02-007-89
PM10/PM2.5 Emission Factor	0.450	lb/ton		SCC 3-02-007-89
Hourly Loading Rate	208	tons/hr		* Emission Factor was adjusted to
Aspiration Rate	63,900	cfm		uncontrolled based on assumption that
% Control PM	98.10%			cyclones provide 90% control
% Control PM10/PM2.5	97.89%			

Hourly throughput is based on maximum extraction process design - weight of oil extracted.

Potential PM emissions

DC Decks No. 1, No. 2, No. 3, and No. 4

		Uncontrolled for each unit		Controlled for each unit
Max Hourly	=	374.40	lbs/hr	7.11 lbs/hr
Max Yearly	=	1639.87	tons/yr	31.16 tons/yr

Potential PM10/PM2.5 emissions

DC Decks No. 1, No. 2, No. 3, and No. 4

		Uncontrolled for each unit		Controlled for each unit
Max Hourly	=	93.60	lbs/hr	1.97 lbs/hr
Max Yearly	=	409.97	tons/yr	8.65 tons/yr

Methodology

Emission factors are from AP 42 Table 9.9.1-1 Particulate Emission Factors for Grain Elevators (3/03).

Potential Emissions (ton/yr) = Throughput (ton/yr)* Emission factor (lb/ton) / 2000 (lbs/ton)

Controlled Potential Emissions (ton/yr) = Throughput (ton/yr) * Emission factor (lb/ton) / 2000 (lbs/ton)* (1-Control Efficiency)

Control efficiency for PM is 99.41% and for PM10 is 99.41% according to the renewal application.

Appendix A: Emission Calculations
Diatomaceous Earth (DE) Storage Bin, Bean Storage Bins No. 2 thru 4, 6, 7, and 8 and
Bean Storage Silos No. 1 thru 2

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

PTE for
Diatomaceous Earth (DE) Storage Bin

PM Emission Factor	0.990	lb/ton	Source:	SCC 3-05-038-13
PM10/PM2.5 Emission Factor	0.160	lb/ton		SCC 3-05-038-13
Hourly Loading Rate	0.088	tons/hr		
Aspiration Rate	Not Provided	cfm		
% Control PM	99.50%			
% Control PM10/PM2.5	99.50%			

Hourly throughput is based on maximum transport system capacity.

Potential PM emissions
Diatomaceous Earth (DE) Storage Bin

		Uncontrolled		Controlled	
Max Hourly	=	0.09	lbs/hr	0.00	lbs/hr
Max Yearly	=	0.38	tons/yr	0.00	tons/yr

Potential PM10/PM2.5 emissions
Diatomaceous Earth (DE) Storage Bin

		Uncontrolled		Controlled	
Max Hourly	=	0.01	lbs/hr	0.00	lbs/hr
Max Yearly	=	0.06	tons/yr	0.00	tons/yr

PTE for
Bean Storage Bin No. 2, 3, 4, 6, 7, and 8 and Bean Storage Silo No. 1 and 5

PM Emission Factor	0.025	lb/ton	Source:	SCC 3-02-005-40
PM10/PM2.5 Emission Factor	0.006	lb/ton		SCC 3-02-005-40
Hourly Loading Rate	277.6	tons/hr		
Aspiration Rate	348,000	cfm		
% Control PM	N/A			
% Control PM10/PM2.5	N/A			

Hourly throughput is based on maximum transport system capacity.

This source utilizes oil suppression. Mineral oil has a control efficiency of 60 to 80%. A control efficiency of 60% is assumed for mineral oil for a conservative estimate.

Potential PM emissions
Bean Storage Bin No. 2, 3, 4, 6, 7, and 8 and Bean Storage Silo No. 1 and 5

		Uncontrolled		Controlled	
Max Hourly	=	6.94	lbs/hr	6.94	lbs/hr
Max Yearly	=	30.40	tons/yr	30.40	tons/yr

Potential PM10/PM2.5 emissions
Bean Storage Bin No. 2, 3, 4, 6, 7, and 8 and Bean Storage Silo No. 1 and 5

		Uncontrolled		Controlled	
Max Hourly	=	1.75	lbs/hr	1.75	lbs/hr
Max Yearly	=	7.66	tons/yr	7.66	tons/yr

Methodology

Emission factors are from AP 42 Table 9.9.1-1 Particulate Emission Factors for Grain Elevators (3/03)

Uncontrolled Emissions (ton/yr) = Throughput (ton/yr) * Emission factor (lb/ton) / 2000 (lbs/ton)

Controlled Potential Emissions (ton/yr) = Throughput (ton/yr) * Emission factor (lb/ton) / 2000 (lbs/ton) * (1-Control Efficiency).

Control efficiency for PM is 99.41% and for PM10 is 99.41% according to the renewal application.

**Appendix A; Emission Calculations
Kaolin Receiving Tank and Hull Overflow Tank**

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

**PTE for
Kaolin Receiving Tank**

PM Emission Factor	0.99	lb/ton	Source:	SCC 3-05-038-13
PM10/PM2.5 Emission Factor	0.16	lb/ton		SCC 3-05-038-13
Hourly Loading Rate	40	tons/hr		* Controlled based on assumption that filters provide
Aspiration Rate	750	cfm		90.5% control
% Control PM	99.50%			
% Control PM10/PM2.5	99.50%			

Hourly throughput based estimation of truck transfer rate.

Potential PM emissions
Kaolin Receiving Tank

			Uncontrolled		Controlled
Max Hourly	=	39.60	lbs/hr	0.20	lbs/hr
Max Yearly	=	173.45	tons/yr	0.87	tons/yr

Potential PM10/PM2.5 emissions
Kaolin Receiving Tank

			Uncontrolled		Controlled
Max Hourly	=	6.40	lbs/hr	0.03	lbs/hr
Max Yearly	=	28.03	tons/yr	0.14	tons/yr

**PTE for
Hull Overflow Tank**

PM Emission Factor	0.025	lb/ton	Source:	SCC 3-02-005-40
PM10/PM2.5 Emission Factor	0.0063	lb/ton		SCC 3-02-005-40
Hourly Loading Rate	7.7	tons/hr		
Aspiration Rate	1,000	cfm		
% Control PM	0.00%			
% Control PM10/PM2.5	0.00%			

Hourly throughput based estimation of truck transfer rate.

Potential PM emissions
PTE for Hull Overflow Tank

			Uncontrolled		Controlled
Max Hourly	=	0.19	lbs/hr	0.19	lbs/hr
Max Yearly	=	0.84	tons/yr	0.84	tons/yr

Potential PM10/PM2.5 emissions
PTE for Hull Overflow Tank

			Uncontrolled		Controlled
Max Hourly	=	0.05	lbs/hr	0.05	lbs/hr
Max Yearly	=	0.21	tons/yr	0.21	tons/yr

Methodology

Emission factors are from AP 42 Table 9.9.1-1 Particulate Emission Factors for Grain Elevators (3/03)

Uncontrolled Emissions (ton/yr) = Throughput (ton/yr)* Emission factor (lb/ton) / 2000 (lbs/ton)

Controlled Potential Emissions (ton/yr) = Throughput (ton/yr) * Emission factor (lb/ton) / 2000 (lbs/ton)* (1-Control Efficiency).

Control efficiency for PM is 99.41% and for PM10 is 99.41% according to the renewal application.

Appendix A: Emissions Calculations
Hexane (VOC) Emissions from Meal Storage and
Desolventized Meal

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

c) Meal Storage

	Original		Increased	
Basis:	148 tons/hr	¹ of meal capacity	198 tons/hr	¹ of meal capacity
	40 lbs/cu.ft.	¹ density of meal	40 lbs/cu.ft.	¹ density of meal
	7400 cu.ft./hr.	of meal production	9900 cu.ft./hr.	of meal production
	200 ppm	hexane conc. in displaced storage air		
	0.218 lb/cu.ft.	density of oil @ 90.8°F		
		average temperature of soy oil temperature from stack test		
Annual Emissions =	0.32 lbs/hr	=	0.43 lbs/hr	
	1.41 tons/yr		1.89 tons/yr	

d) Desolventized meal

	Original		Increased	
	148 tons/hr		198 tons/hr	
<u>Normal Operating Conditions</u>	100 ppm for	8730 hrs/yr		
Solvent Content	29.6 lbs/hr		39.6 lbs/hr	
Solvent Content	129.20 tons/yr		172.85 tons/yr	
<u>Upset Conditions</u>	625 ppm for	30 hrs/yr		
Solvent Content	185 lbs/hr		248 lbs/hr	
Solvent Content	2.78 tons/yr		3.71 tons/yr	
Annual Emissions =	131.98		176.57	

Notes:

¹ Provided by the source in T085-21297-00102, issued on January 24, 2006.

Methodology:

Production (cu.ft./hr) = Capacity (tons/hr) * Meal Density (lb/cu. ft.) * 2000 lb/ton

Annual Emissions (lb/hr) = Production (cu. ft./hr) * Oil Density (lb/cu. ft.) * Concentration (ppm) ÷ 1,000,000

Annual Emissions (tons/yr) = Annual Emissions (lbs/hr) * 8760 hr/yr ÷ 2000 lbs/ton

Appendix A: Emissions Calculations
Fugitive (VOC) Emissions from Soybean Extraction Process

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

f) Plant Startup/Shutdown

Startup Solvent Loss =	8,400	lbs or	1,500	gallons
Shutdown Solvent Loss =	8,400	lbs or	1,500	gallons
Startup Duration =	2	hrs		
Shutdown Duration =	2	hrs		
Total Duration =	4	hrs		
Frequency =	2	times/yr		
Annual Emissions =	4,200	lbs/hr	for	8 hrs/yr
Annual Emissions =	16.80	tons/yr		

g) General due to Equipment Failure, Routine Maintenance, and Leaks

These losses occur throughout the year, and there are no clearly predetermined conditions.
Based on experience at this type of facility, the general emission factor equals 0.28 lbs/ton of beans.

Original Emissions =	0.28	lbs/ton of beans *	192.5	tons/hr *	8760	hr/yr ÷	2000	hr/yr
Original Emissions =	236.08	tons/yr						
Increased Emissions =	0.28	lbs/ton of beans *	264	tons/hr *	8760	hr/yr ÷	2000	hr/yr
Increased Emissions =	323.77	tons/yr						

h) Plant Upsets

Duration =	3	hrs
Frequency =	10	times/yr

When the system loses normal vacuum (negative pressure) condition, VOCs are lost.

Original Air Flow in Flakes =	192.5	tons/hr *	2000	lb/ton ÷	60	min/hr ÷	(75-15) lb/cfm
Original Air Flow in Flakes =	106.9	cfm					
Increased Air Flow in Flakes =	264	tons/hr *	2000	lb/ton ÷	60	min/hr ÷	(75-15) lb/cfm
Increased Air Flow in Flakes =	146.7	cfm					

Assume the amount of VOCs lost to the atmosphere is roughly equal to the air normally pulled in.

Original Emissions =	106.9	cfm *	60	min/hr *	0.2	lbs/cu.ft. ÷	2000	lb/ton *	30	hr/yr
Original Emissions =	19.25	tons/yr								
Increased Emissions =	146.7	cfm *	60	min/hr *	0.2	lbs/cu.ft. ÷	2000	lb/ton *	30	hr/yr
Increased Emissions =	26.40	tons/yr								

Methodology:

These fugitive calculations were provided by the source and are shown in Part 70 Permit No. T085-21297-00102 issued on January 24, 2006.

Appendix A: Emission Calculations
VOC Emissions
Biodiesel Manufacturing Process

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

This process is controlled by a soy oil absorber followed by a water absorber and vents through stack S-5.

Emissions from day tanks, methanol storage tanks, and methoxide (catalyst) storage tank are also controlled by the water absorber and vent through Stack S-5.

Biodiesel storage tanks vent to day tanks.

Process Description	VOC Emission Limit (lb/hr)	Operating Hours (hrs/yr)	Limited PTE of VOC (tons/yr)	Control Efficiency	Unlimited PTE of VOC (tons/yr)
Normal Operation ^{1,2}	0.30	7,736	1.16	99%	116.04
Normal Operation with Methanol Unloading ^{1,2,3}	0.63	1,000	0.32	99%	31.50
Upset Conditions ^{1,4}	29.40	24	0.35	95%	7.06
TOTAL		8,760	1.83		154.60

Notes:

¹ This emission limit was proposed by the permittee and will be verified by stack testing.

² This emission limit and control efficiency are the BACT requirements for the biodiesel manufacturing process pursuant to T085-21297-00102, issued on January 24, 2006.

³ Biodiesel manufacturing process with methanol tank loading is limited to 1,000 hours/yr.

⁴ Biodiesel manufacturing process upset operation is limited to 24 hours/yr.

Methodology:

Limited PTE of VOC (tons/yr) = VOC Emission Limit (lbs/hr) x Operating Hours (hrs/yr) x 1 ton/2000 lbs

Unlimited PTE of VOC (tons/yr) = Limited PTE of VOC ÷ (1 - Control Efficiency)

Appendix A: Emissions Calculations
VOC and HAP Emissions
Biodiesel Distillation

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

This process is controlled by a mineral oil absorber followed by a water absorber and vents through stack S-5.

30,000 lb/hr biodiesel at 120F
The objective is to remove monoglyceride from biodiesel. ASTM has defined two grades of biodiesel - the distilled low monoglyceride biodiesel would be blended with biodiesel containing higher than 0.4% monoglyceride in order to reduce the monoglyceride to less than 0.4%, thereby achieving the grade 1B specification. The distillation process will be designed to process 30,000 lb/hr biodiesel.

Biodiesel Feed Rate	30,000 lb/hr	
Methanol	0.08%	
Monoglycerides	0.45%	
Diglycerides/Triglycerides	0.05%	
Free Glycerin	0.00%	
Free Fatty Acid	0.05%	
NVR	0.05%	
Biodiesel/FAME	99.4%	
Biodiesel Product	29,664 lb/hr	
Monoglycerides	0.05%	
Biodiesel/FAME	99.95%	(Contains 0.07% Methanol in the Biodiesel product)
Acid Value	<0.1	
Free Glycerin	0.002%	
Bottoms Purge	300 lb/hr	
Monoglycerides	45%	
Biodiesel/FAME	44%	
Balance: Diglycerides/Triglycerides, Free Fatty Acid and NVR	11%	
Vent to Scrubber		
Methanol	0.30 lb/hour	
Uncontrolled Emissions =	1.31 tons/year	
Scrubber Control Efficiency	99.5%	
VOC Emissions (Methanol) Controlled	0.002 lbs/hr	6.57E-03 tpy

Note:
The amount of methanol (0.30 lbs/hr) vented to scrubber was provided by the source.
The objective of biodiesel distillation is to remove monoglyceride from biodiesel.
The distillation will be designed to process 30,000 lbs per hour of biodiesel.
Emissions are vented to Mineral Oil Absorber and Water Absorber with a control efficiency of 99.5%.
All VOC emissions are considered HAPs.

Appendix A: Emission Calculations
VOC Emissions
Biodiesel Loading Racks (Rail and Truck)

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

Process Description	Capacity (gal/min)	¹ VOC Emission Factor (lbs/kgal)	² VOC Emission Limit (lbs/kgal)	Max. Throughput (kgal/yr)	Unlimited PTE of VOC (tons/yr)	³ Limited Throughput (kgal/yr)	Limited PTE of VOC (tons/yr)
Loading Rack (Rail)	500	0.03	0.02	262,800	3.94	110,000	1.10
Loading Rack (Truck Rack #1)	430	0.03	0.02	226,008	3.39		
Loading Rack (Truck Rack #2)	430	0.03	0.02	226,008	3.39		

Notes:

¹ The unlimited PTE was calculated using the VOC emission factor for Splash loading - Dedicated normal service of Distillate Oil No. 2 (0.03 lbs/kgal) from Table 5.2-5 AP-42 Chapter 5.2 Transportation And Marketing Of Petroleum Liquids because biodiesel is assumed to be similar to distillate oil no. 2 for the purpose of determining the Maximum PTE.

² These VOC emission limits are pursuant to SPM 085-25147-00102, issued on January 28, 2008.

³ The limited throughput was pursuant to SPM 085-25147-00102, issued on January 28, 2008. Limited emissions shown are based on the limited VOC emission limit (0.02 lbs/kgal) and the limited throughput.

Methdology:

Max. Throughput (kgal/yr) = Capacity (gal/min) * 60 min/hr * 8760 hrs/yr ÷ 1,000 gal/kgal

Uncontrolled PTE of VOC (tons/yr) = VOC Emission Limit (lbs/kgal) * Max. Throughput (kgal/yr) ÷ 2000 lb/ton

Limited PTE of VOC (tons/yr) = VOC Emission Limit (lbs/kgal) * Limited Throughput (kgal/yr) ÷ 2000 lb/ton

Appendix A: Emission Calculations
VOC Emissions
Glycerine Storage Tanks, Biodiesel Wastewater, and Equipment Leaks

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

Process Description	VOC Emission Limit (lbs/hr)	Max. Operating Hours (hrs/yr)	PTE of VOC after Control (tons/yr)	Control Efficiency	PTE of VOC before Control (tons/yr)
Glycerine Tanks #12 and #13	0.0011	8,760	0.0048	0%	0.0048
Biodiesel Wastewater (Fugitive)	0.77	8,760	3.37	0%	3.37
Equipment Leaks (Fugitive)	0.64	8,760	2.80	78%	12.74
Total			6.18		16.12

Notes:

The VOC emission rates for these processes were calculated based on the maximum soy oil process rate of 110 million gallons per year.

The VOC limits were established pursuant to SPM 085-25147-00102.

The control efficiency for equipment leaks is from Protocol for Equipment Leak Emission Estimates, EPA-453/R-95-017, Table 5-2.

Methodology:

$\text{PTE of VOC after Control (tons/yr)} = \text{VOC Emission Rate (lbs/hr)} * \text{Max. Operating Hours (hrs/yr)} \div 2000 \text{ lb/ton}$

$\text{PTE of VOC before Control (tons/yr)} = \text{PTE of VOC after Control (tons/yr)} \div (1 - \text{Control Efficiency})$

Appendix A: Emission Calculations
VOC Emissions
Biodiesel Storage Tanks

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

Unit ID	Emission Unit Description	Storage Capacity	PTE of VOC (lbs/yr/unit)	Number of Units	PTE of VOC (tons/yr)
I140000	Biodiesel Storage Tank #14	735,000 gallons	1,426.55	1	0.71
I040000 and I050000	Biodiesel Storage Tank #4 and Biodiesel Storage Tank #5	725,000 gallons	1,762.00	2	1.76
I060000	Biodiesel Storage Tank #6	360,000 gallons	657.69	1	0.33
I070000, I080000, I090000, I100000, and I110000	Biodiesel Storage Tanks #7, Tank #8, Tank #9, Tank #10, and Tank #11	325,000 gallons	601.76	5	1.50
TOTAL					4.31

Notes:

The PTE of VOC from the tanks were calculated by the Permittee using EPA TANKS software (version 4.09d) and have been verified.

Methodology:

$\text{PTE of VOC (tons/yr)} = \text{PTE of VOC (lbs/yr/unit)} * \text{Number of Units} \div 2000 \text{ lb/ton}$

Appendix A: Emissions Calculations**PM/PM₁₀ Emissions****Noncontact Cooling Towers**

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

Process Description:

Type of ²Cooling Tower: Induced Draft

¹Circulation Flow Rate: 660,000 gal/hr

¹Total Drift: 0.005% of the circulating flow

¹Maximum Total Dissolved Solids: 3,600 ppm* *Average Total Dissolved Solids: 2,400 ppm

Density: 8.345 lbs/gal

Potential to Emit PM/PM₁₀/PM_{2.5}:

Assume all the dissolved solids become PM₁₀ emissions and assume PM emissions are equal to PM₁₀ emissions.

PTE of PM/PM ₁₀ /PM _{2.5} (lb/hr) = 660,000 gal/hr x 0.005% x 8.345 lbs/gal x 3,600 ppm ÷ 1,000,000 ppm =	0.99	lb/hr
PTE of PM/PM ₁₀ /PM _{2.5} (tons/yr) = 0.99 lbs/hr x 8760 hr/yr ÷ 2000 lb/ton =	4.34	tons/yr

Notes:

The information above was provided by the source, pursuant to T085-21297-00102 issued on January 24, 2006.

Calculation based on AP-42 Chapter 13.4. Assume that non VOC biocide utilized; therefore no VOCs included.

The cooling tower is a noncontact cooling tower, and it serves both the extraction plant and the biodiesel plant.

The approximate relative distribution of water to the cooling tower would be around 90% from extraction and 10% from biodiesel.

Fugitive emissions from the cooling tower are counted towards the biodiesel plant for PSD purposes, but not towards the extraction plant.

**Appendix A: Emission Calculations
Soybean Oil Unloading**

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

Determine Emission Factors for Soybean Oil Unloading at Louis Dreyfus Agricultural Industries LLC

Use Equation 1 from AP-42 Section 5.2 Transportation and Marketing of Petroleum Liquids

Pursuant to AP-42, Section 5.2-4, emissions from loading petroleum liquid can be estimated (with a probable error of ± 30 percent) using the following expression:

$$L_L = 12.46 \frac{\text{SPM}}{T}$$

L_L = loading loss, pounds per 1000 gallons (lb/10³ gal) of liquid loaded
 S = a saturation factor (see Table 5.2-1)
 P = true vapor pressure of liquid loaded, pounds per square inch absolute (psia) (see Section 7.1, "Organic Liquid Storage Tanks")
 M = molecular weight of vapors, pounds per pound-mole (lb/lb-mole) (see Section 7.1, "Organic Liquid Storage Tanks")
 T = temperature of bulk liquid loaded, °R (°F+460)
 Soy = volume of soybean oil transferred
 Hexane = volume of hexane transferred

Calculations for Soybean Oil portion of emissions

S = 0.60 Submerged Loading: dedicated normal service
 M = 292 lb/lb-mol (per TANKS calculation)
 Soy = 452,016,000 gal/year (The maximum rate that crude oil may be unloaded is the same as for truck loadout at 430 gallons per minute) x 2 truck racks

T, deg F	T, deg R	¹ P, psia	L_L , lb/10 ³ gal	Emissions from Soy, lb/year	L_L , lb/10 ³ gal with 30% error	Emissions from Soy, lb/year with 30% error
60	519	1.081E-14	4.543E-14	2.053E-08	5.906E-14	2.669E-08
100	559	1.641E-12	6.402E-12	2.894E-06	8.322E-12	3.762E-06

Note 1: P, psia is calculated on the next page

Calculations for Hexane portion of emissions

S = 1.45 Submerged Loading: dedicated normal service
 M = 86.17 lb/lb-mol (AP-42, Section 7.1, Table 7.1-3)
 Hexane = 452,016 gal/year (@1000 ppm (worse case))

T, deg F	T, deg R	² P, psia	L_L , lb/10 ³ gal	Emissions from Hexane, lb/year	L_L , lb/10 ³ gal with 30% error	Emissions from Hexane, lb/year with 30% error
60	519	1.876	5.6233	2,541.82	7.3103	3,304.37
100	559	4.892	13.6145	6,153.96	17.6988	8,000.15

Note 2: P, psia was provided by source

Total Potential to Emit

T, deg F	Total Emissions, lb/year	Total Emissions, ton/year
60	3,304.37	1.65
100	8,000.15	4.00

Annual Average Temp = 58.82F

Appendix A: Emission Calculations
Calculations for Vapor Pressure for Soybean Oil Unloading

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

Instructions: Enter the vapor pressure from the MSDS in B17, and the Celsius temperature for that vapor pressure in C17. Then enter the atmospheric boiling point (deg C) in C19. Values for the condensation spreadsheet calculate automatically.

	mmHg	deg C	K
MSDS vapor pressure data	0.000000001	50	323
MSDS boiling point		300	573

	B	A
Clausius-Clapeyron coefficients	20253	42
ln P = A- B/T		

For condensation worksheet	p, mmHg	T, deg C	T, deg R	T, deg F	P, psia
	3.4E-14	4	499	40.04	6.562E-16
	1.42E-13	10	509	50.02	2.7406E-15
	5.6E-13	16	519	59.98	1.0808E-14
	2.1E-12	21	529	69.95	4.053E-14
	7.6E-12	27	539	80.02	1.4668E-13
	2.6E-11	32	549	90.02	5.018E-13
	8.5E-11	38	559	100.01	1.6405E-12
	100	269	975	516.03	1.93

Saturation vapor pressure: 8.56282E-20 atm

Appendix A: Emission Calculations
VOC Emissions
Other Storage Tanks

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

Unit ID	Equipment Description	Capacity	Control Efficiency	Pot'l VOC Working Loss (lbs/yr)	Pot'l VOC Standing Loss (lbs/yr)	Pot'l VOC Max Emissions (lbs/yr)	Pot'l VOC Max Emissions (tons/year)	Pot'l VOC After Control (tons/year)
I220000	¹ One (1) Soybean Oil Pre-Treat Tank	35,170 gallons	0%	0.00	0.00	0.00	0.01	0.01
	³ 3 Soybean Oil Tanks (Degummed Oil Tanks #1 and #2 and Crude Oil Tank #3)	725,000 gallons each	0%	See Note 3			<= 1.0	<= 1.0
	⁴ 5 Hexane Tanks	20,690 gallons	98%	142.47	661.93	4,022	2.01	0.04
	Diesel/#2 Fuel Oil Storage Tank	44,839 gallons	0%	26.78	0.88	27.66	0.014	0.014

Original Throughput = 110 million gallons per year
Proposed Throughput = 119 million gallons per year

Unit ID	Equipment Description	Capacity	Control Efficiency	Original Pot'l VOC Max Emissions (lbs/yr)	Proposed Pot'l VOC Max Emissions (lbs/yr)	Proposed Pot'l VOC Max Emissions (tons/year)	Proposed Pot'l VOC After Control (tons/year)
I250000, I260000, I270000, I280000, I290000, and I300000	^{2,5} 6 Methanol Tanks	38,850 gallons each	98.35%	8,299.44	8,978.49	4.49	0.07
I230000 and I240000	^{2,5} 2 Sodium Methylate Tanks	38,850 gallons each	98.35%	1,103.59	1,193.88	0.60	0.01

Notes:

¹ Pursuant to 085-27442-00102, issued on January 25, 2010, the tank has potential VOC emissions of less than 0.01 ton/year.

² Stack test result from the existing soy oil absorber and water absorber controlling the biodiesel production plant has a control efficiency of 98.35% for VOC (methanol only). Methanol storage tanks and sodium methylate tanks are controlled by the soy oil absorber.

³ Emissions from the biodiesel storage tanks were calculated by the Permittee using EPA TANKS software (version 4.09d) storing oil with an average of 200 ppmwt hexane content are equal to 0.06 tons/yr per tanks. The source has three tanks and the ppmwt hexane varies in the purchased soy oil. Therefore, to be conservative the PTE from VOC from the purchased soy oil from all three tanks is assumed to be less than or equal to 1 ton of VOC per year.

⁴ Hexane tanks are used at the soy oil extraction plant and are controlled by a mineral oil absorber with assumed control efficiency of 98%.

⁵ The methanol/sodium methylate unloading pump is being replaced with a larger pump, allowing these tanks to increase their unloading rate to 119 million gallons per year.

Methodology:

VOC Max Emissions (tons/yr) = VOC Max Emissions (lb/yr) ÷ 2000 lb/ton

VOC After Control (tons/yr) = VOC Max Emissions (tons/yr) * (1 - Control Efficiency)

Proposed Pot'l VOC Max Emissions (tons/yr) = Original Pot'l VOC Max Emissions (tons/yr) * Proposed Throughput ÷ Original Throughput

Appendix A: Emission Calculations
Fugitive Dust Emissions - Paved Roads
PTE from Truck Traffic

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

Annual fugitive particulate emissions from Trucks on paved roads = VMT * E

Calculate VMT:

		Annual Tonnage	Trips per Year	Trips per Day	Vehicle Weight (tons)	VMT per trip (miles/trip)	VMT per year (miles/yr)	Avg Vehicle Weight (tons)
Bean	Full	1,686,300	64858	178	40	0.314	20,391	
	Empty	---	64858	178	14	0.636	41,273	
Meal	Empty	---	49865	137	14	0.379	18,888	
	Full	1,296,480	49865	137	40	0.572	28,521	
Hull/Pellet	Empty	---	1937	5.3	14	0.379	734	26.1
	Full	50,370	1937	5.3	40	0.572	1,108	
Oil	Empty	---	15000	41.1	15	0.417	6,250	
(110 MM gal)	Full	375,000	15,000	41.1	40	0.534	8,011	
Totals		3,408,150		360.7			125,176	

Calculate E:

Unmitigated Emission Factor, $E_f = [k * (sL)^{0.91} * (W)^{1.02}]$ (Equation 1 from AP-42 13.2.1.3 (01/2011))

Where E = particulate emission factor (lb/VMT)
k = particle size multiplier (lb/VMT)
sL = road surface silt loading (g/m^2)
W = weight average (tons) of vehicles traveling the road

k = 0.011 lb/VMT for PM₃₀ = TSP = PM
k = 0.0022 lb/VMT for PM₁₀
k = 0.00054 lb/VMT for PM_{2.5}
sL = 0.6 grams/ m^2
W = 26.1 tons

$E_{PM} = 0.193$ lbs PM/VMT
 $E_{PM10} = 0.039$ lbs PM₁₀/VMT
 $E_{PM2.5} = 0.009$ lbs PM_{2.5}/VMT

Calculate Emissions:

Grain Elevator Original Capacity	=	1,686,300	tons/yr	Grain Elevator Proposed Capacity	=	2,251,836	tons/yr
Total fugitive PM	=	VMT x E_{PM}		Total fugitive PM	=	VMT x E_{PM}	
	=	24,102	lbs/yr		=	32,185	lbs/yr
	=	2.75	lbs/hr		=	3.67	lbs/hr
	=	12.05	tons/yr		=	16.09	tons/yr
Total fugitive PM ₁₀	=	VMT x E_{PM10}		Total fugitive PM ₁₀	=	VMT x E_{PM10}	
	=	4,820	lbs/yr		=	6,437	lbs/yr
	=	0.55	lbs/hr		=	0.73	lbs/hr
	=	2.41	tons/yr		=	3.22	tons/yr
Total fugitive PM _{2.5}	=	VMT x $E_{PM2.5}$		Total fugitive PM _{2.5}	=	VMT x $E_{PM2.5}$	
	=	1,183	lbs/yr		=	1,580	lbs/yr
	=	0.14	lbs/hr		=	0.18	lbs/hr
	=	0.59	tons/yr		=	0.79	tons/yr

Appendix A: Emission Calculations
Fugitive Dust Emissions - Paved Roads
PTE from Car Traffic

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

Annual fugitive particulate emissions from Cars on paved roads = VMT * E

Calculate VMT:

$$\text{VMT} = (\text{Total number cars}) * (\text{Maximum car mileage onsite}) * (365 \text{ days/yr})$$

Assume Total Number Cars = Number of Employeess

Total Number Cars = 70

Maximum car mileage onsite = 0.5 miles

$$\begin{aligned} \text{VMT} &= \text{Total number of cars} \times \text{Maximum car mileage onsite/day} \times 365 \text{ days/year} \\ &= 12775 \end{aligned}$$

Calculate E:

Unmitigated Emission Factor, $E_f = [k * (sL)^{0.91} * (W)^{1.02}]$ (Equation 1 from AP-42 13.2.1.3 (01/2011))

Where E = particulate emission factor (lb/VMT)
 k = particle size multiplier (lb/VMT)
 sL = road surface silt loading (g/m^2)
 W = weight average (tons) of vehicles traveling the road

$k = 0.011$ lb/VMT for PM₃₀ = TSP = PM

$k = 0.0022$ lb/VMT for PM₁₀

$k = 0.00054$ lb/VMT for PM_{2.5}

$sL = 0.6$ grams/ m^2

$W = 2.0$ tons

$E_{PM} = 0.0140$ lbs PM/VMT

$E_{PM10} = 0.0028$ lbs PM₁₀/VMT

$E_{PM2.5} = 0.0007$ lbs PM_{2.5}/VMT

Calculate Emissions:

$$\begin{aligned} \text{Total fugitive PM} &= \text{VMT} \times E_{PM} \\ &= 179 \text{ lbs/yr} \\ &= 0.02 \text{ lbs/hr} \\ &= 0.09 \text{ tons/yr} \end{aligned}$$

$$\begin{aligned} \text{Total fugitive PM}_{10} &= \text{VMT} \times E_{PM10} \\ &= 36 \text{ lbs/yr} \\ &= 0.004 \text{ lbs/hr} \\ &= 0.02 \text{ tons/yr} \end{aligned}$$

$$\begin{aligned} \text{Total fugitive PM}_{2.5} &= \text{VMT} \times E_{PM2.5} \\ &= 8.79 \text{ lbs/yr} \\ &= 0.001 \text{ lbs/hr} \\ &= 0.004 \text{ tons/yr} \end{aligned}$$

**Appendix A: Emissions Calculations
Combustion Summary**

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

Uncontrolled PTE (tons/year)							
Emission Units	PM	PM ₁₀	PM _{2.5}	SO ₂	NO _x	VOC	CO
Boiler (natural gas)	1.79	7.18	7.18	0.57	132.26	5.20	79.36
Fire Pumps (diesel fuel) 2006*	0.95	0.95	0.95	0.44	13.37	1.08	2.88
Space Heaters (natural gas only)	0.004	0.016	0.016	0.001	0.215	0.012	0.180
Total	2.75	8.14	8.14	1.01	145.84	6.29	82.42

HAPs Emission Summary									
Uncontrolled PTE (tons/year)									
HAPs - Organics									
Total worst case individual HAPs from boiler, fire pumps, space heaters, and generator	Acetaldehyde	Arsenic	Dichlorobenzene	Formaldehyde	Hexane	Toluene	Xylenes	Benzene	Lead
	2.32E-03	--	1.1E-03	0.07	1.70	4.45E-03	8.60E-04	4.80E-03	4.72E-04
	HAPs - Metals								
	Beryllium	Cadmium	Chromium	Manganese	Nickel	Mercury	Selenium	Total PAH	Total
	--	1.04E-03	1.33E-03	3.60E-04	1.99E-03	--	--	5.07E-04	1.80

Appendix A: Emissions Calculations
Natural Gas Combustion Only
MM BTU/HR >100
Boiler B-1

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

Boiler B-1 Heat Input Capacity (MMBtu/hr)	Potential Throughput (MMCF/year)
220	1,889.4

	Pollutant						
Emission Factor (lb/MMCF)	PM* 1.9	PM10* 7.6	PM2.5 7.6	SO ₂ 0.6	NOx 140.0 **see below	VOC 5.5	CO 84.0
Unlimited Potential Emissions (tons/yr)	1.79	7.18	7.18	0.57	132.26	5.20	79.36

*PM emission factor is filterable PM only. PM10 and PM2.5 emission factors are filterable and condensable PM combined.

**Emission Factors for NOx from Large Wall-Fired Boilers: Uncontrolled (Pre-NSPS) = 280, Uncontrolled (Post-NSPS) = 190, Controlled Low NOx Burners = 140, Controlled Flue gas recirculation = 100

***Limited emissions are in order to render the requirements of 326 IAC 2-2 (PSD) not applicable. See TSD for specific limits for each pollutant.

Methodology

All emission factors are based on normal firing.

MMBtu = 1,000,000 Btu

MMCF = 1,000,000 Cubic Feet of Gas

NOx and CO Emission Factors are from AP 42, Chapter 1.4, Table 1.4-1

PM and SO₂ Emission Factors are from AP 42, Chapter 1.4, Table 1.4-2

Potential Throughput (MMCF) = Heat Input Capacity (MMBtu/hr) x 8,760 hrs/yr x 1 MMCF/1,020 MMBtu

Potential Emission (tons/yr) = Potential Throughput (MMCF/yr) x Emission Factor (lb/MMCF)/2,000 lb/ton

	HAPs - Organics				
Emission Factor in lb/MMcf	Benzene 2.1E-03	Dichlorobenzene 1.2E-03	Formaldehyde 7.5E-02	Hexane 1.8E+00	Toluene 3.4E-03
Potential Emission in tons/yr	1.98E-03	1.13E-03	7.09E-02	1.70E+00	3.21E-03

	HAPs - Metals					
Emission Factor in lb/MMcf	Lead 5.0E-04	Cadmium 1.1E-03	Chromium 1.4E-03	Manganese 3.8E-04	Nickel 2.1E-03	Total
Potential Emission in tons/yr	4.72E-04	1.04E-03	1.32E-03	3.59E-04	1.98E-03	1.78

Methodology:

Methodology is the same as above.

Organic HAPs Emission Factors are from AP 42, Chapter 1.4, Table 1.4-3

Metal HAPs Emission Factors are from AP 42, Chapter 1.4, Table 1.4-4

The five highest organic and metal HAPs emission factors are provided above.

Additional HAPs emission factors are available in AP-42, Chapter 1.4, Tables 1.4-3 and 1.4-4

Appendix A: Emissions Calculations
Three (3) Distillate Fuel Oil Fired Emergency Fire Pumps

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

Three (3) Fire Pumps
Capacity in hp

1725.00

SO₂ Emission factor = 0.00205 x S

S = % Sulfur Content = 0.50

	Pollutant						
Emission Factor (lb/hp-hr)	PM* 2.200E-03	PM10* 2.200E-03	PM2.5* 2.200E-03	SO ₂ 1.025E-03	**NO _x 3.100E-02	VOC 2.514E-03 **TOC value	CO 6.680E-03
Unlimited Potential Emissions (tons/yr)	0.95	0.95	0.95	0.44	13.37	1.08	2.88

Note:

*PM and PM_{2.5} emission factors are assumed to be equivalent to PM₁₀ emission factors. No information was given regarding which method was used to determine the factor or the fraction of PM₁₀ which is condensable.

** The VOC value given is total organic compounds (TOC).

Methodology

MMBtu = 1,000,000 Btu

The generators are only emergency generators. Therefore, they will not operate more than 500 hours per year.

Potential Emission (tons/yr) = Hp x Emission Factor (lb/hp-hr)/2,000 lb/ton x 500 hrs/year.

Calculations are based using fuel oil with 0.50% sulfur content, changes in the % sulfur content of fuel oil will affect the actual amount of SO₂ that is emitted.

Emission Factors are from AP42 (Supplement B 10/96), Tables 3.3-1 and 3.3-2

Emission (tons/yr) = Hp x Emission Factor (lb/hp-hr)/2,000 lb/ton x 500 hrs/year.

	HAPs						
Emission Factor in lb/hp-hr	Benzene 6.5E-06	Formaldehyde 8.3E-06	Toluene 2.9E-06	Xylenes 2.0E-06	Acetaldehyde 5.4E-06	Total PAH 1.2E-06	TOTAL HAPs (tons/yr)
Potential Emission in tons/yr	2.816E-03	3.562E-03	1.235E-03	8.603E-04	2.315E-03	5.072E-04	1.130E-02

Methodology:

Methodology is the same as above.

The six highest organic HAPs emission factors are provided above.

Additional HAPs emission factors are available in AP-42, Chapter 3 Table 3.3-2

HAP emission factors converted from lb/MMBtu in Table 3.3-2 to lb/hp-hr using the following method:

Emission Factor in lb/MMBtu x 1 MMBtu/1,000,000 Btu x 7000 Btu/hp-hr = Emission Factor in lb/hp-hr

Conversion factor of 7,000Btu/hp-hr taken from AP-42, Table 3.3-1

Appendix A: Emissions Calculations
Natural Gas Combustion Only
MM BTU/HR <100
Two (2) NG Space Heaters - 0.25 MMBtu/hr, Each

Company Name: Louis Dreyfus Company Agricultural Industries, LLC
Address City IN Zip: 7344 State Road 15 South, Claypool, Indiana 46510
Significant Source Modification No.: 085-39532-00102
Significant Permit Modification No.: 085-39729-00102
Permit Reviewer: Daniel W. Pell

Heat Input Capacity MMBtu/hr	HHV mmBtu mmscf	Potential Throughput MMCF/yr
0.5	1020	4.3

Emission Factor in lb/MMCF	Pollutant						
	PM*	PM10*	direct PM2.5*	SO2	NOx	VOC	CO
	1.9	7.6	7.6	0.6	100 **see below	5.5	84
Potential Emission in tons/yr	0.004	0.016	0.016	0.001	0.21	0.01	0.18

*PM emission factor is filterable PM only. PM10 emission factor is filterable and condensable PM10 combined.

PM2.5 emission factor is filterable and condensable PM2.5 combined.

**Emission Factors for NOx: Uncontrolled = 100, Low NOx Burner = 50, Low NOx Burners/Flue gas recirculation = 32

Methodology

All emission factors are based on normal firing.

MMBtu = 1,000,000 Btu

MMCF = 1,000,000 Cubic Feet of Gas

Emission Factors are from AP 42, Chapter 1.4, Tables 1.4-1, 1.4-2, 1.4-3, SCC #1-02-006-02, 1-01-006-02, 1-03-006-02, and 1-03-006-03

Potential Throughput (MMCF) = Heat Input Capacity (MMBtu/hr) x 8,760 hrs/yr x 1 MMCF/1,020 MMBtu

Emission (tons/yr) = Throughput (MMCF/yr) x Emission Factor (lb/MMCF)/2,000 lb/ton

Emission Factor in lb/MMcf	HAPs - Organics					Total - Organics
	Benzene 2.1E-03	Dichlorobenze 1.2E-03	Formaldehyde 7.5E-02	Hexane 1.8E+00	Toluene 3.4E-03	
Potential Emission in tons/yr	4.509E-06	2.576E-06	1.610E-04	3.865E-03	7.300E-06	4.040E-03

Emission Factor in lb/MMcf	HAPs - Metals					Total - Metals
	Lead 5.0E-04	Cadmium 1.1E-03	Chromium 1.4E-03	Manganese 3.8E-04	Nickel 2.1E-03	
Potential Emission in tons/yr	1.074E-06	2.362E-06	3.006E-06	8.159E-07	4.509E-06	1.177E-05
Total HAPs						4.052E-03
Worst HAP						3.865E-03

Methodology is the same as above.

The five highest organic and metal HAPs emission factors are provided above.

Additional HAPs emission factors are available in AP-42, Chapter 1.4.

**Indiana Department of Environmental Management
Office of Air Quality**

Appendix B – BACT Analysis for Biodiesel Manufacturing Process and
a BACT Analysis for the Soybean Oil Extraction Process
Technical Support Document (TSD)
Prevention of Significant Deterioration (PSD)

Source Background and Description
--

Source Name: Source Location: County: SIC Code: Operation Permit No.: Operation Permit Issuance Date: Significant Source Modification No.: Significant Permit Modification No.: Permit Reviewer:	Louis Dreyfus Company Agricultural Industries LLC 7344 State Road 15 South, Claypool, Indiana 46510 Kosciusko 2075 (Soybean Oil Mills) 2079 (Shortening, Table Oils, Margarine, and Other Edible Fats and Oils, Not Elsewhere Classified) 2869 (Industrial Organic Chemicals, Not Elsewhere Classified) T085-36889-00102 January 5, 2017 085-39532-00102 085-39729-00102 Daniel W. Pell
--	--

Proposed Modification

On January 23, 2018, the Office of Air Quality (OAQ) received an application from Louis Dreyfus Company Agricultural Industries LLC, to establish a 326 IAC 8-1-6 VOC BACT limit for the Biodiesel Manufacturing Operation which uses soybean oil to produce a biodiesel fuel.

The original Title V permit for the biodiesel process that was established to meet the requirements for Best Available Control Technology (BACT), were based on estimates provided by Louis Dreyfus (before plant was constructed and operational) that uncontrolled VOC emissions were more than 25 tons per year. Initial stack test results for the biodiesel manufacturing process recorded VOC emissions at the inlet to the scrubber system to be less than 25 tons per year. Based on these stack test results, IDEM OAQ removed the mostly duplicative VOC BACT requirements from the permit, although the comparable PSD Minor Limits for VOC (pound per hour) and LDAR requirements were retained to keep allowable VOC emissions below PSD major source thresholds. Louis Dreyfus now believes that this was an inaccurate assessment of the uncontrolled VOC emissions to the scrubber system under various operating conditions and rates, and believes that uncontrolled potential VOC emissions exceed the 25 tons per year applicability threshold of 326 IAC 8-1-6. Therefore, the biodiesel manufacturing process is subject to the requirements of 326 IAC 8-1-6 (Best Available Control Technology (BACT)). The affected biodiesel manufacturing process equipment are noted as follows:

Affected Biodiesel Manufacturing Process Equipment
1st Primary Transester Column
1st Secondary Transester Column
2nd Primary Transester Column
2nd Secondary Transester Column
Esterification Reactor
Vacuum group package
Biodiesel Distillation
Methanol Storage Tank #1

Affected Biodiesel Manufacturing Process Equipment
Methanol Storage Tank #2
Methanol Storage Tank #3
Methanol Storage Tank #4
Methanol Storage Tank #5
Methanol Storage Tank #6
Sodium Methylate (catalyst) Storage Tank #1
Sodium Methylate (catalyst) Storage Tank #2

There are no physical modifications being made to the existing Biodiesel Manufacturing Operation.

Summary of the Best Available Control Technology (BACT) Process

BACT is an emission limitation based on the maximum degree of pollution reduction of emissions, which is achievable on a case-by-case basis. BACT analysis takes into account the energy, environmental, and economic impacts on the source. These reductions may be determined through the application of available control techniques, process design, work practices, and operational limitations.

Federal EPA generally requires an evaluation that follows a "top down" process. In this approach, the applicant identifies the best controlled similar source on the basis of controls required by regulation or permit, or controls achieved in practice. The highest level of control is then evaluated for technical feasibility. IDEM evaluates BACT based on a "top down" approach.

The five (5) basic steps of a top-down BACT analysis used by the Office of Air Quality (OAQ) to make BACT determinations are listed below:

Step 1: Identify Potential Control Technologies

The first step is to identify potentially "available" control options for each emission unit and for each pollutant under review. Available options should consist of a comprehensive list of those technologies with a potentially practical application to the emissions unit in question. The list should also include lowest achievable emission rate (LAER) technologies and controls applied to similar source categories.

Step 2: Eliminate Technically Infeasible Options

The second step is to eliminate technically infeasible options from further consideration. To be considered feasible, a technology must be both available and applicable. It is important in this step that any presentation of a technical argument for eliminating a technology from further consideration be clearly documented based on physical, chemical, engineering, and source specific factors related to safe and successful use of the controls. Innovative control means a control that has not been demonstrated in a commercial application on similar units. Innovative controls are normally given a waiver from the BACT requirements due to the uncertainty of actual control efficiency. IDEM evaluates any innovative controls if proposed by the source. Only available and proven control technologies are evaluated. A control technology is considered available when there are sufficient data indicating that the technology results in a reduction in emissions of regulated pollutants.

Step 3: Rank the Remaining Control Technologies by Control Effectiveness

The third step is to rank the technologies not eliminated in Step 2 in order of descending control effectiveness for each pollutant of concern. The ranked alternatives are reviewed in terms of control effectiveness (percent pollutant removed). If the highest ranked technology is proposed as BACT, it is not necessary to perform any further technical or economic evaluation, except, for the environmental analyses and any more stringent limits established from other RBLC Permits.

Step 4: Evaluate the Most Effective Controls and Document the Results

The fourth step begins with an evaluation of the remaining technologies under consideration for each pollutant of concern in regards to energy, environmental, and economic impacts for determining a final control technology. The highest ranked alternative is evaluated for environmental, energy and economic impacts specific to the proposed modification. If the analysis determines that the highest ranked control is not appropriate as BACT, due to any of the energy, environmental, and economic impacts, then the next most effective control is evaluated. The evaluation continues until a technology under consideration cannot be eliminated based on adverse energy, environmental, or economic impacts. If the highest ranked technology is proposed as BACT, it is not necessary to perform any further economic or environmental analysis. In no case can the selected BACT be less stringent than any New Source Performance Standard (NSPS), National Emission Standards for Hazardous Air Pollutants (NESHAP) or Reasonably Available Control Technologies (RACT) standard or emission limit.

Step 5: Select BACT

The fifth and final step is to select as BACT the most effective of the remaining technologies under consideration for each pollutant of concern. For the technologies determined to be feasible, there may be several different limits that have been set as BACT for the same control technology. The permitting agency has to choose the most stringent limit as BACT unless the applicant demonstrates in a convincing manner why that limit is not feasible.

The Office of Air Quality (OAQ) makes BACT determinations by following the five steps identified above.

Volatile Organic Compounds (VOCs) BACT - Biodiesel Manufacturing Process

Step 1: Identify Potential Control Technologies

The volatile organic compounds (VOC) emissions can be controlled by the following methods:

- (1) Wet Scrubbers
- (2) Absorber
- (3) Wet Scrubber and Absorber Operating in Series
- (4) Refrigerated Condenser
- (5) Carbon Adsorption
- (6) Thermal Oxidation
- (7) Catalytic Oxidation
- (8) Flares

Step 2: Eliminate Technically Infeasible Options

The test for technical feasibility of any control option is whether it is both available and applicable in reducing VOC emissions. The control technologies listed in the previous section are discussed and evaluated below for their technical feasibility.

(1) Wet Scrubbers

Wet scrubbing is an effective means of removing soluble or condensable organic vapors in a gas. VOC emissions from the Louis Dreyfus Company Agricultural Industries Biodiesel Manufacturing Process exhaust consists of methanol and hexane emissions. Methanol is soluble in water and a wet scrubber could be used to remove methanol vapors from a gas stream. The general concept behind the use of wet scrubbers is to create contact between the scrubbing liquid and the gas to maximize the mass transfer from the vapor phase to the liquid phase.

Based on the information reviewed for this BACT determination, IDEM, OAQ has determined that the use of a wet scrubber is technically feasible to control methanol, but would not be expected to control hexane emissions as hexane is not water soluble.

(2) Mineral Oil Absorber

A mineral oil absorber may be used to control hexane emissions; however, IDEM, OAQ has determined that the use of a mineral oil absorber would not be expected to control methanol emissions. Methanol is water soluble whereas, hexane is not water soluble.

(3) Wet Scrubber and Mineral Oil Absorber Operating in Series

VOC emissions from the Louis Dreyfus Company Agricultural Industries Biodiesel Manufacturing Process exhaust consists of methanol and hexane emissions. Methanol is soluble in water and a wet scrubber could be used to remove methanol vapors from a gas stream. A mineral oil absorber may be used to control hexane emissions.

Based on the information reviewed for this BACT determination, IDEM, OAQ has determined that the use of a wet scrubber and a mineral oil absorber operating in series is technically feasible for the control of methanol and hexane emissions.

(4) Refrigerated Condenser

A refrigerated condenser works by cooling the gas stream to a temperature level where target compounds will condense and can be collected.

Based on the information reviewed for this BACT determination, IDEM, OAQ has determined that although the use of a refrigerated condenser is technically feasible for the control of methanol and hexane emissions, such systems are most practical for exhaust streams with high concentrations.

(5) Carbon Adsorption

Carbon adsorption works through a surface reaction between activated carbon and the target compound. Activated carbon has a high surface area for these reactions to occur. In addition, activated carbon has a unique affinity to adsorb each gas constituent which is known as the adsorption capacity. When the activated carbon becomes saturated (reaches the adsorption capacity for a given gas constituent) then the activated carbon must be regenerated by desorbing the adsorbed compounds. Desorption is accomplished by injecting steam or applying a pressure shift to the carbon bed.

Based on the information reviewed for this BACT determination, IDEM, OAQ has determined that carbon adsorption is technically feasible and could potentially be used to control methanol and hexane emissions during the Biodiesel Manufacturing Process.

(6) Thermal Oxidation

Thermal oxidizers are manufactured in a variety of different forms that work on the same basic principle. During thermal oxidation, the gas stream is raised to a temperature where destruction of the target VOC will occur.

- (a) Direct thermal oxidation uses an energy source, usually a natural gas-firing source, to raise the temperature of the gas stream.

- (b) Regenerative thermal oxidizers use a heat storage media to store heat and thereby minimize the required energy input to raise the gas temperature and to achieve thermal oxidation of the VOC.
- (c) Recuperative thermal oxidizers employ a heat exchanger to accomplish the same goal; which is to minimize the required energy input to raise the gas temperature.

(7) Catalytic Oxidation

A catalytic oxidizer uses a catalyst to lower the activation energy of the oxidation reaction. The advantage of a catalytic oxidizer over a direct thermal, regenerative, or recuperative oxidizer is that the reaction temperatures are much lower, thus less energy is needed to complete the oxidation reaction. The disadvantages of a catalytic oxidation system are that the catalyst is prone to degradation and needs periodic replacement. Factors that contribute to catalyst degradation include thermal cycling, and erosion, poisoning, and fouling from contact with gas constituents.

(8) Flares

Flares use a direct flame as an oxidizer to combust organic materials contained in the exhaust stream. They operate on the same principle as thermal oxidizers, but utilize no energy recovery.

Based on the information reviewed for this BACT determination, IDEM, OAQ has determined that thermal oxidation, catalytic oxidation, and oxidation provided by flare control systems are feasible for the control of methanol and hexane emissions during the Biodiesel Manufacturing Process.

Step 3: Rank the Remaining Control Technologies by Control Effectiveness

Based on the technical feasibility analysis in Step 2, the control technologies may be ranked as follows for controlling VOC emissions from the Biodiesel Manufacturing Process:

- (1) Wet Scrubber - 98% to 99% Methanol control efficiency
- (2) Mineral Oil Absorber - 98% to 99% Hexane control efficiency
- (3) Thermal or Catalytic Oxidizer - 98% to 99% VOC control efficiency
- (4) Flare - 98% VOC control efficiency
- (5) Carbon Adsorption - 95% to 98% VOC control efficiency
- (6) Refrigerated Condenser - 90% VOC control efficiency (conservative estimate for the VOC emissions from the Biodiesel Manufacturing Process)

Step 4: Evaluate the Most Effective Controls and Document the Results

The following table lists the proposed VOC BACT determination along with the existing VOC BACT determinations for similar plants. All data in the table is based on the information obtained from the permit application submitted by Louis Dreyfus Company Agricultural Industries LLC and the U.S. EPA RACT/BACT/LAER Clearinghouse (RBLCL).

BACT ID / Permit #	Facility	Issued Date	Process Description	Limitation	Control Method
Draft Permit No. 085-	Louis Dreyfus Company Agricultural Industries LLC	Proposed	Biodiesel manufacturing process Normal operation	0.30 lb/hr VOC emission rate; 99% VOC Control	Mineral Oil Absorber and Wet Scrubber

BACT ID / Permit #	Facility	Issued Date	Process Description	Limitation	Control Method
39532-00102 Proposed Limit				Efficiency	
			Biodiesel manufacturing process with methanol tank loading	0.63 lb/hr VOC emission rate (1,000 hours per twelve (12) consecutive months); 99% VOC Control Efficiency	Mineral Oil Absorber and Wet Scrubber
			Biodiesel manufacturing process upset operation	2.94 lb/hr VOC emission rate (24 hours per twelve (12) consecutive months); 95% VOC Control Efficiency	Mineral Oil Absorber and Wet Scrubber
			Biodiesel manufacturing process Normal operation (Fugitive Hexane emissions from Equipment Leaks)	0.64 lb/hr	LDAR in 40 CFR Part 60, VV, (Subpart VV requires using Method 21)
			Biodiesel manufacturing process with methanol tank loading (Fugitive Hexane emissions from Equipment Leaks)		
			Biodiesel manufacturing process upset operation (Fugitive Hexane emissions from Equipment Leaks)		
(IN)/T085-21297-00102	Louis Dreyfus Company Agricultural Industries LLC	1/24/2006	Biodiesel manufacturing process Normal operation	0.30 lb/hr VOC emission rate; VOC control efficiency > 99% during normal operation	A soy oil absorber followed by a water scrubber.
			Biodiesel manufacturing process with methanol tank loading	0.63 lb/hr VOC emission rate (1,000 hours per twelve (12) consecutive months); 99% VOC Control Efficiency	A soy oil absorber followed by a water scrubber.
			Biodiesel manufacturing process upset operation	2.94 lb/hr VOC emission rate (24 hours per twelve (12) consecutive months); VOC Control Efficiency > 95% for upset conditions.	A soy oil absorber followed by a water scrubber.
			Biodiesel manufacturing process Normal operation (Fugitive Hexane emissions from Equipment Leaks)	0.64 lb/hr	LDAR in 40 CFR Part 60, VV, (Subpart VV requires using Method 21)
			Biodiesel manufacturing process with methanol tank loading		

BACT ID / Permit #	Facility	Issued Date	Process Description	Limitation	Control Method
			(Fugitive Hexane emissions from Equipment Leaks)		
			Biodiesel manufacturing process upset operation (Fugitive Hexane emissions from Equipment Leaks)		
TX-0815	TOTAL PETROCHEMICALS & REFINING USA, INC.	01/17/2017	Ethylene Production - Low Pressure Process Vents	99.9% VOC Control Efficiency	Thermal oxidizer
			Ethane Side Cracker (Fugitive VOC Leaks from Equipment)	17.2 tons VOC/year	Leak Detection and Repair (LDAR) Program
LA-0240	FLOPAM INC.	6/14/2010	DADMAC/CM/ADAM/ATBS Plants	99% VOC Control Efficiency	Thermal Oxidizer
			Fugitive VOC Leaks from Equipment	2.0 lbs/hr (maximum); LAER Limitation	Comply with 40 CFR 65 Subpart F
TX-0823	LYONDELL CHEMICAL COMPANY	6/07/2017	Ethers Production Unit	99.8% VOC Control Efficiency	Absorber and Flare
			Ethers Production Unit (Fugitive VOC Leaks from Equipment)	4.36 tons/yr; LAER Limitation	EPA Method 21
IN-0123	Bio-Alternative LLC	3/18/2011	Biodiesel Production Plant	98% Overall VOC Control Efficiency	Final vent condenser and scrubber
TX-0813	REXTAC, LLC - ODESSA PETROCHEMICAL PLANT	11/22/2016	Propylene Process Vent	98% VOC Control Efficiency	Flare
			Propylene Process Vent (Fugitive VOC Leaks from Equipment)	88.52 tons/yr	EPA Method 21
IN-0153	SWIFT FUELS, LLC	12/01/2012	MESITYLENE PROCESSING SKID (EU01)	98% VOC Control Efficiency	Flare
MO-0079	AMERICAN ENERGY PRODUCERS, INC.	1/25/2008	Biodiesel Production	95% VOC Control Efficiency	Condenser, Soy Oil Adsorber, and Water Scrubber
			Biodiesel Production (Fugitive VOC Leaks from Equipment)	Not Stated	Leak Detection and Repair (LDAR) Program that meets the requirements of 40 CFR Part 60, Subpart VV
MO-0081	AMERICAN ENERGY PRODUCERS, INC.	1/22/2009	Biodiesel Production	Not Stated	Leak Detection and Repair (LDAR) for the biodiesel production processes that meet the requirements of 40 CFR Part 60, Subpart VV

BACT ID / Permit #	Facility	Issued Date	Process Description	Limitation	Control Method
			Biodiesel Production (Soybean Processing Operations - Fugitive VOC Leaks from Equipment)	0.145 gal/ton of oilseed	Leak Detection and Repair (LDAR) Program
IN-0179	Ohio Valley Resources	9/25/2013	Nitrogenous Fertilizer Production Plant (Fugitive VOC Leaks from Equipment)	Not Stated	Use of a Leak Detection and Repair (LDAR) Program Using 40 CFR 60, Subpart VVa Procedures.
IN-0180	Midwest Fertilizer Corporation	6/04/2014	Nitrogen Fertilizer Manufacturing Facility (Fugitive VOC Leaks from Equipment)	Not Stated	Leak Detection and Repair (LDAR) Program Using 40 CFR 60, Subpart VVa Procedures.
IN-0200	Eli Lilly and Company - Clinton Laboratories	7/24/2014	Fugitive VOC Leaks from Equipment	Not Stated	Leak Detection and Repair (LDAR) Program
FL-0322	Southeast Renewable Fuels (SRF), LLC	12/23/2010	Sweet Sorghum-to-Ethanol Advanced Biorefinery (Fugitive VOC Leaks)	6.52 tons/yr	Use of a Leak Detection and Repair (LDAR) Program Using 40 CFR 60, Subpart VVa Procedures.
TX-0804	INVISTA S.A R.L (ADN Unit)	7/15/2016	ADN Unit (Fugitive VOC Emissions from Equipment Leaks)	5.41 lbs/hr	LDAR Program

Control Device:

Thermal oxidation systems, flares, condensers, absorbers, and scrubbers were identified as technically feasible add-on control methods.

The most stringent limit has a control efficiency of 99.9% which is provided by a thermal oxidizer and it is used during the production of Ethylene at Total Petrochemicals & Refining USA, Inc. (TX-0815). This Ethylene production is created during a petrochemical process described as "steam cracking" in which steam is added to a mixture of gaseous, saturated hydrocarbons which are broken down into smaller, unsaturated hydrocarbons; one of which is Ethylene. Louis Dreyfus currently produces biodiesel fuel, not Ethylene. Therefore, the products and processes from these two sources are different and not comparable.

The next most stringent limit identified in RBLC is a single determination for Lyondell Chemical (RBLC TX-0823), which is contained in the "Synthetic Organic Chemical Manufacturing Industry (SOCMI) Process Vent" Category with an estimated control efficiency of 99.8% through the use of an absorber that returns VOC to the process followed by a flare. This control method represents a LAER limit for this source.

Regulatory Requirements

The VOC BACT must be no less stringent than any applicable New Source Performance Standards (NSPS) or National Emission Standards for Hazardous Air Pollutants (NESHAP). The control efficiencies of the proposed control method are more stringent than the New Source Performance Standards (NSPS) or National Emission Standards for Hazardous Air Pollutant (NESHAP) control requirements noted below:

NSPS 40 CFR Part 60, Subpart NNN

The distillation unit utilized for biodiesel production is subject to requirements of 40 CFR Part 60, Subpart NNN (Standards of Performance for Volatile Organic Compound Emissions from Synthetic Organic Chemical Manufacturing Industry Distillation Operations). Subpart NNN requires that VOC emissions from distillation operations be controlled by a minimum of 98%, or to 20 ppmv, whichever is less restrictive.

NSPS 40 CFR Part 60, Subpart RRR

The 1st Primary Transester Column, 1st Secondary Transester Column, 2nd Primary Transester Column, 2nd Secondary Transester Column, and the Esterification Reactor are subject to requirements of NSPS 40 CFR Part 60, Subpart RRR (Standards of Performance for Volatile Organic Compound Emissions from Synthetic Organic Chemical Manufacturing Industry Reactor Processes). Subpart RRR requires that VOC emissions from the reactors be controlled by a minimum of 98%, or to 20 ppmv, whichever is less restrictive.

NESHAP 40 CFR Part 63, Subpart FFFF

Emissions from the Biodiesel Operations are also regulated by NESHAP 40 CFR Part 63, Subpart FFFF (National Emission Standards for Hazardous Air Pollutants from Miscellaneous Organic Chemical Manufacturing). Subpart FFFF requires that VOC emissions from the Biodiesel Vent exhaust be controlled by a minimum of 98%, or to 20 ppmv, whichever is less restrictive, or that the Total Resource Effectiveness (TRE) value for the vent be maintained at a TRE above 8.0.

NSPS 40 CFR Part 60, Subpart VV

The source must also control fugitive emissions from system components through NSPS 40 CFR Part 60, Subpart VV, (Standards of Performance for Equipment Leaks of VOC in the Synthetic Organic Chemical Manufacturing Industry).

Proposal: Louis Dreyfus Company Agricultural Industries LLC - Claypool, Indiana

Based on its review of available information, Louis Dreyfus Company Agricultural Industries LLC, concludes that BACT for Biodiesel Manufacturing Processes is control of methanol recovery vent emissions by 99% through the use of a mineral oil absorber and a wet scrubber.

Louis Dreyfus currently operates a mineral oil absorber and a wet scrubber to control emissions. Since the mineral oil absorber and a wet scrubber provide the same control efficiency as a thermal oxidizer or catalytic oxidizer, the purchase and construction of either a thermal oxidizer or a catalytic oxidizer would not be economically feasible.

Leak Detection:

The leak detection and repair (LDAR) program, as required in the current permit (outlined in Condition D.4.3(e)), is an instrument based detection program that follows procedures contained in 40 CFR Part 60, Method 21. Louis Dreyfus Company Agricultural Industries LLC, has requested that the leak detection and repair (LDAR) program contained in the VOC BACT requirement of the permit, be changed from a Method 21 based system to a system that utilizes olfactory, visual, and auditory (OVA) detection. The source has noted that the Method 21 LDAR detection program has identified only one leak during the past six years. Louis Dreyfus Company Agricultural Industries LLC has noted that the Method 21 program is simply an expense that is not providing any meaningful results in regards to detection of leaks. The existing LDAR program results in no reduction in VOC from its soybean oil extraction operation, and provides no benefit to Louis Dreyfus Company Agricultural Industries LLC to achieve the site overall solvent loss ratio. Louis Dreyfus Company Agricultural Industries LLC believes that the OVA based program will identify leaks more effectively.

IDEM, OAQ has reviewed this information, and determined that sufficient justification has not been provided by the source to demonstrate that a change to the existing Leak Detection and

Repair (LDAR) program to use an OVA method would continue to demonstrate the same level of leak detection. Therefore, at this time, this change is not being made.

Step 5: Select BACT

Pursuant to 326 IAC 8-1-6 (New facilities; general reduction requirements), IDEM has established the following as BACT for VOC for the Biodiesel Manufacturing Process:

Process	Control	VOC control efficiency, %	VOC Limit (lbs/hour)	Operation Limit
Biodiesel manufacturing process Normal operation	Mineral Oil Absorber and Water absorber	Overall VOC control efficiency of 99%	0.30	NA
Biodiesel manufacturing process with methanol tank loading	Mineral Oil Absorber and Water absorber	Overall VOC control efficiency of 99%	0.63	1,000 hours per twelve (12) consecutive months
Biodiesel manufacturing process upset operation	Mineral Oil Absorber and Water absorber	Overall VOC control efficiency of 95%	29.4	24 hours per twelve (12) consecutive months
Glycerin storage tanks	None	None	0.0011	NA
Biodiesel wastewater	None	None	0.77	NA
Biodiesel fugitive emissions	None	None	0.64	LDAR as required by 40 CFR 60, Subpart VV



INDIANA DEPARTMENT OF ENVIRONMENTAL MANAGEMENT

We Protect Hoosiers and Our Environment.

100 N. Senate Avenue • Indianapolis, IN 46204

(800) 451-6027 • (317) 232-8603 • www.idem.IN.gov

Eric J. Holcomb
Governor

Bruno L. Pigott
Commissioner

December 13, 2018

Mr. David Selig
Louis Dreyfus Company Agricultural Industries, LLC
7344 State Road 15 South
Claypool, IN 46510

Re: Public Notice
Louis Dreyfus Company Agricultural Industries LLC
Permit Level: Title V Significant Source
Modification and Significant Permit Modification
Permit Number: 085-39532-00102 and
085-39729-00102

Dear Mr. Selig:

Enclosed is a copy of your draft Title V Significant Source Modification and Significant Permit Modification, Technical Support Document, emission calculations, and the Public Notice which will be printed in your local newspaper.

The Office of Air Quality (OAQ) has prepared two versions of the Public Notice Document. The abbreviated version will be published in the newspaper, and the more detailed version will be made available on the IDEM's website and provided to interested parties. Both versions are included for your reference. The OAQ has requested that the Times Union in Warsaw, Indiana publish the abbreviated version of the public notice no later than December 15, 2018. You will not be responsible for collecting any comments, nor are you responsible for having the notice published in the newspaper.

OAQ has submitted the draft permit package to the Warsaw Community Public Library, 310 East Main Street in Warsaw, Indiana. As a reminder, you are obligated by 326 IAC 2-1.1-6(c) to place a copy of the complete permit application at this library no later than ten (10) days after submittal of the application or additional information to our department. We highly recommend that even if you have already placed these materials at the library, that you confirm with the library that these materials are available for review and request that the library keep the materials available for review during the entire permitting process.

Please review the enclosed documents carefully. This is your opportunity to comment on the draft permit and notify the OAQ of any corrections that are needed before the final decision. Questions or comments about the enclosed documents should be directed to Daniel Pell, Indiana Department of Environmental Management, Office of Air Quality, 100 N. Senate Avenue, Indianapolis, Indiana, 46204 or call (800) 451-6027, and ask for extension (317) 234-8532 or dial (317) 234-8532.

Sincerely,

Vivian Haun

Vivian Haun
Permits Branch
Office of Air Quality

Enclosures
PN Applicant Cover Letter 1/9/2017



INDIANA DEPARTMENT OF ENVIRONMENTAL MANAGEMENT

We Protect Hoosiers and Our Environment.

100 N. Senate Avenue • Indianapolis, IN 46204

(800) 451-6027 • (317) 232-8603 • www.idem.IN.gov

Eric J. Holcomb
Governor

Bruno L. Pigott
Commissioner

ATTENTION: PUBLIC NOTICES, LEGAL ADVERTISING

December 12, 2018

Times Union
PO Box 1448
Warsaw, IN 46581-1448

Enclosed, please find one Indiana Department of Environmental Management Notice of Public Comment for Louis Dreyfus Company Agricultural Industries LLC, Kosciusko County, Indiana.

Since our agency must comply with requirements which call for a Notice of Public Comment, we request that you print this notice one time, no later than December 15, 2018.

Please send the invoice, notarized form, clippings showing the date of publication to Bo Liu, at the Indiana Department of Environmental Management, Accounting, Room N1340, 100 North Senate Avenue, Indianapolis, Indiana, 46204.

To ensure proper payment, please reference account # 100174737.

We are required by the Auditor's Office to request that you place the Federal ID Number on all claims. If you have any conflicts, questions, or problems with the publishing of this notice or if you do not receive complete public notice information for this notice, please call Vivian Haun at 800-451-6027 and ask for extension 317-233-6878 or dial 317-233-6878.

Sincerely,

Vivian Haun

Vivian Haun
Permit Branch
Office of Air Quality

Permit Level: Title V Significant Source Modification and Significant Permit Modification
Permit Number: 085-39532-00102 and 085-39729-00102

Enclosure
PN Newspaper.dot 1/9/2017



INDIANA DEPARTMENT OF ENVIRONMENTAL MANAGEMENT

We Protect Hoosiers and Our Environment.

100 N. Senate Avenue • Indianapolis, IN 46204

(800) 451-6027 • (317) 232-8603 • www.idem.IN.gov

Eric J. Holcomb
Governor

Bruno L. Pigott
Commissioner

December 13, 2018

To: Warsaw Community Public Library

From: Jenny Acker, Branch Chief
Permits Branch
Office of Air Quality

Subject: **Important Information to Display Regarding a Public Notice for an Air Permit**

Applicant Name: Louis Dreyfus Company Agricultural Industries LLC
Permit Number: 085-39532-00102 and 085-39729-00102

Enclosed is a copy of important information to make available to the public. This proposed project is regarding a source that may have the potential to significantly impact air quality. Librarians are encouraged to educate the public to make them aware of the availability of this information. The following information is enclosed for public reference at your library:

- Notice of a 30-day Period for Public Comment
- Request to publish the Notice of 30-day Period for Public Comment
- Draft Permit and Technical Support Document

You will not be responsible for collecting any comments from the citizens. Please refer all questions and request for the copies of any pertinent information to the person named below.

Members of your community could be very concerned in how these projects might affect them and their families. **Please make this information readily available until you receive a copy of the final package.**

If you have any questions concerning this public review process, please contact Joanne Smiddie-Brush, OAQ Permits Administration Section at 1-800-451-6027, extension 3-0185. Questions pertaining to the permit itself should be directed to the contact listed on the notice.

Enclosures
PN Library 1/9/2017



INDIANA DEPARTMENT OF ENVIRONMENTAL MANAGEMENT

We Protect Hoosiers and Our Environment.

100 N. Senate Avenue • Indianapolis, IN 46204

(800) 451-6027 • (317) 232-8603 • www.idem.IN.gov

Eric J. Holcomb
Governor

Bruno L. Pigott
Commissioner

Notice of Public Comment

December 13, 2018

**Louis Dreyfus Company Agricultural Industries LLC
085-39532-00102 and 085-39729-00102**

Dear Concerned Citizen(s):

You have been identified as someone who could potentially be affected by this proposed air permit. The Indiana Department of Environmental Management, in our ongoing efforts to better communicate with concerned citizens, invites your comment on the draft permit.

Enclosed is a Notice of Public Comment, which has been placed in the Legal Advertising section of your local newspaper. The application and supporting documentation for this proposed permit have been placed at the library indicated in the Notice. These documents more fully describe the project, the applicable air pollution control requirements and how the applicant will comply with these requirements.

If you would like to comment on this draft permit, please contact the person named in the enclosed Public Notice. Thank you for your interest in the Indiana's Air Permitting Program.

Please Note: *If you feel you have received this Notice in error, or would like to be removed from the Air Permits mailing list, please contact Patricia Pear with the Air Permits Administration Section at 1-800-451-6027, ext. 3-6875 or via e-mail at PPEAR@IDEM.IN.GOV. If you have recently moved and this Notice has been forwarded to you, please notify us of your new address and if you wish to remain on the mailing list. Mail that is returned to IDEM by the Post Office with a forwarding address in a different county will be removed from our list unless otherwise requested.*

Enclosure
PN AAA Cover Letter 1/9/2017



INDIANA DEPARTMENT OF ENVIRONMENTAL MANAGEMENT

We Protect Hoosiers and Our Environment.

100 N. Senate Avenue • Indianapolis, IN 46204

(800) 451-6027 • (317) 232-8603 • www.idem.IN.gov

Eric J. Holcomb
Governor

Bruno L. Pigott
Commissioner

AFFECTED STATE NOTIFICATION OF PUBLIC COMMENT PERIOD DRAFT INDIANA AIR PERMIT

December 13, 2018

A 30-day public comment period has been initiated for:

Permit Number: 085-39532-00102 and 085-39729-00102
Applicant Name: Louis Dreyfus Company Agricultural Industries LLC
Location: Claypool, Kosciusko County, Indiana

The public notice, draft permit and technical support documents can be accessed via the **IDEM Air Permits Online** site at:

<http://www.in.gov/ai/appfiles/idem-caats/>

Questions or comments on this draft permit should be directed to the person identified in the public notice by telephone or in writing to:

Indiana Department of Environmental Management
Office of Air Quality, Permits Branch
100 North Senate Avenue
Indianapolis, IN 46204

Questions or comments regarding this email notification or access to this information from the EPA Internet site can be directed to Chris Hammack at chammack@idem.IN.gov or (317) 233-2414.

Affected States Notification 1/9/2017

Mail Code 61-53

IDEM Staff	VHAUN 12/13/2018 085-39532 and 39729-00102 DRAFT			AFFIX STAMP HERE IF USED AS CERTIFICATE OF MAILING
Name and address of Sender		Louis Dreyfus Company Agricultural Industries LLC Indiana Department of Environmental Management Office of Air Quality – Permits Branch 100 N. Senate Indianapolis, IN 46204	Type of Mail: CERTIFICATE OF MAILING ONLY	

Line	Article Number	Name, Address, Street and Post Office Address	Postage	Handling Charges	Act. Value (If Registered)	Insured Value	Due Send if COD	R.R. Fee	S.D. Fee	S.H. Fee	Rest. Del. Fee	Remarks
1		David Selig Louis Dreyfus Company Agricultural Industries LLC 7344 SR 15 S Claypool IN 465109746 (Source RM)										
2		Bruce Chapin Louis Dreyfus Company Agricultural Industries LLC 4800 Main St Ste 600 Kansas City MO 64112 (RO RM)										
3		Warsaw Community Public Library 310 E Main St Warsaw IN 46580-2882 (Library)										
4		Kosciusko County Board of Commissioners 100 W. Center St, Room 220 Warsaw IN 46580 (Local Official)										
5		David Jordan Environmental Resources Management (ERM) 8425 Woodfield Crossing Blvd., Suite 560-W Indianapolis IN 46240 (Consultant)										
6		Claypool Town Council P.O. Box 6 Claypool IN 46510 (Local Official)										
7		Kosciusko County Health Department 100 W. Center Street, 3rd Floor Warsaw IN 46580-2877 (Health Department)										
8		Lisa Green The Journal Gazette 600 W Main St Fort Wayne IN 46802 (Affected Party)										
9		Jeri Seely The Mail-Journal PO Box 188 Milford IN 46542 (Affected Party)										
10		Christina Seiler The Rochester Sentinel PO Box 260 Rochester IN 46975 (Affected Party)										
11		Mr. Roger Schneider The Goshen News 114 S. Main St Goshen IN 46526 (Affected Party)										
12												
13												
14												
15												

Total number of pieces Listed by Sender 11	Total number of Pieces Received at Post Office	Postmaster, Per (Name of Receiving employee)	The full declaration of value is required on all domestic and international registered mail. The maximum indemnity payable for the reconstruction of nonnegotiable documents under Express Mail document reconstructing insurance is \$50,000 per piece subject to a limit of \$50, 000 per occurrence. The maximum indemnity payable on Express mil merchandise insurance is \$500. The maximum indemnity payable is \$25,000 for registered mail, sent with optional postal insurance. See Domestic Mail Manual R900, S913, and S921 for limitations of coverage on inured and COD mail. See International Mail Manual for limitations o coverage on international mail. Special handling charges apply only to Standard Mail (A) and Standard Mail (B) parcels.
--	--	--	--